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Igor I. Gorban

Randomness and Hyper-randomness

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Preface

One of the most remarkable physical phenomena is the *statistical stability (regularity) of mass phenomena* as revealed by the stability of statistics (functions of samples). There are two theories describing this phenomenon. The first is classical *probability theory*, which has a long history, and the second is *the theory of hyper-random phenomena* developed in recent decades.

Probability theory has established itself as the most powerful tool for solving various statistical tasks. It is even widely believed that any statistical problem can be effectively solved within the paradigm of probability theory. However, it turns out that this is not so.

Some conclusions of probability theory do not accord with experimental data. A typical example concerns the potential accuracy. According to probability theory, when we increase the number of measurement results of any physical quantity, the error in the averaged estimator tends to zero. But every engineer or physicist knows that the actual measurement accuracy is always limited and that it is not possible to overcome this limit by statistical averaging of the data.

Studies of the causes of discrepancies between theory and practice led to the understanding that the problem is related to an *unjustified idealization of the phenomenon of statistical stability*.

Probability theory is in fact a physical-mathematical discipline. The mathematical component is based on A.N. Kolmogorov's classical axioms, while the physical component is based on certain physical hypotheses, in particular the hypothesis of *perfect statistical stability of actual events, variables, processes, and fields* assuming the *convergence of statistics* when the sample size goes to infinity.

Experimental investigations of various processes of different physical kinds over broad observation intervals have shown that the hypothesis of perfect statistical stability *is not confirmed experimentally*. For relatively short temporal, spatial, or spatio-temporal observation intervals, an increase in data volume usually reduces the level of fluctuation in the statistics. However, when the volumes become very large, this tendency is no longer visible, and once a certain level is reached, the fluctuations remain practically unchanged or even grow. This indicates *a lack of convergence for real statistics* (*their inconsistency*).

If the volume of processing data is small, the violation of convergence has practically no influence on the results, but if this volume is large, the influence is very significant. The study of violations of statistical stability in physical phenomena and the development of an effective way to describe the actual world, one which accounts for such violations, has resulted in the construction of the new *physical-mathematical theory of hyper-random phenomena*.

The theory of hyper-random phenomena is also a physical-mathematical theory. Its mathematical component is based on the axioms and statements of the mathematical component of the probability theory, and its physical component is based on hypotheses that differ essentially from the physical hypotheses of probability theory, in particular the hypothesis of limited statistical stability assuming the absence of convergence in the actual statistics. Therefore, for mathematicians the theory of hyper-random phenomena is a *branch of probability theory*, and for physicists, it is a *new physical theory based on a new view of the world*.

There is much literature describing probability theory from various points of view (Kolmogorov 1929, 1956; Mises 1964; Bernshtein 1946, etc.) and oriented toward readerships with different mathematical knowledge (Feller 1968; Loéve 1977; Gnedenko 1988; Angot 1957; Devor 2012; Gorban 2003; Pugachev 1979; Peebles 1987; Tutubalin 1972; Rozhkov 1996; Ventsel 1962, etc.). Quite a few studies have also been published in the area of statistical stability violation and the theory of hyper-random phenomena. Among the latter, we can mention three monographs in Russian (Gorban 2007, 2011, 2014), two monographs in Ukrainian (Uvarov and Zinkovskiy 2011a, b), and a monograph in English (Gorban 2017) devoted to various mathematical, physical, and practical questions.

Probability theory and the theory of hyper-random phenomena give different descriptions of the phenomenon of statistical stability. Until recently, there were no books comparing these theories. However, this gap was closed in 2016 with the monograph in Russian (Gorban 2016). This book is an English version of that.

The aims of the current monograph, like those before it, are:

- *To acquaint the reader with the phenomenon of statistical stability*
- *To describe probability theory and the theory of hyper-random phenomena from a single standpoint*
- *To compare these theories*
- *To describe their physical and mathematical essence at the conceptual level*

This monograph consists of *five parts*. The *first* entitled *The Phenomenon of Statistical Stability* consists of the introductory chapter. Manifestations of this phenomenon and different approaches for its description are described in it. The *second part* entitled *Probability Theory* contains four chapters (Chaps. 2–5) and describes the foundations of probability theory. The *third part* entitled *Experimental Study of the Statistical Stability Phenomenon* contains only Chap. 6, dedicated to a description of the techniques developed to assess statistical stability violations and also the results of experimental investigations of statistical stability violations in actual physical processes of various kinds. The title of the *fourth part* is *Theory of Hyper-random Phenomena*. It includes four chapters (Chaps. 7–10) presenting the foundations of the theory of hyper-random phenomena. The *fifth part* entitled *The Problem of an Adequate Description of the World* includes only Chap. 11, which discusses the concept of world building.

The book aims at a wide readership: from university students of a first course majoring in physics, engineering, and mathematics to engineers, postgraduate students, and scientists researching the statistical laws of natural physical phenomena and developing and using statistical methods for high-precision measurement, prediction, and signal processing over broad observation intervals. To understand the material in the book, it is sufficient to be familiar with a standard first university course on mathematics.

Kiev, Ukraine Igor I. Gorban
25 June 2016

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Introduction

The Phenomenon of Statistical Stability It is hard to find someone who, tossing a coin, has never tried to guess which way it will fall: heads or tails. It is impossible to predict the result accurately. However, repeating the experiment *many times*, one can expect a surprising regularity: the relative frequency of heads (or tails) is virtually independent of the number of experiments.

The relative frequency stability in the game is a manifestation of a *fundamental physical law of nature*, namely, the *phenomenon of statistical stability (regularity)*. Through multiple measurements of physical quantities, it can be established that the variation of the averages is less than the variation of single measurements. This is also a manifestation of the phenomenon of statistical stability. In general, by the phenomenon of statistical stability, we understand the *stability of the averages*: in other words, the *stability of statistics* (i.e., *functions of the sample*).

There are two theories describing this phenomenon: classical *probability theory* with its long history and the relatively new *theory of hyper-random phenomena*.

Although the term "hyper-random phenomenon" was included in the scientific literature only in 2005 (Gorban 2005), the foundations of the theory of hyper-random phenomena were already being laid at the turn of 1970s and 1980s. This book is devoted to the study of the phenomenon of statistical stability and a comparison between the theories describing it.

The Nature of the Theories Both the theories mentioned are of a *physical-mathematical* nature. Each consists of two components: one *mathematical* and one *physical*. The mathematical components use abstract mathematical models, and the physical components operate with actual entities in the world. Note that the physical components play an extremely important role, providing a link between the actual physical world and the abstract world of mathematical models.

Probability Theory The mathematical component of probability theory explores various *random phenomena: random events, variables, processes, and fields*.

The term *random phenomenon* refers to an *abstract mathematical object* (*model*) that satisfies certain mathematical axioms (Kolmogorov's axioms) (Kolmogorov

1929, 1956a). The typical features of a random phenomenon are that it is of a *mass type* (there are multiple realizations) and it is characterized by a *probability measure* (*probability*) interpreted as the *relative frequency of occurrence of possible events for an infinitely large number of occurrences of the phenomenon*. The latter means that the *relative frequency of any event has a limit*, which is interpreted as the *probability* of occurrence of this event.

Note that a *mass phenomenon that does not have a probability measure is not considered to be random*. This is an extremely important point that must be taken into account.

Subject and Scope of Investigation of Probability Theory The subject matter of the *mathematical part of probability theory* is *random phenomena*, and the scope of study is *links between these mathematical objects*. The subject matter and the scope of study of the *physical part of the probability theory*, as well as the whole of this theory, are accordingly the *physical phenomenon of statistical stability* and the *means for describing it using random* (*stochastic*) *models*.

The Problem of an Adequate Description of a Reality *Random* (*stochastic or otherwise probabilistic*) models, like any other, give an approximate description of reality. In many cases, the random models provide acceptable accuracy in the description of actual phenomena (actual events, quantities, processes, and fields), and this explains why they have found such wide application.

However, the random models *do not always adequately reflect the specific realities of the world*. This is manifested especially clearly in the various tasks associated with processing large amounts of data obtained over broad observation intervals, in particular in high-precision measurement tasks based on statistical processing of a large number of measurement results, or the prediction of the way events over broad monitoring intervals, or the solution of a number of similar problems.

The Hypothesis of Perfect Statistical Stability Investigations into what causes the inadequacy of stochastic models of actual phenomena have shown that the *phenomenon of statistical stability has a particularity that is ignored by such models*. Indeed, they are founded on the physical *hypothesis of perfect* (*ideal*) *statistical stability*. It implies the *convergence of any real statistics*, i.e., the *existence of a limit* to which the statistics will tend when the sample size goes to infinity.

For many years, the hypothesis of perfect statistical stability didn't raise any doubts, although some scholars [even the founder of axiomatic probability theory *A.N. Kolmogorov* (Kolmogorov 1956b, 1986) and famous scientists such as *A.A. Markov* (Markov 1924), *A.V. Skorokhod* (Ivanenko 1990), *E. Borel* (Borel 1956), *V. N. Tutubalin* (Tutubalin 1972), and others] noticed that, in the real world, this hypothesis is valid only with certain reservations.

Recent experimental research on various physical processes over long observation intervals has shown that *it is not confirmed experimentally*. For relatively short temporal, spatial, or spatio-temporal observation intervals, an increase in data volume usually reduces the level of fluctuation in the statistics. However, when the volumes become very large, this tendency is no longer visible, and once a certain level is reached, the fluctuations remain practically unchanged or even grow. This indicates a *lack of convergence for real statistics* (their *inconsistency*).

Violation of statistical stability in the real world means that *the probability concept has no physical interpretation. Probability is thus a mathematical abstraction.*

The Hypothesis of Imperfect Statistical Stability The alternative to the hypothesis of ideal (perfect) statistical stability is the *hypothesis of imperfect* (*limited*) *statistical stability*, assuming *no convergence of the actual statistics*.

The development of an effective way to describe the real world, one which accounts for violations of statistical stability, has resulted in the construction of the new *physical-mathematical theory of hyper-random phenomena*.

The Theory of Hyper-random Phenomena The mathematical part of the theory of hyper-random phenomena studies various *hyper-random phenomena: hyper-random events, variables, processes, and fields.*

A *hyper-random phenomenon* is an *abstract mathematical object* (*model*) which represents a set of unlinked random objects (*random events, variables, processes, or fields*) regarded as a comprehensive whole. Each random component of the hyper-random phenomenon is associated with some perfect statistical condition.

As in the case of a random phenomenon, the typical feature of a hyper-random phenomenon is its *mass type* (existence of multiple realizations). However, in contrast to a random phenomenon, a hyper-random phenomenon is not characterized by a given probability measure (probability) but by a *set of measures*. In this way, it is possible to describe not only a mass event, the relative frequency of which has a limit when the number of realizations goes to infinity, but any mass event, *the relative frequency of which does not have a limit.*

Subject and Scope of Investigation of the Theory of Hyper-random Phenomena The subject matter of the *mathematical part of the theory of hyper-random phenomena* is *hyper-random phenomena*, and the scope of study is *links between these mathematical objects*. The subject matter and the scope of study of the *physical part of the theory of hyper-random phenomena*, as well as the whole of this theory, are accordingly the *physical phenomenon of statistical stability* and the *means to describe it using hyper-random models*, taking into account the violation of statistical stability.

Similarities and Differences Between Probability Theory and the Theory of Hyper-random Phenomena The mathematical component of the theory of hyper-random phenomena, like probability theory, is based on the Kolmogorov's axioms and, therefore, from the mathematical point of view is a branch of the latter. However, the physical components of these theories differ significantly.

The physical part of *probability theory* is based on two hypotheses:

- *The hypothesis of perfect statistical stability of real events, quantities, processes, and fields*
- *The hypothesis of an adequate description of physical phenomena by random models*

The physical part of the theory of hyper-random phenomena is based on other hypotheses:

- *The hypothesis of imperfect statistical stability of real events, quantities, processes, and fields*
- *The hypothesis of an adequate description of physical phenomena by hyper-random models*

In fact, probability theory and the theory of hyper-random phenomena are two different paradigms that give different interpretations of the real world. The first leads us to accept a *random* (*stochastic*) *concept of world structure* and the second a *world-building concept based on hyper-random principles*.

Scope of Application of the Various Models Although probability theory and the theory of hyper-random phenomena describe the same phenomenon of statistical stability, their areas of practical application are different.

Probability theory based on the hypothesis of perfect statistical stability is applied when processing *small volumes of statistical data* when one can assume that the *statistical conditions are almost unchanging*.

The theory of *hyper-random phenomena* takes into account the imperfect nature of the phenomenon of statistical stability and there are no restrictions on the volume of data. Theoretically, it can be used for both small and large data volumes, in both the absence and the presence of statistical stability violation.

Random models are usually simpler than hyper-random models, so are preferred when sample sizes are not too large. However, hyper-random models have obvious advantages over random models when the limited statistical character of statistical stability becomes apparent and it is impossible to provide an adequate description of physical phenomena using random models. This is mainly when processing large volumes of real-world data under unpredictable changes in statistical conditions.

Therefore, the primary application of the hyper-random models is to statistical analysis of various physical processes (electrical, magnetic, electromagnetic, acoustic, hydroacoustic, seismic-acoustic, meteorological, and others) of long duration, as well as high-precision measurements of various physical quantities and the forecasting of physical processes by statistical processing of large data sets.

The hyper-random models may also be useful for simulating various physical events, variables, processes, and fields, for which, due to the extremely small size of the statistical material, high-quality estimates of the parameters and characteristics cannot be obtained and it is only possible to estimate bounds within which they are located.

The aim of the book is to acquaint the reader with the phenomenon of statistical stability, to describe probability theory and the theory of hyper-random phenomena from a single standpoint, to compare these theories, and to reveal their physical and mathematical essence at the conceptual level.

We have tried to present the material as simply and clearly as possible, avoiding rarely used or specialized concepts, terms, and formulas. The monograph focuses on issues which:

- *Reveal the physical and mathematical essence of probability theory and the theory of hyper-random phenomen*a
- *Allow the reader to understand the difference between these theories on the physical and mathematical levels*
- *Determine the place of these theories among others*
- *Have the greatest practical interest*

Specific Features of the Book The monograph and its Russian version (Gorban 2016) are based on three other books (Gorban 1998, 2000, 2003) devoted to probability theory and mathematical statistics and also four monographs (Gorban 2007, 2011, 2014, 2017) devoted to investigations of the phenomenon of statistical stability and the theory of hyper-random phenomena.

The monograph has a *physical-technical bias* and is oriented toward a *wide readership*: from university students of a first course majoring in physics, engineering, and mathematics to engineers, postgraduate students, and scientists researching the statistical laws of natural physical phenomena and developing and using statistical methods for high-precision measurement, prediction, and signal processing over broad observation intervals.

Given that not all readers may have the required mathematical and engineering background, a number of basic issues have been included in the book, in particular the main concepts of set theory and measure theory, but also ordinary and generalized limits, ordinary and generalized Wiener–Khinchin transformations, and others.

As a result, to understand the material in the book, it is sufficient to be familiar with a standard first university course on mathematics.

Structure of the Book The monograph consists of *five parts*.

The *first part* entitled *The Phenomenon of Statistical Stability* contains only an introductory chapter which describes this phenomenon.

The *second part* entitled *Probability Theory* includes four chapters (Chaps. 2–5). It contains a description of the foundations of probability theory.

The *third part* entitled *Experimental Study of the Statistical Stability Phenomenon* contains only Chap. 6 describing the techniques developed for evaluation of statistical stability violations and also the results of experimental investigations of statistical stability violations in actual physical processes of various kinds.

The title of the *fourth part* is *Theory of Hyper-random Phenomena*. It includes four chapters (Chaps. 7–10) presenting the foundations of the theory of hyper-random phenomena.

The *fifth part* entitled *The Problem of an Adequate Description of the World* includes just Chap. 11 discussing the concept of world building.

The individual chapters can be summarized as follows.

Chapter 1 Here, we examine the main manifestations of the phenomenon of statistical stability: the statistical stability of the relative frequency and sample average. Attention is drawn to an emergent property of the phenomenon of statistical stability. We discuss the hypothesis of perfect (absolute or ideal) statistical stability, which assumes the convergence of relative frequencies and averages. Examples of statistically unstable processes are presented. We discuss the terms “identical statistical conditions” and “unpredictable statistical conditions.” Hilbert’s sixth problem concerning the axiomatization of physics is then described. The universally recognized mathematical principles of axiomatization of probability theory and mechanics are considered. We propose a new approach for solution of the sixth problem, supplementing the mathematical axioms by physical adequacy hypotheses which establish a connection between the existing axiomatized mathematical theories and the real world. The basic concepts of probability theory and the theory of hyper-random phenomena are considered, and adequacy hypotheses are formulated for the two theories. Attention is drawn to the key point that the concept of probability has no physical interpretation in the real world.

Chapter 2 We discuss the concept of a “random event.” The classical and statistical approaches used to formalize the notion of probability are described, along with the basic concepts of set theory and measure theory. The Kolmogorov approach for axiomatizing probability theory is presented. The probability space is introduced. The axioms of probability theory are presented, together with the addition and multiplication theorems. The notion of a scalar random variable is formalized. We present ways to describe a random variable in terms of the distribution function, probability density function, and moments, including in particular the expectation and variance. Examples of scalar random variables with different distribution laws are presented. Methods for describing a scalar random variable are generalized to a vector random variable. The transformation of random variables and arithmetic operations on them are briefly examined.

Chapter 3 The notion of a stochastic (random) function is formalized, and the classification of these functions is discussed. We present different ways to describe a stochastic process, in terms of a distribution function, a probability density function, and moment functions and in particular the expectation, variance, covariance, and correlation functions. We consider a stationary stochastic process in the narrow and broad sense. We describe the Wiener–Khinchin transformation and generalized Wiener–Khinchin transformation. The spectral approach for describing a stochastic process is presented. The ergodic and fragmentary ergodic processes are considered.

Chapter 4 The concepts of random sampling and statistics of random variables are introduced. We consider estimators of probability characteristics and moments. We discuss the types of convergence used in probability theory, in particular the convergence of a sequence of random variables in probability and convergence in distribution. The law of large numbers and the central limit theorem are described in the classical interpretation. We discuss the statistics of stochastic processes and specific features of samples of random variables and stochastic processes.

Chapter 5 Modern concepts for evaluating measurement accuracy are examined and different types of error are described. We consider the classical determinate–random measurement model, in which the error is decomposed into systematic and random components. The point and interval estimators are described. For random estimators, the concepts of "biased estimator," "consistent estimator," "effective estimator," and "sufficient estimator" are determined. The concept of critical sample size is introduced.

Chapter 6 Here, we formalize the notion of statistical stability of a process. The parameters of statistical instability with respect to the average and with respect to the standard deviation are investigated. Measurement units are proposed for the statistical instability parameters. We specify the concept of an interval of statistical stability of a process. The dependencies of the statistical stability of a process on its power spectral density and its correlation characteristics are established. We then consider various processes described by a power function of the power spectral density and investigate the statistical stability of such processes. For narrowband processes, we present the investigation results of statistical stability violations. Statistically unstable stationary processes are considered. We present experimental results for the statistical stability of a number of actual processes of different physical kinds.

Chapter 7 The notion of a hyper-random event is formulated. The properties of hyper-random events are examined. The concept of a scalar hyper-random variable is specified. We present three ways to describe it: by its conditional characteristics (in particular, conditional distribution functions and conditional moments), by the bounds of the distribution function and their moments, and by the bounds of moments. The concept of a vector hyper-random variable is introduced. The methods that describe the scalar hyper-random variables are extended to these vector hyper-random variables. The issue of transformation of hyper-random variables and arithmetic operations on them are briefly examined.

Chapter 8 The notion of a hyper-random function is formalized. The classification of hyper-random functions is presented. Three ways to describe a hyper-random function are considered: by the conditional characteristics (in particular, conditional distribution functions and conditional moments), by the bounds of the distribution function and their moments, and by the bounds of the moments. The definition of a stationary hyper-random process is given. The spectral method for describing a stationary hyper-random processes is presented. The concepts of an ergodic hyper-random process and a fragmentary-ergodic hyper-random process are formalized. We discuss the effectiveness of the different approaches for describing hyper-random processes.

Chapter 9 The notion of a hyper-random sample and statistics of hyper-random variables are formalized. Estimators of the characteristics of hyper-random variables are examined. The notions of a generalized limit and a spectrum of limit points are introduced. Here, we formalize the notions of convergence of hyper-random sequences in a generalized sense in probability and in distribution. The generalized law of large numbers and generalized central limit theorem are presented and their peculiarities are studied. We present experimental results demonstrating the lack of convergence of the sample means of real physical processes to fixed numbers.

Chapter 10 A number of measurement models are considered. The point determinate–hyper-random measurement model is examined. It is shown that the error corresponding to this model is in general of a hyper-random type that cannot be represented by a sum of random and systematic components. For hyper-random estimators, the notions of "biased estimator," "consistent estimator," "effective estimator," and "sufficient estimator" are introduced. We specify a concept of critical sample size for hyper-random samples. We describe a measurement technique corresponding to the determinate–hyper-random measurement model. It is shown that, under unpredictable changes of conditions, the classical determinate–random measurement model poorly reflects the actual measurement situation, while the determinate–hyper-random model provides an adequate picture.

Chapter 11 We investigate different ways to produce an adequate description of the real physical world. Here, we discuss the reasons for using the random and hyper-random models. We present the classification of uncertainties. We also discuss approach leading to a uniform description of the various mathematical models (determinate, random, interval, and hyper-random) by means of the distribution function. A classification of these models is proposed. We examine the causes and mechanisms at the origin of uncertainty, marking out reasonable areas for practical application of random and hyper-random models.

Every chapter ends with a list of the main references, and the book ends with a list of subsidiary references.

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Contents

About the Author

Igor I. Gorban graduated from the Kiev Polytechnic Institute, USSR, majoring in hydroacoustics. At the MorPhysPribor Central Research Institute, Leningrad, he received a Ph.D. and, at the Institute of Cybernetics of the Academy of Sciences of Ukraine, Kiev, a Dr. Sc. He was awarded the academic rank of Senior Research Associate and then Full Professor.

He worked at the Kiev Research Institute for Hydroequipment, participating in a number of developmental and research programs. He was in charge of algorithms for several sonar systems, was a research adviser for two scientific expeditions to the Pacific to study hydroacoustic signals, was the first deputy to the Chief Designer, and was Chief Designer of the sonar complexes.

Since 1993, he has been working at the Institute of Mathematical Machines and Systems Problems, National Academy of Sciences of Ukraine, as Principal Scientist and Deputy Director for Research.

He was a scientific adviser for several research projects. At the Kiev Air Force Institute for students majoring in applied mathematics, he taught probability theory and mathematical statistics. For 6 years, he was a member of the council of experts at the Higher Attestation Commission of Ukraine.

He supervises the research of postgraduate students, is a member of specialized councils for the defense of Dr. Sc. dissertations, and is a member of the editorial boards of several scientific journals.

Igor I. Gorban is the author of three theories:

- *The theory of space–time processing of hydroacoustic signals under complex dynamic conditions*
- *The theory of fast multichannel processing of hydroacoustic signals*

- *The physical-mathematical theory of hyper-random phenomena that takes into account violations of statistical stability*

The results of his research have been published in more than 200 scientific publications, including 12 books devoted to

– Processing of hydroacoustic signals under complex dynamic conditions (3 monographs)
– Probability theory and mathematical statistics (3 books)
– Investigation of the phenomenon of statistical stability and description of the theory of hyper-random phenomena (6 monographs).

The results of his research have been used in several sonars and also in radar.

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Symbols

Operators

$\mathrm{Cov}[X, Y]$	Covariance of random variables X and Y
$\mathrm{E}[X]$	Expectation of random variable X
$\mathrm{E}_i[X]$, $\mathrm{E}_s[X]$	Lower and upper bounds of expectation of hyper-random variable X
$\mathrm{E}_I[X]$, $\mathrm{E}_S[X]$	Expectations of lower and upper bounds of hyper-random variable X
$\bar{m}$	Parameter m obtained by averaging over time
$P\{A\}$	Probability of condition A
$P(A)$	Probability of event A
$P_I(A)$, $P_S(A)$	Lower and upper bounds of probability of hyper-random event A
T	Transpose operator
$\mathrm{Var}[X]$	Variance of random variable X

Special Mathematical Signs

inf, sup	Lower and upper bounds
$\lim\limits_{N\to\infty} x_N$	Usual limit of number sequence $x_1, \ldots, x_N$
$\lim\limits_{N\to\infty} X_N$	Usual limit of sequence of random variables $X_1, \ldots, X_N$ (almost sure convergence, i.e., with probability one)
$\mathop{\mathrm{LIM}}\limits_{N\to\infty} x_N$	Generalized limit of number sequence $x_1, \ldots, x_N$
$\mathop{\mathrm{LIM}}\limits_{N\to\infty} X_N$	Generalized limit of a sequence of random or hyper-random variables $X_1, \ldots, X_N$ (almost sure convergence, i.e., with probability one)
$\mathrm{med}[X]$	Median of random variable X
$\mathrm{sign}[x]$	Unit step function
$\subset$, $\supset$	Concatenation sign
$\cup$	Union (logical addition)
$\cap$	Intersection (logical multiplication)
$\setminus$	Complement

$\varnothing$	Empty set
$\in$	Membership (adhesion) sign
$\{X\}$	Set (collection) of X
$\tilde{x}$	Tilde above letter indicates many-valuedness of value
θ^*	Estimate of random or hyper-random variable Θ
Θ^*	Estimator of random or hyper-random variable Θ
$\dot{S}$	Point above letter indicates complex character of value
$\overset{*}{S}$	Complex conjugate of $\dot{S}$
$(x_1, \ldots, x_N)$	Vector with components $x_1, \ldots, x_N$
$\{X_1, \ldots, X_N\}$	Set or ordered set with elements $X_1, \ldots, X_N$
X/g or X_g	Random component of hyper-random variable X corresponding to condition g

Parameters and Functions

$D_{ix},\ D_{sx}$	Lower and upper bounds of variance of hyper-random variable X or ergodic hyper-random function $X(t)$
$D_{Ix},\ D_{Sx}$	Variances of lower and upper bounds of distribution function of hyper-random variable X
$\bar{D}_{ix},\ \bar{D}_{sx}$	Lower and upper bounds of variance of ergodic hyper-random function $X(t)$
$f(x)$	Probability density function of random variable X
$f(x/g)$ or $f_{x/g}(x)$	Conditional probability density function of hyper-random variable X under condition g
$f_I(x), f_S(x)$	Probability density functions of lower and upper bounds of distribution function of hyper-random variable X
$F(x)$	Distribution function of random variable X
$F_I(x), F_S(x)$	Lower and upper bounds of distribution function of hyper-random variable X
$F(x/g)$ or $F_{x/g}(x)$	Conditional distribution function of hyper-random variable X under condition g
$F(x/m, D)$	Gaussian distribution function with expectation m and variance D
h_N	Parameter of statistical instability with respect to average
h_{0N}	Measurement unit for parameters of statistical instability h_N, H_N
$h_{0N}^{\pm}$	Borders of confidence interval for the parameters of statistical instability h_N, H_N
H_N	Parameter of statistical instability with respect to SD
$K_{ix}(t_1, t_2), K_{sx}(t_1, t_2)$	Lower and upper bounds of correlation function of hyper-random function $X(t)$

$K_{Ix}(t_1,t_2), K_{Sx}(t_1,t_2)$	Correlation functions of lower and upper bounds of hyper-random function $X(t)$
$K_{x/g}(t_1,t_2)$	Conditional correlation function of hyper-random function $X(t)$
$\bar{K}_x(\tau)$	Autocorrelation function of random function $X(t)$
$\bar{K}_{ix}(\tau)$, $\bar{K}_{sx}(\tau)$	Lower and upper bounds of autocorrelation function of ergodic hyper-random function $X(t)$
m_{ix}, m_{sx}	Lower and upper bounds of expectation of hyper-random variable X
m_{Ix}, m_{Sx}	Expectations of lower and upper bounds of distribution function of hyper-random variable X
$m_{x/g}$	Conditional expectation of hyper-random variable X
$m_{x/g}(t)$	Conditional expectation of hyper-random function $X(t)$
$\bar{m}_x$	Time average of realization of random function $X(t)$
$\bar{m}_{ix}$, $\bar{m}_{sx}$	Lower and upper average bounds of ergodic hyper-random function $X(t)$
$r_{Ix}(t_1,t_2), r_{Sx}(t_1,t_2)$	Normalized covariance functions of lower and upper bounds of hyper-random function $X(t)$
$R_{ix}(t_1,t_2), R_{sx}(t_1,t_2)$	Lower and upper bounds of covariance function of hyper-random function $X(t)$
$R_{Ix}(t_1,t_2), R_{Sx}(t_1,t_2)$	Covariance functions of lower and upper bounds of hyper-random function $X(t)$
$\bar{R}_x(\tau)$	Autocovariance function of random function $X(t)$
$\bar{R}_{ix}(\tau)$, $\bar{R}_{sx}(\tau)$	Lower and upper bounds of autocovariance function of ergodic hyper-random function $X(t)$
$S_{ix}(f), S_{sx}(f)$	Lower and upper bounds of power spectral density of hyper-random function $X(t)$
$S_{Ix}(f), S_{Sx}(f)$	Power spectral densities of upper and lower bounds of hyper-random function $X(t)$
$\dot{S}_{x/g}(f)$	Instantaneous spectrum of hyper-random function $X(t)$ under condition g
$S_{x/g}(f)$	Conditional power spectral density of hyper-random function $X(t)$
γ_N	Parameter of statistical instability with respect to average
γ_{0N}	Measurement unit for parameters of statistical instability γ_N, Γ_N
$\gamma_{0N}^{\pm}$	Borders of confidence interval for parameters of statistical instability γ_N, Γ_N
Γ_N	Parameter of statistical instability with respect to SD
$\delta(t)$	Dirac delta function
μ_N	Parameter of statistical instability with respect to average
μ_{0N}	Measurement unit for parameters of statistical instability μ_N, M_N

$\mu_{0N}^{\pm}$	Borders of confidence interval for parameters of statistical instability μ_N, M_N
M_N	Parameter of statistical instability with respect to SD
$\Phi(x)$	Gaussian distribution function with zero expectation and unit variance
$\aleph$	Cardinal number of real numbers (continuum)
$\aleph_0$	Cardinal number of countable set

Part I
The Phenomenon of Statistical Stability

The argument over a card game (Steen Jan, 1660s, Art Gallery, Berlin-Dahlem, Source: https://www.wikiart.org/en/jan-steen/argument-over-a-card-game)

David Hilbert, German mathematician (1862–1943) (Source: https://media1.britannica.com/eb-media/91/124791-004-B2DE3FCC.jpg)

D. Hilbert:

> In parallel with research on the foundations of geometry, one could approach the problem of an axiomatic construction, along the same lines, of the physical sciences in which mathematics play an exclusive role, and in particular, probability theory and mechanics. (Aleksandrov 1969).

G.A. Korn, T.M. Korn:

> Statistical description and probability models apply to physical processes exhibiting the following empirical phenomenon. Even though individual measurements of a physical quantity x cannot be predicted with sufficient accuracy, a suitably determined function $y = y(x_1, x_2, \ldots)$ of a set (*sample*) of repeated measurements $x_1, x_2, \ldots$ of x can often be predicted with substantially better accuracy, and the prediction of y may still yield useful decisions. Such a function y of a set of sample values is called a *statistic*, and the incidence of increased predictability is known as *statistical regularity*. Statistical regularity, in each individual situation, is an *empirical physical law* which, like the law of gravity or the induction law, is ultimately derived from experience and *not* from mathematics. Frequently a statistic can be predicted within creasing accuracy as the size n of the sample $(x_1, x_2, \ldots, x_n)$ increases (physical laws of large numbers). The best-known statistics are *statistical relative frequencies* and *sample averages*. (Korn and Korn 2000)

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Chapter 1
The Physical Phenomenon of Statistical Stability

Abstract Here we examine the main manifestations of the phenomenon of statistical stability: the statistical stability of the relative frequency and sample average. Attention is drawn to an emergent property of the phenomenon of statistical stability. We discuss the hypothesis of perfect (absolute or ideal) statistical stability, which assumes the convergence of relative frequencies and averages. Examples of statistically unstable processes are presented. We discuss the terms 'identical statistical conditions' and 'unpredictable statistical conditions'. Hilbert's sixth problem concerning the axiomatization of physics is then described. The universally recognized mathematical principles of axiomatization of probability theory and mechanics are considered. We propose a new approach for solution of the sixth problem, supplementing the mathematical axioms by physical adequacy hypotheses which establish a connection between the existing axiomatized mathematical theories and the real world. The basic concepts of probability theory and the theory of hyper-random phenomena are considered, and adequacy hypotheses are formulated for the two theories. Attention is drawn to the key point that the concept of probability has no physical interpretation in the real world.

1.1 Manifestation of the Phenomenon of Statistical Stability

One of the most surprising physical phenomena is the *phenomenon of statistical stability* (*regularity*), comprising *stability of relative frequencies* of mass events, *sample averages*, and other *functions of the sample* called *statistics*.

The first to draw attention to the phenomenon of statistical stability was the cloth merchant *J. Graunt* (1939) in 1662. Information about research on statistical stability is fragmentary for the period from the end of the seventeenth century to the end of the nineteenth century, e.g., by *J. Bernoulli*, *S. D. Poisson*, *I. J. Bienayme*, *A. A. Cournot*, *L. A. J. Quetelet*, *J. Venn*, etc. (Scheinin 2009; Chaykovskiy 2004).

Systematic study of statistical stability began at the end of the nineteenth century. In 1879, the German statistician *W. Lexis* made the first attempt to link

The chapter is based on material from the books (Gorban 2003, 2014, 2016, 2017).

I.I. Gorban, *Randomness and Hyper-randomness*, Mathematical Engineering,
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the concept of statistical stability of the relative frequency with the dispersion (Scheinin 2009). At the turn of the century and in the early twentieth century, statistical stability was studied by *C. Pearson*, *A. A. Chuprov*, *L. von Bortkiewicz*, *A. A. Markov*, *R. E. von Mises*, and others (Scheinin 2009; Chaykovskiy 2004).

Let us consider the various manifestations of this phenomenon, starting with the statistical stability of the relative frequency of events.

1.1.1 Statistical Stability of the Relative Frequency of Events

Let us recall the game which consists in tossing a coin and guessing which way it will come up. It may fall heads or tails (Fig. 1.1a).

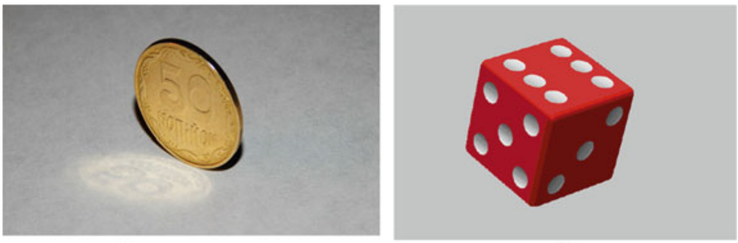
It is impossible to predict accurately which way it will fall. But taking into account the similarity between the two results, it seems natural to assume that, repeating the experiment *many times*, it will fall heads in about 50% cases and tails in about 50% cases. In other words, the *relative frequencies* of these two events, viz.,

$$p_n(H) = n_H/n, \;\; p_n(T) = n_T/n$$

will be approximately equal to 0.5, where n_H and n_T are respectively the numbers of heads and tails and n is the number of experiments.

Experimental studies of *relative frequencies* in the coin-tossing game were carried out by many famous scientists including *P. S. de Laplace*, *G. L. L. de Buffon*, *C. Pearson*, *R. P. Feynman*, *A. de Morgan*, *W. S. Jevons*, *V. I. Romanovskiy*, *W. Feller*, and others. At first glance, this rather trivial task per se was not so simple in their view.

Table 1.1 and Fig. 1.2a present some of the results of their experiments (Gnedenko 1988; Feynman et al. 1963; Rozhkov 1996). Table 1.2 and Fig. 1.2b show the results described in (Mosteller et al. 1961), involving ten runs of the same experiment in which each run consists of 1000 tosses. The tables and figures show that, for a large number of tosses, the relative frequency of heads (or tails) really is close to 0.5.



a b

Fig. 1.1 Coin (**a**) and dice (**b**)

Table 1.1 Results of coin-tossing experiments carried out by different scientists

Experiment	Researcher	Number of tosses	Number of heads	Relative frequency of heads
1	Buffon	4040	2048	0.508
2	Pearson	12,000	6019	0.5016
3	Pearson	24,000	12,012	0.5005
4	Feynman	3000	1492	0.497
5	Morgan	4092	2048	0.5005
6	Jevons	20,480	10,379	0.5068
7	Romanovskiy	80,640	39,699	0.4923
8	Feller	10,000	4979	0.4979

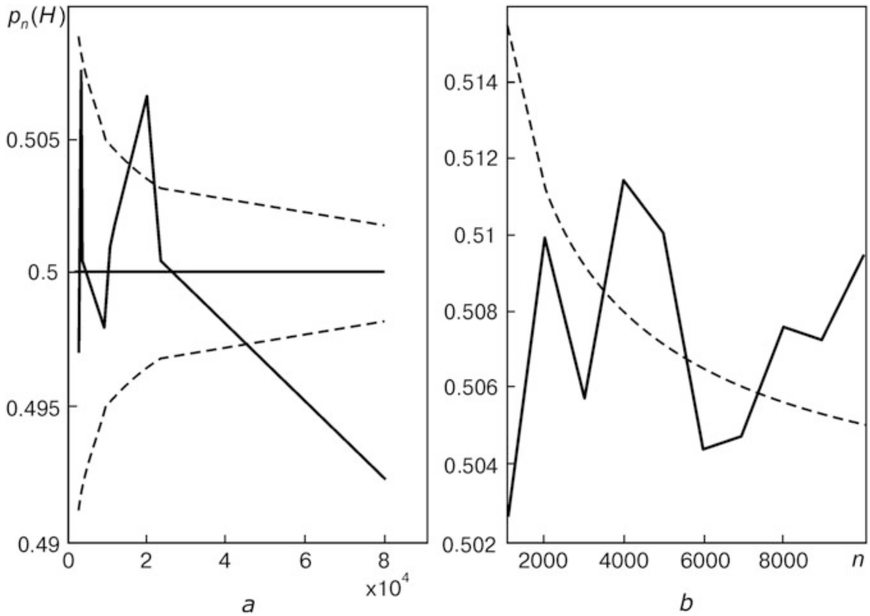


Fig. 1.2 Dependence of the relative frequency of heads $p_n(H)$ on the number of tosses n in the experiments presented in Tables 1.1 (**a**) and 1.2 (**b**). *Dotted lines* represent one standard deviation from the expected value 0.5

Table 1.2 Results of coin-tossing experiments described in (Mosteller et al. 1961)

Run	1	2	3	4	5	6	7	8	9	10
Number of heads	502	518	497	529	504	476	507	528	504	529

The really intriguing result from these experiments is not even the fact that, for a large number of tosses, the relative frequencies are approximately equal to 0.5, but the *stability of the relative frequencies* manifested in a *weak dependence on the number of experiments n*.

In experiments with multiple repeated tosses of a symmetrical dice (Fig. 1.1b), the relative frequency of any particular number on the dice also shows the stability property. Its value is close to 1/6.

Experimental studies of other actual mass physical events show that, in a series of tests, the relative frequencies of events stabilize. This points to the *fundamental nature of the phenomenon of statistical stability*.

This phenomenon has the *emergent property* or *system effect*, a concept that arises in systems theory and synergetics.

Coming from ancient Greek, the word '*system*' refers to 'a whole consisting of separate parts'. Wikipedia interprets this word to mean a set of interacting or interdependent components forming a complex/intricate whole.

Emergence is a process whereby new entities and regularities arise. It is a manifestation of the *law of transition from quantity to quality*. The idea of emergence has been known since ancient times. *Aristotle*, for instance, understood that the *whole was greater than the sum of the parts*.

Examples of systems with emergent properties are a shoal of fish or a flock of birds, whose behavior differs significantly from the behavior of its constituent individuals. Emergent properties are also manifested by the behavior of groups of people in society. The difference between the properties of a chemical substance and those of the chemical elements making up the substance is another manifestation of emergence.

The statistical stability of the relative frequency is a property of *mass* (*multiple*) *events*. This property is not inherent in a single event, but is inherent in their collection. Therefore the *statistical stability of relative frequency can be regarded as an emergent property*.

The mechanism by which emergence comes into being is not always clear. If, for example, the specific behavior of a shoal of fish, or indeed the difference between the properties of a chemical substance and the properties of its constituent chemical elements, can be explained by the presence of certain bonds between the elements of the system, the statistical stability of the relative frequency of events in the absence of any clear relationship between the individual events seems a rather mysterious phenomenon.

The nature of the phenomenon of statistical stability remains somewhat unclear. Attempts to clarify the situation over the past centuries have not so far led to any positive result. Explanations of this phenomenon remain on the same level as explanations of other basic physical phenomena, such as the existence of electromagnetic fields and gravitational fields, the inertia of material bodies, etc. In the framework of certain physical or mathematical models, we can look for and even find explanations for these phenomena, but these explanations do not go beyond the limitations and assumptions adopted in the construction of the models. The true entity of the physical phenomena remains hidden.

However, the lack of a clear understanding of the nature of a physical phenomenon is not an obstacle for constructing of *phenomenological theories*, often very useful for the solution of practical tasks. Classic examples of such theories are the theoretical mechanics based on Newton's laws, Maxwell's phenomenological theory describing the electromagnetic field, Einstein's relativity theory, and many others.

It should be noted, furthermore, that *all theories of natural science*, including *physical theories*, are *phenomenological* ones. They are based on some physical phenomena that are not explained by the theory, but are taken as *undeniable truths*. The question of the causes of these phenomena is not asked in the framework of such theories. Attention is focused primarily on the *manifestations of the phenomena in the real world* and on the *adequate description of the phenomena using mathematical tools*.

This observation is fully applicable to theories describing the phenomenon of statistical stability, viz., *probability theory* and the *theory of hyper-random phenomena*.

1.1.2 Statistical Stability of Statistics

The phenomenon of statistical stability is manifested not only in the relative frequency stability of mass events, but also in the stability of the *average* $y(t)$ of the process $x(t)$, or its *sample mean*

$$y_n = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i, \tag{1.1}$$

where $x_1, \ldots, x_n$ are discrete samples of the process.

Note that the phenomenon of statistical stability is observed by averaging processes of various types: *random, determinate, and actual physical processes*.

For an illustration, Fig. 1.3a, c present the realization of a random noise and a fragment of a periodic determinate process, while Fig. 1.3b, d shows the corresponding sample average, calculated over the interval from zero to the current value of the argument t.

As can be seen from Fig. 1.3b, d, increasing the interval t, the averaged values of the random realization and the determinate process tend to stabilize, and levels of fluctuation decrease.

The initial fragment of a 60-h recording of the mains voltage fluctuation $x(t)$ and the corresponding average $y(t)$ are presented in Fig. 1.4a, b.

As can be seen from the figures, during the observation interval, the voltage $x(t)$ fluctuates over the range from 228 to 250 V, and the average value $y(t)$ changes smoothly. The initial fluctuation $x(t)$ is clearly an undamped process, and the average value $y(t)$ tends to a certain value (around 234 V). During the first

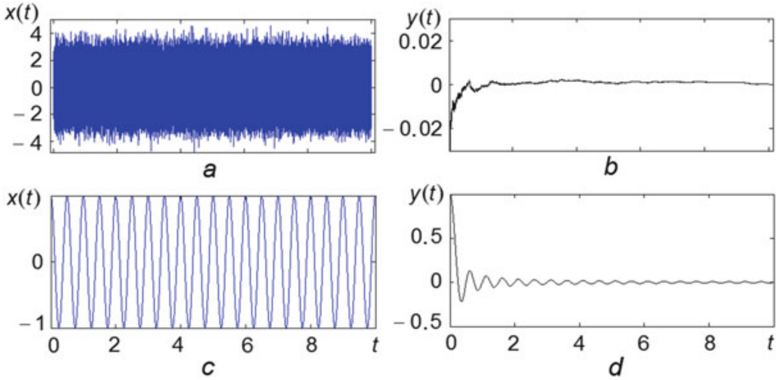


Fig. 1.3 Realization of *white* Gaussian noise (**a**) and harmonic oscillation (**c**), together with the dependence of the corresponding sample mean on the average interval (**b**, **d**)

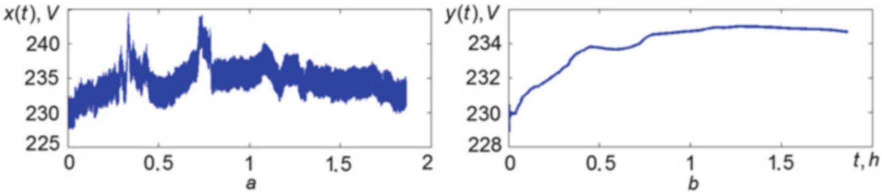


Fig. 1.4 Dependence of the mains voltage (**a**) and the corresponding average (**b**) on time over 1.8 h

30 min of observation, the process $y(t)$ changes from 228 to 234 V (the difference is 6 V), and during the last 30 min it changes from 235 to 234.7 V (the difference is 0.3 V). Thus, by increasing the averaging time, the range of the average decreased significantly (by a factor of 20) and the average stabilized.

To investigate the statistical stability of air temperature fluctuations in the Kiev area, observation data collected over 112 years (from 1881 to 1992) were used (Weather 2009). The changes in the daily minimum temperature $x(t)$ and corresponding average $y(t)$ over time t are presented in Fig. 1.5a, b, respectively, for the first 22 years of observations.

As can be seen from the figures, the seasonal fluctuations in the daily minimum temperature present in the original process $x(t)$ appear in the averaged process $y(t)$, albeit in a weaker form. At the beginning of the averaging interval, the range of the average is close to 11°, while at the end, it is less than 0.5°. Thus, the range of the average values decreases by a factor of more than 20 and the average value acquires stability.

The phenomenon of statistical stability is revealed not only in averages, but also in other statistics, and in particular, in the *sample standard deviation* z_n, defined as the square root of the average squared deviation of samples $x_1, \ldots, x_n$ from the sample mean y_n:

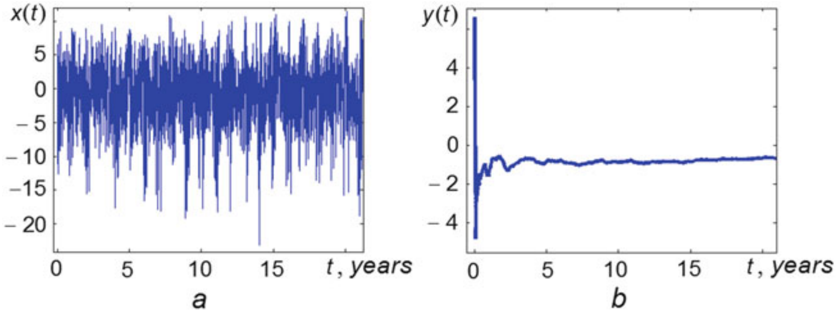


Fig. 1.5 Fluctuation in the daily minimum air temperature in the Kiev area (**a**) together with the oscillations in the corresponding average (**b**) over the first 22 years of observation

$$z_n = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - y_n)^2} \quad (n = 2, 3, \ldots). \tag{1.2}$$

Investigation of a whole range of different processes shows that:

The phenomenon of statistical stability is manifested in calculations of statistics of *various random, determinate, and actual physical processes*, and this indicates *its fundamental nature*.

1.2 Interpretations of the Phenomenon of Statistical Stability

1.2.1 Perfect Statistical Stability

Analyzing Tables 1.1 and 1.2 and taking into account the statistical stability of the actual physical events, *it seems natural to assume* that, if the number of tests n increases indefinitely, the level of fluctuation of the relative frequency $p_n(A)$ of *any actual event A* will tend to zero. Similarly, analyzing the functions shown in Figs. 1.3b, 1.4b, and 1.5b, *it seems natural to* assume that, by increasing the sample size n (increasing the observation time t) without limit, the level of fluctuation of the sample mean y_n of any *random or real physical oscillation* $x(t)$ will also tend to zero.

In other words, we may *hypothesize* that there is a *convergence* of the sequence of relative frequencies $p_1(A), p_2(A), \ldots$ of *any actual event A* to some *determinate value* $P(A)$ and that there is a *convergence* of the sequence of averages $y_1, y_2, \ldots$ of *any actual process* to a *determinate value m*, implying that the difference between

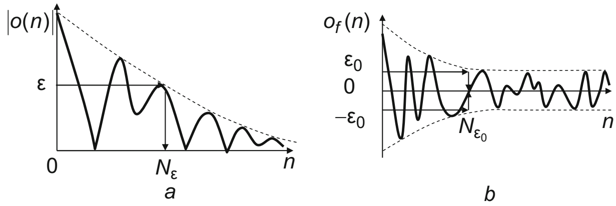


Fig. 1.6 Illustration of an infinitesimal variable $o(n)$ (**a**) and a finitesimal variable $o_f(n)$ (**b**)

the relative frequency $p_n(A)$ and the value $P(A)$, as well as between the average y_n and the value m, is equal to an *infinitesimal* $o(n)$:

$$p_n(A) - P(A) = o(n), \tag{1.3}$$

$$y_n - m = o(n). \tag{1.4}$$

In classical mathematical analysis (see, for instance, Fikhtengolts 1958), an *infinitesimal* $o(n)$ is (Fig. 1.6a) a *variable* depending on n whose *absolute value*, starting from some $n = N_\varepsilon$, becomes and remains *less than any arbitrarily small preassigned positive number* ε.

When n tends to infinity, the infinitesimal $o(n)$ goes to zero, i.e., its *limit*, denoted by $\lim\limits_{n\to\infty} o(n)$, is zero.

It follows from (1.3) and (1.4) that, *in the case of convergence*, the limit of the relative frequency is

$$\lim_{n\to\infty} p_n(A) = P(A), \tag{1.5}$$

and the limit of the average is

$$\lim_{n\to\infty} y_n = m. \tag{1.6}$$

Modern probability theory is based on the *hypothesis of perfect statistical stability* or, in other words, on the *assumption of convergence of statistics*.

The value $P(A)$ is interpreted as the *probability* of the event A, and the value m is regarded as the *expectation* of the process $x(t)$.[1]

For many years it *was believed that the hypothesis of perfect statistical stability adequately reflects the reality*. Although some scholars (even the founder of axiomatic probability theory *A. N. Kolmogorov* (1986), and such famous scientists as *A. A. Markov* (1924), *A. V. Skorokhod* (Ivanenko and Labkovsky 1990), *E. Borel* (1956), *V. N. Tutubalin* (1972 (1)), and others) noticed that *in the real world, this hypothesis is valid only with certain reservations*. They suspected that probability

[1]A more correct definition of these notions is given in Chaps. 2 and 3.

theory does not describe the physical phenomenon of statistical stability entirely adequately.

Real life is not organized as probability theory describes it. Thinking about this we may recall a picture by a certain Renaissance painter who lived far from the problems of physics and mathematics, but acutely sensed the complexity and diversity of the factors that influence the course and development of actual events in the world (see the reproduction of the picture by *Jan Steen* on p. 1).

1.2.2 Imperfect Statistical Stability

Note that the possibility of describing the relative frequency of events and the sample mean of processes by the expressions (1.3)–(1.6) are no more than a *guess*. It *does not follow* from any experiments or logical reasoning.

The experimental results, in particular, concerning coin-tossing, *do not suggest that there is a convergence of the relative frequency to any number* (in this case to 0.5). Moreover, the graphs in Figs. 1.2 indicate rather a lack of convergence than the presence of convergence.

It is easy to check that not all processes, even oscillating processes, have the statistical stability property. For example, Fig. 1.7a, c show two determinate processes $x(t)$, while Fig. 1.7b, d shows their sample means $y(t)$. As can be seen from the figures in both cases, the sample means $y(t)$ are not stabilizing, i.e., the fluctuations $x(t)$ are statistically unstable.

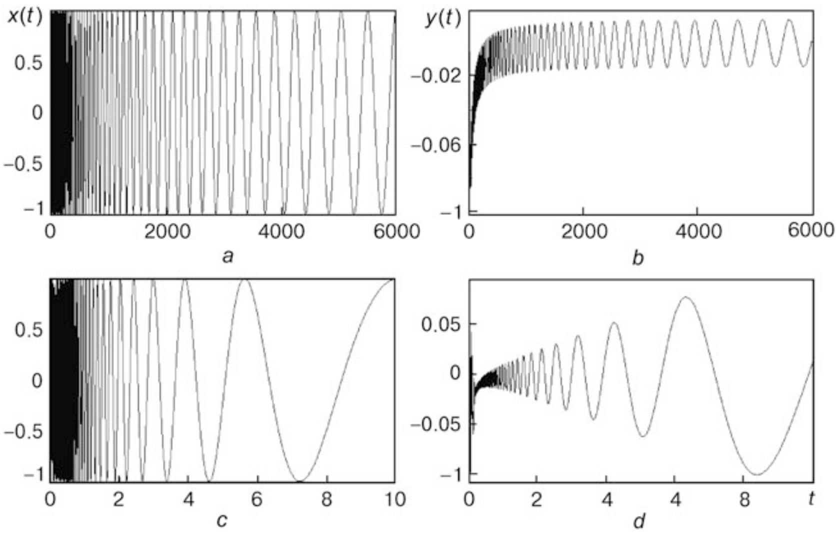


Fig. 1.7 Statistically unstable processes (**a**, **c**) and their corresponding averages (**b**, **d**)

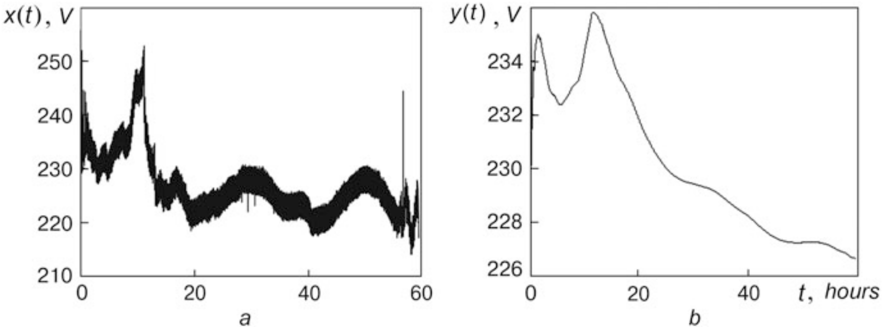


Fig. 1.8 Dependence of the mains voltage (**a**) and the corresponding average (**b**) on time over 60 h

The results of many experimental investigations of actual physical phenomena for large data volumes show that the sequences $p_1(A)$, $p_2(A), \ldots$ and y_1, $y_2, \ldots$ do not instantiate the trend to convergence, whence the equalities (1.3)–(1.6) are not justified.[2]

Although in the case of relatively small volumes of data a tendency of the relative frequency of events and averages to stabilize is usually clearly visible, *in the case of large data volumes, this tendency is not fixed*. This indicates that *the phenomenon of statistical stability is not perfect*.

The non-perfect statistical stability of actual processes is illustrated in Fig. 1.8. Figure 1.8a presents mains voltage fluctuations $x(t)$ over 2.5 days, the initial fragment over 1.8 h being imaged in Fig. 1.4a. Figure 1.8b presents the dependence of the corresponding average.

The processes in Figs. 1.4b and 1.8b are essentially different. In the short fragment of the process (see Fig. 1.4b), one can see a stabilization trend in the average, whereas in the much longer process (see Fig. 1.8b), there is not even a hint of stabilization.

Consider another example (Fig. 1.9), concerning the daily minimum temperature in the Kiev region over a period that is five times longer than for the fluctuations imaged in Fig. 1.5a, viz., the period from 1881 to 1992.

As can be seen from Figs. 1.5b and 1.9b, if we monitor the changes in the average temperature over two decades, this does not reveal a violation of statistical stability; but doing so over a much longer period of time shows a clear instability with respect to the average.

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[2] A description of the investigations leading to this statement is presented in Chap. 6.

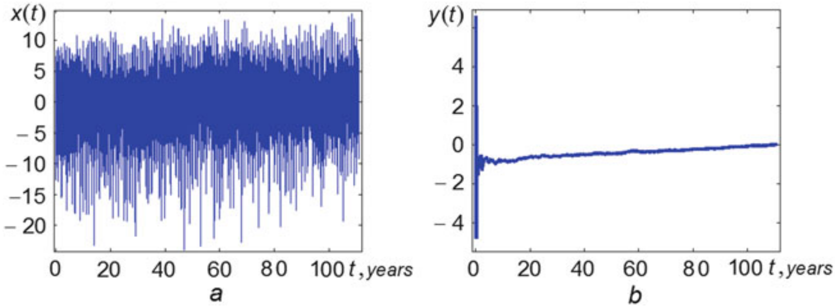

Fig. 1.9 Fluctuation in the daily minimum air temperature in the Kiev area (**a**) together with the oscillations in the corresponding average (**b**) over 112 years of observation

The imperfect statistical stability of actual events and averages of actual processes means that the difference between the relative frequency $p_n(A)$ and some fixed value $P(A)$, as well as between the average y_n and some fixed value m, is equal to a *finitesimal* (*variable*) $o_f(n)$:

$$p_n(A) - P(A) = o_f(n), \tag{1.7}$$

$$y_n - m = o_f(n). \tag{1.8}$$

The *finitesimal* (*restricted small*) *variable* $o_f(n)$ of order ε_0 ($\varepsilon_0 > 0$) is a *variable* (Fig. 1.6b) depending on n,

- The values of which, starting from some $n = N_{\varepsilon_0}$, lie *in the interval* $[-\varepsilon_0, \varepsilon_0]$, and
- When n goes from N_{ε_0} to infinity, the values $-\varepsilon_0$ and ε_0 *are repeated infinitely often*.

This means that, when n goes to infinity, the finitesimal $o_f(n)$ does not approach zero, but remains in the interval $[-\varepsilon_0, \varepsilon_0]$. In other words,

A *finitesimal has no ordinary limit*. This is its *fundamental difference* with the infinitesimal, which has a limit when $n \to \infty$.

In contrast to probability theory, the *theory of hyper-random phenomena* takes into account the violation of convergence of statistics, i.e., it *proceeds from the assumption that the statistical stability of statistics is restricted*.

Thus, the difference between probability theory and the theory of hyper-random phenomena consists in *different interpretations of the phenomenon of statistical stability*.

1.3 Identical and Statistically Unpredictable Conditions

When explaining the phenomenon of statistical stability in the framework of probability theory, it is usually stipulated that tests must be conducted under *identical statistical conditions*. However, the concept of 'identical statistical conditions' is not as trivial as it may seem at first glance, and indeed the set expression 'statistical conditions' requires some explanation.

For instance, in coin-tossing experiments, statistical conditions may refer to the way of flipping the coin, the height and strength of the throw, the speed and direction of the air flow, the roughness of the surface on which the coin falls, etc. In addition, each of these items can be detailed. Considering, for example, the method of flipping, one may distinguish flat flipping, tossing with rotation, tossing over the shoulder, tossing on the floor or on the table, tossing with control of the coin position before flipping, etc. If the aggregate of all experimental conditions were completely reproduced in each experiment, the results would always be the same. But it is impossible to reproduce all conditions in this way.

'Identical statistical conditions' cannot be understood literally. In experimental studies, some of the conditions will vary from test to test in a way that cannot be controlled by the experimenter. This means that the results (outcomes) of experiments cannot be accurately predicted. The changing conditions from test to test may lead (although not necessarily) to a result that differs from the previous one.

When there is a relatively small number of experiments, the average (in particular, the relative frequency of events) depends essentially on the number n of trials, the conditions under which each experiment is carried out, and the sequence according to which these conditions change. In the presence of convergence, when the number of experiments is increased, the average depends less and less on the conditions and the sequence according to which they change. It even ceases to depend on the conditions under which *any limited number of tests* is carried out.

When one talks about experiments carried out under identical statistical conditions, one does not mean that they are carried out under constant (unchanging) conditions. One means that they are carried out with possibly changing conditions under which the *convergence of the relevant average to a certain limit is ensured*.

The possibility of convergence implies *the potential to obtain* (at least theoretically) *a perfectly accurate statistical forecast* (*an estimate with zero error*) of the average for the case of infinite sample size. Thus, the same statistical conditions guarantee the convergence of the averages and the possibility of ideally accurate forecasting.

As a consequence, the widely used term 'identical statistical conditions' is not particularly well chosen. It does not mean what it is usually assumed to mean.

Note that for different statistics, e.g., the sample mean (1.1) and the sample standard deviation (1.2), the conditions under which convergence is ensured *may differ*. Therefore, when one says that the tests are carried out under identical statistical conditions, *one must specify with respect what statistic these conditions are constant*.

Instead of identical statistical conditions, one can consider unpredictable statistical conditions. When one says that the experiments are conducted under *unpredictable statistical conditions*, one focuses on the fact that the conditions change in such way that the *relevant statistic diverges*.

1.4 Hilbert's Sixth Problem

1.4.1 The Essence of the Problem

The Second International Congress of mathematicians took place in Paris in 1900, and it was on this occasion that *David Hilbert* made his speech entitled 'Mathematical problems' (Aleksandrov 1969). Here he formulated what he considered to be the 23 most important problems whose study could significantly stimulate the further development of science. *The sixth problem was the mathematical description of the axioms of physics*.

In the part of his presentation relating to this problem, Hilbert noted that, in parallel with research on the foundations of geometry, one could approach the problem of an axiomatic construction, along the same lines, of the physical sciences in which mathematics play an exclusive role, and in particular, probability theory and mechanics.

Hilbert paid great attention to the axiomatization of science. In the lecture he gave at the meeting of the Swiss Mathematical Society in 1917, he said (Hilbert 1970) that, with the further development of any science, it becomes increasingly necessary to make a purposeful selection of its underlying assumptions, in their purest form, treating them as axioms that would form the foundation of this area of knowledge. He also said that this axiomatic method would lead to a deeper foundation of knowledge.[3]

Note that *Hilbert considered probability theory as a physical discipline and not a mathematical one*. Apparently, he perceived it rather as a branch of physics in which one studies the phenomenon of statistical stability.

It is interesting to recall a remark made by *B. V. Gnedenko* concerning Hilbert's sixth problem (Aleksandrov 1969), one that was representative of the view held by

[3]It should be said that not all scientists shared Hilbert's view on the axiomatization question. For example, the mathematician *V. I. Arnold* (1999) considered mathematics as a part of physics and keenly criticized any attempts to create a closed description of disciplines in a strictly axiomatic form.

many mathematicians. He wrote: “For Hilbert, probability theory is a branch of physics, in which mathematical methods play a prominent role. Now, this view is not as common as it was at the turn of the century, because the mathematical content of probability theory has since been identified. Now there is no doubt that the concepts and research methods that have been produced in it, and also the results that have been obtained, have common scientific significance, going far beyond physics and even natural science.”

We thus see that there are different views on probability theory. Without doubt B. V. Gnedenko was right to say that a lot of concepts, methods, and research results developed within the framework of probability theory have common scientific significance, going far beyond physics. However, we must not forget that mathematics deals with abstract models that give only an approximate description of the actual physical world, and therefore there is always a danger of obtaining mathematical results that are not in conformity with observation results. In this respect, Hilbert’s position, based as it was on the physics of phenomena, was of course closer to the realities of life.

1.4.2 Approaches to Axiomatizing Probability Theory

Many scientists have responded to Hilbert’s appeal. Various approaches to axiomatizing probability theory have been proposed by *G. Bohlmann* (1908), *S. N. Bernstein* (1917), *R. von Mises* (1918), *A. Lomnicki* (1923) (based on the ideas of *E. Borel*), *A. N. Kolmogorov* (1929), and others (Aleksandrov 1969; Prokhorov and Shiryyaev 2014; Shiryaev 2000; Krengel 2011). Some scholars, in particular, R. von Mises, considered the problem from the standpoint of natural science, while others, such as A. N. Kolmogorov approached it from a more mathematical angle.

The axiomatic approach proposed by A. N. Kolmogorov (1929, 1956) is now favoured in probability theory. It is based on the concepts of set theory and measure theory.[4] This approach has even been elevated to the rank of a standard (ISO 2006). We shall stick with it throughout the rest of the book.

The basic notions in probability theory are the *abstract notions* of a *random event, random variable, and random* (*stochastic*) *function* (*process*). Precise definitions of them are given in Chaps. 2 and 3. Here we note only their main common feature, namely that *any random event, any value of a random variable, and any value of the random function at a fixed point* are characterized by a non-negative number not exceeding unity, called a *probability* [*probability measure* (*distribution function*[5])]. In other words:

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[4]The basic notions of these theories are presented in Chap. 2.

[5]The distribution function is defined in Chap. 2.

Mathematical objects that do not have a probability measure are not considered to be random (*stochastic*).

Note that probability theory tends to be regarded by mathematicians as a *mathematical discipline*. However, we should not forget that this theory, as well as other formal theories now regarded as purely mathematical, but at the same time widely used to describe physical phenomena, are *inextricably associated with the physical particularities of the actual world*. Therefore, in the context of axiomatization, it seems necessary to consider these links and to regard these, not as mathematical, but as *physical-mathematical* disciplines*, in which the physical bases play a no less significant role than the mathematical ones*.

1.4.3 How to Solve Hilbert's Sixth Problem

In many modern theories, physical objects and research subjects are replaced by abstract mathematical objects and their dependencies in mathematical models. This approach greatly facilitates the solution of physical tasks and provides a way to obtain solutions in general form. But at the same time, it breaks the connection with reality. As a result, *the possibility of apprehending the physical nature of the investigated phenomena is limited. The subject matter and scope of research are no longer real physical phenomena and physical laws, but the corresponding abstract mathematical models*.

For example, in classical probability theory as a mathematical discipline, *the subject of study is an abstract probability space* and the *scope of research is the mathematical relationships between its elements*. The physical phenomenon of statistical stability of the relative frequency of events which constitutes the foundation of this discipline would not then appear to play any role, although this is not so in reality, of course.

A more constructive approach to axiomatization of the physical disciplines is based on another principle within which *the subject matter is the actual physical world and the scope of research comprises methods for adequate description of the world*.

This concerns not only probability theory, but also other branches of physics. Now there are many different axiomatized mathematical theories that are useful for solving practical tasks. To use them correctly, it suffices to supplement the systems of mathematical axioms by physical assumptions (hypotheses) establishing the links between the abstract theories and the real world.

Apart from *consistency* and *independence*, the basic requirement for such physical hypotheses [*adequacy axioms* (Gorban 2011, 2014, 2017)] is *to take into account the physical effects of the ambient world* (which are experimentally verifiable) defining the scope of study, and to ensure that one has an *adequate description of these effects using the mathematical models* of the relevant theory. By adopting the corresponding physical hypotheses, the abstract *mathematical theory* is transformed into a *physical-mathematical theory* within which it is possible to give a logically correct description of reality.

1.5 Adequacy Axioms

1.5.1 Description of the Phenomenon of Statistical Stability in the Framework of Probability Theory

In order to treat probability theory as a *physical-mathematical* theory, Kolmogorov's system of *mathematical axioms*, which underlie its mathematical part, must be supplemented by *physical hypotheses*. In fact, the following physical hypotheses may be used as adequacy axioms (Gorban 2011, 2014, 2017):

Hypothesis 1 *For mass phenomena occurring in the real world, the relative frequency of an event has the property of ideal statistical stability, i.e., when the sample size increases, the relative frequency converges to a constant value.*

Hypothesis 2 *Mass phenomena are adequately described by random models which are exhaustively characterized by distribution functions.*

When solving practical tasks of a probabilistic nature these hypotheses are usually accepted implicitly as self-evident. Moreover, it is often assumed that the hypothesis of perfect statistical stability is valid for any physical mass phenomenon. In other words, a *random* (*stochastic*) *concept of world structure is accepted*.

> The subject matter of the mathematical part of probability theory is abstract random phenomena and the scope of research is links between these mathematical objects. The subject matter and scope of research of the physical part of probability theory, and also of probability theory as a physical-mathematical discipline, are correspondingly actual physical phenomena and methods for adequately describing them using random models.

1.5.2 Description of the Phenomenon of Statistical Stability in the Framework of the Theory of Hyper-random Phenomena

In Sect. 1.2, attention was drawn to the fact that the experimental study of real physical phenomena over broad observation intervals does not confirm the hypothesis of ideal statistical stability (Hypothesis 1). However, over less broad observation intervals, the incomplete compliance of this hypothesis with reality does not usually lead to significant losses, and the, application of probability theory is undoubtedly justified. Nevertheless, over longer observation intervals, the imperfect nature of statistical stability plays a significant role and it is impossible to ignore this fact.

For a correct application of classical probability theory in this case, it is sufficient in principle to replace Hypothesis 1 by the following (Gorban 2011, 2014, 2017):

Hypothesis 1′ *For real mass phenomena, the relative frequency of an event has the property of limited statistical stability, i.e., when the sample size increases, the relative frequency does not converge to a constant value.*

The replacement of Hypothesis 1 by Hypothesis 1′ leads to considerable mathematical difficulties due to the *violation of convergence*. There are different ways to overcome them. The development of one of these led to the *physical-mathematical theory of hyper-random phenomena* (Gorban 2007, 2011, 2014, 2016, 2017).

In classical probability theory, the basic mathematical entities are random events, random variables, and random functions, exhaustively characterized by distribution functions. In the theory of hyper-random phenomena, the analogues of these basic entities are *hyper-random events*, *hyper-random variables*, and *hyper-random functions*, which are sets of non-interconnected random events, random variables, and stochastic functions, respectively, each regarded as a comprehensive whole.

It is essential to understand that the hyper-random events, variables, and functions are *many-valued objects exhaustively characterized by sets of probability measures*.

For correct use of the theory of hyper-random phenomena, one must also adopt the following hypothesis, in addition to Hypothesis 1′ (Gorban 2011, 2014, 2017):

Hypothesis 2′ *Mass phenomena are adequately described by hyper-random models which are exhaustively characterized by sets of distribution functions.*

So *the mathematical part of the theory of hyper-random phenomena is based on the classical axioms of probability theory, and the physical part on Hypotheses 1′ and 2′.*

The assumption that these hypotheses are valid for a wide range of mass phenomena leads to *a world-building concept based on hyper-random principles*.

The subject matter of the mathematical part of the theory of hyper-random phenomena is abstract hyper-random phenomena and the scope of research is links between these mathematical objects. The subject matter and scope of research of the physical part of the theory of hyper-random phenomena and also the theory of hyper-random phenomena as the physical-mathematical discipline are correspondingly actual physical phenomena and methods for adequately describing them by hyper-random models.

The comparison of the subject matter and scope of research of probability theory and the theory of hyper-random phenomena show that the subject matter is the same for both theories, but the scope of the research is different.

Since the theory of hyper-random phenomena uses the system of mathematical axioms for probability theory, it is from the mathematical standpoint a *branch of classical probability theory*. But from the physical point of view, the theory of hyper-random phenomena is a *new physical theory based on new physical hypotheses*.

In general, *the theory of hyper-random phenomena can be regarded as a new physical-mathematical theory* constituting a complete solution of Hilbert's sixth problem in the context of probability theory.

1.6 Is Probability a 'Normal' Physical Quantity?

In the journal Physics-Uspekhi (Advances in Physical Sciences), the article (Alimov and Kravtsov 1992) was published with the intriguing title given now to this section. The authors drew attention to the fact that "an essential element, implicit in the physical interpretation of probability, is a system of hypotheses, agreements, and conjectures, formulated in a complicated way, that would naturally or traditionally be subsumed under the formal apparatus of probability theory, but which are indeed independent hypotheses requiring verification". In other words, without making other, more precise definitions, it is impossible to give a correct answer to this question.

According to the arguments presented above, the answer may in fact be obtained in the following way. First of all, note that the notion of *physical quantity* is a standardized one. According to the standard (GOST 1970), a physical quantity is a *feature* that is, in a qualitative respect, common to many physical objects (physical systems, their states, and the processes occurring in them), but in a quantitative respect, specific to each object.

From this point of view, *probability* as considered in the framework of axiomatic mathematical probability theory *is not* formally *a physical quantity*. It is instead *a mathematical abstraction that bears no relation to actual physical phenomena*.

If we adopt the additional Hypotheses 1 and 2, the notions of relative frequency limit and probability become equivalent. By measuring the relative frequency, it is thus possible, within a certain error, to estimate the probability. As the sample size tends to infinity, the error will tend to zero, and the relative frequency to its probability.

If a 'normal' physical quantity is understood as a physical quantity that can theoretically be measured with zero error at infinite sample size, then by accepting Hypotheses 1 and 2, the probability does indeed turn out to be a 'normal' physical quantity.

But since Hypothesis 1 is not confirmed experimentally, it must be replaced by Hypothesis $1'$. By accepting this new hypothesis, it is asserted that an event will have no well defined relative frequency limit. Therefore the abstract mathematical concept of the probability of an event cannot be identified with any physical quantity, and in this case, the concept of probability *has no physical interpretation*.

Of course, by measuring the relative frequency of an event, one obtains a rough estimate of its probability. However, since the error does not tend to zero when the sample size is increased, the *probability cannot be interpreted as a 'normal' physical quantity*.

Thus *the concept of probability is a mathematical abstraction that has no physical interpretation*.

Note that adopting Hypotheses 1 and 2 in probability theory amounts to assuming the existence of the probability as a number characterizing the possibility of an occurrence.

In the axiomatized variant of probability theory proposed in 1917 by *S. N. Bernstein*, we may read the following (Bernshtein 1934): "The basic assumption of probability theory (*the postulate of the existence of mathematical probability*) is that there are complex conditions β which (theoretically at least) can be recreated an unlimited number of times, under which *the occurrence of fact A* in the current experiment *has a certain probability expressed by a mathematical number*."

Other well-known variants of axiomatization, in particular the one proposed by R. von Mises in 1918 (Mises 1919, 1964) and a recognized alternative axiomatization proposed by A. N. Kolmogorov in 1929 (Kolmogorov 1929, 1956), are also based on this postulate.

In the theory of hyper-random phenomena, accepting the limited nature of statistical stability (in particular, the lack of convergence of the relative frequency) means *rejecting the postulate of the existence of probability*.

Thus, the *postulate of existence of the probability serves as a watershed separating the probability theory and the theory of hyper-random phenomena*.

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Part II
Probability Theory

Dice

Andrey Nikolaevich Kolmogorov, Russian mathematician, Academician of the Academy of Sciences of the USSR (1903–1987) (Source: http://www.kolmogorov.info/images/kolmogorov-bse.jpg)

A.N. Kolmogorov:

> The cognitive importance of probability theory comes from the fact that mass random phenomena in a comprehensive action create strict laws. The very concept of mathematical probability would be fruitless, if it were not realized in the form of a frequency of occurrence of some result under repetition of similar conditions. Therefore, the work of *Pascal* and *Fermat* can only be regarded as a background to probability theory, while its real history begins with the law of large numbers developed by J. Bernoulli and the normal approximation to the binomial distribution found shortly afterwards by *DeMoivre*. (Bernoulli 1986, p. 4).

A.N. Kolmogorov:

> Under certain conditions, which we will not discuss more deeply here, it can be assumed that for some of the events A that may or may not occur as a result of setting up complex conditions σ, there correspond certain real numbers $P(A)$ with the following properties:
>
> - One can be almost sure that if a set of complex conditions σ is repeated a large of number times n and m is the number of cases in which the event A occurs, then the ratio m/n will differ only slightly from the number $P(A)$.
> - If the number $P(A)$ is very small, one can be almost sure that, in a single realization of the conditions σ, the event A will not occur. (Kolmogorov 1974, p. 12, 13).

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Chapter 2
Basis of Probability Theory

Abstract We discuss the concept of a 'random event'. The classical and statistical approaches used to formalize the notion of probability are described, along with the basic concepts of set theory and measure theory. The Kolmogorov approach for axiomatizing probability theory is presented. The probability space is introduced. The axioms of probability theory are presented, together with the addition and multiplication theorems. The notion of a scalar random variable is formalized. We present ways to describe a random variable in terms of the distribution function, probability density function, and moments, including in particular, the expectation and variance. Examples of scalar random variables with different distribution laws are presented. Methods for describing a scalar random variable are generalized to a vector random variable. The transformation of random variables and arithmetic operations on them are briefly examined.

2.1 The Concept of Random Phenomena

There are many different interpretations of the concept of a random phenomenon. Currently, there is no consensus on this issue, even among scientists. In everyday language the term *random phenomenon* usually refers to a *mass phenomenon* for which the results of observation cannot be predicted. In this context, one distinguishes *unpredictable events, values, processes, and fields.*

A more exact interpretation of this concept leads to probability theory. A key role is played by the concept of an *event*, that is, an already occurred or possible result (outcome) of an experiment or test. Note that one sometimes sees a distinction between an experiment and a test. An *experiment* is any observation, while a *test* is an observation under *controlled* (or *partly controlled*) conditions (situations). However, in the following we shall use these words synonymously.

A random event is any event observed under partly controlled conditions and *characterized by a probability*. A random event is an event that *has not happened yet*, but which *can occur with a certain probability*. An event that *has already*

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occurred is not considered as random. This remark also applies to any other random phenomenon.

In *probability theory*, a *random phenomenon* is any *mass phenomenon* (event, variable, process, or field) observed under partially controlled conditions and having a *probability measure*. Mass phenomena that *do not have a probability measure* are *not considered as random*.

2.2 Options for the Definition of Probability

Probability is a concept that is not as trivial as it seems at first glance. There are many options for its definition, and even several mathematical options. The best known variants are the classical, statistical, and axiomatic definitions.

2.2.1 Classical Approach

Apparently, the classical approach is historically the first. It is based on the idea of the incompatibility and equal possibility of a *finite number* of elementary events. We need to explain what is meant by ‘elementary’, ‘incompatible’, and ‘equally possible’ events.

Among events there are events that make up other events. They are called the *elementary events*. We denote the set of all elementary events by Ω. An event A occurs when any elementary event ω belonging to some set Ω_A occurs. Concerning elementary events ω from the set Ω_A, one says that these events *favor* (*foster*) the event A.

Let us consider a simple example. In a standard pack of playing cards there are thirty-six cards of four suits (hearts, diamonds, clubs, and spades). Each suit contains nine cards of different denominations (six, seven, eight, nine, ten, jack, queen, king, and ace). The result of choosing one of the thirty-six cards can be considered as an elementary event ω. In this case, the number of elementary events equals thirty-six. These elementary events form the set Ω.

A set of random events is not confined only to these elementary events. A random *non-elementary event*, for example, would be the result of choosing cards of a certain suit from the pack, for example, hearts. The nine hearts cards favor this event A, forming a set Ω_A. The event A occurs when one chooses any card from this set.

Incompatible events are ones that cannot occur together in the experiment. An example of incompatible events would be to choose a card of two different suits or two different denominations. *Compatible* (*joint*) *events* are ones that can occur simultaneously in the experiment. An example of compatible events would be to choose a card of a definite suit and a definite denomination. These events can occur simultaneously in the experiment.

Equal possibility (*likelihood*) *of elementary events* means that none has priority over the others. So, in a well-shuffled pack of cards, the possibility of choosing any particular card is the same.

Another example of *incompatible elementary events* would be two numbers coming out together when tossing a dice with numbers from one to six on its sides. In the case of a symmetric (balanced) dice, the events consisting of particular numbers coming up are *equally possible elementary events*, whereas in an asymmetric dice, they are *not equally possible events*. An even (or odd) number in the roll of the dice is an example of a *non-elementary event*.

The *classical probability* $P(A)$ of an event A is the ratio of the number L_A of *incompatible equally possible elementary events* ω favorable to the event A to the total number L of elementary events: $P(A) = L_A/L$. In other words, in this case, the probability $P(A)$ of an event A is the ratio of the number L_A of elementary events belonging to the set Ω_A to the total number L of elementary events which belong to the set Ω.

In the example with cards, the total number of elementary events is $L = 36$, the number of elementary events favorable to choosing hearts (event A) is $L_A = 9$, and the probability of choosing hearts is $P(A) = 9/36 = 0.25$.

In the example with a perfectly balanced dice, the total number of elementary events is $L = 6$, the number of elementary events favorable to an even number is $L_A = 3$, and the probability of getting an even number is $P(A) = 3/6 = 0.5$.

The disadvantage with the classical definition of probability is the difficulty in selecting equally possible elementary events in asymmetric cases and the impossibility of extending this approach to the case of an infinite number of events.

2.2.2 Statistical Approach

A statistical approach avoids the difficulty in selecting equally possible elementary events. In this approach, the probability is determined on the basis of observation results. Such a probability is called statistical probability. The *statistical probability* $P(A)$ of a mass event A is a limit of the relative frequency $p_n(A)$ of the event when the number of tests n tends to infinity.

The statistical approach is of a physical nature and has a simple clear interpretation, and is thus widely used among physicists, engineers, and specialists in other applied fields. A zealous supporter of the statistical approach was *R. von Mises*, who proposed, on this basis, a variant axiomatization of probability theory (Mises 1919, 1964). Among mathematicians, the statistical approach has not found wide acceptance.

A significant disadvantage of this approach is that, in practice, the number of tests n is always limited, so it is in principle impossible to calculate the statistical probability simply by analyzing experimental results. One can only obtain a more or less accurate assessment.

When testing is carried out under the same statistical conditions, the accuracy of the assessment increases with the number of tests. However, this occurs only when the statistical experimental conditions remain unchanged. In real life, conditions do change, so it is impossible to achieve high accuracy when estimating the statistical probability.

It is possible to avoid these difficulties by defining the concept of probability without using statistical data. From a mathematical point of view, the most correct axiomatic definition of the notion of probability was suggested by *A. N. Kolmogorov* (1929, 1956, 1974). This approach is based on *set theory* and *measure theory*. Let us briefly consider the main concepts of these theories.

2.2.3 Main Concepts of Set Theory

A *set* is a collection of objects of an arbitrary nature. These objects are called *set members* or *set elements*. In mathematics, the concept of the set is the initial, *strictly undefinable* concept, *introduced axiomatically*.

A set can be assigned by listing its elements, for instance $a\,, b\,, \ldots\,, x$. In this case ones write $\{a, b, \ldots, x\}$. Often a set is assigned using some rule or contingency $R(x)$. Then one writes $\{x | R(x)\}$.

A *subset* A of the set Ω is a set all elements of which are elements of the set Ω. In this case, using *concatenation signs* $\subset$, $\supset$, we write $A \subset \Omega$ or $\Omega \supset A$. Note that the set Ω is itself a subset of the set Ω. If the set member x is contained in the set A, then using a *membership* (*adhesion*) *sign* $\in$, we write $x \in A$. By *definition*, the *empty set* $\varnothing$ is a subset of *any set*. When counting the number of elements of a set, the empty set is *not taken into account*.

In the example of cards, the thirty-six cards in a pack form the set Ω. The subsets of it include for example the set Ω, six cards of a given suit, four cards of a given denomination, and so on.

Two sets A and B are said to be *equal* if they consist of the same elements (i.e., $A \subset B$ and $B \subset A$). For example, the set A of red suit cards and the set B of non-black suit cards are equal. Note that the sets $\{a, b, c\}$ and $\{c, a, b, c\}$ are considered equal because both consist of the same elements $a\,, b\,, c$.

If the number of set members is finite, then the set is said to be finite, while if the number is infinite, the set is said to be *infinite*. In the latter case, if the number of set members can be counted, then the set is said to be *countable*, and if the number of set members cannot be counted, the set is said to be *uncountable*. A *discrete set* is a set consisting of a finite or countable number of elements.

To characterize the number of elements of a set, the concept of *cardinal number* is used. *Two sets* have the same *cardinal number* if a *one-to-one correspondence*, can be established among all their members, i.e., to every element of one set we can associate a unique element of the other set, and vice versa.

The cardinal number of a finite set is equal to the number of its elements. The *cardinal numbers of all countable sets* are the same (denoted by $\aleph_0$). The *cardinal number of all real numbers* is called the *continuum* (denoted by $\aleph$). Note that there are uncountable sets whose cardinal numbers are greater than the continuum.

Here are a few examples. An example of a finite set is a set whose members are the numeral 1, the number π, some geometric figure (for instance, a triangle), and the sign $*$. This set consists of four elements, so its cardinal number equals four. The sets of all integer numbers (from minus infinity to plus infinity) and positive integer numbers (from one to plus infinity) are countable sets. From the standpoint of set theory, the numbers of elements in these sets are identical. The cardinal numbers of these sets are $\aleph_0$.

The cardinal numbers of points on the segments of any finite length, on an infinite line, and on the plane are the same. The cardinal numbers of all these sets equal to the continuum. These strange and, at first glance, unexpected statements are easy to prove. Let us show, for example, the statement that on two segments A_1B_1 and A_2B_2 of *different lengths* the numbers of points are *equal*.

To do this, we arrange these segments in parallel one above the other (Fig. 2.1) and draw the lines A_1A_2 and B_1B_2. Their point of intersection is denoted by the letter O.

Then an arbitrary line OC_1 passing through the point O and intersecting the segment A_1B_1 intersects the segment A_2B_2, and conversely, an arbitrary line OC_2 passing through the point O and intersecting the segment A_2B_2 intersects the segment A_1B_1. Thus, we find a one-to-one correspondence between the points C_1 and C_2 of the segments A_1B_1 and A_2B_2. If there is such a correspondence, the number of points on the segments are equal. Other parts of the above claim are proven similarly.

We now introduce the *union* (*sum* (*logical addition*)) $\cup$, *intersection* (*logical multiplication*) $\cap$, and *complement* $\setminus$ *operations*. The union $A_1 \cup A_2$ of sets A_1 and A_2 is a set A that consists of all elements included in at least one of the sets A_1 and A_2 (Fig. 2.2). The intersection $A_1 \cap A_2$ of sets A_1 and A_2 is a set B containing precisely those elements included simultaneously in both A_1 and A_2. The complement $\Omega \setminus A$ of

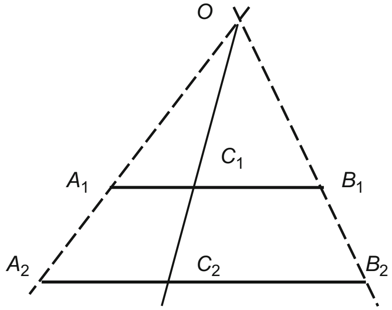

Fig. 2.1 One-to-one correspondence between the points C_1 and C_2 of the segments A_1B_1 and A_2B_2

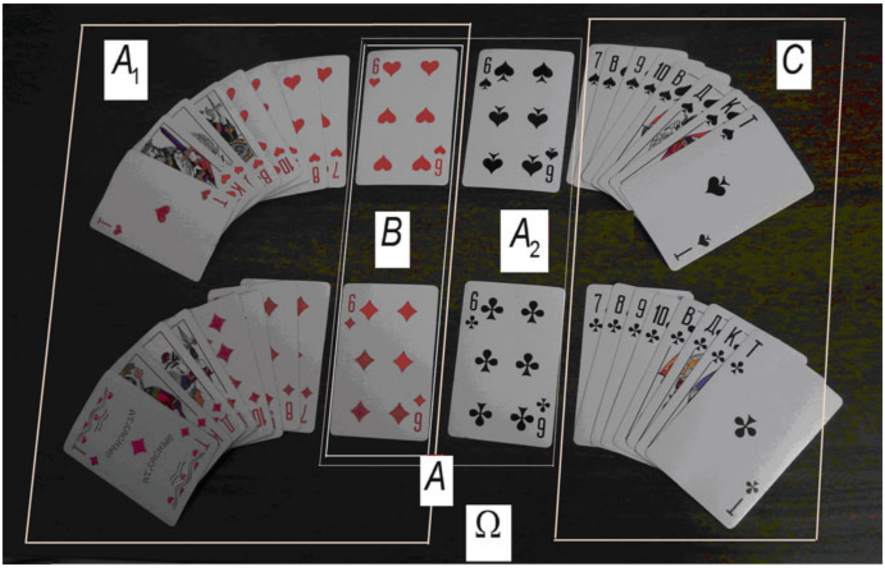

Fig. 2.2 Illustration of the union, intersection, and complement operations in the card example

the set $A \subset \Omega$ is the set $C = \bar{A}$ containing precisely those elements of the set Ω that do not belong to the set A.

For example, the union $A_1 \cup A_2$ of eighteen cards of red suit A_1 and four sixes A_2 (Fig. 2.2) forms the set A of twenty cards, namely, eighteen cards of red suit and two sixes of black suit. The intersection $A_1 \cap A_2$ of eighteen cards of red suit A_1 and four sixes A_2 forms a set B that consists of two sixes of red suit. The complement $\Omega \backslash A = C$ is a set of sixteen cards containing all cards of black suit, except two sixes.

Sets A_1 and A_2 are *disjoint* (*mutually exclusive*) if their intersection is the empty set: $A_1 \cap A_2 = \varnothing$. An *algebra of sets* is a class of subsets $\Im$ of the set Ω with the operations of *union, intersection, and complement* such that the following three conditions are satisfied:

(1) The set Ω belongs to the set $\Im$, i.e., $\Omega \in \Im$;
(2) If a set $A \in \Im$, then its complement also belongs to the set $\Im$, i.e., $\bar{A} \in \Im$;
(3) If sets A_1 and A_2 belong to the set $\Im$ ($A_1 \in \Im$ и $A_2 \in \Im$), their union and intersection also belong to the set $\Im$, i.e., $A_1 \cup A_2 \in \Im$ and $A_1 \cap A_2 \in \Im$.

Thus, in an algebra of sets, the operations of union, intersection, and complement *do not take one outside the set* $\Im$, i.e., the set $\Im$ is *closed* under these operations.

In the card example the set $\Im$ can be formed from any combinations of cards.

A set Ω may be either finite (as is the case of the cards) or countable. If Ω is countable, the set $\Im$ is also countable.

In addition to properties (1–3), a *countable set* $\Im$ may have the following property:

(4) If sets $A_n \in \Im$ for all $n = 1, 2, \ldots,$ then the union of a *countable number* of these sets also belongs to $\Im$, i.e., $\cup_{n=1}^{\infty} A_n \in \Im$.

Then the algebra is called a *σ-algebra* (*sigma-algebra*). Note that the fourth property, which plays an important role in measure theory and probability theory, *is not a consequence* of the third property.

2.2.4 Main Concepts of Measure Theory

A measure is a mathematical concept that generalizes the notion of length, area, volume, mass, energy, and other quantities that characterize a subset of a set *quantitatively*. A *measure* $\mu(A)$ is a function of the subset A of a non-empty set Ω which has the following properties:

(1) The measure $\mu(A)$ has a non-negative value for all sets A on which it is defined, i.e., $\mu(A) \geq 0$. For the empty set $\varnothing$, the measure equals zero, i.e., $\mu(\varnothing) = 0$;
(2) The measure of the set formed by union of a *countable number* of *disjoint subsets* A_n $(n = 1, 2, \ldots)$ *is equal to the sum* of the measures of the subsets A_n, i.e., $\mu\left(\bigcup\limits_n A_n\right) = \sum\limits_n \mu(A_n)$.

The latter condition is called the *axiom of countable additivity*.

For any measure μ, we require that the measure be specified *on all subsets of a σ-algebra* (i.e., for all elements of the set $\Im$). If this condition is satisfied, we say that the triple $(\Omega, \Im, \mu)$ forms a *space with measure*.

The measure is said to be *normalized* if the measure of the set Ω is equal to unity (i.e., $\mu(\Omega) = 1$). For a normalized measure, $\mu(A) \leq 1$ for all sets $A \in \Im$.

2.2.5 Axiomatic Definition of Probability

In the *axiomatic definition of probability*

(1) There is a space Ω of elementary events $\omega \in \Omega$;
(2) There is a *σ-algebra* $\Im$[1] of subsets, called *events*;
(3) For any *event* A, there is a normalized measure $P(A)$, called a *probability measure*, or simply a *probability*.

[1]More correctly, the smallest σ-algebra, called the *Borel σ-algebra*.

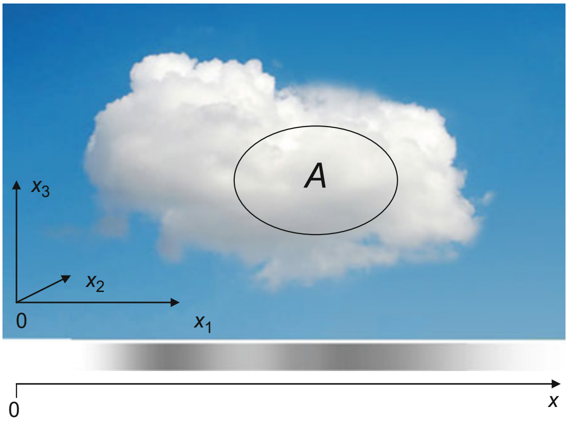

Fig. 2.3 Figurative representation of a probability space by a cloud

A space with a probability measure is called a *probability space*. Like any measure space, the probability space is given by a triple $(\Omega, \Im, P)$. The probability space $(\Omega, \Im, P)$ can be represented figuratively by a cloud consisting of a countable set Ω of not necessarily identical water drops (Fig. 2.3).

Every elementary event ω is represented by a water drop, the set Ω by the cloud, and the σ-algebra $\Im$ by the set of all subsets formed from cloud droplets. The probability $P(A)$ of the subset A corresponds to the mass of water in the relevant part of the cloud, divided by the total mass of water in the cloud.

A *persistent* (*certain*) *event* I is an event taking place for all the elementary events $\omega \in \Omega$. An *impossible event* is an event that does not occur for any $\omega \in \Omega$.

Events A_1 and A_2 are *disjoint* (*mutually exclusive*) if their intersection is the empty set, i.e., $A_1 \cap A_2 = \varnothing$. Events $A_1, A_2, \ldots, A_I$ are *pairwise disjoint* if the intersection of any two different events is the empty set, i.e., if $A_i \cap A_j = \varnothing$ for any $i \neq j$ $\left(i, j = \overline{1, I}\right)$.

The *probability* $P(A)$ is defined as a normalized measure by the following three *axioms*:

(1) The probability of any event A is a non-negative number, i.e., $P(A) \geq 0$;
(2) For pairwise disjoint events $A_1, A_2, \ldots$ (both finite and *countable*), the probability of their union is the sum of the probabilities of the events, i.e., $P\left(\bigcup_n A_n\right) = \sum_n P(A_n)$;
(3) The probability of the event Ω is equal to unity (i.e., $P(\Omega) = 1$).

2.2.6 Random Events

Events that satisfy the axioms of Sect. 2.2.5 are called *random events*. It follows from these axioms that the probability of any random event lies in the range from zero to unity (i.e., $0 \leq P(A) \leq 1$), and the probability of an empty event is equal to zero (i.e., $P(\varnothing) = 0$). Note that if the probability of an event A is unity ($P(A) = 1$), it *does not generally follow* that this event is persistent, and if the probability of an event B is zero ($P(B) = 0$), it *does not follow* that it is an impossible event.

For general joint events A_1 and A_2, the probability $P(A_1 \cap A_2)$ of the intersection $A_1 \cap A_2$ is equal to the product of the probability $P(A_1)$ of the event A_1 and the conditional probability $P(A_2/A_1)$ of the event A_2 under the condition that event A_1 has occurred[2]:

$$P(A_1 \cap A_2) = P(A_1)P(A_2/A_1). \tag{2.1}$$

This statement is known as the *multiplication theorem*.

Let us return to the card example (Fig. 2.2). Assume that the probability of choosing a certain card from the pack does not depend on its suit and denomination. Then this probability is 1/36. Let event A_1 be the choice of any card of red suit and event A_2 the choice of a six. Then the event A_2/A_1 is the choice of a six from a half pack containing only red suit cards, and the event $A_1 \cap A_2 = B$ is the choice of a six of red suit. In this case $P(A_1) = 18 \times (1/36) = 1/2$, $P(A_2) = 4 \times (1/36) = 1/9$, and $P(A_2/A_1) = 2 \times (1/18) = 1/9$. According to (2.1), the probability of choosing a six of red suit is $P(B) = P(A_1 \cap A_2) = (1/2) \times (1/9) = 1/18$.

Random events A_1 and A_2 are said to be *independent* (*independent in probability*) if the probability of their intersection equals the product of their probabilities, i.e., $P(A_1 \cap A_2) = P(A_1)P(A_2)$. It follows from this definition that events A_1 and A_2 are independent if the appearance of one of them does not cause changes in the probability of occurrence of the other.

Let A_1 and A_2 be possibly *dependent events*. Then the probability of the sum of these events is the sum of the probabilities of the events A_1 and A_2, minus the probability of their intersection:

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2). \tag{2.2}$$

This easily proved formula is known as the *summing theorem*. Continuing with the card example, the probability of choosing a red card or any six $P(A_1 \cup A_2)$ can be calculated by (2.2): $P(A_1 \cup A_2) = 1/2 + 1/9 - 1/18 = 5/9$.

Using the simple rules above, one can solve not only trivial tasks such as those described above, but also some very difficult combinatorial problems.

[2]It is assumed that $P(A_1) \neq 0$. Otherwise, the probability $P(A_2/A_1)$ is not determined.

2.3 Random Variables

2.3.1 Basic Definitions

A *random variable* X is a measurable function defined on the space Ω of elementary events ω. The *value of a random variable* X can be represented by the function $x = \psi(\omega)$, where $\omega \in \Omega$. The set of values of the random variable forms the *value space of the random variable* . Before a test, it is *not known* what kind of elementary event will occur and therefore it is *not known* what value x a random variable X will take.

A random variable is given not only by the space of values of the random variable, but also characteristics which characterize the probabilities of these values. If a random variable is defined as the space of random values and the probabilities of these values, then we say that the random variable is *probabilistically defined*.

Random variables may be *scalar* or *vector*. Returning to the 'cloud' representation of the probability space (Fig. 2.3), a three-dimensional random variable $\vec{X} = (X_1, X_2, X_3)$ can be thought of as a random three-dimensional vector whose specific components x_1, x_2, x_3 describe the location of a particular water droplet in space, while a scalar random variable X is thought of as a scalar magnitude, describing the location of a particular water droplet on the axis x.

In the following, random variables will be denoted by *capital letters*, and their values by *small letters*. Random variables are described by different characteristics. The most complete description provides probability characteristics and a less complete description numerical characteristics (parameters).

2.3.2 Probabilistic Characteristics of a Scalar Random Variable

A comprehensive description of a scalar random variable X gives the *distribution function* (*cumulative distribution function*)[3] $F_x(x)$ representing the probability that a random variable is less than x, i.e.,

$$F_x(x) = P\{X < x\}. \tag{2.3}$$

Examples of distribution functions of a *discrete random variable* X (taking discrete values x_i, $i = \overline{1, I}$) and continuous random variables are shown in Figs. 2.4a and 2.5a.

[3]If it is clear from the text which random variable the distribution function concerns, the subscript on the symbol is often omitted.

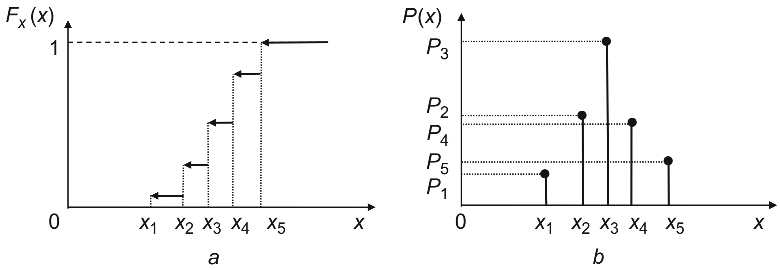

Fig. 2.4 The distribution function (**a**) and the probability mass function (**b**) of a discrete random variable

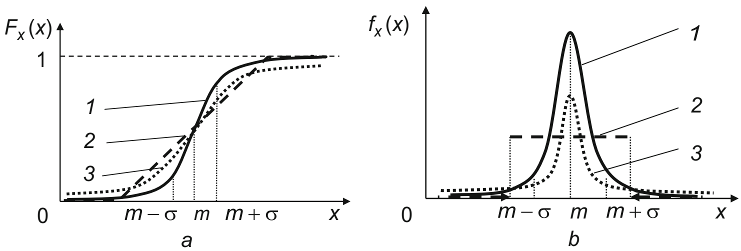

Fig. 2.5 Distribution functions (**a**) and probability densities (**b**) of continuous random variables: Gaussian distributed (*1*), uniformly distributed (*2*), and Cauchy distributed (*3*)

Each random variable is *uniquely described* by its distribution function $F_x(x)$. However, to a fixed distribution function there correspond in general a number of random variables.

The distribution function $F_x(x)$ takes *non-negative values*, lying in the range from zero to unity. This function is non-decreasing and continuous on the left (see Figs. 2.4a and 2.5a).

A random variable is *continuous* if its distribution function is continues and has piecewise continuous derivative. Comprehensive description of such random variable also gives the *probability density function* (*probability distribution*)[4] $f_x(x)$ (Fig. 2.5b). The probability distribution $f_x(x)$ and the distribution function $F_x(x)$ are *uniquely related* by

$$f_x(x) = \frac{dF_x(x)}{dx}, \tag{2.4}$$

$$F_x(x) = \int_{-\infty}^{x} f_x(x)\, dx. \tag{2.5}$$

[4] If it is clear from the text which random variable the distribution function concerns, the subscript on the symbol is often omitted.

Like the distribution function, the probability distribution takes only non-negative values. The area under the entire curve $f_x(x)$ equals unity (*normalization to unity*). The probability that a random variable takes a value in the interval $[x_1,\ x_2]$ is equals to the area under the curve $f_x(x)$ corresponding to this interval. This property is called the *additive property of the probability distribution* .

In the 'cloud' interpretation (Fig. 2.3), the probability distribution of a random variable X can be understood as the mass density distribution of water along the axis x. This distribution is imaged in the bottom part of the figure by a strip of showing the different densities on a grey scale.

Note that not all mathematical functions can be distribution functions or probability density functions. Those functions that have the properties listed above can be distribution functions and probability density functions of random variables, and only those.

2.3.3 Probabilistic Characteristics of a Discrete Random Variable

Since the distribution function of a discrete random variable is discontinuous, strictly speaking, a discrete random variable does not have a probability density function in the classical framework. The role of the probability density function in this case is played by the *probability mass function* (*probability distribution*) P_i, whose domain is all the x_i, $i = \overline{1, I}$ (Fig. 2.4b). Note that $\sum_{i=1}^{I} P_i = 1$. Using the generalized *Dirac* delta function for a discrete random variable, we can introduce the concept of probability density function.

The *Dirac delta function* (*delta function*) $\delta(x - x_0)$ is a function that possesses an infinite value at the point $x = x_0$ and zero at all other points on the real axis (Fig. 2.6a). Formally, it is a derivative of the unit step function at the point $x = x_0$ (Fig. 2.6b):

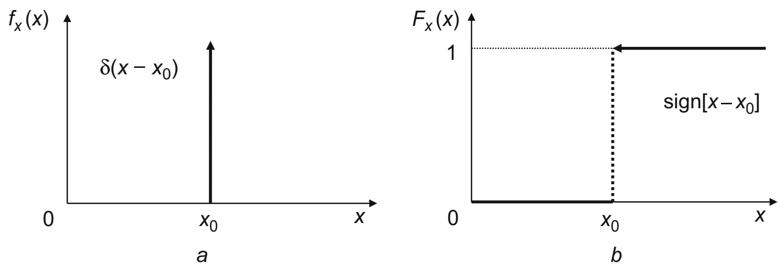

Fig. 2.6 Dirac delta function (**a**) and unit step function (**b**)

$$\operatorname{sign}[x - x_0] = \begin{cases} 0 & \text{if} \quad x \le x_0, \\ 1 & \text{if} \quad x > x_0. \end{cases} \tag{2.6}$$

An important property of the δ-function is that, for any function $\psi(x)$ whose domain is the set of all real numbers, we have

$$\int_{-\infty}^{\infty} \psi(x)\delta(x - x_0)\, dx = \psi(x_0).$$

The probability distribution of a discrete random variable taking the values $x_1, \ldots, x_I$ can be represented by the expression

$$f_x(x) = \left\{P_i \ \delta(x - x_i) | i = \overline{1, I}\right\}.$$

The use of the δ-function not only opens up the possibility of a unified description of discrete and continuous random variables, but also provides a way to describe determinate and random variables from a common position, namely viewing a determinate value x_0 *approximately* as a random variable X whose probability distribution $f_x(x)$ is a δ-function at the point x_0 (Fig. 2.6a) and whose distribution function $F_x(x)$ is the unit step function at this point (Fig. 2.6b).

2.3.4 Examples of Random Variables

Hundreds of different types of distributions have been investigated (Gubarev 1981, 1992; Muller et al. 1979), but in practice, as a rule, only a few of them are actually used. The most frequently used is the *Gaussian* (*normal*) *distribution*, whose probability density function $f_x(x)$ is described by a symmetric bell-shaped curve (continuous curve 1 in Fig. 2.5b):

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left[-\frac{(x - m_x)^2}{2\sigma_x^2}\right], \ \sigma_x > 0. \tag{2.7}$$

This is defined by two parameters m_x and σ_x. The parameter m_x characterizes the position of the peak of the curve on the axis, and the parameter σ_x the width of the bell and its maximum value: the greater the value of σ_x, the wider the bell and the smaller its maximum. Note that the parameter m_x is the *expectation* and the parameter σ_x the *standard deviation*.[5]

The Gaussian distribution function $F_x(x)$ cannot be expressed in terms of elementary functions. However, it can be calculated using the tabulated function called *Laplace function* or *probability integral*:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-z^2/2\right) dz, \tag{2.8}$$

which describes the distribution when $m_x = 0$ and $\sigma_x = 1$.

The expectation m_x characterizes the position of the curve $F_x(x)$ on the x axis and the standard deviation σ_x its slope: the smaller the standard deviation, the steeper the curve of the distribution function.

A *uniformly distributed* probability distribution $f_x(x)$ is often used. This is constant on an interval $[a, b]$ and zero outside this interval (dashed line 2 in Fig. 2.5b):

$$f_x(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } \; x \in [a, b], \\ 0 & \text{if } \; x \notin [a, b]. \end{cases}$$

The distribution function $F_x(x)$ of such a random variable is a linear function within the interval $[a, b]$, increasing from zero to unity (dotted line 2 in Fig. 2.5a). The slope of this line segment is determined by the *range of the random variable*, that is, the difference between the upper b and lower a boundaries of its distribution. The smaller the range, the steeper the line.

The *Cauchy distribution* (*Student's*[6] *distribution of the first order*) has interesting properties. We shall have more to say about these properties. But for now we note only that the distribution function $F_x(x)$ (curve 3 in Fig. 2.5a) and probability distribution $f_x(x)$ (curve 3 in Fig. 2.5b) of such a random variable are defined by two parameters, viz., the shift parameter x_0 and the scale parameter $\gamma > 0$:

$$F_x(x) = \frac{1}{\pi} \operatorname{arctg}\left(\frac{x - x_0}{\gamma}\right) + \frac{1}{2}, \tag{2.9}$$

$$f_x(x) = C[x_0, \gamma] = \frac{1}{\pi}\left[\frac{\gamma}{(x - x_0)^2 + \gamma^2}\right]. \tag{2.10}$$

These parameters in the Cauchy distribution play the same role as the expectation m_x and the standard deviation σ_x in the Gaussian distribution.

[5] These concepts are defined in the next subsection.

[6] *Student* is an alias of *W. S. Gosset*.

2.3.5 Numerical Parameters of Scalar Random Variables

In addition to the probabilistic characteristics, different *numerical characteristics* (*parameters*) can be used to provide an image of the distribution and, if necessary, to approximate it by other distributions. The best known among these are the *moments*, in particular the *expectation* m_x (the *moment of the first order*), the *variance* D_x (the *central moment of the second order*), and other parameters of the random variable.

The moments are defined by the concept of *mathematical expectation of a function*. By the expectation $\mathrm{E}[\varphi(X)]$ of the function $\varphi(X)$ of a random variable X we mean the average (taking into account the probability distribution $f_x(x)$) of the function $\varphi(X)$:

$$\mathrm{E}[\varphi(X)] = \int_{-\infty}^{\infty} \varphi(x) f_x(x) dx. \tag{2.11}$$

Note that the expectation of a function is a *functional*, the value of which is *determinate*, and $\mathrm{E}[\cdot]$ in (2.11) is the *expectation operator* that acts on the function $\varphi(X)$.

Let us recall the concepts of *function, functional,* and *operator*. A rule assigning to each object x of a class A an object y of a class B is called a transformation (mapping) of class A to class B. The class A is the *applicable domain* and the class B the *actual range*.

If classes A and B are sets of numbers, then y is called the *function* of argument x, and if class A is a set of functions and class B is a set of numbers, then y is called a *functional*. Finally, if both classes A and B represent a set of functions, the transformation is called an *operator*. The most general concept is an operator. The concepts of function and functional are special cases of operators. The functional can be interpreted as a 'function of a function'.

The *expectation* m_x *of a random variable* X is the expectation of the function $\varphi(X) = X$. In this way, $m_x = \mathrm{E}[X]$. The *variance* D_x *of a random variable* X is the mathematical expectation of the function $\varphi(X) = (X - m_x)^2$, centered on the expectation m_x, i.e., $D_x = \mathrm{E}[(X - m_x)^2] = \mathrm{Var}[X]$, where $\mathrm{Var}[X]$ is the variance operator. The *standard deviation* (*SD*) σ_x *of a random variable* X is the square root of the variance D_x, i.e., $\sigma_x = \sqrt{D_x}$.

The expectation m_x of a random variable X with the probability density $f_x(x)$ characterizes the *average value* of the variable, just as the variance D_x and standard deviation σ_x characterize the *scattering* of the random variable around the expectation m_x.

Sometimes, instead of the expectation and standard deviation one uses so-called *robust parameters*, such as the median and median absolute deviation. The *median* $e_x = \mathrm{med}[X]$ of the random variable X is that value of the variable that divides the area under the probability density function $f_x(x)$ in half, i.e., it is a solution of the equation

$$F_x(x) = 0.5.$$

If the distribution function is strictly increasing, this equation has a unique solution; otherwise it has a set of solutions. If the distribution is discrete, then one usually takes the *average of the two middle values* as the median.

The *median absolute deviation* is the median of the modulus of the random variable X relative to the median e_x:

$$s_x = \mathrm{med}\left[|X - e_x|\right].$$

Note that there are many other useful parameters for the description of random variables, such as the *crude* and *central moments of order* ν defined by $m_\nu = \mathrm{E}[X^\nu]$ and $\mu_\nu = \mathrm{E}[(X - m_x)^\nu]$, respectively, the *coefficients of asymmetry*, *excess*, *variations*, etc., *quantiles*, *cumulants*, and others. They will not be used in the following, so we shall not dwell further on them.

2.3.6 Numerical Parameters of Various Random Variables

It follows from (2.7) that the Gaussian distribution is entirely determined by the expectation m_x and standard deviation σ_x (or variance D_x). The *odd central moments of this distribution are equal to zero, and the even ones are expressed in terms of the variance* D_x. The expectation of the *uniform distribution* is $m_x = (a + b)/2$, and the variance is $D_x = (b - a)^2/12$. A random variable that approximates the *determinate value* x_0 has expectation m_x equal to x_0 and variance D_x equal to zero.

When the probability density function has *one maximum* (*unimodal*) the expectation and standard deviation are *particularly informative*. Let us consider an example. Gaussian random models often give an adequate description of actual measurement results of different physical quantities. The Gaussian distribution is unimodal. In this case, the estimate $m_x^* = y_n$ of the expectation m_x calculated by

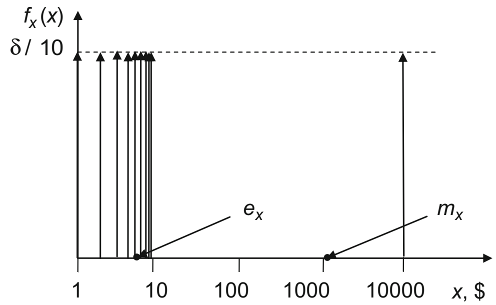

Fig. 2.7 Probability density function of the bottle prices

averaging measurement results $x_1, \ldots, x_n$ [see (1.1)] is close to the measurand and the estimate $\sigma_x^* = z_n$ of the standard deviation σ_x described by (1.2) adequately characterizes the dispersion of the measurement results.

However, if the probability density function is not unimodal, these parameters can be *spurious*. Let us consider an example which, although not typical for probability theory, will allow us to illustrate the possibility of applying the theory to solve, not only probabilistic, but also determinate tasks, and also to demonstrate the usefulness of the robust parameters.

Imagine a grocer's that sells nine sorts of ordinary wine, with prices of \$1, . . . ,\$9 a bottle and also a collection wine, priced at \$10,000 a bottle. Let us ask what is the average price of a bottle. Note that the price is a *determinate value* and the concept of average price is *not specified*, so the average can be regarded, for example, as the arithmetic mean, geometric mean, median, etc.

Solving the task by the methods of probability theory, the prices of the bottles can be described by a probability density function, represented by the ten δ-functions normalized by ten (Fig. 2.7). Calculation of the parameters gives the following results: the expectation is $m_x = \$1004.5$, the standard deviation $\sigma_x = \$3160.7$, the median $e_x = \$5.5$, and the median absolute deviation $s_x = \$2.5$.

Obviously, in this example, the expectation m_x and standard deviation σ_x are completely uninformative. At the same time, the robust parameters e_x and s_x have a clear interpretation: the median e_x characterizes the average price of the bottles which are in the middle of the series of prices (the middle of \$5 and \$6), and the median absolute deviation s_x characterizes the variation in prices in the region of this middle value. Note that, if the price of the collection wine bottle exceeds \$9, the median value and median absolute deviation are independent of this price.

Note also that *not all distributions have moments*. For instance, the Cauchy distribution does not have any.[7] If the distribution has moments, the moments of the first two orders, i.e., the expectation and variance (or standard deviation), are often used instead of the distribution function and probability density function. If the distribution has no moments or, as in the latter example, they are not informative, the robust parameters are used.

2.4 Vector Random Variables

The statements of the previous section can be generalized to *vector random variables* (*system of random variables*). For simplicity, we begin our considerations with a two-dimensional variable represented by a column vector $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ with scalar random components X_1 and X_2 (system $\{X_1, X_2\}$ of scalar random variables), where T is the *transpose operator*.

[7] More correctly, for the Cauchy distribution, there is the principal value integral, which describes the first moment (first order moment). The value of this integral is x_0.

2.4.1 Probabilistic Characteristics of a System of Two Random Variables

To describe a two-dimensional random variable $\vec{X} = (X_1, X_2)^{\mathrm{T}}$, we use a *two-dimensional distribution function* (*cumulative distribution function*) $F_{\vec{x}}(x_1, x_2)$ representing the probability of the inequalities $\{X_1 < x_1,\ X_2 < x_2\}$:

$$F_{\vec{x}}(x_1, x_2) = P\{X_1 < x_1, X_2 < x_2\}, \tag{2.12}$$

and associated with it the *two-dimensional probability density function* $f_{\vec{x}}(x_1, x_2)$:

$$\frac{f_{\vec{x}}(x_1, x_2) = \partial^2 F_{\vec{x}}(x_1, x_2)}{\partial x_1 \partial x_2.} \tag{2.13}$$

When solving practical tasks one often applies the conditional probability. The *conditional* (*marginal*) *distribution function of a random variable* X_2 *with the proviso* $X_1 = x_1$ is a *one-dimensional* distribution function $F_{x_2/x_1}(x)$ defined with the proviso that the random variable X_1 has adopted a specific value x_1.

According to (2.1), the probability distribution $f_{\vec{x}}(x_1, x_2)$ of a two-dimensional random variable $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ is equal to the product of the probability distribution $f_{x_1}(x_1)$ of the random variable X_1 and the conditional probability distribution $f_{x_2/x_1}(x_2)$ of the random variable X_2:

$$f_{\vec{x}}(x_1, x_2) = f_{x_1}(x_1) f_{x_2/x_1}(x_2). \tag{2.14}$$

Random variables X_1 and X_2 are said to be *independent* if their joint probability distribution $f_{\vec{x}}(x_1, x_2)$ is equal to the product of the one-dimensional probability distributions $f_{x_1}(x_1)$ and $f_{x_2}(x_2)$ of the variables X_1 and X_2, respectively:

$$f_{\vec{x}}(x_1, x_2) = f_{x_1}(x_1) f_{x_2}(x_2), \tag{2.15}$$

i.e., the conditional probability distribution $f_{x_2/x_1}(x_2)$ is equal to the unconditional probability distribution $f_{x_2}(x_2)$.

For independent random variables X_1 and X_2, and only for these, the two-dimensional distribution function $F_{\vec{x}}(x_1, x_2)$ equals the product of the one-dimensional distribution functions $F_{x_1}(x_1)$ and $F_{x_2}(x_2)$:

$$F_{\vec{x}}(x_1, x_2) = F_{x_1}(x_1) F_{x_2}(x_2). \tag{2.16}$$

The independence of random variables X_1 and X_2 does not mean that these variables are in no way related. They may be linked, but on the probability level, their relationship does not become apparent.

2.4.2 Numerical Parameters of a System of Two Random Variables

The *expectation* $\mathrm{E}[\varphi(X_1, X_2)]$ *of a function* $\varphi(X_1, X_2)$ *of random variables* X_1 and X_2 that have joint probability distribution $f_{\vec{x}}(x_1, x_2)$ is the average of the function $\varphi(X_1, X_2)$:

$$\mathrm{E}\left[\varphi(X_1, X_2))\right] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \varphi(x_1, x_2) f_{\vec{x}}(x_1, x_2)\, dx_1 dx_2. \tag{2.17}$$

The *mathematical expectation* $\vec{m}_{\vec{x}} = \left(m_{x_1}, m_{x_2}\right)^{\mathrm{T}}$ *of a two-dimensional random variable* $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ is a vector, the components m_{x_1} and m_{x_2} of which are the expectations of the random variables X_1 and X_2, i.e., $m_{x_1} = \mathrm{E}[X_1]$ and $m_{x_2} = \mathrm{E}[X_2]$.

The *variance* $\vec{D}_{\vec{x}} = \left(D_{x_1}, D_{x_2}\right)^{\mathrm{T}} = \mathrm{D}\left[\vec{X}\right]$ *of a two-dimensional random variable* $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ is a vector whose components D_{x_1} and D_{x_2} are equal to the variances of the random variables X_1 and X_2.

The *standard deviation* $\vec{\sigma}_{\vec{x}} = \left(\sigma_{x_1}, \sigma_{x_2}\right)^{\mathrm{T}}$ *of a two-dimensional random variable* $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ is a vector whose components σ_{x_1} and σ_{x_2} are equal to the square root of the corresponding variances D_{x_1} and D_{x_2} of the random variables X_1 and X_2.

The expectation vector $\vec{m}_x$ characterizes the average value of the vector $\vec{X}$. The variance vector $\vec{D}_x$ and the standard deviation vector $\vec{\sigma}_{\vec{x}}$ characterize the scattering of the components of the vector $\vec{X}$ along the x_1 and x_2 axes relative to the corresponding components of the vector $\vec{m}_x$.

The relationship between the random variables X_1 and X_2 is characterized by the *product correlation* (*crude second order*) *moment*

$$K_{x_1 x_2} = \mathrm{E}[X_1 X_2]$$

and the *product covariance* (*central second order*) *moment*

$$R_{x_1 x_2} = \mathrm{Cov}[X_1, X_2] = \mathrm{E}[(X_1 - m_{x_1})(X_2 - m_{x_2})].$$

The product covariance moment $R_{x_1 x_2}$ normalized by the standard deviations σ_{x_1} and σ_{x_2} is called the *correlation coefficient*:

$$r_{x_1 x_2} = \frac{R_{x_1 x_2}}{\sigma_{x_1}\sigma_{x_2}}. \tag{2.18}$$

The product covariance moment $R_{x_1 x_2}$ is associated with the product correlation moment and the expectations m_{x_1} and m_{x_2} by the simple relation

$$R_{x_1 x_2} = K_{x_1 x_2} - m_{x_1} m_{x_2}. \tag{2.19}$$

The product covariance moment $R_{x_1x_2}$ characterizes the linear and only the linear relationship between the given random variables. If there is no linear connection, the product covariance moment $R_{x_1x_2}$ and the correlation coefficient $r_{x_1x_2}$ are both equal to zero. In this case, the random variables X_1 and X_2 are said to be *uncorrelated* (*linearly independent*).

Note that the noncorrelatedness and independence of random variables are *different concepts*. The independence of random variables *implies* their noncorrelatedness. However, in general, noncorrelatedness *does not imply* independence. These concepts differ when there is a *nonlinear relationship* between the random variables. The concepts of independence and noncorrelatedness coincide only in special cases, such as when the random variables X_1 and X_2 are jointly Gaussian.

The random variables X_1 and X_2 are said to be *orthogonal* if the product correlation moment $K_{x_1x_2}$ equals zero. The concepts of noncorrelatedness and orthogonality are *different notions*. But if at least one of the expectations m_{x_1}, m_{x_2} equals zero, then the orthogonality of the random variables implies their noncorrelatedness and noncorrelatedness implies their orthogonality [see (2.19)].

2.4.3 System of Two Jointly Gaussian Random Variables

As an example of a system of two random variables we consider a *system of two jointly Gaussian random variables* described by a *two-dimensional Gaussian* (*normal*) *distribution law*. The probability density function of such a system of jointly Gaussian random variables X_1 and X_2 has a bell-shaped form (Fig. 2.8) and is defined by five parameters, viz., the mathematical expectations m_{x_1}, m_{x_2} of the

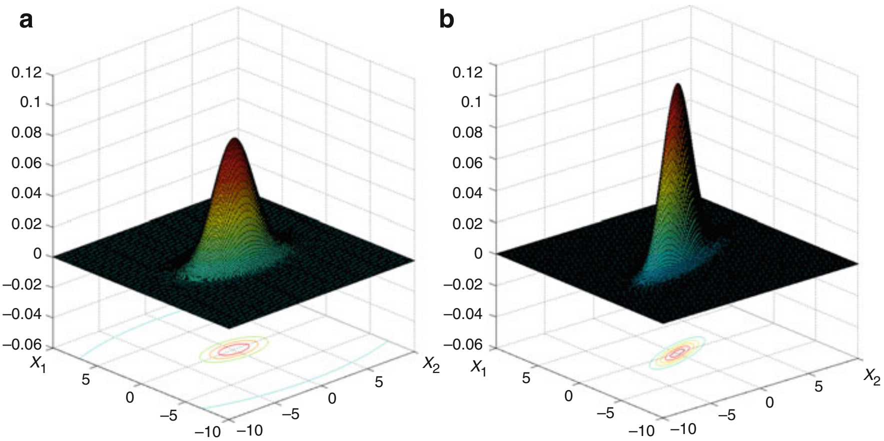

Fig. 2.8 Probability density function of a system of jointly Gaussian random variables X_1 and X_2 when the correlation coefficient $r_{x_1x_2} = 0$ (**a**) and $r_{x_1x_2} = 0.7$ (**b**) ($\sigma_{x_1} = 1$, $\sigma_{x_2} = 2$)

variables X_1 and X_2, their standard deviations σ_{x_1}, σ_{x_2}, and their correlation coefficient $r_{x_1x_2}$:

$$f_{\vec{x}}(x_1, x_2) = \frac{1}{2\pi\sqrt{|R_{\vec{x}}|}} \exp\left(-\frac{1}{2}(\vec{x} - \vec{m}_{\vec{x}})^{\mathrm{T}} R_{\vec{x}}^{-1} (\vec{x} - \vec{m}_{\vec{x}})\right), \tag{2.20}$$

where $\vec{x} = (x_1, x_2)^{\mathrm{T}}$ is a column vector whose components are the values of the random variables X_1 and X_2; $\vec{m}_{\vec{x}} = (m_{x_1}, m_{x_2})^{\mathrm{T}}$ is a column vector of whose components are the expectations of the variables X_1 and X_2; $R_{\vec{x}}$ is a square covariance matrix

$$R_{\vec{x}} = \left\| \begin{matrix} \sigma_{x_1}^2 & r_{x_1x_2}\sigma_{x_1}\sigma_{x_2} \\ r_{x_1x_2}\sigma_{x_1}\sigma_{x_2} & \sigma_{x_2}^2 \end{matrix} \right\|;$$

$R_{\vec{x}}^{-1}$ is the inverse of the matrix $R_{\vec{x}}$; and $|R_{\vec{x}}|$ is the determinant of the matrix $R_{\vec{x}}$.

The cut set of the probability density function on any fixed level C is the ellipse (*scattering ellipse*). Its center is at the point $(m_{x_1}, \ m_{x_2})$ and its sizes are determined by the standard deviations σ_{x_1} and σ_{x_2}, the correlation coefficient $r_{x_1x_2}$, and the value of the parameter C. Varying the parameter C, one can obtain scattering ellipses of different sizes, although the center and orientation of their axes coincide (Fig. 2.8).

If $m_{x_1} = m_{x_2} = 0$, the centers of the scattering ellipses are located at the coordinate origin. If the random variables are uncorrelated $(r_{x_1x_2} = 0)$, the axes of the ellipses are oriented along the coordinate axes. If, in addition, the standard deviations are equal ($\sigma_{x_1} = \sigma_{x_2} = \sigma$), the ellipses degenerate into a circle of radius σC.

By a linear transformation of the coordinate axes, a system of correlated jointly Gaussian random variables X_1, X_2 can be reduced to a system of uncorrelated (independent) jointly Gaussian random variables Y_1, Y_2 with zero expectations. Note that a jointly Gaussian system of random variables *remains jointly Gaussian* under any *linear transformation* of the coordinates.

2.4.4 Characteristics and Parameters of a System of more than Two Random Variables

The above relations for a system of two random variables can be generalized to a system of $N > 2$ random variables. The *N-dimensional distribution function* (*cumulative distribution function*) *of an N-dimensional vector random variable* (*system of random variables*) is described by the expression

$$F_{\vec{x}}(\vec{x}) = F_{\vec{x}}(x_1, \ldots, x_N) = P\{X_1 < x_1, \ldots, X_N < x_N\}$$

and the corresponding *multidimensional probability density function* (*probability distribution*) by the expression

$$f_{\vec{x}}(\vec{x})=f_{\vec{x}}(x_1,\ldots,x_N)=\frac{\partial^N F_{\vec{x}}(x_1,\ldots,x_N)}{\partial x_1\ldots\partial x_N}.$$

The dependence of the probability distribution $f_{\vec{x}}(\vec{x})$ on the value $\vec{x}=(x_1,x_2,x_3)^{\mathrm{T}}$ of the three-dimensional random vector $\vec{X}=(X_1,X_2,X_3)^{\mathrm{T}}$ can be represented figuratively by a cloud (see Fig. 2.3).

A *special case* of a multidimensional vector random variable $\vec{X}$ is the *multidimensional Gaussian random variable* whose components have a *jointly Gaussian distribution*. In the N-dimensional case the probability distribution is

$$f_{\vec{x}}(\vec{x})=\frac{1}{(2\pi)^{N/2}|R_{\vec{x}}|^{1/2}}\exp\left[-\frac{1}{2}(\vec{x}-\vec{m}_{\vec{x}})^{\mathrm{T}}R_{\vec{x}}^{-1}(\vec{x}-\vec{m}_{\vec{x}})\right], \tag{2.21}$$

where $\vec{x}=(x_1,\ldots,x_N)^{\mathrm{T}}$ is a column vector that describes the values of the random vector $\vec{X}=(X_1,\ldots,X_N)^{\mathrm{T}}$; $\vec{m}_{\vec{x}}=(m_{x_1},\ldots,m_{x_N})^{\mathrm{T}}$ is an expectation column vector of the vector $\vec{X}$; $R_{\vec{x}}$ is an $N\times N$ square *covariance matrix*:

$$R_{\vec{x}}=\begin{Vmatrix} R_{x_1x_1} & \ldots & R_{x_1x_N}\\ \ldots & \ldots & \ldots\\ R_{x_Nx_1} & \ldots & R_{x_Nx_N}\end{Vmatrix},$$

$R_{x_nx_m}$ is the covariance moment of the variables X_n and X_m ($n,m=\overline{1,N}$).

Note that the diagonal elements of the matrix $R_{\vec{x}}$ are the variances of the random variables $X_1,\ldots,X_N$ ($R_{x_nx_n}=D_{x_n}$, $n=\overline{1,N}$).

2.5 Operations on Random Variables

The distribution function is an exhaustive characteristic of the random variable. On this basis, one often assumes that the *random variables* X_1, X_2 *are equal*, if their distribution functions $F_{x_1}(x)$, $F_{x_2}(x)$ *coincide*, viz. $F_{x_1}(x)=F_{x_2}(x)$.

Note that from a mathematical point of view, such a definition of the equality of random variables is *not correct*. This is so because the random variable is uniquely described by the distribution function, but in general, the distribution function does not uniquely describe the random variable. As an example, *two different random variables* differing in sign have the same probability density function if it is symmetrical with respect to zero. In practice, one is not usually interested in random variables themselves, but in their *probabilistic and numerical characteristics*. Therefore, this mathematical incorrectness *can be ignored* when solving practical tasks.

Transformation of random variables leads to new random variables. Their probabilistic and numerical characteristics will in general differ from those of the original variables. From knowledge of the transformation function and the characteristics of the original random variables, one can calculate the characteristics of random variables obtained as a result of the conversion. The formulas are generally quite bulky, so we shall not present them here, but only give the formulas describing arithmetic operations. Let the initial operands be the random variables X_1 and X_2 described by a two-dimensional probability density function $f_{\vec{x}}(x_1, x_2)$ and the result of the operation be the random variable Y described by the probability density function $f_y(y)$. For *summation* of the variables, the probability density function is described by the expression

$$f_y(y) = \int_{-\infty}^{\infty} f_{\vec{x}}(y - x_2, x_2)\, dx_2, \tag{2.22}$$

for their *subtraction*,

$$f_y(y) = \int_{-\infty}^{\infty} f_{\vec{x}}(y + x_2, x_2)\, dx_2, \tag{2.23}$$

for their *multiplication*,

$$f_y(y) = \int_{-\infty}^{\infty} f_{\vec{x}}\left(\frac{y}{x_2}, x_2\right) \frac{dx_2}{|x_2|}, \tag{2.24}$$

and for their *division*,

$$f_y(y) = \int_{-\infty}^{\infty} |x_2| f_{\vec{x}}(y x_2, x_2)\, dx_2. \tag{2.25}$$

The expectation m_y and variance D_y of the *sum* of random variables X_1 and X_2 with expectations m_{x_1}, m_{x_2}, variances D_{x_1}, D_{x_2}, and covariance moment $R_{x_1 x_2}$ are given by the expressions

$$m_y = m_{x_1} + m_{x_2}, \;\; D_y = D_{x_1} + D_{x_2} + 2R_{x_1 x_2},$$

while the expectation and variance of the *difference* of these random variables are given by the expressions

$$m_y = m_{x_1} - m_{x_2}, \;\; D_y = D_{x_1} + D_{x_2} - 2R_{x_1 x_2}.$$

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Chapter 3
Stochastic Functions

Abstract The notion of a stochastic (random) function is formalized and the classification of these functions is discussed. We present different ways to describe a stochastic process, in terms of a distribution function, a probability density function, and moment functions, and in particular the expectation, variance, covariance, and correlation functions. We consider a stationary stochastic process in the narrow and broad senses. We describe the Wiener–Khinchin transformation and generalized Wiener–Khinchin transformation. The spectral approach for describing a stochastic process is presented. The ergodic and fragmentary ergodic processes are considered.

3.1 Main Concepts

A *stochastic* (*random*) *function* $X(t)$ is a many-valued numerical function of an independent argument t, whose value for any fixed value $t \in T$ (where T is the domain of the argument) is a random variable, called a *cut set*. The set of all cut sets of the random function defines the *state space* S (*phase space* or *actual range*).

The i-th *realization of the stochastic function* $X(t)$ (Fig. 3.1) is a determinate function $x_i(t)$ which, for a fixed experiment $i \in I$, assigns to each $t \in T$ one of the values $x \in S$.

The stochastic function has features inherent in both a random variable and a determinate function: fixing the value of the argument t turns it into a random variable, and fixing the experiment i transforms it into a determinate function.

The number I of realizations of the stochastic function may be finite, countably infinite, or uncountable.

The dimension L of the applicable domain T may take different values. If $L = 1$, then the argument t is often regarded as a time and the stochastic function $X(t)$ is interpreted as a *stochastic process*. If $L > 1$, then the argument t is a vector. In this case, the function $X(t)$ is called a *stochastic* (*random*) *field*.

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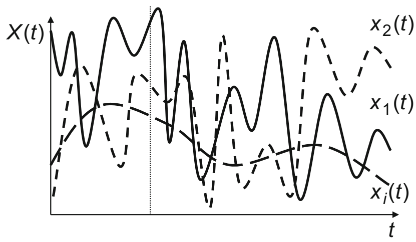

Fig. 3.1 Realizations of a stochastic function $X(t)$

If the state space is one-dimensional, then the stochastic function is *scalar*, and if the dimension of the state space is greater than one, then the stochastic function is *vectorial*. Hereafter, we shall consider scalar stochastic functions whose actual range is *real*.

3.2 Description of Stochastic Processes

A stochastic process $X(t)$ can be viewed as a *vector random variable*. At the same time, like any N-dimensional vector random variable, it can be described by an N-dimensional *distribution function* $F_{\vec{x}}(\vec{x};\vec{t}) = F_{\vec{x}}(x_1,\ldots,x_N;t_1,\ldots,t_N)$ or an N-dimensional *probability density function* $f_{\vec{x}}(\vec{x};\vec{t}) = f_{\vec{x}}(x_1,\ldots,x_N;t_1,\ldots,t_N)$.

In these expressions, we specify a set of cut sets $x_1, \ldots, x_N$ (vector $\vec{x}$), and after the semicolon, we specify the set of values $t_1, \ldots, t_N$ (vector $\vec{t}$) to which these cut sets refer.

The moments of a stochastic process depend on the vector $\vec{t}$. Therefore, in the description of the moments of a stochastic process (and other numerical characteristics), we also give the arguments indicating the cut sets to which these moments refer. In this case the moments are interpreted as *moment functions* of arguments $t_1, \ldots, t_N$.

For example,

- The *expectation of the function* $\varphi(X(t_1), \ldots, X(t_N))$ *of the cut sets of a stochastic process* $X(t)$ is represented by the function $\mathrm{E}[\varphi(X(t_1), \ldots, X(t_N))]$;
- The *expectation of a stochastic process* $X(t)$, by the function $m_x(t) = \mathrm{E}\,[X(t)]$;
- The *variance*, by the function $D_x(t) = \mathrm{E}[(X(t) - m_x(t))^2]$;
- The *correlation function*, by the function $K_x(t_1,t_2) = \mathrm{E}[X(t_1)X(t_2)]$;
- The *covariance function*, by the function

$$R_x(t_1,t_2) = \mathrm{Cov}[X(t_1),X(t_2)] = \mathrm{E}\left[\Big(X(t_1) - m_x(t_1)\Big)\Big(X(t_2) - m_x(t_2)\Big)\right].$$

The *cut sets* t_1, t_2 *of a stochastic process* $X(t)$ are *independent* if $f_{\vec{x}}(x_1,x_2;t_1,t_2) = f_{x_1}(x_1;t_1) f_{x_2}(x_2;t_2)$, *uncorrelated* if the covariance function $R_x(t_1,t_2) = 0$, and *orthogonal* if the correlation function $K_x(t_1,t_2) = 0$.

If the cut sets are uncorrelated, then $K_x(t_1, t_2) = m_x(t_1)m_x(t_2)$, and if they are orthogonal, then $R_x(t_1, t_2) = -m_x(t_1)m_x(t_2)$.

If the cut sets of the stochastic process are *correlated*, they are *dependent*. The converse is not always true. If the cut sets are *independent*, they are *uncorrelated*. If the cut sets are *orthogonal* they can be either *dependent* or *independent*, either *correlated* or *uncorrelated*. If the expectations of at least one of the two given cut sets are equal to zero, then *orthogonality* of the cut sets implies *noncorrelatedness*, and *noncorrelatedness* implies *orthogonality*.

3.3 Gaussian Stochastic Process

Part of the theory of stochastic functions in which stochastic (random) functions are described only with the moment functions of first and second orders is called the *correlation theory of stochastic functions* (*processes*). Description of stochastic functions on the basis of the moment functions of the first two orders is convenient and hence often used to solve practical tasks. However, as a rule, such a description is not complete. Exceptions are *Gaussian* (*normal*) *stochastic functions*.

A *stochastic process* is *Gaussian* (*normal*) if, for any finite set of its cut sets $X(t_1), \ldots, X(t_N)$, a joint probability density function is described by the expression

$$f_{\vec{x}}(\vec{x}; \vec{t}) = \frac{1}{\sqrt{(2\pi)^N |R_x|}} \exp\left(-\frac{1}{2}(\vec{x} - \vec{m}_x)^{\mathrm{T}} R_x^{-1}(\vec{x} - \vec{m}_x)\right),$$

where $\vec{x} = (x(t_1), \ldots, x(t_N))^{\mathrm{T}}$ is an N-dimensional column vector whose n-th component is the value of a random variable $X(t_n)$; $\vec{m}_x = (m_x(t_1), \ldots, m_x(t_N))^{\mathrm{T}}$ is an N-dimensional column vector of expectations whose n-th component is the expectation of a random variable $X(t_n)$; R_x is the $N \times N$ covariance matrix whose elements are the covariance moments $R_x(t_n, t_m)$ of random variables $X(t_n), X(t_m)$, $n, m = \overline{1, N}$; R_x^{-1} is the inverse covariance matrix of R_x; T is the transposition operator; and $|R_x|$ is the determinant of the matrix R_x.

Gaussian stochastic processes have a number of important properties distinguishing them from other stochastic processes. First of all, a Gaussian stochastic process is completely determined by the expectation $m_x(t)$ and covariance function $R_x(t_1, t_2)$. Thus the correlation theory gives an exhaustive description of such processes. All higher-order moments of a Gaussian stochastic process can be expressed in terms of its moments of the first two orders.

For Gaussian stochastic processes, *noncorrelatedness of cut sets is identical to their independence*, and *independence is identical to their noncorrelatedness*.

A Gaussian stochastic process $X(t)$ with correlated (dependent) cut sets can be reduced by the coordinate transformation to a Gaussian stochastic process with uncorrelated (independent) cut sets.

Any linear transformation of a Gaussian stochastic process by a linear operator leads to another Gaussian stochastic process.

This feature is called the *stability property of a Gaussian stochastic process under linear transformations*. Note that under a nonlinear transformation, a Gaussian stochastic process is transformed to a non-Gaussian stochastic process.

3.4 Stationary Stochastic Processes

Of particular interest are the so-called stationary stochastic processes. There are two types of stationary processes: those that are stationary in a narrow sense and those that are stationary in a broad sense.

3.4.1 Stochastic Processes That Are Stationary in the Narrow Sense

A *stochastic process* $X(t)$ is *stationary in the narrow sense* if its N-dimensional probabilistic characteristics for any N depend only on the magnitude of the time intervals $t_2 - t_1, \ldots, t_N - t_1$ and do not depend on the position of these intervals on the t axis. Stochastic processes that do not belong to the class of stationary processes are said to be *non-stationary*. The main property of a stationary process is the invariance of its probabilistic characteristics of any N-th dimension under a shift along the t axis.[1]

Therefore, one-dimensional probabilistic characteristics of a stationary random process $X(t)$, in particular, the distribution function $F_x(x;t) = F_x(x)$ and the probability density function $f_x(x;t) = f_x(x)$, are independent of the argument t. Two-dimensional probabilistic characteristics depend only on the difference between the values $\tau = t_2 - t_1$ of the argument t, in particular, the distribution

[1]This relates, in particular, to the N-dimensional probability density function:

$$f_{\vec{x}}(x_1, \ldots, x_N; t_1, \ldots, t_N) = f_{\vec{x}}(x_1, \ldots, x_N; t_1 + \tau, \ldots, t_N + \tau),$$

where τ is an arbitrary number.

function $F_{\vec{x}}(x_1, x_2; t_1, t_2) = F_{\vec{x}}(x_1, x_2; \tau)$ and the probability density function $f_{\vec{x}}(x_1, x_2; t_1, t_2) = f_{\vec{x}}(x_1, x_2; \tau)$.

The expectation and variance of a stationary process are independent of time t: $m_x(t) = m_x = \text{const}$, $D_x(t) = D_x = \text{const}$. The covariance and correlation functions depend on the difference between the values of the argument: $R_x(t_1, t_2) = R_x(t_2 - t_1) = R_x(\tau)$, $K_x(t_1, t_2) = K_x(t_2 - t_1) = K_x(\tau)$.

3.4.2 Stochastic Processes That Are Stationary in the Broad Sense

A *stochastic process* $X(t)$ is *stationary in the broad sense* if the expectation is constant ($m_x(t) = m_x = \text{const}$) and the covariance function depends only on the difference between the values τ of the argument t: $R_x(t_1, t_2) = R_x(t_2 - t_1) = R_x(\tau)$. Processes that do not satisfy these requirements are called *non-stationary in the broad sense*.

The concepts of stationarity in the narrow sense and in the broad sense are *not always identical*. These concepts coincide, in particular, for Gaussian stochastic processes.

Hereinafter, by a stationary stochastic process we shall understand (unless otherwise stated) a process that is *stationary in the broad sense*.

In the area of largest maximum of the covariance function of a stationary stochastic process, achieved at the point $\tau = 0$, when the absolute value τ increases, the value of the covariance function gradually decreases (Fig. 3.2a) or decreases with oscillation (Fig. 3.2b).

The correlation interval of the stochastic process τ_c is the value of the argument τ of its covariance function $R_x(\tau)$, beyond which the *values of this function* (or the *value of its envelope*) are so small that they can be neglected.

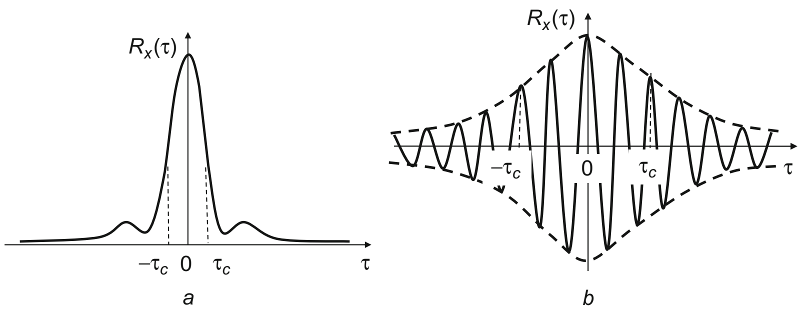

Fig. 3.2 Covariance function of broadband (**a**) and narrowband (**b**) stochastic processes

3.5 Spectral Description of Stochastic Processes

3.5.1 Wiener-Khinchin Transformation

It is convenient to represent stochastic processes in the spectral area. Consider the process $X_T(t)$ defined over the interval $t \in [-T/2, T/2]$. It can be represented in an equivalent form by the *complex spectrum* $\dot{A}_{X_T}(f)$, where f is frequency and the dot indicates that the given value is complex.

The process $X_T(t)$ and its complex spectrum are related by *Fourier transformation*:

$$X_T(t) = \int_{-\infty}^{\infty} \dot{A}_{X_T}(f) \exp(j2\pi f t) df, \tag{3.1}$$

$$\dot{A}_{X_T}(f) = \int_{-T/2}^{T/2} X_T(t) \exp(-j2\pi f t) dt. \tag{3.2}$$

An important characteristic of the process is its *power spectrum* $S_{X_T}(f)$ associated with a complex spectrum $\dot{A}_{X_T}(f)$ by the expression

$$S_{X_T}(f) = \frac{1}{T} \left|\dot{A}_{X_T}(f)\right|^2. \tag{3.3}$$

Note that the complex spectrum specifies the process uniquely, but the power spectrum only ambiguously. To any dependence of the power spectrum on the frequency, there corresponds a set of processes with different instantaneous spectra. If the process $X_T(t)$ is stochastic, then the variables $\dot{A}_{X_T}(f)$ and $S_{X_T}(f)$ at fixed frequency f are random.

The *power spectral density* (*PSD*) of the process $X_T(t)$ *over a finite observation interval T* is the power spectrum $S_{X_T}(f)$ averaged over the population:

$$S_{x_T}(f) = \mathrm{E}[S_{X_T}(f)] = \mathrm{E}\left[\frac{1}{T}\left|\dot{A}_{X_T}(f)\right|^2\right]. \tag{3.4}$$

When $T \to \infty$, the power spectral density $S_{x_T}(f)$ tends to the power spectral density $S_x(f) = \lim_{T\to\infty} S_{x_T}(f)$ of the process $X(t)$ defined for an *infinite observation interval*. Note that these relations are valid *for both stationary and nonstationary processes*.

If the process is *stationary*, then its power spectral density and its correlation function are related by the well known *Wiener–Khinchin transformation*:

$$S_x(f) = \int_{-\infty}^{\infty} K_x(\tau)\exp(-j2\pi f\tau)d\tau, \tag{3.5}$$

$$K_x(\tau) = \int_{-\infty}^{\infty} S_x(f)\exp(j2\pi f\tau)df. \tag{3.6}$$

It follows from (3.6) that the variance is described by the expression

$$D_x = K_x(0) = \int_{-\infty}^{\infty} S_x(f)df,$$

from which it follows that the PSD $S_x(f)$ characterizes the *power distribution in frequency*.

3.5.2 Narrowband and Broadband Processes

The spectrum is usually uneven. Its components are often concentrated in a continuous frequency band (*effective bandwidth*) Δf. Depending on the ratio between the band Δf and the frequency f_0 at which the PSD reaches its maximum value, one distinguishes broadband and narrowband processes. A *narrowband process* is one for which $\Delta f/f_0 \ll 1$. Other processes are called *broadband*. Typical covariance functions of broadband and narrowband stochastic processes are shown in Fig. 3.2.

The value of the frequency band Δf is associated with the correlation interval τ_c by the expression $\Delta f = 1/\tau_c$. It follows from this formula that the wider the effective bandwidth, the lower the correlation interval, and also that the greater the correlation interval, the narrower the bandwidth.

A well known example of broadband processes is white noise. *White noise* is an idealized stationary stochastic process $N(t)$ with zero expectation and constant power spectral density: $S_x(f) = N_0/2 = \text{const}$. The covariance and correlation functions of white noise are described by a δ-function: $R_x(\tau) = K_x(\tau) = \delta(\tau)N_0/2)$. This means that cut sets of such a process that are arbitrarily close to each other are uncorrelated, i.e., in this case the correlation interval $\tau_c = 0$.

The *effective frequency band* Δf of white noise and its *variance* are *infinite*. Therefore, white noise is *not physically realizable*. This is just a convenient mathematical abstraction, approximately describing stochastic processes with small changes of PSD over a wide frequency range.

3.5.3 Generalized Wiener-Khinchin Transformation

The Wiener-Khinchin transformation which relates the correlation function of a *stationary stochastic process* $X(t)$ to its power spectral density $S_x(f)$ can be generalized to the case of *non-stationary stochastic processes*. Consider the non-stationary stochastic process $X(t)$ defined on the whole real axis. The correlation function of the process can be represented by

$$K_x(t, \tau) = \mathrm{E}[X(t)X(t - \tau)]. \tag{3.7}$$

The average of the function $K_x(t, \tau)$ over t gives the averaged correlation function $\bar{K}_x(\tau)$ described by the expression

$$\bar{K}_x(\tau) = \lim_{T \to \infty} \frac{1}{T} \int\limits_{-T/2}^{T/2} K_x(t, \tau)\, dt. \tag{3.8}$$

The spectrum of the averaged correlation function $\bar{K}_x(\tau)$ can be written as

$$\bar{S}_x(f) = \int\limits_{-\infty}^{\infty} \bar{K}_x(\tau) \exp(-j2\pi f \tau) d\tau. \tag{3.9}$$

It is easily seen that the spectrum $\bar{S}_x(f)$ of the averaged correlation function $\bar{K}_x(\tau)$ equals the PSD $S_x(f)$ of the process $X(t)$: $\bar{S}_x(f) = S_x(f)$. Therefore, the averaged correlation function of the process can be found from its PSD:

$$\bar{K}_x(\tau) = \int\limits_{-\infty}^{\infty} S_x(f) \exp(j2\pi f \tau) df. \tag{3.10}$$

The pair of relations (3.9) and (3.10) present the *generalized Wiener-Khinchin transformation*, which is valid for both stationary and non-stationary processes.

3.6 Ergodic Stochastic Processes

Some *stationary stochastic processes* $X(t)$, called *ergodic processes*, have a specific feature: any of their realizations $x(t)$ contains all the information about the stochastic process. Therefore, to calculate the characteristics of such a stochastic process *we do not require a population of its realizations*, but *only a single realization*, and we stress that it can be *any realization*. The calculation of the characteristics and parameters of an ergodic stochastic process can be carried out, not by averaging over a population of realizations, but by *averaging the data of a single realization*.

There are several possible definitions of an ergodic stochastic process, but we shall confine ourselves to just one of them. A *stochastic process* $X(t)$ *that is stationary in the broad sense* is *ergodic in the broad sense* if its expectation m_x coincides with the *time average*

$$\bar{m}_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)dt \tag{3.11}$$

of any of its realizations $x(t)$, and the covariance function $R_x(\tau)$ with its *autocovariance function*

$$\bar{R}_x(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} (x(t+\tau) - \bar{m}_x)(x(t) - \bar{m}_x)dt. \tag{3.12}$$

On this basis, the expectation, the correlation function, and the covariance function can be calculated by the following formulas: $m_x = \bar{m}_x$, $K_x(\tau) = \bar{K}_x(\tau)$, $R_x(\tau) = \bar{R}_x(\tau)$, where $\bar{K}_x(\tau)$ is the *autocorrelation function*:

$$\bar{K}_x(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t+\tau)x(t)dt.$$

Some *non-stationary stochastic processes* manifest stationarity and ergodicity properties over finite intervals. Such processes are said to be *fragmentary-ergodic*. They consist of *almost ergodic fragments* of certain duration T (Fig. 3.3).

An *almost ergodic fragment of the non-stationary stochastic process* is a fragment for which the expectation and covariance function can be calculated with negligible error using a single realization of this fragment.

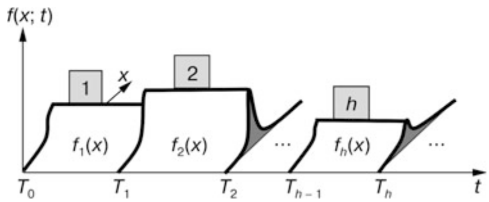

Fig. 3.3 One-dimensional probability density function $f(x; t)$ of a fragmentary-ergodic stochastic process $X(t)$ with almost ergodic fragments described by one-dimensional probability density functions $f_h(x)$, $T_{h+1} - T_h = T$, $h = 1, 2, \ldots$

In solving practical tasks an important problem is to determine the duration T of the almost ergodic fragments, in other words, to determine the *ergodicity interval of the non-stationary stochastic process*.

3.7 Transformation of Stochastic Processes

A transformation of a stochastic process generates a new stochastic process whose characteristics are defined by the characteristics of the initial process and the characteristics of the transformation operator. We distinguish inertialess and inertial transformations, and linear and non-linear ones.

In an *inertialess transformation*, the value of the output process $Y(t)$ at time t depends only on the value of the initial process $X(t)$ at the same time t, while in an *inertial transformation*, the output process at time t depends on the set of values of the initial process, not only at time t, but at other times too.

Since a stochastic function can be interpreted as a vector random variable, the calculation of probabilistic and numerical characteristics of a process obtained as a result of a (linear or non-linear) inertialess transformation is carried out in the same way as the calculation of similar characteristics of the vector variable obtained as the result of such a transformation.

For an inertial transformation, the calculation of the characteristics of a stochastic process is more complicated (Levin 1974; Gorban 2003). We shall not dwell in detail on this issue. We shall merely write down the expression which will be needed later on:

$$S_y(f) = \left|\dot{K}(f)\right|^2 S_x(f). \tag{3.13}$$

In this formula, $S_y(f)$ is the power spectral density of the stochastic process $Y(t)$ obtained as the result of a *linear transformation* of the process $X(t)$, $\dot{K}(f)$ is the *complex transfer function of the transformation operator* that characterizes the operator, and $S_x(f)$ is the power spectral density of the initial stochastic process $X(t)$.

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Chapter 4
Fundamentals of the Mathematical Statistics of Probability Theory

Abstract The concepts of random sampling and statistics of random variables are introduced. We consider estimators of probability characteristics and moments. We discuss the types of convergence used in probability theory, in particular the convergence of a sequence of random variables in probability and convergence in distribution. The law of large numbers and the central limit theorem are described in the classical interpretation. We discuss the statistics of stochastic processes and specific features of samples of random variables and stochastic processes.

Actual data samples are always of limited size, and the laws of their distribution may differ widely. So naturally there are many questions about the correct description and representation of real data.

Among the main tasks here are the following:

- Construction of mathematical models adequately representing actual data,
- Evaluation of various characteristics and parameters characterizing actual data,
- Establishment of links between the characteristics and parameters corresponding to an infinitely large amount of data and evaluations of the same characteristics and parameters for a limited amount of data,
- Calculation of the accuracy of assessments of different parameters.

These and many other issues are studied in the special part of probability theory called *mathematical statistics*.

4.1 Statistics of Random Variables

4.1.1 A Random Sample

Statistical information is any information about random phenomena (events, variables, functions). In this section we confine ourselves to statistical information relating to *random variables*.

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Random variables can be both scalar and vectorial. Further, in the interest of simplicity, we assume that the random variables are scalar. However, we should not forget that many of the statements will also be valid for vector variables and stochastic processes.

The *general population* (*whole assembly*) *of the random variable* X described by the distribution function $F_x(x)$ [see (2.3)] is the *infinite set of all its determinate realizations* (*sample elements or components*). This set is generally considered to be countable.

The final set of members of the general population $x_1, \ldots, x_N$ obtained in a number N of experiments is called a *sample of the population*, or simply a *sample*, and its elements $x_1, \ldots, x_N$ forming a vector $\vec{x}$ are called *realizations* (*sample values*). Note that the elements $x_1, \ldots, x_N$ are *determinate values* (*numbers*). The sample $x_1, \ldots, x_N$ is taken to belong to the random variable X if it is obtained from the general population described by the distribution function $F_x(x)$.

The infinite set of samples $\vec{x}$ of size N formed from a general population is represented by an N-dimensional random vector $\vec{X} = (X_1, \ldots, X_N)$ called a *random sample*. Note that the *random sample* represented by a random vector $\vec{X}$ and a *determinate sample* (simple *sample*) $\vec{x}$ which is the realization of a random sample are *different vectors*. The components X_n $(n = \overline{1, N})$ of a random sample $\vec{X}$ described by the same distribution function $F_x(x)$ (or probability distribution $f_x(x)$, Fig. 4.1a) are often considered to be independent.

Then the N-dimensional distribution function $F_{\vec{x}}(\vec{x})$ of the random sample $\vec{X}$ is the product of the one-dimensional distribution functions $F_x(x_n)$ of the components X_n:

$$F_{\vec{x}}(\vec{x}) = \prod_{n=1}^{N} F_x(x_n).$$

Note that, in addition to the widely used *non-dependent homogeneous random sample* just described, a different model is sometimes used, in which random sample components are also independent, but have *different distribution laws* (Fig. 4.1b). The sample in which the elements have different distribution laws is called a *heterogeneous random sample*.

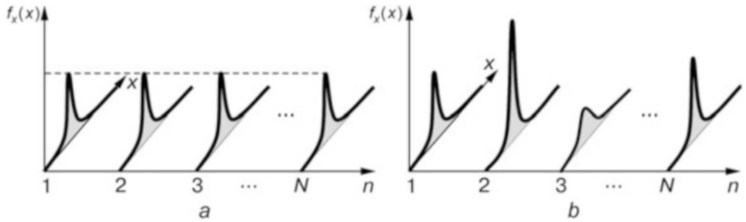

Fig. 4.1 Probability distributions of the components of a random sample $\vec{X} = (X_1, X_2, X_3, \ldots, X_N)$: homogeneous (**a**) and heterogeneous (**b**)

A *statistic* is any function of any sample. A function $Y = \varphi(X_1, \ldots, X_N)$ of the random sample $\vec{X}$ is a *statistic of the random sample*; the function $y = \varphi(x_1, \ldots, x_N)$ of the determinate sample $\vec{x}$ is the *statistic of the determinate sample*. The statistic $Y = \varphi(X_1, \ldots, X_N)$ is a *random variable*, and the statistic $y = \varphi(x_1, \ldots, x_N)$ is a *determinate value* (*number*).

A sample is said to be *representative* if it allows us to describe the properties of the random variable with the required accuracy.[1] To obtain a representative sample, the size must be sufficiently large. The specific statistic is an *assessment*.

From the general population of the random variable X, one can calculate the exact probability and numerical characteristics, such as the distribution function $F_x(x)$, its expectation m_x, variance D_x, and other moments. Using the random sample, one can calculate approximate *random estimators* of the same characteristics, in particular the *random estimator* of the distribution function $F_x^*(x)$, *random estimator* of the expectation m_x^*, *random estimator* of the variance D_x^*, and *random estimators* of other moments. If the sample is determinate, one can calculate approximate *determinate estimates* of the same characteristics, in particular a *determinate estimate* of the distribution function $F_x^*(x)$, a *determinate estimate* of the expectation m_x^*, a *determinate estimate* of the variance D_x^*, and a *determinate estimates* of other moments.

The *variation* (*statistical*) *series* is the vector formed from the elements of the *determinate sample* $\vec{x}$ *sorted in ascending or descending order*, and the *ranked series* is the vector formed from the elements of the *determinate sample* $\vec{x}$ *sorted in descending order*. The *sample range* is the difference between the maximum and minimum elements of the *determinate sample*.

4.1.2 Assessments of Probability Characteristics

Consider a discrete random variable X taking discrete values $x_1, \ldots, x_J$. The *relative frequency* ω_j^* *of the sample value* x_j is the ratio of the number N_j of identical values x_j appearing in the sample to the sample size $N : \omega_j^* = N_j/N$. The relative frequency ω_j^* is a determinate value, if a determinate sample is used, and a random variable, if a random sample is used. In both cases, the relative frequency ω_j^* is an analogue of the probability p_j that the random variable X takes the value x_j.

The *empirical mass function* is a table of sample values x_j and corresponding relative frequencies $\omega_j^* : \left(x_j, \omega_j^*\right),\ j = \overline{1, J}$. Use of the empirical mass function is meaningful only if the number of different values J is small. For continuous random variables as well as for discrete random variables with a large number of different values of the discrete random variable another approach is used. The entire range of sample values is divided into a series of intervals (class intervals) and then one

[1] The issue of accuracy is discussed in the next chapter.

counts how many values fall into each of them. After dividing the resulting values by the sample size N we obtain a table $\left(x_i, p_i^*\right),\ \ i=\overline{1,I}$, which is analogous to the previous one, where x_i is the middle of the i-th class interval, p_i^* is the corresponding relative frequency, and I is the number of intervals.

The *relative frequency density* f_i^* is the ratio of the relative frequency p_i^* to the length Δx_i of the corresponding interval: $f_i^* = p_i^*/\Delta x_i$. In this way we obtain the table $\left(x_i, f_i^*\right),\ \ i=\overline{1,I}$. If the sample is determinate, the density f_i^* is an estimate of the probability distribution $f_x(x)$, and if it is random, the density f_i^* is the estimator.

The number of intervals should not be too large. Practice shows that in most cases it is advisable to select a number of intervals in the region of 10–20.

For pictorial representation of the tables $\left(x_i, p_i^*\right),\ \ i=\overline{1,I}$ and $\left(x_i, f_i^*\right),\ \ i=\overline{1,I}$ *polygons* and *histograms* are used. *Relative frequency polygons* and *relative frequency density polygons* are two graphs that present the sample distribution by lines that connect the points $\left(x_i, p_i^*\right),\ \ i=\overline{1,I}$ and $\left(x_i, f_i^*\right),\ \ i=\overline{1,I}$. *Relative frequency histograms* and *relative frequency density histograms* are analogous to stem-and-leaf diagrams in which the data points falling within each class interval are listed in order. Examples of relative frequency histograms and relative frequency polygons are shown in Fig. 4.2a, b.

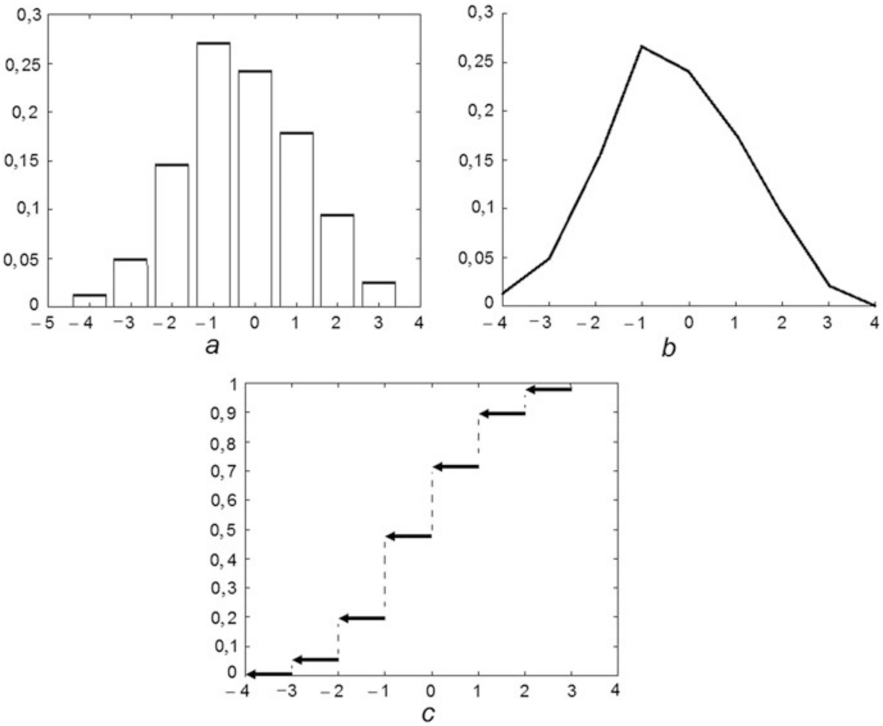

Fig. 4.2 Examples of a relative frequency histogram (**a**), a relative frequency polygon (**b**), and an empirical distribution function (**c**) obtained for the same sample

The *empirical distribution function* is a function $F_x^*(x)$ which, for each value x of the random variable X, determines the relative frequency of the event $X < x$. The empirical distribution function is an analog of the distribution function $F_x(x)$ called in mathematical statistics the *true* (*theoretical*) *distribution function*. In the case of determinate sampling the empirical distribution function $F_x^*(x)$ is described by a non-decreasing step function of its argument x. It has the same properties as the theoretical distribution function. An example of such a function is shown in Fig. 4.2c.

4.1.3 Assessment of Moments

Assessments of moments are *sample moments* in particular the sample mean, sample variance, sample standard deviation, etc.[2]

The *sample mean* (*average*) *of a random sample* $X_1, \ldots, X_N$ is a random variable

$$m_x^* = \frac{1}{N}\sum_{n=1}^{N} X_n. \tag{4.1}$$

The *sample variance of a random sample* is a random variable

$$D_x^* = \frac{1}{N-1}\sum_{n=1}^{N}\left(X_n - m_x^*\right)^2, \tag{4.2}$$

and the *sample standard deviation* is a random variable $\sigma_x^* = \sqrt{D_x^*}$.

The *sample correlation moment of random samples* $X_1, \ldots, X_N$ and $Y_1, \ldots, Y_N$ is a random variable

$$K_{xy}^* = \frac{1}{N-1}\sum_{n=1}^{N} X_n Y_n,$$

and the *sample covariance moment* is a random variable

$$R_{xy}^* = \frac{1}{N-1}\sum_{n=1}^{N}\left(X_n - m_x^*\right)\left(Y_n - m_y^*\right),$$

where m_x^* and m_y^* are the sample expectations of random samples $X_1, \ldots, X_N$ and $Y_1, \ldots, Y_N$.

[2]The following determinations of the sample moments are not the only ones possible.

4.2 Convergence of Sequences of Random Variables

For further discussion the concept of convergence of a sequence of random variables is required. This concept is analogous to the concept of convergence of a numerical sequence. When discussing the concept of infinitesimal variable in Sect. 1.2.1, a definition of the concept of convergence of a numerical sequence was given. Let us recall yet another definition of this concept.

The number a is the *limit of a numerical sequence* $x_1, \ x_2, \ldots$ (otherwise, the *numerical sequence* $x_1, \ x_2, \ldots$ *converges* to a number a and we write $\lim\limits_{n\to\infty} x_n = a$) if for each arbitrarily small positive number ε there exists a number N such that for all $n > N$ the inequality $|x_n - a| < \varepsilon$ is satisfied (Fig. 4.3).

The idea behind this definition (as given in Sect. 1.2.1) is that when $n \to \infty$ the magnitude x_n tends to the number a. It can be shown that if the limit of the sequence exists then it is *unique*.

Thus, in the framework of classical mathematical analysis, the *numerical sequence can converge only to one number* or *not converge at all* (*diverge*).

In probability theory, different variants of the convergence of random sequences are used, in particular, the convergence of a sequence of random variables to a random (or determinate) value. The following four types of convergence are the ones most often used: *convergence in distribution* (*in the Bernoulli sense*), *in mean-square*, *almost surely* (*with probability one*), and *in probability* (*in measure*). For most practical tasks the type of convergence does not play an important role. Therefore we will not dwell in detail on this issue, and look only at the two commonly used types of convergence: convergence in probability and in distribution.

Let $\mathrm{X} = \{X_1, \ldots, X_n\}$ be a sequence of random variables and X a random variable. Suppose that, for all $X_1, \ldots, X_n$ and X, the distribution functions $F_1(x), \ldots, F_n(x)$ and $F(x)$ are defined. Then the sequence X

(1) *Converges to* X *in probability* if $P\{|X_n - X| > \varepsilon\} \to 0$ for any $\varepsilon > 0$ and $n \to \infty$;
(2) *Converges to* X *in distribution* (*in the Bernoulli sense*) if at every point x, where $F(x)$ is continuous, $F_n(x) \to F(x)$ when $n \to \infty$.

Convergence in distribution is weaker than convergence in probability, i.e., sequences that converge in probability converge in distribution too. The converse is *not always true*.

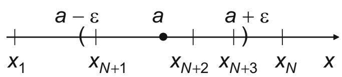

Fig. 4.3 Illustration of the convergence of a numerical sequence

In the general case the magnitude X is the random variable and in the particular case it could be a number.

Therefore, if a random sequence converges, it may converge to a *number* or to a *random variable*.

In future references to the convergence of a random sequence we shall mean *convergence in probability*, unless otherwise stated. *Probability theory* studies exclusively *convergent random sequences*. *Divergent sequences are studied in the theory of hyper-random phenomena*, discussed in the second half of the book.

4.3 The Law of Large Numbers

It can be shown that under unrestricted growth in the size of a convergent random sample, the *empirical distribution function converges to the theoretical distribution function*. This is the essence of the *Glivenko–Cantelli theorem*, the fundamental theorem of mathematical statistics in probability theory. This theorem is based on the *law of large numbers* originally proven by Jakob Bernoulli and published in his posthumous work in 1713 (Bernoulli 1986). This law is the theoretical basis of the sampling method.

There are several variants of the law of large numbers. Let us focus on one of them, formulated and proved by *P.L. Chebyshev* in 1867.

Chebyshev Theorem Let $X_1, \ldots, X_N$ be a sequence of pairwise independent random variables with expectations $m_1, \ldots, m_N$ and *bounded variances*. Then, when the sample size N goes to infinity, the average of the sample values $X_1, \ldots, X_N$ tends in probability to the average of the expectations $m_1, \ldots, m_N$:

$$\lim_{N\to\infty} P\left\{\left|\frac{1}{N}\sum_{n=1}^{N} X_n - \frac{1}{N}\sum_{n=1}^{N} m_n\right| > \varepsilon\right\} = 0 \ldots (\varepsilon > 0). \tag{4.3}$$

If the average of the expectations $m_{xN} = \frac{1}{N}\sum_{n=1}^{N} m_{x_n}$ has a limit m_x (which is assumed in the framework of probability theory), $\lim_{N\to\infty} m_{xN} = m_x$. Then denoting $m_{xN}^* = \frac{1}{N}\sum_{n=1}^{N} X_n$, (4.3) can be written as

$$\lim_{N\to\infty} P\left\{\left|m_{xN}^* - m_x\right| > \varepsilon\right\} = 0. \tag{4.4}$$

It follows from (4.4) that *theoretically* the accuracy of estimation of the expectation, which is calculated by computing the sample mean, increases with increasing sample size, and when $N \to \infty$, *increases to infinity*.

Unfortunately, this optimistic theoretical result from probability theory *contradicts the experimental data*. We shall return in Chap. 10 to the question of the actual accuracy of the estimates, considering it in the framework of the theory of hyper-random phenomena.

4.4 Central Limit Theorem

We now turn to the issue of the distribution law of estimates. As the sample size increases to infinity, the limiting distribution law is determined by the *central limit theorem*, which is widely used in practice. Its formulation and proof have a long history, associated with the names of *Abraham de Moivre*, *P. S. Laplace*, *P. L. Chebyshev*, *A. A. Markov* and, of course, *A. M. Lyapunov*. There are many variants of this theorem. We shall consider one taken with some simplification from the textbook (Gnedenko 1988).

Lindeberg–Feller Theorem Let $X_1, \ldots, X_N$ be, in general, a non-uniform random sample with mutually independent terms described by distribution functions $F_{x_n}(x)$ with expectations m_{x_n} and variances D_{x_n} $(n = \overline{1, N})$. We assume a not very restrictive condition called the *Lindeberg condition* (Gnedenko 1988). Then the distribution function $F_{m^*_{xN}}(x)$ of the sample mean m^*_{xN} *converges uniformly* to a *Gaussian distribution function*

$$F(x/m_{xN}, D_{xN}) = \Phi\left(\frac{x - m_{xN}}{\sqrt{D_{xN}}}\right) \tag{4.5}$$

with expectation $m_{xN} = \frac{1}{N}\sum_{n=1}^{N} m_{x_n}$ and variance $D_{xN} = \frac{1}{N^2}\sum_{n=1}^{N} D_{x_n}$, viz.,

$$\lim_{N\to\infty} F_{m^*_{xN}}(x) = \lim_{N\to\infty} F(x/m_{xN}, D_{xN}), \tag{4.6}$$

where $\Phi(x)$ is the tabulated *Laplace function* described by (2.8).

According to (4.6), when the sample size increases, the random variable m^*_{xN} becomes a *Gaussian random variable*.

If the average variance ND_{xN} of the random variables X_n, $n = \overline{1, N}$ is limited, the variance D_{xN} of the sample mean m^*_{xN} is *inversely proportional to the sample volume* N. Therefore, when $N \to \infty$ the *variance* D_{xN} *tends to zero*.

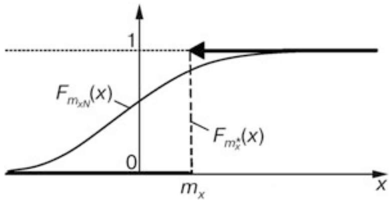

Fig. 4.4 Limiting distribution function $F_{m_x^*}(x)$ of the sample mean of the random variable when $N \to \infty$

Thus, when the size N of a random sample increases, the distribution function $F_{m_{xN}^*}(x)$ *tends to a Gaussian distribution function* $F(x/m_{xN}, D_{xN})$, the *variance of the sample mean* m_{xN}^* *tends to zero*, and *its expectation* m_{xN}*goes to the expectation* m_x. The *limiting distribution function* $F_{m_x^*}(x)$ *of the distribution function* $F_{m_{xN}^*}(x)$ *is equal to the limiting Gaussian distribution function* $F(x/m_x, 0)$, which is a unit step function at the point m_x (Fig. 4.4).

The limiting probability density function corresponding to the limiting distribution function is the δ-function at the point m_x.

A lot of experimental research in *actual processes* shows that when data volumes are not too large the sample mean tends to the Gaussian distribution. However, this trend disappears when data volumes become very large. In Chap. 9 this effect will be analyzed in detail.

4.5 Statistics of Stochastic Processes

The concepts and statements of mathematical statistics relating to random variables can be generalized to stochastic processes. Here we briefly discuss the basic concepts of the mathematical statistics of stochastic processes, but without going into the details.

The *general population* (*whole assembly*) *of the stochastic process* $X(t)$ is an infinite number of its realizations $x_1(t), x_2(t), \ldots$. An *element* (*member*) *of the population of the stochastic process* is any one of its realizations $x_n(t)$. The finite set of members $x_n(t)$ $\left(n = \overline{1,N}\right)$ of the general population obtained in a finite number of experiments, i.e., the vector $\vec{x}(t) = (x_1(t), \ldots, x_N(t))$, is called a *sample of the population* or just a *sample*. In practice, the duration of the experiments is limited. Therefore, the sample of the stochastic process $X(t)$ is a finite set of determinate processes $x_n(t)$ of finite duration T. A sample set of volume N formed from the general population is an N-dimensional vector stochastic process $\vec{X}(t) = (X_1(t), \ldots, X_N(t))$ called a *random sample*. Note that the *random sample* $\vec{X}(t)$ should be distinguished from a *sample* $\vec{x}(t)$, which is a realization of a random sample $\vec{X}(t)$. One usually supposes that the elements $X_n(t)$ $\left(n = \overline{1,N}\right)$ of a random sample $\vec{X}(t)$ are *independent*.

A *statistic of the stochastic process* $X(t)$ is any function $Y(t) = \varphi(X_1(t), \ldots, X_N(t))$ of the random sample $\vec{X}(t)$, as well as any function $y(t) = \varphi(x_1(t), \ldots, x_N(t))$ of the determinate sample $\vec{x}(t)$. In general, the statistic $Y(t) = \varphi(X_1(t), \ldots, X_N(t))$ is a *stochastic process*, and the statistic $y(t) = \varphi(x_1(t), \ldots, x_N(t))$ is a *determinate process*. If $Y(t)$ does not depend on the argument t, then $Y(t) = Y$ is a *random variable* and $y(t) = y$ is a *determinate value* (*number*). A specific statistic is an *assessment*.

Using the general population, it is theoretically possible to calculate the *exact* probabilistic and numerical characteristics, for instance, a distribution function $F_x(x; t)$ of the stochastic process $X(t)$ and its expectation $m_x(t)$, variance $D_x(t)$, correlation function $K_x(t_1, t_2)$, covariance function $R_x(t_1, t_2)$, and so forth.

Using a sample one can calculate evaluations of the same characteristics, for example an assessment of the distribution function $F_x^*(x; t)$, an assessment of the expectation $m_x^*(t)$, an assessment of the variance $D_x^*(t)$, an assessment of the correlation function $K_x^*(t_1, t_2)$, an assessment of the covariance function $R_x^*(t_1, t_2)$, and other assessments. If the *sample is random*, the assessments are *random* (they are called *estimators*), and if the *sample is determinate*, the assessments are *determinate* (they are called *estimates*).

4.6 Particularities of Samples of Random Variables and Stochastic Processes

We should pay attention to the similarities and differences between samples of a random variable X and samples of a stochastic process $X(t)$. In a random sample $\vec{X} = (X_1, \ldots, X_N)$ of a random variable X of fixed size N, the *elements* X_n *are disordered*. In a random sample $\vec{X}(t) = (X_1(t), \ldots, X_N(t))$ of a stochastic process $X(t)$ of a fixed size N, the *elements* $X_n(t)$ *are disordered* too. Therefore, in both cases, these elements *do not form a sequence of samples*.

However, a sample of a stochastic process has a feature not inherent in a sample of a random variable, namely, the set of values of any n-th element $X_n(t)$ of the random sample $\vec{X}(t)$, which *correspond to different values of the argument t*, *are in order*. They form a *sequence of random variables* $X_n(t_1), X_n(t_2), X_n(t_3), \ldots$ $(t_1 < t_2 < t_3 < \ldots)$. The ordering of the values of any element $X_n(t)$ of a random sample $\vec{X}(t)$ may lead (though not necessarily) to a dependence of the statistics $Y(t)$ and $y(t)$ on the argument t, and in particular to a dependence of the assessments on the argument t.

Any actual data $x_1, \ldots, x_N$ are the observation results of the physical phenomenon at different times and in different points of space, for example, the results of repeated measurements of the size of a machine element, weight load, voltage, current, or any other physical quantity. Taking into account the dependence of the result of observation on the time and point in space, it would be more correct to consider the resulting data as a function of time and space coordinates. However, it does not always make sense to do so.

Empirical data can often be assumed to be *independent or practically independent* of space-time coordinates.[3] Then the sample elements may be considered *disordered* and we may use the *model of a random variable*. The use of such a model is fully justified even when the empirical data *strongly depends on the space-time coordinates*, but this fact is *not important* for the given statistic.

In practice, it is usually rather simple statistics that are of interest, primarily the sample mean and sample variance (or the sample standard deviation). For these statistics, the question of ordering the data is not relevant.

When the assumption of independence of the space-time coordinates is unacceptable for a given statistic, a more complicated *model of the stochastic process* must be used, taking into account the *order of the data*. A statistic in this case may be either a random value or a stochastic process. An example of the first type of statistic could be a sampling function characterizing the correlation interval of a noise process, and an example of the second type of statistic might be a sampling function characterizing the shape of the pulse in an electronic device or the power spectral density of a noise process.

> Special attention should be paid to the fact that for *any statistic* the main characteristic is its *statistical stability*. In practice, a useful result can be obtained only if the *statistic is stable*. This means that *changing the sample size over a wide range does not lead to a major change in the values of the statistic*. *Unstable statistics are useless*.

A perfect (*ideal*) *statistic* is a function of a sample that is *perfectly statistical stable*. This means that, under an unlimited increase in the data volume, the statistic tends to a certain number (or to a determinate process), viz., it *has the convergence property*. A well known example of ideal statistics is provided by the *consistent estimators* discussed in the next chapter.

Unfortunately, ideal statistics, ideal statistical stability, convergence of statistics, and consistent estimators are *beautiful mathematic abstractions* which cannot be realized in practice and do not show up in real life. We must therefore look for ways to build and use *imperfect, but almost stable statistics* which provide a way to obtain objective information about the given phenomena.

[3]This hypothesis is widely used in metrology for measurement of various physical quantities.

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Chapter 5
Assessing Measurement Accuracy on the Basis of Probability Theory

Abstract Modern concepts for evaluating of measurement accuracy are examined and different types of error are described. We consider the classical determinate—random measurement model, in which the error is decomposed into systematic and random components. The point and interval estimators are described. For random estimators, the concepts of 'biased estimator,' 'consistent estimator,' 'effective estimator,' and 'sufficient estimator' are determined. The concept of critical sample size is introduced.

5.1 Descriptive Principles of Measurement Accuracy

Methods of mathematical statistics are used in various applications, such as *metrology*, which is the science of measurement.

Measurement is a set of operations that consists in comparing the *measurand* (*measured physical quantity*) with a uniform quantity *taken as a unit*. Measurement of a physical quantity does not only mean obtaining a number characterizing the measurand quantitatively. One must still *specify the accuracy of the measurement*.

Measurement accuracy is a quality category, characterized quantitatively by *measurement error* or *measurement uncertainty*. These parameters correspond to two different approaches used for the description of accuracy. The first is based on the *concept of measurement error* and the second on the *concept of measurement uncertainty*.

5.1.1 The Concept of Measurement Error

The foundations of the concept of error were laid by *Galileo Galilei* (Galilei 1948) near four centuries ago, who introduced the *concepts of systematic and random errors*.

In metrology three similar ideas are used in the definition of error: the *ideal value of the measurand*, its *actual* (*conventional true*) *value*, and the *measurement result*.

This chapter is based on material from the books (Gorban 2003, 2016)

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The *ideal value* of the measurand is the value of the measured physical quantity that perfectly reflects the properties of the object in quantitative and qualitative relations. The ideal value is an absolute truth that cannot be reached. Note that the ideal value is understood as a *determinate*, *invariant*, and *unique* value. In practice, the abstract notion of "ideal value" is replaced by the term "conventional true value". The *actual* (*conventional true*) *value* is the value of the measurand obtained experimentally, which is close to the ideal one and differs from it by an amount which is considered negligible for the given purpose. The *measurement result* is an approximate estimate of the ideal value obtained by the measurement. The *error in the measurement result* is the difference between the measurement result and the ideal (or actual) value of the measurand.

The accuracy varies with time. The dependence of the error on time is a wideband process.

Errors are often divided into *systematic, random, progressive,* and *blunder* (*slip*) errors.

This classification was introduced for convenience and arose from attempts to describe different parts of the frequency spectrum of the error. This division is conventional. It is important to realize that it was introduced only for convenience.

Systematic error (*bias*) is error which, in repeated measurements, *remains constant* or *varies according to a certain law*. *Random error* is error which, in repeated measurements, varies *randomly*. Random error is usually associated with random temporal and (or) spatial changes in the various influencing factors, while systematic error is associated with deviations of the parameters and (or) measurement conditions from the ideal. Systematic error reflects the particularities of the spectrum at zero frequency (or nearby).

Random error can be reduced by statistical processing of a number of measurement results, and systematic error by taking into account certain known dependencies of the measurement results on influencing parameters. In some cases, systematic error can be partially compensated by special measurement methods that allow one, *without determining it exactly*, to reduce its influence on the final result. A number of such methods are known: the substitution method, error compensation on sign, the opposition method, symmetric observations, and others.

If the systematic error does not change from one measurement to another (this is usually taken as the default, unless otherwise stated), then *it coincides with the expectation of the total error*. So the *expectation of the random error is equal to zero*.

Progressive error is an *unpredictable error*, varying slowly in time. *It reflects the particularities of the error spectrum at low and infra-low frequencies*. This error *cannot be represented* by systematic error and random error with any specific distribution law. From a mathematical point of view, progressive error arises due to the *unpredictable variability* of the distribution law in time, in particular, the expectation, variance, and other parameters.

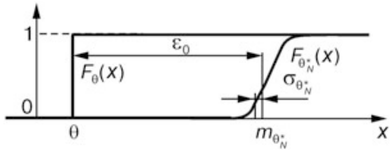

Fig. 5.1 The classical determinate—random measurement model

Progressive error is usually associated with aging or wear processes affecting the measuring device or its components: the discharge of power supplies, aging of radioelements, deformation and changes of elasticity in mechanical elements, oxidation, corrosion, and so forth. These processes are very slow. Noticeable changes in the error often occur only after months, years, or even decades. For a long time metrologists did not manifest any special interest in progressive error. However, it is precisely the *progressive error that limits the potential accuracy of actual measurements*.

A *blunder* is a *random error* in a separate observation, which essentially differs from other results obtained under fixed conditions. Blunders usually occur through mistakes, incorrect operator actions, or abrupt short-term changes in the measurement conditions.

Thus, a set of measurement errors is usually regarded (with a few exceptions in the form of progressive errors) as a random process, as a rule, stationary, that is represented by systematic and random components (with zero mean).

When constructing physical models of measurement values and their estimators, it is usually suggested that the *measurand has a determinate character, while its estimator is random*. Determinate mathematical models are used to describe measurement values, and random (stochastic) models with specific distribution laws are used to describe their estimators. Modern classical measurement theory is based on this description.

When measuring a scalar quantity, the measurand θ can be represented by a unit step distribution function $F_\theta(x)$ and the measurement result Θ_N^* by a distribution function $F_{\theta_N^*}(x)$ (Fig. 5.1). Such a measurement model may naturally be referred to as *determinate—random* (Gorban 2007, 2011, 2014).

5.1.2 The Uncertainty Concept

In the framework of the concept of uncertainty, two types of evaluation are considered (JCGM 2008; Uncertainty 2009): type A and type B standard uncertainty. *Type* A *uncertainty evaluation* is a method of evaluating uncertainty by the statistical analysis of series of observations, while *type* B *uncertainty evaluation* is a method of estimating uncertainty by means other than the statistical analysis of series of observations (JCGM 2008).

There are many possible sources of uncertainty in measurement, including (Uncertainty 2009):

(a) Incomplete definition of the measurand;
(b) Imperfect realization of the definition of the measurand;
(c) Nonrepresentative sampling—the sample measured may not represent the defined measurand;
(d) Inadequate knowledge of the effects of environmental conditions on the measurement or imperfect measurement of environmental conditions;
(e) Personal bias in reading analog instruments;
(f) Finite instrument resolution or discrimination threshold;
(g) Inexact values of measurement standards and reference materials;
(h) Inexact values of constants and other parameters obtained from external sources and used in data-reduction algorithms;
(i) Approximations and assumptions incorporated in the measurement method and procedure;
(j) Variations in repeated observations of the measurand under apparently identical conditions.

These sources are not necessarily independent, and some of the sources (a)–(i) may contribute to the source (j).

The measurement uncertainty of the measurand θ is characterized by the *type* A *evaluation* $u_{A\theta}$, *type* B *evaluation* $u_{B\theta}$, the *combined standard uncertainty* $u_\theta = \sqrt{u_{A\theta}^2 + u_{B\theta}^2}$, and the *expanded uncertainty* $U_\theta = ku_\theta$ (where k is a *coverage factor*).

Thus the classification of errors into random and systematic is determined by the *different nature of their origination and manifestation* in the measuring process, and the classification of the uncertainties into type A and type B evaluations is determined by the *different methods for obtaining them.*

Recently, metrologists have moved toward the opinion that the uncertainty concept is more progressive than the error concept. We can agree with this point of view in part, but not completely. On the one hand, *in the uncertainty concept, the real nature of the measurement inaccuracy does not play an essential role*. It opens the possibility in principle of taking into account, not only the random sources of inaccuracy, but others too, so this concept should be welcomed. On the other hand, a weak aspect of the uncertainty concept is that it ignores the nature of the origination and manifestation of the inaccuracy. As a result, *the uncertainty concept becomes detached from the physical realities.*

One cannot ignore the physics of the given phenomena or take into account only the random sources of measurement inaccuracy. In our opinion, further conceptual development of measurement theory should be based either on modernizing the error concept by extending the measurement model to take into account more than just random factors, or on modernizing the uncertainty concept to account for the origination and manifestation of the measurement inaccuracy.

In the following, we shall follow the first line.

5.2 Point Estimators

5.2.1 Basic Concepts

Within the framework of probability theory, the measurement of a physical quantity is interpreted as evaluation of the parameter θ of a *random variable* X represented by the sample $X_1, \ldots, X_N$. It is usually assumed that the estimated parameter θ (measurand) is a *single-valued determinate scalar quantity whose value is not changed during the measurement*. In addition the measurement results are of *random type* and adequately described by a *random sample* $X_1, \ldots, X_N$ (usually homogeneous with independent elements). At the same time the results of concrete measurements $x_1, \ldots, x_N$ are *determinate realizations* of the random sample.

Probability theory offers two options for estimating parameters. The first is based on formation of point estimators and the second on using interval estimators. We begin with the first option.

The *point random estimator* Θ_N^*, or simply *estimator*, of the parameter θ is a statistic (i.e., a function of the sample $X_1, \ldots, X_N$). The sample is random, so the estimator Θ_N^* is random too. The *estimate* θ_N^* of the estimator Θ_N^* is the determinate value regarded as an approximate value of the parameter θ.

The deviation of the estimator Θ_N^* from the real value of the parameter θ is characterized by the *measurement error* $Z_N = \Theta_N^* - \theta$ (random variable), and the deviation of the concrete estimate θ_N^* from θ by the *measurement error* $z_N = \theta_N^* - \theta$ (determinate value).

Errors are described in various ways. The error Z_N is often characterized by the *systematic error* ε_0 and the *standard deviation* (SD) $\sigma_{\theta_N^*}$ of the estimator Θ_N^* (or the *variance* $\sigma_{\theta_N^*}^2$ of the estimator). Numerical parameters characterizing the error Z_N are also the *mean square error*

$$\Delta_{z_N}^2 = \mathrm{E}\left[\left|\Theta_N^* - \theta\right|^2\right] = \varepsilon_0^2 + \sigma_{\theta_N^*}^2 \tag{5.1}$$

and the square root Δ_{z_N} of this parameter.

Estimators can be represented by different statistics. Important characteristics of the estimator are *unbiasedness*, *consistency*, *efficiency*, and *sufficiency*.

5.2.2 Biased and Unbiased Estimators

The *estimator* Θ_N^* of the parameter θ is *unbiased* if the expectation $m_{\theta_N^*}$ of the random variable Θ_N^* calculated on the population of samples of *any final size N* equals the estimated parameter, i.e., for all N, one has the equality $m_{\theta_N^*} = \theta$. Otherwise, the *estimator* is *biased* and is characterized by the *bias* $\varepsilon_{0N} = m_{\theta_N^*} - \theta$.

It is often assumed that the expectation $m_{\theta_N^*}$ and the bias ε_{0N} do not depend on the sample size N. Then $m_{\theta_N^*} = m_{\theta^*}$, $\varepsilon_{0N} = \varepsilon_0$.

In the framework of the determinate—random measurement model the error Z_N is represented as the sum of two components: a *systematic error* and a *random error with zero expectation*. The systematic error is characterized by the *determinate bias* ε_0 and the random error V_N by the *error variance* $\sigma_{z_N}^2$ (which coincides with the estimator variance $\sigma_{\theta_N^*}^2$) or the *standard deviation of the error* σ_{z_N} (which coincides with the estimator standard deviation $\sigma_{\theta_N^*}$) (Fig. 5.1).

As can be seen from the figure, the error

$$z_N \in \left[\varepsilon_0 - k\sigma_{\theta_N^*}, \varepsilon_0 + k\sigma_{\theta_N^*}\right], \tag{5.2}$$

and the estimating parameter

$$\theta \in \left[\ \theta_N^* - \varepsilon_0 - k\sigma_{\theta_N^*}, \theta_N^* - \varepsilon_0 + k\sigma_{\theta_N^*}\right], \tag{5.3}$$

where k is a constant determining the degree of confidence (usually $k \in [1, 3]$).

The presence of estimator bias and the magnitude of the bias ε_0 depend on the distribution law of the random variable X. In some laws the estimator can be biased, and in others unbiased. One usually tries to use estimators that are unbiased for any distribution.

In fact, the bias depends on a priori information about other distribution parameters, for instance, the expectation or variance. Estimators that are unbiased with a priori information may be biased in its absence, and vice versa.

An *unbiased estimator of the expectation* of the random variable X is the estimator $m_x^* = \frac{1}{N}\sum_{n=1}^{N} X_n$. If the *expectations* of the random variables X and Y *are not known a priori*, *unbiased estimators* of the variance, standard deviation, and cross-covariance moment are the estimators

$$D_x^* = \frac{1}{N-1}\sum_{n=1}^{N}\left(X_n - m_x^*\right)^2,$$

$$\sigma_x^* = k_N\sqrt{\frac{1}{N-1}\sum_{n=1}^{N}\left(X_n - m_x^*\right)^2},$$

$$R_{xy}^* = \frac{1}{N-1}\sum_{n=1}^{N}\left(X_n - m_x^*\right)\ \left(Y_n - m_y^*\right);$$

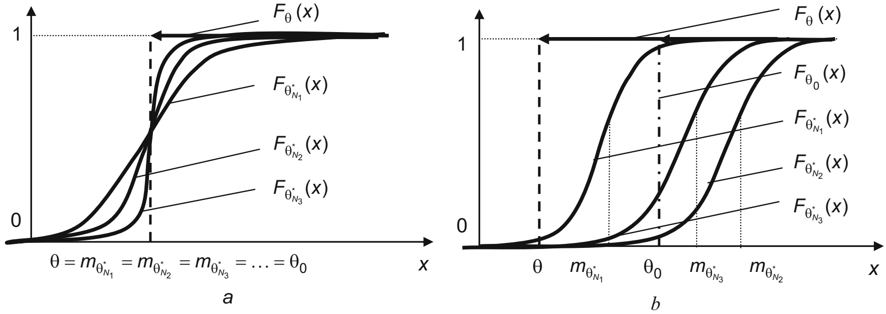

Fig. 5.2 Examples of an unbiased consistent estimator (**a**) and a biased inconsistent estimator which converges to the value $\theta_0 \neq \theta$ (**b**)

and if the *expectations* of these random variables *are known a priori*, then *unbiased estimators* are

$$D_x^* = \frac{1}{N}\sum_{n=1}^{N}(X_n - m_x)^2,$$

$$\sigma_x^* = k_{N+1}\sqrt{\frac{1}{N}\sum_{n=1}^{N}(X_n - m_x)^2},$$

$$R_{xy}^* = \frac{1}{N}\sum_{n=1}^{N}(X_n - m_x)\ \left(Y_n - m_y\right),$$

where with increasing sample size the coefficient k_N quickly approaches unity.[1]

5.2.3 Consistent and Inconsistent Estimators

An important characteristic of an estimator is its consistency. A *random estimator* Θ_N^* *of the parameter* θ is *consistent* if, for increasing sample size N, it converges in probability to this parameter:

$$\lim_{N\to\infty} P\left\{\left|\Theta_N^* - \theta\right| > \varepsilon\right\} = 0,$$

where $\varepsilon > 0$ is any small number. If the estimator Θ_N^* does not converge to any value or *converges to the value* $\theta_0 \neq \theta$, it is called an *inconsistent estimator*. In *probability theory*, *consistent estimators* are investigated, while inconsistent estimators are considered to be ‘bad’. Examples of an unbiased consistent estimator and a biased inconsistent estimator are shown in Fig. 5.2.

[1]The formula for calculating the value of this coefficient in the case of the Gaussian distribution, and the results of the calculation are given, for example, in (Gorban 2003).

As shown by experiment (see Chap. 6), real estimators do not have the convergence property, i.e., they are *inconsistent*. Research on inconsistent estimators goes beyond classical probability theory. They are studied in the *theory of hyper-random phenomena*. We shall come back to the examination of inconsistent estimators in Chap. 10.

5.2.4 Efficient Estimators

An important characteristic of an estimator is its efficiency. A *random estimator* Θ^*_{eN} *of the parameter* θ is *efficient* if the corresponding mean square error $\Delta^2_{z_{eN}}$ is smaller than the mean square error $\Delta^2_{z_{iN}}$ of any other estimator Θ^*_{iN} of the same sample size. In other words

$$\mathrm{E}\left[\left(\Theta^*_{eN}-\theta\right)^2\right] < \mathrm{E}\left[\left(\Theta^*_{iN}-\theta\right)^2\right], \quad i = 1, 2, \ldots .$$

Note that there is another definition of estimator efficiency based on the *Cramer–Rao inequality* (Van Trees 2004; Gorban 2003, 2016).

One usually tries to use estimators close to efficient estimators. The losses in efficiency caused by using an inefficient estimator Θ^*_N are described by the *parameter of relative effectiveness* $l = \mathrm{E}\left[\left(\Theta^*_{eN}-\theta\right)^2\right] / \mathrm{E}\left[\left(\Theta^*_{N}-\theta\right)^2\right]$ which lies in the range [0, 1]. The parameter l for an efficient estimator is equal to unity.

5.2.5 Sufficient Estimators

A *random estimator* Θ^*_N is said to be *sufficient* if the N-dimensional conditional distribution density $f_{\vec{x}/\theta^*}(x_1, \ldots, x_N)$ of the sample of the random variable X is independent of the parameter θ being evaluated. The distribution density $f_{\vec{x}/\theta^*}$ $(x_1, \ldots, x_N)$ for a sufficient estimator does not contain information about the parameter θ, i.e., a *sufficient estimator contains all useful information about the parameter that is in the sample*.

For a sample $X_1, X_2, \ldots, X_N$ of size N, an example of a sufficient estimator is the estimator of the expectation $m^*_x = \dfrac{1}{N}\displaystyle\sum_{n=1}^{N} X_n$, and an example of an insufficient estimator is the estimator $m^*_x = \frac{1}{N-1}\displaystyle\sum_{n=1}^{N-1} X_n$ in which the sample X_N is not used.

If the estimator is efficient, it is necessarily sufficient. The converse is not true. Sufficient estimators do not always exist. If there is a sufficient estimator, it is *not uniquely determined*. Any estimator which has a mutually unique relationship with a sufficient estimator is also a sufficient estimator.

Table 5.1 The properties of estimators of the parameters of the random variable in the absence of a priori information

Estimated parameter	Statistic used to estimate the parameter	Unbiasedness	Consistency	Effectiveness[a]	Sufficiency
Expectation m_x	$m_x^* = \frac{1}{N} \sum\limits_{n=1}^{N} X_n$	+	+	+	+
Variance D_x	$D_x^* = \frac{1}{N} \sum\limits_{n=1}^{N} \left(X_n - m_x^*\right)^2$	–	+	–	+
	$D_x^* = \frac{1}{N-1} \sum\limits_{n=1}^{N} \left(X_n - m_x^*\right)^2$	+	+	–	+
Probability of a random event A	$p^* = \frac{N_A}{N}$	+	+	+	+

[a]Efficiency for a Gaussian distribution

The properties of some estimators of the parameters of the random variable in the absence of a priori information about the expectation are given in Table 5.1.

5.3 Direct Statistical Measurement

One of the most widespread types of measurement is *direct statistical measurement*, consisting in direct measurement of a physical quantity and statistical processing of the resulting data. There are many measurement techniques taking into account the different specificities of the measurement conditions.

The estimator Θ_N^* is often the average of the measurement results $X_1, \ldots, X_N$: $\Theta_N^* = \frac{1}{N}\sum_{n=1}^{N} X_n$. Then the determinate estimate θ_N^* formed on the basis of a set of *concrete measurements* $x_1, \ldots, x_N$ is given by $\theta_N^* = \frac{1}{N}\sum_{n=1}^{N} x_n$. It is generally assumed that the measurement results $X_1, \ldots, X_N$ are *independent*, have *identical but unknown* distribution law, *unknown expectation*, and *unknown variance* D_x. Then the standard deviation $\sigma_{\theta_N^*}$ of the estimator Θ_N^* and the variance D_x are related by $\sigma_{\theta_N^*} = \sqrt{D_x/N}$. Since the variance D_x is unknown, instead of the variance, its estimate is used. The expectation is unknown, so the variance estimate is $D_x^* = \frac{1}{N-1}\sum_{n=1}^{N}\left(x_n - \theta_N^*\right)^2$.

On the basis of (5.2) and (5.3), the bounds of the interval in which the parameter θ is presumed to lie (the *confidence interval*) are given by the expressions

$$\theta_{iN} = \theta_N^* - \varepsilon_0 - k\sigma_{\theta_N^*}^*, \quad \theta_{sN} = \theta_N^* - \varepsilon_0 + k\sigma_{\theta_N^*}^*, \tag{5.4}$$

and the bounds of the measurement error by the expressions

$$z_{iN} = \varepsilon_0 - k\sigma_{\theta_N^*}^*, \quad z_{sN} = \varepsilon_0 + k\sigma_{\theta_N^*}^*, \tag{5.5}$$

where $\sigma_{\theta_N^*}^* = \sqrt{D_x^*/N}$ is the estimate of the standard deviation of the estimate θ_N^*.

It follows from this that, in the case of independent measurement results $X_1, \ldots, X_N$, the measurement technique should involve the following:

(1) Measure N times the parameter θ and form the sample $(x_1, \ldots, x_N)$;

(2) Calculate the estimate $\theta_N^* = \frac{1}{N}\sum_{n=1}^{N} x_n$ of the parameter θ;

(3) Calculate the estimate $\sigma_{\theta_N^*}^* = \sqrt{\frac{1}{N(N-1)}\sum_{n=1}^{N}\left(x_n - \theta_N^*\right)^2}$ of the standard deviation of the estimate θ_N^*;

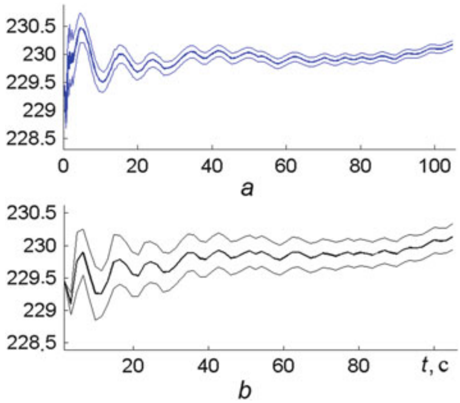

Fig. 5.3 Dependence of the mains voltage estimate on the value of the averaging interval without taking into account the correlation of the samples (**a**) and taking into account this correlation (**b**)

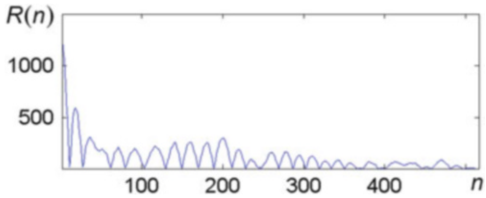

Fig. 5.4 Covariance function of the process presented in Fig. 1.4a

(4) Use (5.4) and (5.5) to calculate the bounds θ_{iN}, θ_{sN} of the interval in which the parameter θ is presumed to lie, and the bounds z_{iN}, z_{sN} of the interval of measurement error z_N.[2]

Note that, if the measurement results are dependent, the technique should take this into consideration. Ignoring this fact can lead to considerable losses in the accuracy of estimation. To illustrate the above, Fig. 5.3 shows an example for the results presented in Fig. 1.4a, viz., evaluation of the mains voltage estimate over the first 100-s observation interval.

The bold lines correspond to the estimate obtained for increasing amounts of data, and the thin lines to the bounds of the interval in which the measured parameter is presumed to lie. Both figures are obtained by the described method, but Fig. 5.3a uses all samples of the data, while Fig. 5.3b uses every eighth sample. Decimation is introduced to take into account the correlation between the samples (see Fig. 5.4).

Comparing Figs. 5.3a, b shows that ignoring the correlation of the samples leads to a essential distortion in the evaluation of the sample mean (especially when the amount of data is small) and to underestimation of the measurement error (by almost a factor of three).

[2]In practical calculations the constant k in (5.4) and (5.5) is given by the investigator and the systematic error ε_0 is taken to be the value specified in the documents accompanying the measurement device.

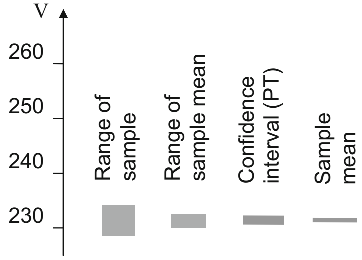

Fig. 5.5 Calculation results for parameters characterizing the mains voltage on a 100-s observation interval

The results of the calculation of different parameters characterizing the mains voltage on a 100-s observation interval are shown in Fig. 5.5. The parameters shown in the figure (the *range of the sample*, the *range of the sample mean*, the *confidence interval* obtained using the techniques of probability theory, and the *sample mean*) are calculated from the data shown in Figs. 1.4a, b, and 5.3b.

The classic measurement technique presented above and others using the determinate—random measurement model are based on the assumption that the sample elements and the estimator are adequately described by random models. When this assumption is not valid, the errors are large. This issue will be discussed in Chap. 10.

5.4 Critical Size of a Random Sample

The mean square error $\Delta^2_{z_N}$ is determined by the bias ε_0 and the standard deviation of the estimator $\sigma_{\theta^*_N}$ [see (5.1)]. With increasing sample size N, this magnitude tends to the square of the bias ε_0^2.

Let the estimator Θ^*_N be the average of the sample $(X_1, \ldots, X_N)$. The sample elements are independent and have identical variance D_x. Then the variance of the estimator

$$\sigma^2_{\theta^*_N} = D_x/N \tag{5.6}$$

and the root of the mean square error is described by the expression

$$\Delta_{z_N} = \sqrt{\varepsilon_0^2 + D_x/N}. \tag{5.7}$$

The dependence of the magnitude Δ_{z_N} on the defining parameters is shown in Fig. 5.6. The thicker line in the figure corresponds to the larger variance D_x.

It is clear from the figure that, when $N \to \infty$, the root mean square error tends to the bias ε_0 (the systematic error). If the bias ε_0 is negligible, the magnitude Δ_{z_N} is in inverse proportion to the root of the sample size N. It follows from this that, theoretically, by increasing N, the accuracy of the measurement can grow without

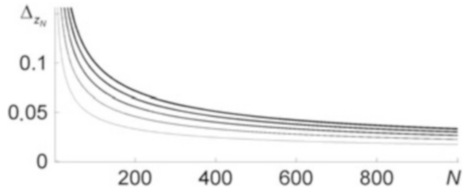

Fig. 5.6 Dependence of the magnitude Δ_{z_N} on the sample size N and the variance D_x. The bias $\varepsilon_0 = 0.01$, $D_x = 0.2;\ \ 0.4;\ \ 0.6;\ \ 0.8;\ \ 1$

limit, and as $N \to \infty$, it should become infinitely large. However, this contradicts the experimental data.[3]

If the bias ε_0 is small but cannot be neglected, the following question arises: how many items should be taken in the sample *to minimize the error*? The answer seems obvious: the more samples, the better the estimator. But note that, even with a very large amount of data, as can be seen from (5.7), the error will not be less than the bias ε_0. So it makes sense to put the question another way: how many elements is it *advisable* to use in the sample, taking into account the fact that the cost of obtaining additional measurement results and processing them increases with increasing sample size? The answer to this question is that it makes sense to choose a number of samples N_0 in which the variance $\sigma^2_{\theta^*_N}$ of the estimator becomes negligible in comparison with the square of the bias ε_0^2.

According to the standard (GOST 2006), the random component of the error can be neglected[4] if $\varepsilon_0 > 8\sigma_{\theta^*_N}$. Then

$$\varepsilon_0^2/\sigma^2_{\theta^*_{N_0}} \simeq 64. \tag{5.8}$$

The value N_0 corresponding to (5.8) may be called the *critical sample size*. From (5.6) and (5.8) the critical sample size is

$$N_0 \simeq 64 D_x/\varepsilon_0^2. \tag{5.9}$$

If for instance the variance $D_x = \varepsilon_0^2$ then the critical sample size $N_0 = 64$.

The expression (5.9) can serve as a guide in choosing the sample size. Note that the above formula can be used only when the square of the bias ε_0^2 is not considerably smaller than the variance D_x. If $\varepsilon_0^2 \ll D_x$, then according to (5.9), $N_0 > 1000$. In practice, it is rarely possible to reduce the measurement error significantly by averaging the data when the number of units exceeds several hundred, or worse, several thousand.

[3]This contradiction between theory and practice was the starting point of an in-depth study of the phenomenon of statistical stability and led to the development of the theory of hyper-random phenomena.

[4]According to the same standard, it is possible to ignore the systematic component of the error if $\varepsilon_0 < 0{,}8\sigma_{\theta^*_N}$.

Probability theory does not give a satisfactory explanation as to why, at low bias, an ultra-high measurement accuracy cannot be achieved by statistical processing of a large number of real data. The explanation of this effect will be presented in Chap. 10.

5.5 Interval Estimator

Consider the classic interval approach for parameter estimation. Let the measured parameter θ be a *determinate value*, the estimator Θ_N^* an *unbiased random variable*, and the random error $Z_N = \Theta_N^* - \theta$ described by the probability density $f_z(z)$.

The probability that the modulus of the error does not exceed the value ε, viz.,

$$\gamma = P\left\{\left|\Theta_N^* - \theta\right| \leq \varepsilon\right\} \tag{5.10}$$

is called the *confidence level* (*confidence coefficient*).

The expression (5.10) represents the probability that the true value of the parameter θ lies in the interval

$$I_\gamma = \left[\Theta_N^* - \varepsilon,\ \ \Theta_N^* + \varepsilon\right] \tag{5.11}$$

(Fig. 5.7). This interval is called the *confidence interval of the parameter θ*.

The length of the confidence interval is 2ε. Its mid-point on the θ axis is the random estimator Θ_N^*. Therefore, the confidence interval is a *random interval*. Note that, not only the location of the confidence interval on the axis of the parameter θ, but also its *length*, i.e., the value 2ε, may be a random variable. The bounds of the confidence interval are called the *confidence bounds* (*lower and upper confidence bounds*). The probability γ can be interpreted as the probability that the interval I_γ covers the point $\theta: \gamma = P(\theta \in I_\gamma)$.

Fig. 5.7 Confidence interval

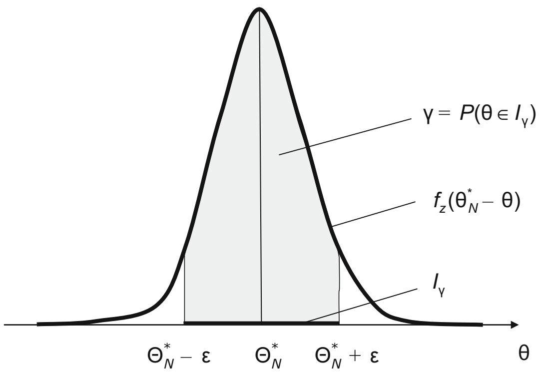

Table 5.2 The dependence of the parameter k on the parameter γ for the Gaussian distribution law

γ	0.80	0.87	0.94	0.95	0.96	0.97	0.98	0.99	0.9973
k	1.282	1.513	1.880	1.960	2.053	2.169	2.325	2.576	3.000

Obviously, the approach described here is also applicable when the estimator is biased on ε_0. Then the confidence interval is described by the expression

$$I_\gamma(p) = \left[\Theta_N^* - \varepsilon_0 - \varepsilon,\ \ \Theta_N^* - \varepsilon_0 + \ \varepsilon\right],$$

and the confidence coefficient by the expression

$$\gamma = P\left\{\left|\Theta_N^* - \varepsilon_0 - \theta\right| \le \varepsilon\right\}.$$

Note that the random error is sometimes characterized by an interval $[-\varepsilon,\ \ \varepsilon]$, *without mention of the confidence coefficient*. There are different interpretations of such a description. The first is that the error has a *definite but unknown distribution law* depending on the essence of the task, and the confidence coefficient is *close to unity* (without specifying its exact value). The second assumes that the *distribution law of the random error is uniform* on the interval $[-\varepsilon,\ \ \varepsilon]$ and the corresponding confidence coefficient is equal to unity. A third interpretation (not usually discussed in the literature) assumes that the interval $[-\varepsilon,\ \ \varepsilon]$ represents the set of possible values of the error without specifying the probability density. This means that the error has an *interval nature* and hence can be interpreted as random only in a figurative sense. The fourth interpretation takes into consideration the dependence [through the confidence coefficient k (see 5.2)] of the error value on the confidence level. It should be noted that the confidence level γ depends not only on the confidence coefficient k, but also on the distribution law. The relationship between the parameters γ and k for a Gaussian distribution law is given in the Table 5.2.

* * *

In our survey of the foundations of the probability theory and in particular, in the description of the basic principles of mathematical statistics of probability theory, we have emphasized more than once that some of the results of the probability theory *do not agree with the results of experimental studies of real physical variables and processes*. The reason for this lies in the *imperfect nature of the physical phenomenon of statistical stability* manifested in *the absence of the tendency to convergence of actual statistics*.

The next chapter presents the results of experimental investigations into violations of statistical stability in actual processes. These investigations served as the basis for the physical-mathematical theory of hyper-random phenomena which focuses on the formation of statistically stable statistics.

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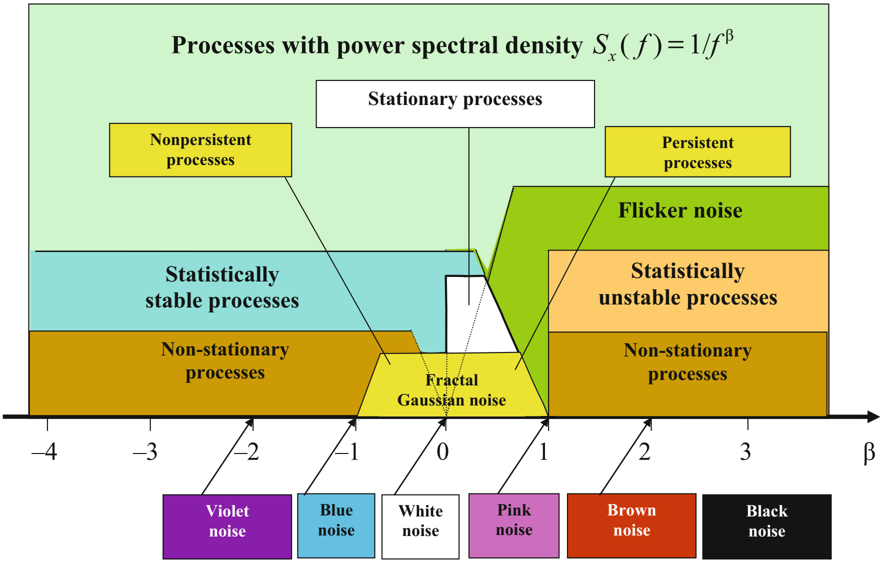
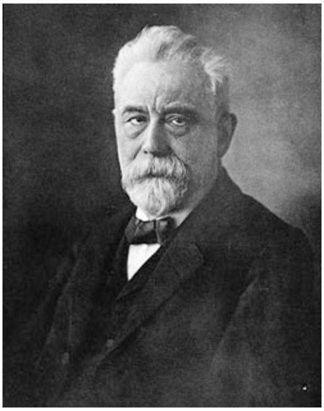
Part III
Experimental Study of the Statistical Stability Phenomenon



Processes with power spectral density described by a power function



Wilhelm Lexis, German statistician, economist, and social scientist (1837–1914) (Source: https://upload.wikimedia.org/wikipedia/commons/0/08/Voit_172_Wilhelm_Lexis.jpg)

> Lexis was aware that many series were not stable. For non-stable series, he imagined that the underlying probabilities varied over time, being affected by what he called "physical" forces (as opposed to the random "non-essential" forces that would cause an observed rate to be different than the underlying probability). (Lexis 2016).

A.V. Skorochod:

> The most fully developed concept of uncertainty uses probabilistic randomness . . . I note that the assumption that a certain sequence, for example, of numbers, was obtained by independent observations of a random variable (whether or not its distribution is known) imposes on the sequence very severe restrictions, which are hardly realistic for many actual phenomena. (Ivanenko and Labkovskiy 1990).

V.N. Tutubalin:

> Scientific conscientiousness requires each investigator to use all the available methods to check for statistical stability, but this availability is hard to guarantee. . . . All conceivable experiments can be divided into three groups. The first group consists of good experiments for which the test results are ensured. The second group consists of the worse experiments for which there is not full stability, but there is statistical stability. The third group includes very bad experiments for which there is not even statistical stability. In the first group, everything is clear without probability theory, and in the third group, this theory is useless. The second group is the real scope of probability theory, but we can hardly ever be quite sure that that the given experiment relates rather to the second group than the third. (Tutubalin 1972).

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Chapter 6
Methodology and Results from Investigation of the Statistical Stability of Processes

Abstract Here we formalize the notion of statistical stability of a process. The parameters of statistical instability with respect to the average and with respect to the standard deviation are investigated. Measurement units are proposed for the statistical instability parameters. We specify the concept of an interval of statistical stability of a process. The dependencies of the statistical stability of a process on its power spectral density and its correlation characteristics are established. We then consider various processes described by a power function of the power spectral density and investigate the statistical stability of such processes. For narrowband processes, we present the investigation results of statistical stability violations. Statistically unstable stationary processes are considered. We present experimental results for the statistical stability of a number of actual processes of different physical kinds.

6.1 Formalization of the Statistical Stability Concept

Curiously enough, the concept of statistical stability was not formalized until recently. First of all, it should be noted that a process that is statistically stable with respect to a given statistic may be unstable with respect to other statistics (see Sect. 1.3). This means that statistical stability is an attribute not only of a process, but also of the statistic. Therefore,

> The *concept of statistical stability concerns the combined 'process—statistic' complex.*

The chapter is based on material from the books (Gorban 2011, 2014, 2016a, 2017)

I.I. Gorban, *Randomness and Hyper-randomness*, Mathematical Engineering,
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6.1.1 Statistically Unstable Sequences and Processes with Respect to the Average and Standard Deviation

The concept of statistical stability can be formalized on the basis of an *analysis of changes in the sample variances of different statistics.*

For example, to define the *concept of statistical stability of the process with respect to the average*, we can take the analysis of *changes in the sample variance*

$$\bar{D}_{Y_N} = \frac{1}{N-1} \sum_{n=1}^{N} (Y_n - \bar{m}_{Y_N})^2 \tag{6.1}$$

of the *fluctuations in the average*

$$Y_n = \frac{1}{n} \sum_{i=1}^{n} X_i \left(n = \overline{1,N}\right), \tag{6.2}$$

where $\bar{m}_{Y_N} = \frac{1}{N} \sum_{n=1}^{N} Y_n$ is the sample mean of the average fluctuations; and to define the *concept of statistical stability of the process with respect to the standard deviation*, we can put the *analysis of changes in the sample variance*

$$\bar{D}_{Z_N} = \frac{1}{N-2} \sum_{n=2}^{N} (Z_n - \bar{m}_{Z_N})^2 \tag{6.3}$$

of the *fluctuations in the sample standard deviation*

$$Z_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (X_i - Y_n)^2 \left(n = \overline{2,N}\right)}, \tag{6.4}$$

where $\bar{m}_{Z_N} = \frac{1}{N-1} \sum_{n=2}^{N} Z_n$ is the average of the sample standard deviations.

For a *random sample* the simplest variants in the formulation may be the following (Gorban 2011).

A random sample (*sequence of random variables*) $X_1, X_2, \ldots$ *is statistically stable with respect to the average* if, when the sample size N goes to infinity, the expectation of the sample variance (6.1) goes to zero.

A random sample (*sequence of random variables*) $X_1, X_2, \ldots$ *is statistically stable with respect to the standard deviation* if, when the sample size N goes to infinity, the expectation of the sample variance (6.3) goes to zero.

Random samples (*sequences of random variables*) that do not satisfy these definitions are considered to be *statistically unstable with respect to the average and standard deviation*, respectively.

The type of convergence in this case is not important. However, to make the definitions with the necessary mathematical rigor, we shall refer to convergence in probability here.

A reduction of the variance of the sample mean with increasing data volume can be caused, not only by stabilization of the average, but also by *decreasing dispersion in the initial process*. To mitigate this effect, it seems reasonable to redefine the concept of statistical stability as follows.

A random sample (*sequence of random variables*) $X_1, X_2, \ldots$ *is statistically stable with respect to the average* if, when the sample size N goes to infinity, the *parameter of statistical instability with respect to the average*, viz.,

$$\gamma_N = \frac{\mathrm{E}[\bar{D}_{Y_N}]}{\mathrm{E}[\bar{D}_{X_N}]} \tag{6.5}$$

tends to zero; and it is *statistically stable with respect to the standard deviation* if the *parameter of statistical instability with respect to the standard deviation*, viz.,

$$\Gamma_N = \frac{\mathrm{E}[\bar{D}_{Z_N}]}{\mathrm{E}[\bar{D}_{X_N}]}. \tag{6.6}$$

tends to zero, where

$$\bar{D}_{X_N} = \frac{1}{N-1}\sum_{n=1}^{N}(X_n - Y_N)^2 \tag{6.7}$$

is the sample variance of the process.

Random samples (*sequences of random variables*) for which the *parameters of statistical instability* γ_N and Γ_N do not go to zero are considered to be *statistically unstable with respect to the average and standard deviation*, respectively.

Definitions concerning the statistical stability/instability of sequences of random variables apply to *random (stochastic) processes*. The parameters of the statistical instability of the random process $X(t)$ are described by the same formulas (6.1)–(6.7). In this case the sequence $X_1, X_2, \ldots, X_N$ is the sequence of values of the process $X(t)$ at successive times $t_1, t_2, \ldots, t_N$.

An approximately determinate value x_0 can be interpreted as a degenerate random variable, whose distribution function is a unit step function $F(x) = \mathrm{sign}[x - x_0]$ at the point x_0. Therefore, these definitions apply to a *sequence of determinate variables*, and *determinate processes*.

Sequences and processes for which the both parameters of statistical instability γ_N and Γ_N tend to zero are said to be *statistical stable in the broad sense* and those for which at least one of these parameters does not tend to zero are said to be *statistically unstable in the broad sense*.[1]

[1] Along with the concept of statistical stability in the broad sense, there is a concept of *statistical stability in the narrow sense* (Gorban 2014). It is not used in our description, so we do not dwell on it.

Instead of the parameters of statistical instability γ_N and Γ_N, it is sometimes more convenient to use the *parameters* μ_N *and* M_N defined as follows:

$$\mu_N = \sqrt{\gamma_N/(1+\gamma_N)}, \tag{6.8}$$

$$\mathrm{M}_N = \sqrt{\Gamma_N/(1+\Gamma_N)}. \tag{6.9}$$

The actual range of the parameters γ_N, Γ_N is $[0, \infty)$ and that of the parameters μ_N, M_N is $[0, 1)$. The smaller the values of the parameters γ_N, μ_N and Γ_N, M_N, the more stable the sequence with respect to the average and standard deviation, respectively. Small values for large sample sizes N point to high statistical stability of the sequence, and large values point to statistical instability.

Any measurement procedure consists in the comparison of the measurement result with some unit. In order to characterize the degree of instability quantitatively, we require *measurement units* and comparison with them would allow us to judge the degree of instability with respect to the average and standard deviation.

6.1.2 Measurement Units for Statistical Instability Parameters

For *the parameters of statistical instability* γ_N and Γ_N, the role of *measurement unit* can be played by a variable γ_{0N} which is the parameter γ_N calculated for the *standard statistically stable sequence of uncorrelated samples of white Gaussian noise*; and for the parameters μ_N and M_N, this role can be played by the variable μ_{0N} associated with the variable γ_{0N} by the expression $\mu_{0N} = \sqrt{\gamma_{0N}/(1+\gamma_{0N})}$.

The peculiarity of *measurement units* γ_{0N} and μ_{0N} is their *dependence on the sample size N*.

The absolute level of statistical instability with respect to the average and standard deviation in units γ_{0N} characterizes the *parameters of statistical instability* h_N *and* H_N:

$$h_N = \gamma_N/\gamma_{0N}, \tag{6.10}$$

$$H_N = \Gamma_N/\gamma_{0N}. \tag{6.11}$$

The actual range of these parameters is $[0, \infty)$. Their measurement unit is the number $h_{0N} = 1$, which does not depend on the sample size.

Values of the parameters h_N and H_N close to unity for large sample sizes N point to high statistical stability of the process, while higher values point to its statistical instability.

6.1.3 Statistical Stability Intervals

When solving practical tasks, the behavior of statistics on an infinite observation interval is not usually important, although this is part of the formal definition of statistical stability. More important is the behavior of the statistics on the *actual observation interval*: the presence or absence of *trends* indicating a violation of statistical stability. If on the observation interval these trends are not visible, the process can be considered as statistically stable, otherwise as statistically unstable.

As a rule, various statistics and processes have different *statistical stability intervals*. The concepts of the interval of statistical stability with respect to the average τ_{sm} and the interval of statistical stability with respect to the standard deviation τ_{sd} can be formulated by defining the *statistical stability borders of the confidence intervals*.

For the statistical instability parameters γ_N and Γ_N, the upper statistical stability border of the confidence interval is given by

$$\gamma_{0N}^{+} = \gamma_{0N} + \varepsilon\sigma_{\gamma_{0N}^{*}}, \tag{6.12}$$

where ε is the confidence parameter that determines the width of the confidence interval and $\sigma_{\gamma_{0N}^{*}}$ is the standard deviation of the variable $\gamma_{0N}^{*} = \bar{D}_{Y_N}/\mathrm{E}[\bar{D}_{X_N}]$ calculated for the standard statistically stable sequence of samples of white Gaussian noise.

For the statistical instability parameters μ_N and M_N, the upper statistical stability border of the confidence interval is described by the expression $\mu_{0N}^{+} = \sqrt{\gamma_{0N}^{+}/\left(1+\gamma_{0N}^{+}\right)}$, and for the parameters h_N and H_N, by the expression $h_{0N}^{+} = \gamma_{0N}^{+}/\gamma_{0N}$.

The *criteria of statistical stability violation with respect to the average and with respect to the standard deviation* (which determine the *intervals of statistical stability* τ_{sm} and τ_{sd}) can be the exiting, respectively, of the parameters γ_N and Γ_N outside the border γ_{0N}^{+}, the parameters μ_N and M_N outside the border μ_{0N}^{+}, or the parameters h_N and H_N outside the border h_{0N}^{+}.

Note that other criteria can be used which give *more accurate assessments of the intervals of statistical stability*, for example, the following (Gorban 2016c). One considers some class of mathematical models of the processes depending on a key parameter β (for instance, the processes with power spectral density discussed in Sect. 6.2.3). The processes of this class with parameter $\beta < \beta_0$ are statistically stable and those with parameter $\beta \geq \beta_0$ are statistically unstable. Then the process with the parameter β_0 is the *limiting unstable process*.

The criterion of statistical stability violation with respect to the average (or with respect to the standard deviation) can be the exiting of the corresponding parameters of statistical instability $\gamma_N,\ \mu_N,\ h_N$ (or $\Gamma_N,\ \mathrm{M}_N,\ H_N$) calculated for the limiting statistically unstable process with parameter β_0.

Assessments of intervals of statistical stability obtained using this approach are higher than assessments obtained using the previous one. In practice, it is important that the estimate should not exceed the real interval of statistical stability. Therefore, preference is usually given to the previous approach.

Note that the interval of statistical stability generally depends on the order in which samples are taken.

Therefore, in calculations, one *cannot violate the natural order of the samples in the sequence or in the process.*

6.1.4 Estimates of Statistical Instability Parameters

In practical work, due to the limited amount of data, instead of the parameters of statistical instability γ_N, μ_N, h_N and Γ_N, M_N, H_N, we have to accept to use the appropriate estimates γ_N^*, μ_N^*, h_N^* and Γ_N^*, M_N^*, H_N^*.

If we can form only one realisation $x_1, x_2, \ldots, x_N$ (determinate sample) from the actual process, the estimates γ_N^*, Γ_N^* can be calculated by the formulas

$$\gamma_N^* = \bar{D}_{y_N} / \bar{D}_{x_N}, \quad \Gamma_N^* = \bar{D}_{z_N} / \bar{D}_{x_N},$$

in which $\bar{D}_{y_N}$, $\bar{D}_{x_N}, \bar{D}_{z_N}$ are values $\bar{D}_{Y_N}$, $\bar{D}_{X_N}, \bar{D}_{Z_N}$ calculated using (6.1)–(6.4), (6.7) for the given realization.

If we can form a number of realizations, the estimates γ_N^*, Γ_N^* can theoretically be calculated using the same formulas for every realization and then averaging the result. Note that the latter approach is rarely used in practice, because the main interest is not the parameters of statistical instabilities themselves, but the intervals of statistical stability determined by them. Therefore, it is *better to estimate the values of the intervals of statistical stability using single realizations and then average the results* (Gorban 2016b).

The estimates μ_N^*, M_N^* and h_N^*, H_N^* are calculated as follows:

$$\mu_N^* = \sqrt{\gamma_N^* / (1 + \gamma_N^*)}, \quad \mathrm{M}_N^* = \sqrt{\Gamma_N^* / (1 + \Gamma_N^*)},$$
$$h_N^* = \gamma_N^* / \gamma_{0N}, \quad H_N^* = \Gamma_N^* / \gamma_{0N}.$$

Estimates of the parameters of statistical instability γ_N^*, μ_N^*, h_N^* and Γ_N^*, M_N^*, H_N^* are *physical quantities* characterizing the process. Typically, in order to measure a physical quantity, *physical standard in the form of a measurement unit* is required. Note that in this case the measurement units γ_{0N}, μ_{0N}, h_{0N} *do not require physical standards*, because they are mathematical functions defined only by the sample size N. For a fixed value of N, they may theoretically be calculated exactly without error.

In physics, a small number of physical constants such as the speed of light in vacuum and some others are exactly set by definition (with zero error) (Fundamental 2016). The values γ_{0N}, μ_{0N}, h_{0N} also have zero error, but this is not due to the fact that they rely on such a definition, but because they are mathematical constants at a fixed value N.

6.2 Statistical Stability of Stochastic Processes

6.2.1 *Dependence of the Statistical Stability of a Stochastic Process on Its Spectrum-Correlation Characteristics*

Studies show that the statistical stability of a stochastic process with respect to the average and standard deviation is determined by its spectrum-correlation characteristics.

In particular, for the discrete stochastic process $X_1, X_2, \ldots, X_N$ (random sample) with zero expectation and power spectral density $S_{x_N}(k)$, the *parameter of statistical instability with respect to the average* γ_N when $N \to \infty$ is described by the following asymptotic formula (Gorban 2014, 2017):

$$\gamma_N = \frac{\sum_{k=2}^{N/2} \frac{1}{(k-1)^2}\left[\frac{\pi^2}{4} + (C + \ln(2\pi(k-1)))^2\right] S_{x_N}(k)}{4\pi^2 \sum_{k=2}^{N/2} S_{x_N}(k)}, \tag{6.13}$$

where k is the spectral sample number ($k = \overline{1, N}$), C is the *Euler–Mascheroni constant* ($C \approx 0.577216$).

If the expectation of the process equals zero and the second-order moments are finite (Gorban 2015b), then as $N \to \infty$, the parameter γ_N is given by the expression

$$\gamma_N = \frac{1}{N(q_N - Q_N)} \sum_{n=1}^{N} Q_n,$$

and the lower bound of the parameter of statistical instability with respect to the standard deviation by the inequality

$$\Gamma_N \geq \frac{1}{N(N-2)(q_N - Q_N)} \times \\ \times \left[(N-1)\sum_{n=2}^{N} \frac{n}{n-1}(q_n - Q_n) - \left(\sum_{k=2}^{N} \sqrt{\frac{k}{k-1}(q_k - Q_k)}\right)^2\right], \tag{6.14}$$

where

$$q_n = \frac{1}{n} \sum_{i=1}^{n} \mathrm{E}\left[X_i^2\right] \tag{6.15}$$

is the average of the variances of the samples, and

$$Q_n = \mathrm{E}\left[Y_n^2\right] = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathrm{E}\left[X_i X_j\right] \tag{6.16}$$

is the average of the correlation moments $K_{x_i x_j} = \mathrm{E}\left[X_i X_j\right]$ of the initial fragment, whose sample size is n.

6.2.2 *Physical Processes with Power Spectral Density Described by Power Functions*

In many cases, actual noise is well approximated by random processes whose power spectral density is described by a power function $1/f^{\beta}$ for various values of the shape parameter β. Such noise is sometimes called *color noise*.

Colour Noise
One thus speaks of violet, blue (cyan), white, pink, brown (red), and black noise (Table 6.1).

As the frequency increases, the power spectral density of the noise decreases for $\beta > 0$ and increases for $\beta < 0$.

Flicker noise and *fractal* (*self-similar*) *processes* are other examples of processes whose power spectral density is described by power functions.

Flicker Noise *Flicker noise* is such that its power spectral density increases with decreasing frequency. Research has shown that flicker noise is an extremely widespread phenomenon, typical of many electrical, magnetic, electromagnetic, acoustic, hydroacoustic, hydrophysical, astrophysical, and other processes (Zhigalskiy 2003; Kogan 1985). It has been found in metals, semimetals, semiconductors, gases, fluids, electrolytes, radio-electronic equipment, uniform and non-uniform conductors at high and low temperatures, films and contacts, animate and inanimate objects, etc. (Zhigalskiy 2003; Kogan 1985).

One thus distinguishes *equilibrium* and *nonequilibrium flicker noise*. As a rule, the power spectral density of the first is described by a law close to $1/f$, while that of the second is close to $1/f^{\beta}$, with a spectral shape parameter $\beta \geq 2$ (Zhigalskiy 2003).

Table 6.1 The color of the noise and the value of the shape parameter

No	Color of the noise	Shape parameter β
1	Violet	-2
2	Blue (cyan)	-1
3	White	0
4	Pink	1
5	Brown (red)	2
6	Black	>2

Fractal (Self-Similar) Processes By a *fractal random process $X(t)$ in the broad sense*, we shall mean a process whose correlation function $K_x(t_1, t_2)$ is equal up to a multiplier a^r of the correlation function of the process, compressed a times:

$$K_x(t_1, t_2) = \mathrm{E}[X(t_1)X(t_2)] = a^r \mathrm{E}[X(at_1)X(at_2)] = a^r K_x(at_1, at_2),$$

where r is the *self-similarity parameter*. Fractal processes are divided into *persistent processes* with positive correlation between the samples and *antipersistent processes* with negative correlation between them. The negative correlation is observed when $\beta \in (-1, 0)$, and the positive correlation when $\beta \in (0, 1)$.

6.2.3 Statistical Stability of Stochastic Processes with Power Spectral Density Described by a Power Function

Taking into account the prevalence of the processes with power spectral density described by a power function, much research has been carried out to investigate their statistical stability.

Studies with (6.13) and (6.14) show that the parameters of statistical instability with respect to the average γ_N and with respect to standard deviation Γ_N tend to zero only when the form factor of the spectrum $\beta < 1$.

> This means that the *process with power spectral density described by a power function is statistically stable with respect to the average and with respect to the standard deviation if $\beta < 1$ and statistically unstable if $\beta \geq 1$.*

In confirmation of the correctness of this conclusion, Fig. 6.1a, b show the results of calculations of the parameters γ_N and μ_N of statistical instability with respect to the average (solid lines). For comparison, the dashed lines show similar curves for standard white noise.

Since the state of statistical stability of the process changes at the point $\beta = 1$, the process with this particular parameter value can be regarded as a *limiting unstable process with respect to the average and standard deviation* (*limiting instability in the broad sense*).

For the limiting unstable process and $N = 1024$, the values of the parameters of statistical instability with respect to the average are $\gamma_N = 0.054$, $\mu_N = 0.23$, $h_N = 1.3$. For larger sample sizes, they are approximately at the same level. Therefore, if the sample volume $N > 1024$, we can use these values as upper borders of the confidence interval γ_{0N}^+, μ_{0N}^+, h_{0N}^+, instead of the ones calculated using (6.12).

Investigations show that *if $\beta \leq 0$ the process is more stable with respect to the average than with respect to the standard deviation* ($\gamma_N < \Gamma_N$), and *if $\beta > 0$ on the contrary it is less stable* ($\gamma_N > \Gamma_N$).

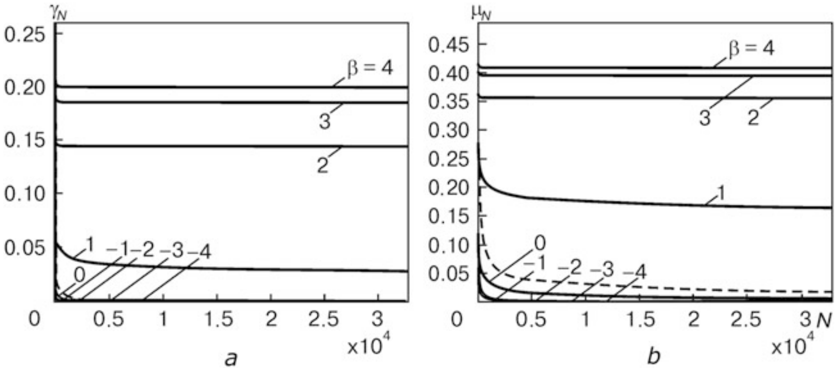

Fig. 6.1 Dependencies of the instability parameters γ_N (**a**) and μ_N (**b**) on the sample size N

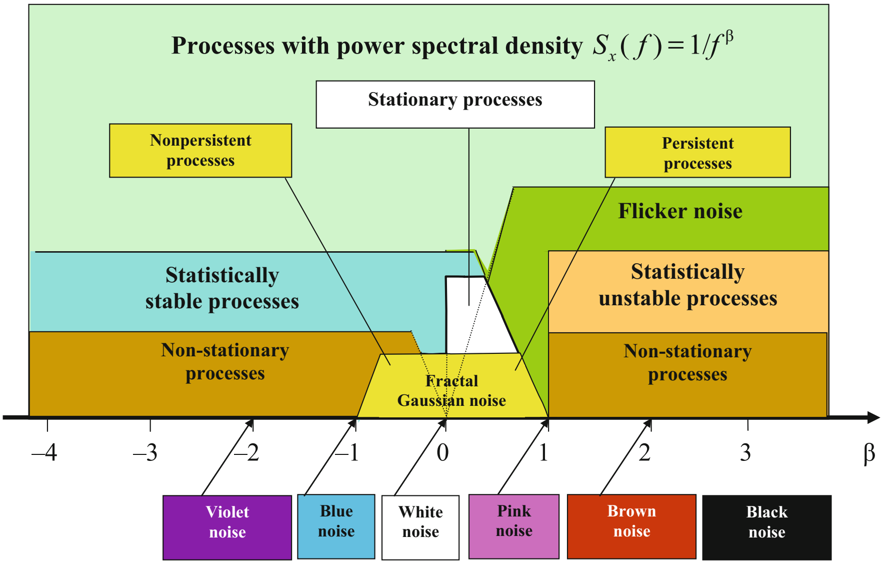

Fig. 6.2 Processes with power spectral density described by a power function

Summarizing the results of Sects. 6.2.2–6.2.3, we can make the following remarks (see Fig. 6.2):

– *Statistically stable with respect to the average and standard deviation* (*stable in the broad sense*) are stationary processes, some non-stationary processes, so-called fractal Gaussian noise, some flicker noise, and also violet, blue, and white noise;
– *Statistically unstable with respect to the average and standard deviation* (*unstable in the broad sense*) are some non-stationary processes, some flicker noise, and pink, brown, and black noise.

6.2.4 Dependence of the Statistical Stability of a Stochastic Process on Its Correlation Characteristics

Investigating the dependence of the statistical stability of a process on the correlation between samples shows that

A *positive correlation* between samples leads to a *decrease* in the statistical stability, and a *negative correlation,* to an *increase*.

The results presented in Fig. 6.3 demonstrate this effect. In the figure, the curves 1 are obtained for standard white Gaussian noise, curves 2 for positively correlated noise, even samples of which are repeats of the previous odd samples, and curves 3 for negatively correlated noise, even samples of which are repeats of the previous odd samples, but with the opposite sign.

6.2.5 Statistical Stability of Narrowband Stochastic Processes

Statistically unstable stochastic processes described by the power spectral density with shape parameter $\beta \geq 1$ are processes of *low-frequency* type. Studies show (Gorban 2015a) that:

Violations of statistical stability occur not only in the case of low-frequency processes, but also for *narrowband stochastic processes*.

This statement follows from simulation research. Figure 6.4a, c show the estimates of the statistical instability parameters with respect to the average and

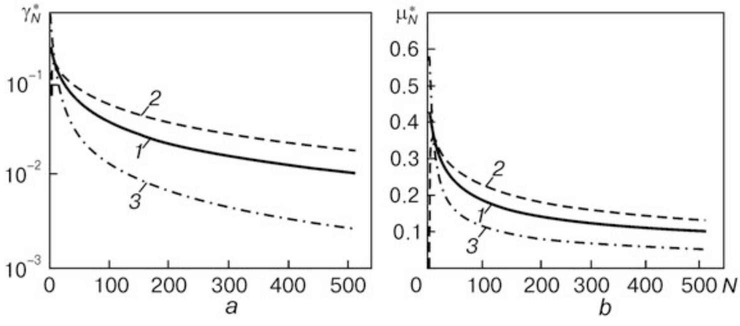

Fig. 6.3 Dependence of the statistical instability parameters γ_N^* (**a**) and μ_N^* (**b**) on the sample size N for different correlation types

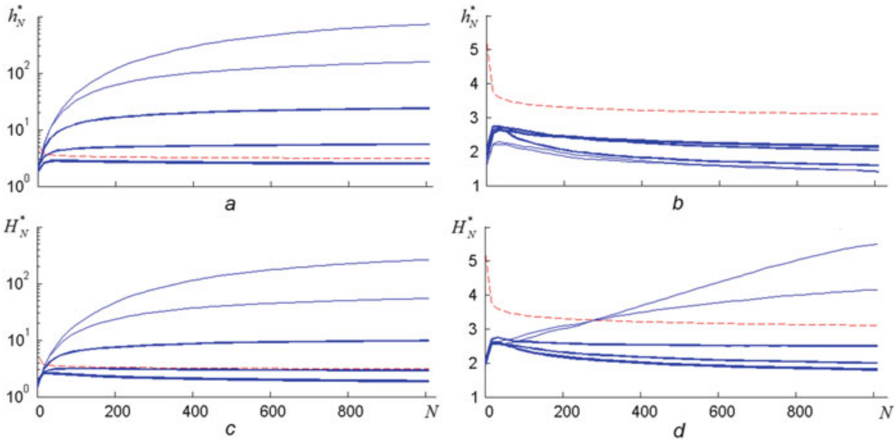

Fig. 6.4 Dependence of the estimate of the statistical instability parameter with respect to the average h_N^* (**a**, **b**) and with respect to the standard deviation H_N^* (**c**, **d**) on the number of samples N for low-frequency noise (**a**, **c**) and band noise (**b**, **d**)

the standard deviation, respectively, for the noise formed by passing white Gaussian noise through an RC-link. Figure 6.4b, d show the estimates of the same parameters for the noise formed by passing the same white noise through a single resonant circuit (SRC). The single resonant circuit had resonance frequency $f_0 = k_0/T$ characterized by the discrete spectral sample number $k_0 = 128$.

The different solid lines in Fig. 6.4 represent the results obtained for the different bandwidths $\Delta f = f_2 - f_1 = \Delta k/T$ of the filters. In the case of the RC-circuit, $\Delta k = 1, 4, 16, 64, 256$ and in the case of the SRC, $\Delta k = 2, 8, 32, 128, 512$ (in both cases, the thicker lines correspond to higher values of Δk). Dotted lines represent the upper bound h_{0N}^+ of the stability confidence interval of the parameter h_N corresponding to the upper bound $\gamma_{0N}^+ = \gamma_{0N} + \varepsilon\sigma_{\gamma_{0N}^*}$ of the stability confidence interval of the parameter γ_N, where $\varepsilon = 3$. The number of realizations was 512.

Analysis of the curves in Fig. 6.4 and others shows that:

- Violation of statistical stability with respect to the *average* occurs only when the noise is *low-frequency type* and its *spectrum* is concentrated in a *narrow band*. Violation of statistical stability with respect to the *standard deviation* occurs for both *low-frequency noise* and *band noise*.
- Violation of statistical stability is much *stronger* for *low-frequency* noise than for *band noise*.
- If there is a violation of statistical stability, *increasing the sample size* leads to a *decrease in stability* with respect to the *average* and also with respect to the *standard deviation*, and *expansion of the bandwidth* leads to an *increase in stability*.
- *Increasing the correlation interval* of *low-frequency noise* leads to a *reduction* in its statistical stability with respect to the *average*. *Increasing the correlation*

interval of the noise leads to a *decrease* in its statistical stability with respect to the *standard deviation,* for both low-frequency and band noise.
- In the case of *low-frequency narrowband* noise, violation of statistical stability with respect to the *standard deviation* appears at *smaller sample sizes* than in the case of *band* noise with the same operating bandwidth.
- The *location of the operating band* of the band noise *does not* strongly *affect* the degree of violation of statistical stability with respect to the *standard deviation*.

6.2.6 Statistically Unstable Stationary Stochastic Processes

Not only the non-stationary stochastic processes, but also those that are *stationary in a narrow sense* can be statistically unstable in a broad sense. Such, for example, are stationary stochastic processes whose cut sets are described by distributions that do not have any moments or do not have moments higher than the first. Examples of such distributions are the Cauchy distribution (Fig. 2.5b), and for certain parameter values, the *Pareto distribution*, the *Fisher–Snedecor distribution* (*F distribution*), *Frechet distribution*, and others.

If there is *no expectation* or variance, as a rule, the probability density function and distribution function have *'heavy tails'* (*'fat tails'*). Such distributions describe actual physical phenomena in which events occur that are *rare, but significantly influence the statistics*.

The *Cauchy distribution function* is given by (2.9), and the probability density function by (2.10). The probability density function of the *Pareto distribution* is

$$f(x) = \frac{\alpha}{(x - x_0)^{\alpha+1}},$$

where the parameter $\alpha > 0$.

In the Cauchy distribution, moments of all orders are absent (though there is an integral in the sense of the principal value which describes the first order moment). In the Pareto distribution, there are no moments of any order if $\alpha \le 1$, and no moments starting from the second order if $1 < \alpha \le 2$.

If a cut set X of a stochastic process has no moment m_ν of order ν, its assessment m_ν^* has no limit (it diverges), i.e., it is *statistically unstable* (*inconsistent*), and therefore a random variable X, its distribution, and corresponding sample $X_1, \ldots, X_N$ can be considered *statistically unstable with respect to the given assessment* m_ν^*.

Therefore, the distribution does not have any moment such that it is statistically unstable with respect to any assessment of it.

If a random variable X or stochastic process $X(t)$ has no variance, it is impossible to use the estimates described by the expressions

$$\gamma_N^* = \bar{D}_{y_N}/\bar{D}_{x_N}, \quad \Gamma_N^* = \bar{D}_{z_N}/\bar{D}_{x_N}. \tag{6.17}$$

And nor can we use the estimates μ_N^*, h_N^* and M_N^*, H_N^* calculated from them. These difficulties can be avoided by replacing the sample variance $\bar{D}_{x_N}$ in the expression (6.17) by a *robust statistic*.

For the Cauchy and Pareto distributions the statistic $Ns_{x_N}^{*2}$ gives a good result, where $s_{x_N}^* = \mathrm{med}\left[\left\{\left|x_n - m_{x_N}^*\right|\middle|\ n = \overline{1,N}\right\}\right]$ is the median absolute deviation and $m_{x_N}^* = \mathrm{med}\left[\left\{x_n\middle|\ n = \overline{1,N}\right\}\right]$ is the median bias. Simulation results for estimates h_N^*, H_N^* using these statistics are presented in Fig. 6.5 (solid lines). Dotted lines in the figure show for comparison the upper statistical stability border of the confidence interval corresponding to the parameter $\varepsilon = 3$. In these simulations, 100 realizations are used.

According to the simulation results the process described by the Cauchy distribution is statistically unstable with respect to the average and standard deviation at any value of the parameter γ. The process described by the Pareto distribution is statistically unstable with respect to the average when $\alpha = 0.75$ and stable when $\alpha > 1$. With respect to the standard deviation, it is statistically unstable when $\alpha = 0.75\,;1.25\,;1.75$ and statistically stable when $\alpha > 2$. These simulation results entirely agree with the theory.

Note that the described approach is *oriented toward processes that have no variance*. If the variance exists, applying it *may not always be justified*. Simulation,

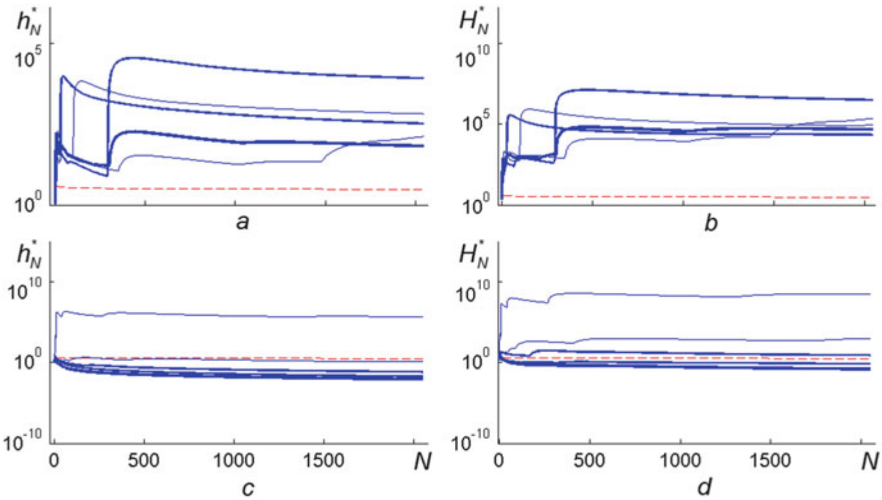

Fig. 6.5 Statistical instability parameters h_N^*, H_N^* for the Cauchy distribution (**a**, **b**) and Pareto (**c**, **d**). The values of the Cauchy parameter γ were 1; 2; 3; 4; 5 and those of the Pareto parameter α were 0.75; 1.25; 1.75; 2.25; 2.75 (*thick line* in the figures, which increases as the value of the parameter increases)

for example, for statistically stable Gaussian noise, shows that although the estimates h_N^*, H_N^* calculated according to the considered approach do not exceed the upper statistical stability border of the confidence interval (i.e., fix the existence of statistical stability), they are lower than the theoretical values. This circumstance *limits the scope for practical application of robust statistics*. Apparently, it only makes sense to use them if *the process contains rare samples with a significant influence on the statistics*.

Let us now consider real physical processes.

6.3 Results of Experimental Investigations of the Statistical Stability of Actual Processes of Various Physical Kinds

To find out whether actual processes are statistically stable or not, and if they are unstable, over what observation interval they can be considered as stable, various actual physical processes have been studied over very long observation intervals. Among the subjects of investigation we find fluctuations in the mains voltage of a city, the Earth's magnetic field, the height and period of waves on the surface of the sea, the temperature and speed of sound in the ocean, the air temperature and atmospheric precipitation in different cities, exchange rates, the X-ray intensity of astrophysical objects, etc. (Gorban 2014, 2016a, 2017). The results of some these investigations are presented below.

6.3.1 Mains Voltage

The parameters of the supply-line voltage are continually changing. To study the statistical stability of such voltage oscillations, a simple research setup was assembled. This consists of a step-down transformer, a matching device (voltage divider), and a computer with 16-bit sound card. The signal was inputted in the computer with a sampling frequency of 5 kHz. The active (effective) voltages were calculated for every 1024 samples and recorded in the computer memory. Recording sessions were conducted over 2 months, with breaks of a few days. The duration of each session was about 60 h. During the session, nearly a million samples of the active voltage ($N \approx 2^{20}$) were recorded.

The processing of the recordings showed that the voltage was changing continually. In different sessions, the changes were of different kinds. Figure 1.8 shows typical dependencies of the voltage and sample mean on time (in hours). The analysis of the experimental data brought out an important feature of the voltage variations: the *undamped nature of their sample means*. The estimates of the statistical instability parameters γ_N^* and μ_N^* with respect to the average calculated for four sessions are shown in Fig. 6.6a, b.

It follows from the figures that, for long observation times, the instability parameters do not show any tendency to fall to zero. For all the recordings obtained

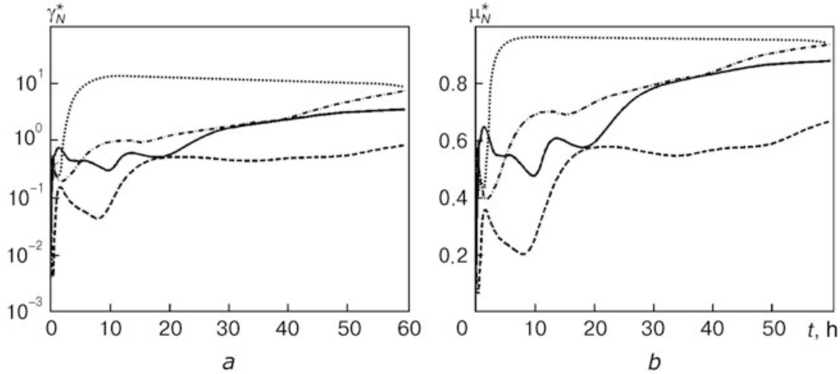

Fig. 6.6 Variations in the estimates γ_N^* (**a**) and μ_N^* (**b**) of the statistical instability parameters during a 60-h observation of the mains voltage

(not only those shown in Fig. 6.6), the values of the statistical instability parameters γ_N^* and μ_N^* are high at the end of the sessions. It follows from this that the *mains voltage is statistically unstable*. The interval over which the statistical instability parameters take high values starts from a few hours and extends to the end of the recording. Consequently, the *statistical stability interval* τ_{sm}*with respect to the average of the mains voltage is approximately an hour.*

6.3.2 Earth's Magnetic Field

The magnetic field of the Earth varies in time and space. Its variations are systematically monitored for many years at different points of the Earth. Initial data used to calculate the statistical instability parameter μ_N^* (Fig. 6.7) was obtained in the Moscow area at the N. V. Pushkov Institute of Terrestrial Magnetism, Ionosphere, and Radio Wave Propagation, RAS (IZMIRAN 2010).

Analysis of the figure shows that, on the whole, the magnetic field of the Earth is statistically unstable, although there are intervals of relative stability. The statistical stability interval τ_{sm} of the fluctuations of the magnetic field is approximately equals to a few months. The statistical prediction of the magnetic induction over this time is problematic, and over several years, almost impossible.

6.3.3 Height and Period of Sea Surface Waves

A specific research project was carried out to assess the statistical stability of the height and period of such waves. We used measurements of the wave parameters obtained by the P. P. Shirshov Institute of Oceanology, RAS, over 15 months of

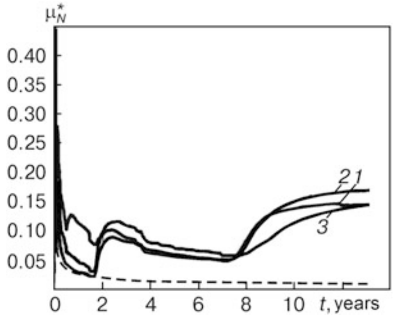

Fig. 6.7 Dependencies of the statistical instability parameters μ_N^* on the observation time for the x-, y-, and z-components of the magnetic field (*solid lines* 1, 2, 3), as well as for standard white Gaussian noise (*dashed line* without a number)

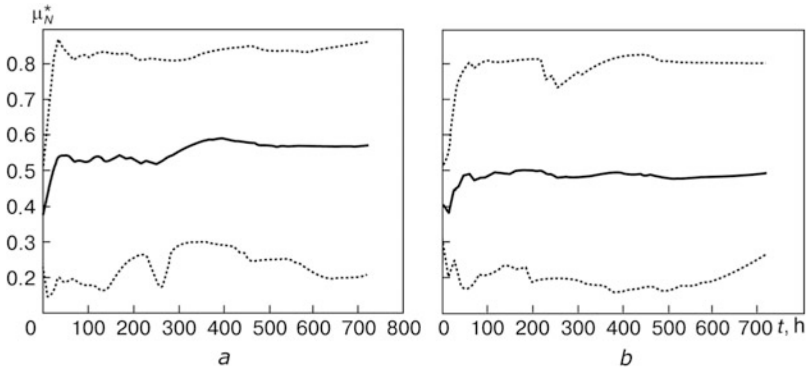

Fig. 6.8 Dependencies of the statistical instability parameter μ_N^* averaged over 15 months (*solid curves*) and of the bounds of this parameter (*dotted curves*) on the observation time: (**a**) for the maximum wave height, (**b**) for the period between wave maxima

observation in Novorossiysk, on the Black Sea (from September 2001 to December 2003) (ESIMO 2010). The data were recorded by the wave station with intervals from one to several hours.

During the observation period, the state of the sea varied significantly. The collected data was used to calculate the estimators μ_N^* of the statistical instability parameters of the wave height and period (Fig. 6.8). It is clear from the figures that the statistical instability parameter μ_N^* takes large values everywhere. This means that the fluctuations in the height and the repetition period of the waves are clearly statistically unstable. The statistical stability interval τ_{sm} of the fluctuations in the height and period of the waves is approximately 12 h. Statistical prediction of these parameters over this time interval is virtually impossible.

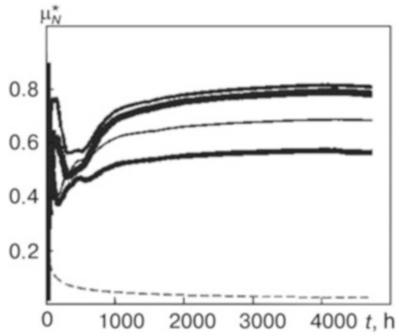

Fig. 6.9 Dependencies of the statistical instability parameter μ_N^* on time for four temperature sensors (*solid lines*), as well as for standard white Gaussian noise (*dashed line*)

6.3.4 Water Temperature in the Ocean

To assess the statistical stability of the oscillations in the water temperature over a long period of time, we used the data obtained by the V. I. Illichev Pacific Oceanological Institute in the Pacific Ocean over the period from 23.10.2010 to 11.05.2011. The temperature measurements were made in the Poset Gulf by two pairs of temperature sensors.

The calculation results of the statistical instability parameter μ_N^* for these sensors are presented in Fig. 6.9. To reduce the influence of seasonal oscillations, low-frequency oscillations were rejected from the data.

It can be seen from the curves that the temperature variations are highly statistically unstable. Statistical stability is lost very quickly.

6.3.5 Air Temperature and Precipitation

The weather and climate are constantly changing under a set of factors. The temperature, precipitation, wind speed, and other meteorological parameters have been monitored daily for many years at different points of the Earth. To investigate the statistical stability of fluctuations in the air temperature and precipitation in the Moscow and Kiev areas, the data collected respectively over 43 years (from 1949 to 1992) and 112 years (from 1881 to 1992) were used (Weather 2009).

Estimates of the statistical instability parameter h_N^* for the daily minimum and maximum air temperatures as well as for the daily precipitation are presented in Fig. 6.10. To reduce the influence of seasonal oscillations, a special preliminary correction of the data was used.

It follows from the figure that the temperature fluctuations are statistically unstable. Significant violations of stability begin after a few weeks of observation. Fluctuations in the precipitation are significantly more stable. They remain stable for decades.

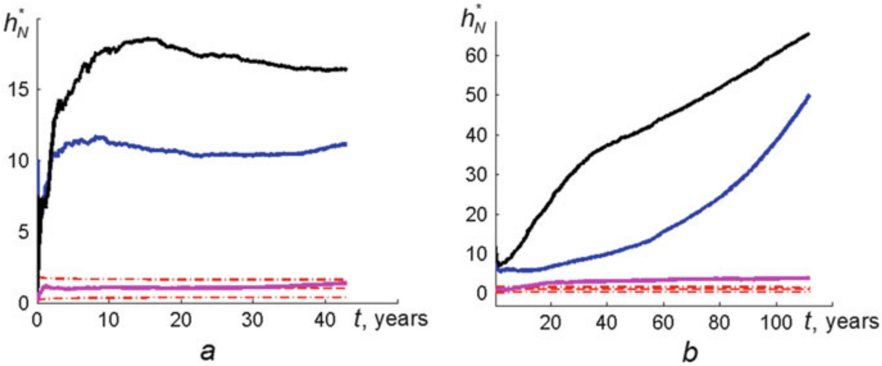

Fig. 6.10 Statistical instability parameter h_N^* for the daily minimum (*upper solid lines*) and maximum (*average solid lines*) air temperatures, and also for the daily precipitation (*lower solid lines*) in the Moscow (**a**) and Kiev (**b**) areas. The *dashed lines* show the statistical instability parameter for standard white Gaussian noise, and *dash*-and-*dotted lines* show the standard deviations from them

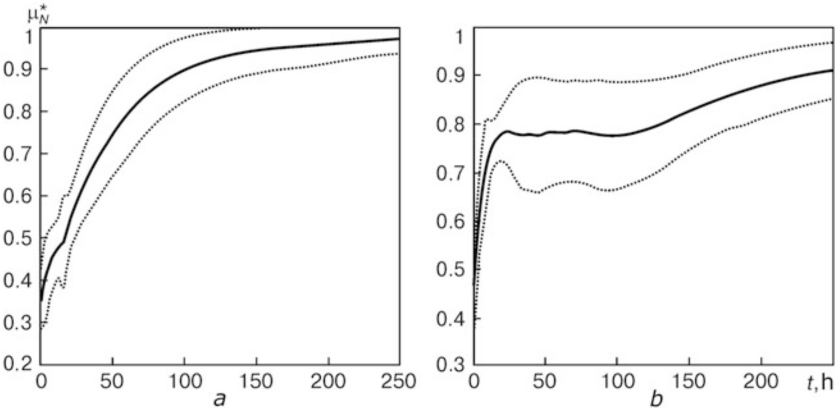

Fig. 6.11 Time dependencies of the statistical instability parameter μ_N^* averaged over 16 decades (*solid curves*) and corresponding boundaries of the one standard deviation confidence interval (*dotted curves*) for the currency fluctuations of the Australian dollar (AUD) with respect to the US dollar (USD): (**a**) in 2001, (**b**) in 2002

6.3.6 Currency Fluctuations

The statistical instability of currency fluctuations is illustrated by the curves in Fig. 6.11, obtained from FOREX data (FOREX 2009). It can be seen from the curves that the statistical instability parameter takes large values from the first hours of observation and is constantly increasing. Thus, the *statistical stability interval* τ_{sm} *of the currency fluctuations is around 1–2 h*. Any *statistical forecast* over this duration is practically impossible.

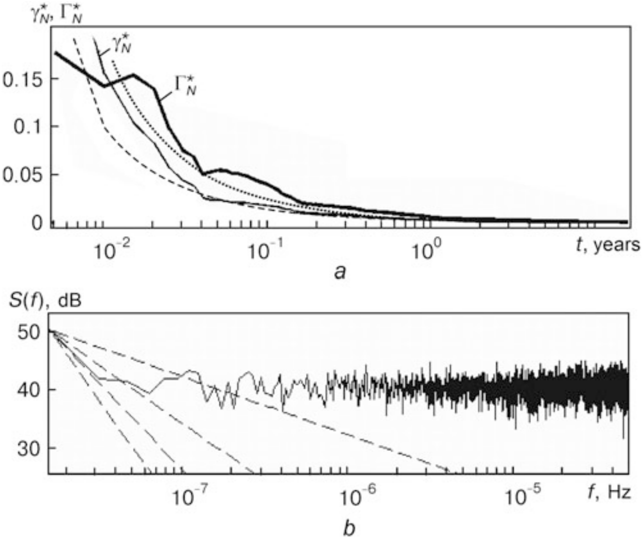

Fig. 6.12 Dependencies on time of the statistical instability parameters with respect to the average γ_N^* (*thin solid line*) and with respect to the standard deviation Γ_N^* (*bold solid line*) for the pulsar PSRJ 1012+5307 (**a**) as well as the corresponding power spectrum density of emission (*solid line*) (**b**)

6.3.7 Astrophysical Objects

Figures 6.12 and 6.13 present the research results of the radiation fluxes from two different types of astrophysical accreting X-ray sources PSRJ 1012+5307 and GRS 1915+105. In Figs. 6.12a and 6.13a, the dashed lines show the parameter of statistical instability γ_{0N} for standard white Gaussian noise, and the dotted lines show the bound of the one standard deviation confidence interval for the standard process $\gamma_{0N}^+ = \gamma_{0N} + \sigma_{\gamma_{0N}^*}$. The dashed lines in Figs. 6.12b and 6.13b represent the graphs of the power functions $1/f^\beta$, $\beta = \overline{1,4}$, shifted along the vertical axis to the level of the first spectral sample.

It clearly follows from the curves in Figs. 6.12a and 6.13a that fluctuations of the source GRS 1915+105 are statistically unstable. These began to emerge at the level of the sample mean and sample standard deviation within a month of observation. Fluctuations of the source PSRJ 1012+5307 are much more stable: at the level of the sample mean, violations of statistical stability do not show up during the entire period of observation, while at the level of the sample standard deviation, they appear a few weeks after the beginning of observation.

For the pulsar PSRJ 1012+5307, the inequality $\gamma_N^* < \Gamma_N^*$ is valid almost over the entire range of monitoring, and for the source GRS 1915+105, the inequality

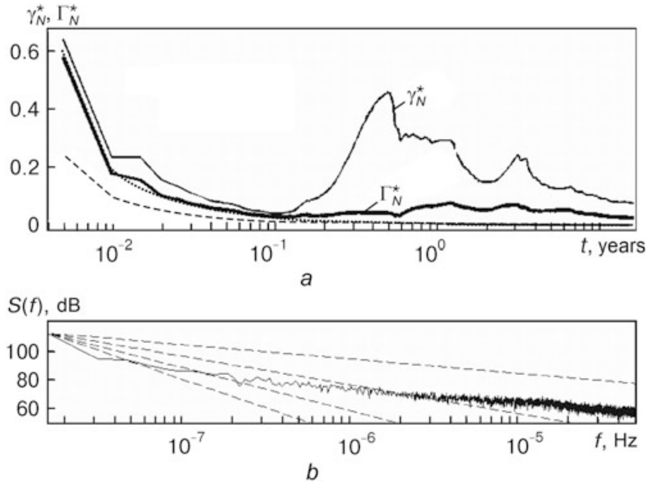

Fig. 6.13 Dependencies on time of the statistical instability parameters with respect to the average γ_N^* (*thin solid line*) and with respect to the standard deviation Γ_N^* (*bold solid line*) for the source GRS 1915+105 (**a**) and the corresponding power spectrum density of emission (*solid line*) (**b**)

$\gamma_N^* > \Gamma_N^*$. This result can be explained by the different natures of the emission spectra (see Sect. 6.2.3). The emission of the pulsar PSRJ 1012+5307 is close to white noise (Fig. 6.12b) and the emission of the source GRS 1915+105 is strongly correlated noise, which in the most energy-intensive part of the spectrum depends on the frequency f approximately according to the law $1/f^3$ (Fig. 6.13b).

Summarizing the above results, we should pay attention to the fact that:

All investigated processes are statistically unstable in a broad sense.

This indicates that:

All actual physical values and processes are in fact statistically unstable (Gorban 2007, 2011, 2014, 2016a, 2017).

The only exceptions may perhaps be the world's physical constants, such as the speed of light in vacuum.

For statistically unstable events, values, and processes, *the concepts of probability, mathematical expectation, standard deviation, distribution function, and probability density function do not have a physical interpretation*. Therefore, for an adequate description of such phenomena, the *classical methods and approaches of probability theory are unsuitable*. The question thus arises: *how can we describe* *actual events, values, and processes, taking into account statistical stability violations*? And the answer is that we may use the *physical-mathematical theory of hyper-random phenomena*, specially developed for this purpose. The next four chapters are devoted to a presentation of the main features of this theory.

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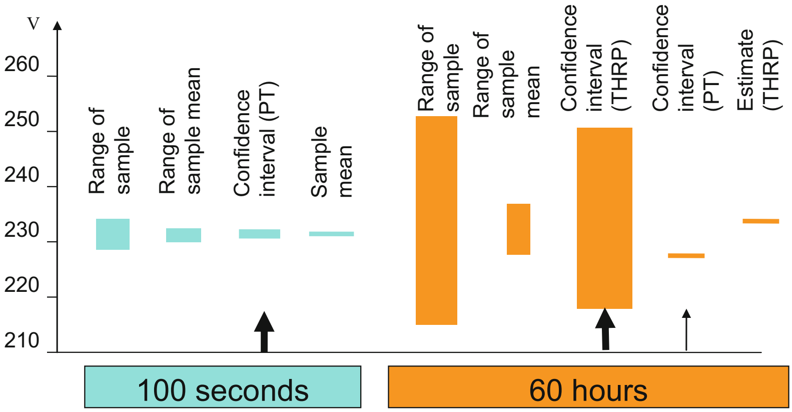
Part IV
The Theory of Hyper-random Phenomena



Calculation results for the parameters characterizing the city mains voltage over 100-s and 60-h observation intervals, using calculation techniques based on probability theory (PT) and the theory of hyper-random phenomena (THRP)

Félix Edouard Justin Émile Borel, French mathematician and politician (1871–1956)
(Source: https://academictree.org/photo/001/cache.010603.%c3%89mile_Borel.jpg)

E. Borel:

> If in case of a very large number of tests the frequency does not tend to a limit, and more or less fluctuates between some limits, it is necessary to say that the probability p is not a constant, but varies over the tests. This occurs, for example, for human mortality over the centuries, as the progress in the medicine and hygiene lead to an increase in the average duration of life. Therefore, the probability p that a newborn child will reach the age of 60 years has a tendency to grow. This empirical point of view is acceptable for the statistician, studying the demographic phenomenon, since here in the absence of other scientific means we have to use infinite observations for prediction (Borel 1956).

A.N. Kolmogorov:

> Speaking about randomness in the ordinary sense of this term, we mean the phenomena in which we do not find laws that can allow us to predict their behavior. In general, there is no reason to assume that a random phenomenon in this sense is subjected to any probability laws. Consequently, one must distinguish randomness in this broad sense and stochastic randomness which is the scope of study of probability theory (Kolmogorov 1986).

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Chapter 7
Basis of the Theory of Hyper-random Phenomena

Abstract The notion of a hyper-random event is formulated. The properties of hyper-random events are examined. The concept of a scalar hyper-random variable is specified. We present three ways to describe it: by its conditional characteristics (in particular, conditional distribution functions and conditional moments), by the bounds of the distribution function and their moments, and by the bounds of moments. The concept of a vector hyper-random variable is introduced. The methods that describe the scalar hyper-random variables are extended to these vector hyper-random variables. The issue of transformation of hyper-random variables and arithmetic operations on them are briefly examined.

The first thing to say when we describe the *physical-mathematical theory of hyper-random phenomena* is that this theory consists of two parts: *physical* and *mathematical*. Its *physical part* is based on the results of the experimental investigations described in the previous chapter, which indicate a lack of convergence of the actual statistics, and the *physical hypothesis* that the actual events, quantities, processes, and fields are adequately described by hyper-random mathematical models, in particular hyper-random events, variables, and functions which take into account violations of statistical stability.

Hyper-random events, *hyper-random variables*, and *hyper-random functions* are *sets of non-interconnected random events, random variables*, *and stochastic functions*, respectively. Each element of these sets (*random phenomena*) is associated with some statistical conditions.

The logic used to construct the *mathematical part* of the theory of hyper-random phenomena closely follows the logic used to construct axiomatic probability theory.

The chapter is based on material from the books (Gorban 2007, 2011, 2014, 2016, 2017)

I.I. Gorban, *Randomness and Hyper-randomness*, Mathematical Engineering,
DOI 10.1007/978-3-319-60780-1_7

7.1 Hyper-random Events

7.1.1 Definition of the Concept of a Hyper-random Event

By analogy with the axiomatic definition of the concept of a random event (see Sect. 2.2.5), the concept of a hyper-random event is specified as follows. A *hyper-random event* considered as a mathematical object is given analytically by the tetrad $(\Omega, \Im, G, P_g)$, where Ω is the space of elementary events $\omega \in \Omega$, $\Im$ is a σ-algebra of subsets of the events (Borel field), and P_g is a probability mass (measure) under fixed conditions $g \in G$.

The hyper-random event A can be represented by a set of random events A/g depending on the conditions $g \in G$. The probabilistic measure $P_g(A) = P(A/g)$ is defined for each random event corresponding to conditions g included in this set, but a *measure for the conditions g is not defined*.

A hyper-random event A is characterized, not by a single probability, but by a *set of conditional probabilities* $\tilde{P}(A) = \{P(A/g) |\, g \in G\}$.[1] This set provides a comprehensive description of the hyper-random event. The hyper-random event A may be characterized in a less complete way by the *upper* $P_S(A)$ and *lower* $P_I(A)$ *probability bounds* (Fig. 7.1) given by the expression

$$P_S(A) = \sup_{g \in G} P(A/g), \quad P_I(A) = \inf_{g \in G} P(A/g). \tag{7.1}$$

If the set of conditions comprises only one element ($g = \text{const}$), the bounds coincide. Then the hyper-random event degenerates into a random event. In this case, the value $P(A) = P_S(A) = P_I(A)$ is the probability of the random event.

Using *a statistical approach*, the hyper-random event A can be interpreted as an event whose relative frequency $p_N(A)$ *does not stabilize* (*or converge*) as the number of experiments N increases, and *does not tend to any limit* when $N \to \infty$.

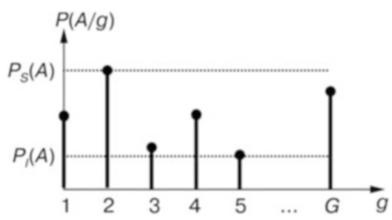

Fig. 7.1 The conditional probabilities $P(A/g)$ (marked by *points*) and the probability bounds $P_S(A)$, $P_I(A)$ (marked by *dotted lines*) for the hyper-random event A

[1]Hereafter, a tilde under a letter is used to indicate that the given object is or may be many-valued.

7.1.2 Properties of Hyper-random Events

On the basis of the axioms of probability theory (see Sect. 2.2.5), it can be shown that:

(1) $P_S(A) \geq 0$, $P_I(A) \geq 0$; (7.2)

(2) for pairwise disjoint events

$$P_S\left(\underset{n}{\cup} A_n\right) \leq \sum_n P_S(A_n), \quad P_I\left(\underset{n}{\cup} A_n\right) \geq \sum_n P_I(A_n); \tag{7.3}$$

(3) $P_S(\Omega) = P_I(\Omega) = 1.$ (7.4)

It follows from (7.1)–(7.4) that

$$0 \leq P_S(A) \leq 1, \quad 0 \leq P_I(A) \leq 1, \quad P_S(\varnothing) = P_I(\varnothing) = 0.$$

For hyper-random events $A_1, A_2, \ldots$, the following formulas are valid:

(4) if $A_m \subset A_{m+1}$, $m \geq 1$, then

$$P_S\left(\overset{M}{\underset{m=1}{\cup}} A_m\right) = P_S(A_M), \quad P_I\left(\overset{M}{\underset{m=1}{\cup}} A_m\right) = P_I(A_M), \tag{7.5}$$
$$P_S\left(\overset{\infty}{\underset{m=1}{\cup}} A_m\right) = \lim_{M\to\infty} P_S(A_M);$$

(5) if $A_{m+1} \subset A_m$, $m \geq 1$, then

$$P_S\left(\overset{M}{\underset{m=1}{\cap}} A_m\right) = P_S(A_M), \quad P_I\left(\overset{M}{\underset{m=1}{\cap}} A_m\right) = P_I(A_M), \tag{7.6}$$
$$P_I\left(\overset{\infty}{\underset{m=1}{\cap}} A_m\right) = \lim_{M\to\infty} P_I(A_M).$$

For hyper-random events A_1 and A_2, one has the inequalities

$$P_S(A_1 \cup A_2) \leq P_S(A_1) + P_S(A_2) - P_I(A_1 \cap A_2), \tag{7.7}$$
$$P_I(A_1 \cup A_2) \geq P_I(A_1) + P_I(A_2) - P_S(A_1 \cap A_2), \tag{7.8}$$

which are similar to the *addition theorem for random events*.

Note that, when the events A_1 and A_2 are inconsistent then $P_S(A_1 \cap A_2) = 0$, $P_I(A_1 \cap A_2) = 0$, and according to (7.7) and (7.8),

$$\begin{aligned} &P_S(A_1 \cup A_2) \le P_S(A_1) + P_S(A_2), \\ &P_I(A_1 \cup A_2) \ge P_I(A_1) + P_I(A_2). \end{aligned} \tag{7.9}$$

When $A_1 \subset A_2$, according to (7.5),

$$P_S(A_1 \cup A_2) = P_S(A_2), \quad P_I(A_1 \cup A_2) = P_I(A_2).$$

In the general case, for the *hyper-random events* A_1 and A_2, one has the inequalities

$$\begin{aligned} &P_S(A_1 \cap A_2) \le P_S(A_1)P_S(A_2/A_1), \quad (P_S(A_1) \ne 0), \\ &P_I(A_1 \cap A_2) \ge P_I(A_1)P_I(A_2/A_1), \quad (P_I(A_1) \ne 0) \end{aligned} \tag{7.10}$$

which are similar to the expression

$$P(A_1 \cap A_2) = P(A_1) \;\, P(A_2/A_1)$$

of the *multiplication theorem* for *random events* when $P(A_1) \ne 0$. In this case, $P_S(A_2/A_1)$ and $P_I(A_2/A_1)$ denote, respectively, the upper and lower bounds of the probability of the event A_2 when the event A_1 has occurred.

Hyper-random events A_1 and A_2 are *independent* if the probability bounds of the intersection of the events factorize:

$$\begin{aligned} &P_S(A_1 \cap A_2) = P_S(A_1)P_S(A_2), \\ &P_I(A_1 \cap A_2) = P_I(A_1)P_I(A_2). \end{aligned} \tag{7.11}$$

The sense of (7.11) is that, for independent hyper-random events A_1 and A_2, the probability bounds of the intersection of the events are determined only by the probability bounds of the event A_1 and the probability bounds of the event A_2. It makes no difference whether or not the event A_1 has happened when we wish to consider the probability bounds of the event A_2, and whether or not the event A_2 has happened before considering the probability bounds of the event A_1. The result is the same.

Hyper-random events A_1 and A_2 are *independent under all conditions* if, for all $g \in G$, the conditional probability of the intersection of the events factorizes:

$$P(A_1 \cap A_2/g) = P(A_1/g)P(A_2/g). \tag{7.12}$$

Independent hyper-random events and hyper-random events independent under all conditions are different concepts. Independence of hyper-random events under all conditions *does not imply* their independence, and neither does independence of hyper-random events *imply* their independence under all conditions.

7.2 Scalar Hyper-random Variables

A *scalar hyper-random variable* X is a numerical function defined on the space Ω of elementary events ω, for which, under fixed observation conditions $g \in G$, a probability measure is defined, but *a probability measure is not defined for the observation conditions*.

As in the case of a random variable, the value x of the hyper-random variable can be obtained from the adjuvant function $x = \psi(\omega)$, where $\omega \in \Omega$. The set of the values of the hyper-random variables for all $g \in G$ forms the *value space* (*actual range*) *of the hyper-random variable* .

The hyper-random variable X can be represented by a *set of random variables* $X_g = X/g$: $X = \{X_g | g \in G\} = \{X/g \in G\}$. It is related to random variables in the same way as a vector quantity is related to scalar quantities: the vector can be represented by a set of scalar quantities, and the hyper-random variable can be characterized by a set of random variables. A special case of a vector is a scalar; a special case of a hyper-random variable is a random variable.

7.2.1 Conditional Characteristics and Parameters

To describe the hyper-random variable X, we use various probabilistic characteristics of the *conditional random variables* X_g $(g \in G)$, and in particular the *conditional distribution functions* (Fig. 7.2)

$$F_{x/g}(x) = P\{X_g < x\}, \tag{7.13}$$

where $P\{X_g < x\} = P\{X < x/g\}$ is the probability of satisfying the inequality $X < x$ under condition g, and the *conditional probability density functions*[2]

$$f_{x/g}(x) = \frac{dF_{x/g}(x)}{dx}. \tag{7.14}$$

The most complete description of the hyper-random variable X gives its *distribution function* $\tilde{F}_x(x)$, which is a set of conditional distribution functions $F_{x/g}(x)$ for all $g \in G$: $\tilde{F}_x(x) = \{F_{x/g}(x) | g \in G\}$. The distribution function $\tilde{F}_x(x)$ can be interpreted as a *many-valued function* whose branches are the conditional distribution functions.

Note that hereafter *many-valued* functions and *many-valued* variables, like $\tilde{F}_x(x)$, are designated by letters with a tilde over them.

[2] It is assumed here and below that all the above distribution functions are continuous or piecewise continuous.

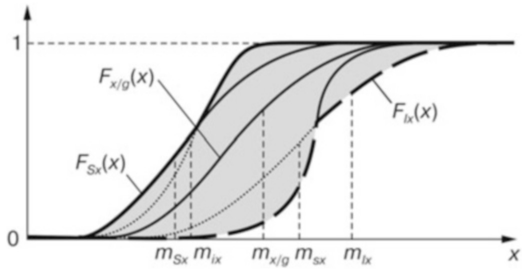

Fig. 7.2 A set of conditional distribution functions $F_{x/g}(x)$ (*thin lines*) and the bounds of the distribution function $F_{Sx}(x)$, $F_{Ix}(x)$ (*bold lines*) of the hyper-random variable X

A less complete description of the hyper-random variable X supplies the *conditional expectations* $\mathrm{E}[\varphi(X_g)]$ *of the various functions* $\varphi(X_g)$ *of the random variables* X_g $(g \in G)$. By the conditional expectation $\mathrm{E}[\varphi(X_g)]$ of the function $\varphi(X_g)$, we understand the mean value of this function (taking into account the conditional probability density function $f_{x/g}(x) = f(x/g)$):

$$\mathrm{E}\left[\varphi\left(X_g\right)\right] = \int_{-\infty}^{\infty} \varphi(x) f_{x/g}(x)dx.$$

Using this expression the *conditional crude* and *central moments* of the *hyper-random variable* $X = \{X_g | g \in G\}$ (*conditional moments*) can be defined, and in particular, the *conditional expectation*

$$m_{x/g} = \mathrm{E}\left[X_g\right] = \int_{-\infty}^{\infty} x f_{x/g}(x)\, dx, \tag{7.15}$$

the *conditional variances*

$$D_{x/g} = \mathrm{Var}[X_g] = \mathrm{E}\left[\left(X_g - m_{x/g}\right)^2\right], \tag{7.16}$$

the *conditional standard deviations* $\sigma_{x/g} = \sqrt{D_{x/g}}$, among others.

In this interpretation, the *expectation*, *variance*, and *standard deviation of the hyper-random variable* $X = \{X_g,\ g \in G\}$ are *many-valued*, described analytically as follows:

$$\begin{aligned} \tilde{m}_x &= \left\{m_{x/g} | g \in G\right\}, \\ \tilde{D}_x &= \left\{D_{x/g} | g \in G\right\}, \\ \tilde{\sigma}_x &= \left\{\sigma_{x/g} | g \in G\right\}. \end{aligned}$$

Other characteristics and parameters discussed in the next section can also be used to describe a hyper-random variable X.

7.2.2 Bounds of the Distribution Function and Moments of the Bounds

A general view of the hyper-random variable X is given by the functions

$$\begin{aligned} F_{Sx}(x) &= \sup_{g\in G} P\{X_g < x\} = \sup_{g\in G} F_{x/g}(x), \\ F_{Ix}(x) &= \inf_{g\in G} P\{X_g < x\} = \inf_{g\in G} F_{x/g}(x), \end{aligned} \tag{7.17}$$

where $F_{Sx}(x)$ and $F_{Ix}(x)$ are respectively the *upper* and *lower bounds* of the probability that $X < x$, without specifying the conditions $g \in G$ (*bounds of the distribution function*) (see Fig. 7.2).

A function $F(x)$ can be the distribution function of some *random variable* X if and only if it is non-decreasing for all x, is left-continuous, and has limiting values $F(-\infty) = 0$, $F(+\infty) = 1$.

The conditional distribution functions $F_{x/g}(x)$ $(g \in G)$ of the *hyper-random variable* X satisfy these conditions. The bounds of the distribution function $F_{Sx}(x)$, $F_{Ix}(x)$ also satisfy these conditions. Therefore, the *bounds of the distribution function can be considered as the distribution functions of some virtual random variables*.

In addition, we have $F_S(x) \geq F_I(x)$, and at the minimum point (if it exists), the bounds of the hyper-random variable coincide and are equal to zero, while at the maximum point (if it exists), the bounds also coincide and are equal to one. Between the bounds of the distribution function is the *uncertainty area* (shaded area in Fig. 7.2).

Just as a determinate value can be regarded approximately as a degenerate random variable (see Sect. 2.3.3), the *random variable* can be interpreted as a *degenerate hyper-random variable*.

A *hyper-random variable* X is *continuous* if the bounds of its distribution function are continuous and have piecewise continuous derivatives.

For a continuous hyper-random variable, the analogues of the probability density function of the *random variable* are the *probability density functions of the bounds, viz.,*

$$f_{Sx}(x) = \frac{dF_{Sx}(x)}{dx}, \quad f_{Ix}(x) = \frac{dF_{Ix}(x)}{dx}, \tag{7.18}$$

which are the derivatives of the upper and lower bounds of the distribution function.

By using generalized functions, in particular the Dirac delta function, the probability densities of the bounds can be determined, not only in the case of continuous hyper-random variables, but also in the case when the bounds of the distribution function are described by *piecewise continuous functions*. Note that the probability densities of the bounds $f_{Sx}(x)$ and $f_{Ix}(x)$ have the same properties as the probability density of a random variable.

Note that the probability densities of the bounds $f_{Sx}(x)$ and $f_{Ix}(x)$ define the uncertainty area. However, they do not characterize it as clearly as the bounds of the distribution function $F_{Sx}(x)$ and $F_{Ix}(x)$.

To describe a hyper-random variable, we may use the *moments of the bounds,* in particular, the expectations, variances, and standard deviations of the bounds, and so on.

The *expectations of the bounds* $\mathrm{E}_S[\varphi(X)]$, $\mathrm{E}_I[\varphi(X)]$ *of the function* $\varphi(X)$ of the hyper-random variable X described by the probability densities of the bounds $f_{Sx}(x)$, $f_{Ix}(x)$ are the integrals

$$\begin{aligned} \mathrm{E}_S[\varphi(X)] &= \int_{-\infty}^{\infty} \varphi(x) f_{Sx}(x)dx, \\ \mathrm{E}_I[\varphi(X)] &= \int_{-\infty}^{\infty} \varphi(x) f_{Ix}(x)dx. \end{aligned} \tag{7.19}$$

The expectations of the bounds do not always exist. Indeed, they exist only when the integrals (7.19) exist in the sense of absolute convergence.

It follows from (7.19) that the *expectations of the bounds* m_{Sx}, m_{Ix} of the hyper-random variable X , which are the expectations of the bounds of the function $\varphi(X) = X$, are described by the formulas

$$\begin{aligned} m_{Sx} &= \mathrm{E}_S[X] = \int_{-\infty}^{\infty} x f_{Sx}(x)dx, \\ m_{Ix} &= \mathrm{E}_I[X] = \int_{-\infty}^{\infty} x f_{Ix}(x)dx \end{aligned} \tag{7.20}$$

(see Fig. 7.2).

For a *real* hyper-random variable X the *variances of the bounds* D_{Sx}, D_{Ix} are defined by

$$\begin{aligned} D_{Sx} &= \mathrm{E}_S\left[(X - m_{Sx})^2\right], \\ D_{Ix} &= \mathrm{E}_I\left[(X - m_{Ix})^2\right], \end{aligned} \tag{7.21}$$

and the *standard deviations of the bounds* by

$$\sigma_{Sx} = \sqrt{D_{Sx}}, \quad \sigma_{Ix} = \sqrt{D_{Ix}}. \tag{7.22}$$

The expectations m_{Sx} and m_{Ix} of the bounds of the hyper-random variable X characterize the average values of X calculated for the upper and lower bounds of the distribution $\tilde{F}_x(x)$. The variances D_{Sx} and D_{Ix} of the bounds of the variable X, and also the standard deviations σ_{Sx} and σ_{Ix} of the bounds, characterize the dispersion of the variable X relative to the expectations m_{Sx} and m_{Ix}.

The expectations of bounds are related by the inequality $m_{Sx} \leq m_{Ix}$. Equality holds in particular if the hyper-random variable X degenerates into a random variable. The variance of the upper bound D_{Sx} may be greater than the variance of the lower bound D_{Ix}, equal to it, or lower than it.

A general view of a hyper-random variable X is given by the *bounds of the moments*.

7.2.3 Bounds of the Moments

The upper and lower bounds of the expectation of the function $\varphi(X)$ of the hyper-random variable X are the values

$$\begin{aligned} \mathrm{E}_s[\varphi(X)] &= \sup_{g \in G} \int_{-\infty}^{\infty} \varphi(x) f_{x/g}(x)dx, \\ \mathrm{E}_i[\varphi(X)] &= \inf_{g \in G} \int_{-\infty}^{\infty} \varphi(x) f_{x/g}(x)dx. \end{aligned} \tag{7.23}$$

The upper and lower bounds of the expectation of the hyper-random variable X are the values

$$\begin{aligned} m_{sx} = \mathrm{E}_s[X] &= \sup_{g \in G} \int_{-\infty}^{\infty} x f_{x/g}(x)dx, \\ m_{ix} = \mathrm{E}_i[X] &= \inf_{g \in G} \int_{-\infty}^{\infty} x f_{x/g}(x)dx. \end{aligned} \tag{7.24}$$

(see Fig. 7.2).

The upper and lower bounds of the variance of the hyper-random variable X are the values

$$D_{sx} = \sup_{g \in G} D_{x/g}, \quad D_{ix} = \inf_{g \in G} D_{x/g}. \tag{7.25}$$

The roots $\sigma_{sx} = \sqrt{D_{sx}}$, $\sigma_{ix} = \sqrt{D_{ix}}$ of these values are the *bounds of the standard deviation*.

7.2.4 Interconnection Between Bounds of Moments and Moments of Bounds

In general, the operators $E_s[\cdot]$, $E_i[\cdot]$ do not coincide with the operators $E_S[\cdot]$, $E_I[\cdot]$, and the bounds of the expectation and variance m_{sx}, m_{ix}, D_{sx}, D_{ix} do not coincide with the expectations and variances of the bounds m_{Sx}, m_{Ix}, D_{Sx}, D_{Ix}.

Note that neither the bounds of the probability density $f_{sx}(x) = \sup_{g \in G} f_{x/g}(x)$ and $f_{ix}(x) = \inf_{g \in G} f_{x/g}(x)$ nor the bounds of the moments carry information about the bounds of the distribution function $F_{Sx}(x)$ and $F_{Ix}(x)$; rather they carry information about the changing range of the corresponding characteristics when the condition g is changed within the set G. The bounds of the probability density and the probability densities of bounds are different characteristics, just as the bounds of moments and the moments of bounds are different parameters representing the hyper-random variable in different ways.

To explain the reasons why the bounds of the characteristics differ from the corresponding characteristics of the bounds, a few examples of the distribution functions of the hyper-random variable X are presented in Fig. 7.3.

It follows from the figures that the conditional distribution functions may not intersect (Fig. 7.3a, b) or they may overlap (Fig. 7.3c, d). In cases (a) and (b), the bounds of the first two moments coincide with the moments of the bounds, while in cases (c) and (d), respectively, there is partial and complete noncoincidence of the corresponding characteristics.

If the expectation of the hyper-random variable X has minimum and maximum values, the expectations of bounds m_{Sx}, m_{Ix} are related to the bounds m_{ix}, m_{sx} of the expectation by the inequality

$$m_{Sx} \leq m_{ix} \leq m_{sx} \leq m_{Ix}.$$

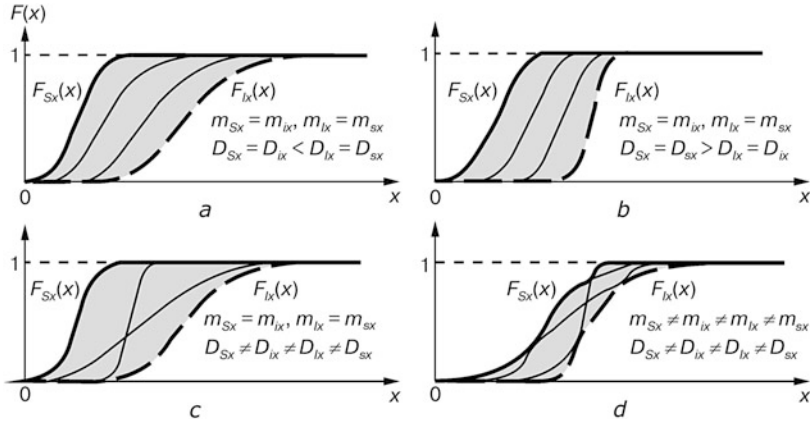

Fig. 7.3 Different types of distribution functions. *Thin lines* represent the conditional distribution functions $F_{x/g}(x)$ and *bold lines* the bounds of the distribution function $F_{Sx}(x)$, $F_{Ix}(x)$

7.3 Vector Hyper-random Variables

In Sect. 7.2 we presented a basic description of scalar hyper-random variables via conditional probability characteristics and conditional moments, bounds of the distribution function and moments of these bounds, as well as moments of bounds. These approachs are applied to vector hyper-random variables. A *vector hyper-random variable* is a vector, each component of which is a scalar hyper-random variable. The N-dimensional vector hyper-random variable $\vec{X} = (X_1, \ldots, X_N)^{\mathrm{T}}$ can be considered as a set of N-dimensional vector random variables $\{\vec{X}_g | g \in G\}$, or as a vector consisting of N scalar hyper-random components X_n $\left(n = \overline{1, N}\right)$.

7.3.1 Conditional Characteristics and Parameters

To describe a vector hyper-random variable $\vec{X}$, different probabilistic characteristics of the *conditional random variables* $\vec{X}_g = \vec{X}/g$ $(g \in G)$ are used, in particular the *conditional distribution function*

$$F_{\vec{x}/g}(\vec{x}) = F_{\vec{x}/g}(x_1, \ldots, x_N) = P\{X_{1g} < x_1, \ldots, X_{Ng} < x_N\}, \tag{7.26}$$

where $P\{X_{1g} < x_1, \ldots, X_{Ng} < x_N\}$ is the probability of satisfying the inequalities $X_1 < x_1, \ldots, X_N < x_N$ under the condition g, and also the *conditional probability density function*

$$f_{\vec{x}/g}(\vec{x}) = \frac{\partial^N F_{\vec{x}/g}(\vec{x})}{\partial x_1 \ldots \partial x_N}. \tag{7.27}$$

The set of any of these conditional characteristics for all $g \in G$ gives the most complete description of the vector hyper-random variable in particular the *distribution function* $\tilde{F}(\vec{x}) = \{F(\vec{x}/g) | g \in G\}$.

The *vector hyper-random variables* $\vec{X}$ and $\vec{Y}$ are said to be *independent under all conditions* $g \in G$ if all conditional probability densities of the vector $\vec{Z} = \left(\vec{X}, \vec{Y}\right)^{\mathrm{T}}$ factorize for all $g \in G$, viz.,

$$f_{\vec{z}/g}(\vec{x}, \vec{y}) = f_{\vec{x}/g}(\vec{x}) f_{\vec{y}/g}(\vec{y}).$$

For independent variables $\vec{X}$ and $\vec{Y}$ under all conditions $g \in G$, it is not only all conditional probability densities that factorize, but also all conditional distribution functions, i.e.,

$$F_{\vec{z}/g}(\vec{x}, \vec{y}) = F_{\vec{x}/g}(\vec{x}) F_{\vec{y}/g}(\vec{y}).$$

Note that the independence of the hyper-random variables under all conditions *does not mean that there are no connections between these variables*. It just means that, at the level of the given characteristics, as for random variables (see Sect. 2.4.1), they are not determined.

The vector hyper-random variable $\vec{X}$ can be characterized by the different numerical characteristics of its random components $\vec{X}_g$, in particular *conditional moments*. For instance, the *two-dimensional hyper-random variable* $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ under conditions $g \in G$ is characterized by the *vector conditional expectations*

$$\vec{m}_{\vec{x}/g} = \left(m_{x_1/g}, m_{x_2/g}\right)^{\mathrm{T}}, \tag{7.28}$$

vector conditional variances

$$\vec{D}_{\vec{x}/g} = \left(D_{x_1/g}, D_{x_2/g}\right)^{\mathrm{T}} = \mathrm{D}\left[\vec{X}_g\right], \tag{7.29}$$

vector conditional standard deviations

$$\vec{\sigma}_{\vec{x}/g} = \left(\sigma_{x_1/g}, \sigma_{x_2/g}\right)^{\mathrm{T}} = \left(\sqrt{D_{x_1/g}}, \sqrt{D_{x_2/g}}\right)^{\mathrm{T}} \tag{7.30}$$

and other parameters, where $m_{x_1/g}$, $m_{x_2/g}$ are expectations of the random variables X_{1g} and X_{2g}, $D_{x_1/g}$, $D_{x_2/g}$ are variances of these variables, and $\sigma_{x_1/g}$, $\sigma_{x_2/g}$ are their standard deviations.

For fixed conditions $g \in G$, the *conditional correlation moment of the hyper-random variables* X_1 and X_2 is given by

$$K_{x_1x_2/g} = \mathrm{E}\left[X_{1g}X_{2g}\right], \tag{7.31}$$

the *conditional covariance moment* by

$$R_{x_1x_2/g} = \mathrm{Cov}\left[X_{1g}, X_{2g}\right] = \mathrm{E}\left[(X_{1g} - m_{x_1/g})(X_{2g} - m_{x_2/g})\right], \tag{7.32}$$

and the *conditional correlation coefficient* by

$$r_{x_1x_2/g} = \frac{R_{x_1x_2/g}}{\sigma_{x_1/g}\sigma_{x_2/g}}. \tag{7.33}$$

The conditional covariance moments $R_{x_1x_2/g}$, the conditional correlation moments $K_{x_1x_2/g}$, and the conditional expectations $m_{x_1/g}$, $m_{x_2/g}$ of the hyper-random variables X_1 and X_2 are related to one another by the relations

$$R_{x_1x_2/g} = K_{x_1x_2/g} - m_{x_1/g}m_{x_2/g}. \tag{7.34}$$

7.3.2 Distribution Bounds and Moments of Bounds

The *bounds of the distribution function of a vector hyper-random variable* $\vec{X} = (X_1, \ldots, X_N)^{\mathrm{T}}$ are defined by

$$F_{S\vec{x}}(\vec{x}) = \sup_{g \in G} F_{\vec{x}/g}(\vec{x}), \quad F_{I\vec{x}}(\vec{x}) = \inf_{g \in G} F_{\vec{x}/g}(\vec{x}), \tag{7.35}$$

while the *probability density functions of the bounds* are defined by

$$f_{S\vec{x}}(\vec{x}) = \frac{\partial^N F_{S\vec{x}}(\vec{x})}{\partial x_1 \ldots \partial x_N}, \quad f_{I\vec{x}}(\vec{x}) = \frac{\partial^N F_{I\vec{x}}(\vec{x})}{\partial x_1 \ldots \partial x_N}. \tag{7.36}$$

The pairs of characteristics (7.35) and (7.36) have properties that are intrinsic respectively to the distribution function and the probability density function of the vector random variable, as well as properties that are typical for the corresponding pairs of characteristics of the scalar hyper-random variable. In particular, $F_S(\vec{x}) \geq F_I(\vec{x})$, and when the components of the vector $\vec{x}$ go to minus or plus infinity, the bounds tend to each other.

The joint probability density functions of bounds $f_{S\vec{z}}(\vec{x}, \vec{y})$, $f_{I\vec{z}}(\vec{x}, \vec{y})$ of the system of hyper-random variables $\vec{Z} = \left(\vec{X}, \vec{Y}\right)^{\mathrm{T}}$ are related to the conditional probability density functions of the bounds $f_{S\vec{y}/\vec{x}}(\vec{y}), f_{I\vec{y}/\vec{x}}(\vec{y})$ of the hyper-random variable $\vec{Y}$ and the probability density functions of the bounds $f_{S\vec{x}}(\vec{x}), f_{I\vec{x}}(\vec{x})$ of the hyper-random variable $\vec{X}$ by the inequalities

$$\begin{aligned} f_{S\vec{z}}(\vec{x}, \vec{y}) &\leq f_{S\vec{x}}(\vec{x}) f_{S\vec{y}/\vec{x}}(\vec{y}), \\ f_{I\vec{z}}(\vec{x}, \vec{y}) &\geq f_{I\vec{x}}(\vec{x}) f_{I\vec{y}/\vec{x}}(\vec{y}), \end{aligned}$$

which follow from (7.10).

The *vector hyper-random variables* $\vec{X}$ and $\vec{Y}$ are *independent* if the probability densities of the bounds $f_{S\vec{z}}(\vec{x}, \vec{y}), f_{I\vec{z}}(\vec{x}, \vec{y})$ factorize:

$$\begin{aligned} f_{S\vec{z}}(\vec{x}, \vec{y}) &= f_{S\vec{x}}(\vec{x}) f_{S\vec{y}}(\vec{y}), \\ f_{I\vec{z}}(\vec{x}, \vec{y}) &= f_{I\vec{x}}(\vec{x}) f_{I\vec{y}}(\vec{y}). \end{aligned}$$

For independent variables $\vec{X}$ and $\vec{Y}$, not only do the probability densities of the bounds factorize, but so also do the bounds of the distribution function:

$$\begin{aligned} F_{S\vec{z}}(\vec{x}, \vec{y}) &= F_{S\vec{x}}(\vec{x}) F_{S\vec{y}}(\vec{y}), \\ F_{I\vec{z}}(\vec{x}, \vec{y}) &= F_{I\vec{x}}(\vec{x}) F_{I\vec{y}}(\vec{y}). \end{aligned}$$

Note that the independence of the hyper-random variables and their independence under all conditions are *different concepts*.

The bounds of the distribution function of the hyper-random variable can be characterized by various numerical characteristics, in particular, the *bounds of the crude and central moments*.

For a two-dimensional hyper-random variable $\vec{X} = (X_1, X_2)^{\mathrm{T}}$ characterized by probability densities of the bounds $f_{S\vec{x}}(x_1, x_2)$ and $f_{I\vec{x}}(x_1, x_2)$, the *expectations of the bounds* $\vec{m}_{S\vec{x}}$ and $\vec{m}_{I\vec{x}}$ are given by the vectors

$$\begin{aligned} \vec{m}_{S\vec{x}} &= (m_{Sx_1}, m_{Sx_2})^{\mathrm{T}} = \mathrm{E}_S\left[\vec{X}\right], \\ \vec{m}_{I\vec{x}} &= (m_{Ix_1}, m_{Ix_2})^{\mathrm{T}} = \mathrm{E}_I\left[\vec{X}\right], \end{aligned} \tag{7.37}$$

the *variances of the bounds* $\vec{D}_{S\vec{x}}$, $\vec{D}_{I\vec{x}}$ by the vectors

$$\begin{aligned} \vec{D}_{S\vec{x}} &= (D_{Sx_1}, D_{Sx_2})^{\mathrm{T}} = \mathrm{D}_S\left[\vec{X}\right], \\ \vec{D}_{I\vec{x}} &= (D_{Ix_1}, D_{Ix_2})^{\mathrm{T}} = \mathrm{D}_I\left[\vec{X}\right], \end{aligned} \tag{7.38}$$

and the *standard deviations of the bounds* $\vec{\sigma}_{S\vec{x}}$, $\vec{\sigma}_{I\vec{x}}$ by the vectors

$$\begin{aligned} \vec{\sigma}_{S\vec{x}} &= (\sigma_{Sx_1}, \sigma_{Sx_2})^{\mathrm{T}} = \left(\sqrt{D_{Sx_1}}, \sqrt{D_{Sx_2}}\right)^{\mathrm{T}}, \\ \vec{\sigma}_{I\vec{x}} &= (\sigma_{Ix_1}, \sigma_{Ix_2})^{\mathrm{T}} = \left(\sqrt{D_{Ix_1}}, \sqrt{D_{Ix_2}}\right)^{\mathrm{T}}, \end{aligned} \tag{7.39}$$

where m_{Sx_1}, m_{Ix_1} and m_{Sx_2}, m_{Ix_2} are the expectations of the bounds, respectively, of the hyper-random variables X_1 and X_2, D_{Sx_1}, D_{Ix_1} and D_{Sx_2}, D_{Ix_2} are the variances of the bounds of these variables, and $\sigma_{Sx_1}, \sigma_{Ix_1}$ and $\sigma_{Sx_2}, \sigma_{Ix_2}$ are the standard deviations of these bounds.

The *correlation moments of the bounds* $K_{Sx_1x_2}, K_{Ix_1x_2}$ are given by

$$\begin{aligned} K_{Sx_1x_2} &= \mathrm{E}_S[X_1X_2], \\ K_{Ix_1x_2} &= \mathrm{E}_I[X_1X_2], \end{aligned} \tag{7.40}$$

the *covariance moments of the bounds* $R_{Sx_1x_2}, R_{Ix_1x_2}$ by

$$\begin{aligned} R_{Sx_1x_2} &= \mathrm{E}_S[(X_1 - m_{Sx_1})(X_2 - m_{Sx_2})], \\ R_{Ix_1x_2} &= \mathrm{E}_I[(X_1 - m_{Ix_1})(X_2 - m_{Ix_2})], \end{aligned} \tag{7.41}$$

and the *correlation coefficients of the bounds* $r_{Sx_1x_2}, r_{Ix_1x_2}$ by

$$r_{Sx_1x_2} = \frac{R_{Sx_1x_2}}{\sigma_{Sx_1}\sigma_{Sx_2}}, \qquad r_{Ix_1x_2} = \frac{R_{Ix_1x_2}}{\sigma_{Ix_1}\sigma_{Ix_2}}. \tag{7.42}$$

The covariance moments $R_{Sx_1x_2}$, $R_{Ix_1x_2}$ of the bounds, the correlation moments $K_{Sx_1x_2}, K_{Ix_1x_2}$ of the bounds, and the expectations $m_{Sx_1}, m_{Sx_2}, m_{Ix_1}, m_{Ix_2}$ of the bounds of the hyper-random variables X_1 and X_2 are related by

$$
\begin{aligned}
R_{Sx_1x_2} &= K_{Sx_1x_2} - m_{Sx_1}m_{Sx_2}, \\
R_{Ix_1x_2} &= K_{Ix_1x_2} - m_{Ix_1}m_{Ix_2},
\end{aligned}
\tag{7.43}
$$

which is similar to the relation (2.19) for random variables.

The *hyper-random variables* X_1 and X_2 are said to be *uncorrelated* if the covariance moments of their bounds are equal to zero, i.e., $R_{Sx_1x_2} = R_{Ix_1x_2} = 0$. In this case $r_{Sx_1x_2} = r_{Ix_1x_2} = 0$, and according to (7.43), the correlation moments of the bounds $K_{Sx_1x_2}$, $K_{Ix_1x_2}$ are related to the expectations of the bounds by

$$
K_{Sx_1x_2} = m_{Sx_1}m_{Sx_2}, \quad K_{Ix_1x_2} = m_{Ix_1}m_{Ix_2}.
$$

The *hyper-random variables* X_1 and X_2 are said to be *orthogonal* if the correlation moments of their bounds are equal to zero, i.e., $K_{Sx_1x_2} = K_{Ix_1x_2} = 0$. In this case, according to (7.43), the covariance moments of the bounds $R_{Sx_1x_2}$ and $R_{Ix_1x_2}$ are related to the expectations of the bounds by

$$
R_{Sx_1x_2} = -m_{Sx_1}m_{Sx_2}, \quad R_{Ix_1x_2} = -m_{Ix_1}m_{Ix_2}.
$$

It is easy to check that the independence of the hyper-random variables X_1 and X_2 implies that they are not correlated. The converse is not true, in general.

7.3.3 Bounds of Moments

The bounds of moments are also used to describe vector hyper-random variables. For a two-dimensional hyper-random variable $\vec{X} = (X_1, X_2)^{\mathrm{T}}$, the *bounds of the expectation* are described by the vectors

$$
\begin{aligned}
\vec{m}_{s\vec{x}} &= \mathrm{E}_s\left[\vec{X}\right] = \left(m_{sx_1}, m_{sx_2}\right)^{\mathrm{T}}, \\
\vec{m}_{i\vec{x}} &= \mathrm{E}_i\left[\vec{X}\right] = \left(m_{ix_1}, m_{ix_2}\right)^{\mathrm{T}},
\end{aligned}
\tag{7.44}
$$

the *bounds of the variance* by the vectors

$$
\vec{D}_{s\vec{x}} = \left(D_{sx_1}, D_{sx_2}\right)^{\mathrm{T}}, \quad \vec{D}_{i\vec{x}} = \left(D_{ix_1}, D_{ix_2}\right)^{\mathrm{T}},
\tag{7.45}
$$

and the *bounds of the standard deviation* by the vectors

$$
\begin{aligned}
\vec{\sigma}_{s\vec{x}} &= \left(\sqrt{D_{sx_1}}, \sqrt{D_{sx_2}}\right)^{\mathrm{T}}, \\
\vec{\sigma}_{i\vec{x}} &= \left(\sqrt{D_{ix_1}}, \sqrt{D_{ix_2}}\right)^{\mathrm{T}},
\end{aligned}
\tag{7.46}
$$

where m_{sx_1}, m_{ix_1} and m_{sx_2}, m_{ix_2} are the bounds of the expectations corresponding to the hyper-random variables X_1 and X_2, and D_{sx_1}, D_{ix_1} and D_{sx_2}, D_{ix_2} are the bounds of the variances of these variables.

The *bounds of the correlation moment* $K_{sx_1x_2}$, $K_{ix_1x_2}$ *of the hyper-random variables* X_1, X_2 are given by

$$K_{sx_1x_2} = \sup_{g\in G} K_{x_1x_2/g}, \quad K_{ix_1x_2} = \inf_{g\in G} K_{x_1x_2/g}, \tag{7.47}$$

the *bounds of the covariance moment* $R_{sx_1x_2}$, $R_{ix_1x_2}$ by

$$R_{sx_1x_2} = \sup_{g\in G} R_{x_1x_2/g}, \quad R_{ix_1x_2} = \inf_{g\in G} R_{x_1x_2/g}, \tag{7.48}$$

and the *bounds of the correlation coefficient* $r_{sx_1x_2}$, $r_{ix_1x_2}$ by

$$\begin{aligned} r_{sx_1x_2} &= \sup_{g\in G} \frac{R_{x_1x_2/g}}{\sigma_{x_1/g}\sigma_{x_2/g}}, \\ r_{ix_1x_2} &= \inf_{g\in G} \frac{R_{x_1x_2/g}}{\sigma_{x_1/g}\sigma_{x_2/g}}. \end{aligned} \tag{7.49}$$

The bounds of the moment are obtained by selecting the extremal values from the set of values corresponding to different conditions $g \in G$. In this case, the bounds of different moments can correspond to different conditions g. Therefore, in general,

$$R_{sx_1x_2} \neq K_{sx_1x_2} - m_{sx_1}m_{sx_2}, \quad R_{ix_1x_2} \neq K_{ix_1x_2} - m_{ix_1}m_{ix_2}.$$

The *hyper-random variables* X_1 and X_2 are said to be *uncorrelated under all conditions* if the bounds of the covariance moment $R_{sx_1x_2}$ and $R_{ix_1x_2}$ are equal to zero.

The *hyper-random variables* X_1 and X_2 are said to be *orthogonal under all conditions* if the bounds of the correlation moment $K_{sx_1x_2}$ and $K_{ix_1x_2}$ are equal to zero.

If the hyper-random variables X_1 and X_2 are uncorrelated under all conditions and the conditional distributions are Gaussian, the axes of all *dispersion ellipses* are oriented along the axes of the coordinates.

The independence of the hyper-random variables X_1 and X_2 under all conditions implies that they are uncorrelated under all conditions. The converse is not true, in general.

The concepts of noncorrelatedness and orthogonality under all conditions differ from the concepts of noncorrelatedness and orthogonality associated respectively with the vanishing of the covariance and correlation moment of the bounds.

The bounds of the moment do not use information about the bounds of the distribution function, so their calculation usually involves lower computational costs than calculation of the moments of the bounds.

7.4 Transformations of Hyper-random Variables

The *scalar hyper-random variables* X_1 and X_2 described respectively by the distribution functions $\tilde{F}_{x_1}(x) = \{F_{x_1/g}(x)|g \in G\}$ and $\tilde{F}_{x_2}(x) = \{F_{x_2/g}(x)|g \in G\}$ are said to be *equal under all conditions*, if indeed under all conditions $g \in G$ for the same g their conditional distribution functions coincide: $F_{x_1/g}(x) = F_{x_2/g}(x)$ (see Sect. 2.5).

The *scalar hyper-random variables* X_1 and X_2 described respectively by the distribution functions $\tilde{F}_{x_1}(x)$ and $\tilde{F}_{x_2}(x)$ are said to be *equal* if the upper and lower bounds of their distribution functions coincide, i.e., $F_{Sx_1}(x) = F_{Sx_2}(x)$, $F_{Ix_1}(x) = F_{Ix_2}(x)$. *Equality under all conditions of vector hyper-random variables* and *equality of vector hyper-random variables* are formulated in a similar way.

Parameters and characteristics of hyper-random variables change under linear and nonlinear transformations. The parameters and characteristics of the transformed hyper-random variables depend in a complicated way on the parameters and characteristics of the original hyper-random variables. We shall not dwell on this issue, and note only the main results of the analysis:

- *All approaches* (based on conditional distribution functions (conditional probability densities) and their moments, the bounds of the distribution functions and their moments, and the bounds of the moments) *can be effectively used* to describe the transformation of *scalar hyper-random variables*;
- To describe the transformation of *vector hyper-random variables*, it is convenient to use the *conditional distribution functions and conditional moments, as well as the bounds of the moments*;
- Use of the *bounds of the distribution function and their moments* to describe the transformation of *vector hyper-random variables is limited*, being hindered by significant difficulties in the calculation of these characteristics and parameters.

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Chapter 8
Hyper-random Functions

Abstract The notion of a hyper-random function is formalized. The classification of hyper-random functions is presented. Three ways to describe a hyper-random function are considered: by the conditional characteristics (in particular, conditional distribution functions and conditional moments), by the bounds of the distribution function and their moments, and by the bounds of the moments. The definition of a stationary hyper-random process is given. The spectral method for describing a stationary hyper-random processes is presented. The concepts of an ergodic hyper-random process and a fragmentary-ergodic hyper-random process are formalized. We discuss the effectiveness of the different approaches for describing hyper-random processes.

8.1 Main Concepts

The approaches presented in the previous chapter can be used to describe hyper-random functions. A *hyper-random function* $X(t)$ is a *many-valued function* of an independent argument t, whose value for any fixed value $t \in T$ (where T is the *applicable domain* of the argument) is a hyper-random variable called the *cut set*. The set of all cut sets of the hyper-random function forms the *state space* S (*phase space* or *actual range*).

The i-th *realization of the hyper-random function* $X(t)$ is a determinate function $x_{ig}(t) = x_i(t)/g = x_i(t; g)$ which, for a fixed experiment $i \in I$, assigns to each $t \in T$ and condition $g \in G$ one of the values $x \in S$.

The hyper-random function can be represented by the *set of random (stochastic) functions* $X_g(t) = X(t)/g = X(t; g)$: $X(t) = \{X_g(t) | g \in G\}$ (Fig. 8.1).

A hyper-random function has features inherent in both a hyper-random variable and a determinate function: fixing the value of the argument t turns it into a hyper-random variable, and fixing the experiment i and the condition g transforms it into a determinate function.

This chapter is based on material from the books (Gorban 2007, 2011, 2014, 2016, 2017)

I.I. Gorban, *Randomness and Hyper-randomness*, Mathematical Engineering,
DOI 10.1007/978-3-319-60780-1_8

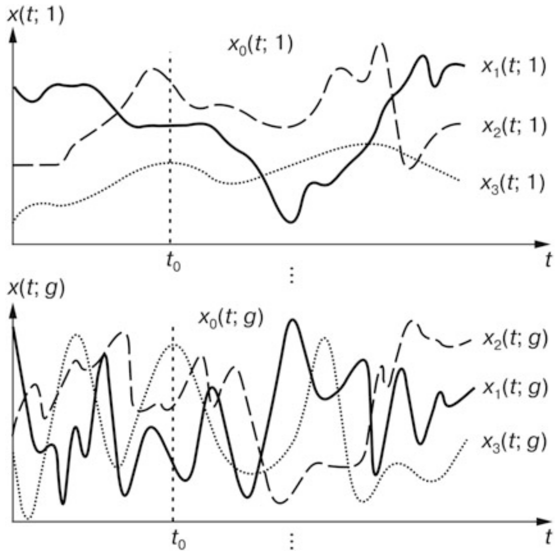

Fig. 8.1 Realizations of a hyper-random function $X(t)$

The number I of realizations of the hyper-random function may be finite, countably infinite, or uncountable.

The dimension L of the applicable domain T may take different values. If $L = 1$, then the argument t is often regarded as a time and the hyper-random function $X(t)$ is interpreted as a *hyper-random process* consisting of a set of random processes $X_g(t)$. If $L > 1$, then the argument t is a vector. In this case, the function $X(t)$ is called a *hyper-random field*. Such a hyper-random function can be represented by a set of random fields $X_g(t)$.

If the state space is one-dimensional, then the hyper-random function is *scalar*, and if the dimension of the state space is greater than one, then the hyper-random function is *vectorial*. In the first case, the hyper-random function is represented by a set of random scalar functions, and in the second case, it is represented by a set of random vector functions.

Hereafter, we shall consider scalar hyper-random processes whose actual range is *real*.

8.2 Hyper-random Processes

8.2.1 Conditional Characteristics and Moments

Just as a *random process* can be viewed as a *random variable vector*, a *hyper-random process* $X(t)$ can be interpreted as a *vector of hyper-random variables*.

Like any N-dimensional vector hyper-random variable, it can be described by a set of N-dimensional *conditional distribution functions* $F_{\vec{x}/g}(\vec{x};\vec{t}) = F_{\vec{x}/g}(x_1,\ldots,x_N;t_1,\ldots,t_N)$ or N-dimensional *conditional probability density functions* $f_{\vec{x}/g}(\vec{x};\vec{t}) = f_{\vec{x}/g}(x_1,\ldots,x_N;t_1,\ldots,t_N)$, where $g \in G$. In these expressions, as for the description of the probability characteristics of random processes, after specifying a set of cut sets $x_1, \ldots, x_N$ (vector $\vec{x}$), the set of values $t_1, \ldots, t_N$ (vector $\vec{t}$) to which these cut sets belong is indicated after the semicolon.

Conditional distribution functions $F_{\vec{x}/g}(\vec{x};\vec{t})$ can be characterized by the crude and central moments of the stochastic processes $X_g(t)$ ($g \in G$), in particular, the *conditional expectations*

$$m_{x/g}(t) = \mathrm{E}\left[X_g(t)\right],$$

conditional variances

$$D_{x/g}(t) = \mathrm{D}[X_g(t)] = \mathrm{Var}[X_g(t)] = \mathrm{E}\left[\left(X_g(t) - m_{x/g}(t)\right)^2\right],$$

conditional correlation functions

$$K_{x/g}(t_1,t_2) = \mathrm{E}\left[X_g(t_1)X_g(t_2)\right],$$

conditional covariance functions

$$\begin{aligned} &R_{x/g}(t_1,t_2) = \mathrm{Cov}\left[X_g(t_1),X_g(t_2)\right] = \\ &\mathrm{E}\left[\left(X_g(t_1) - m_{x/g}(t_1)\right)\left(X_g(t_2) - m_{x/g}(t_2)\right)\right], \end{aligned}$$

and so on.

To describe a hyper-random process $X(t)$, the bounds of the distribution and their moments are used too.

8.2.2 Bounds of the Distribution and Their Moments

The probability characteristics of a hyper-random process $X(t)$ are the *bounds of the distribution function* of the process $F_{S\vec{x}}(\vec{x};\vec{t})$, $F_{I\vec{x}}(\vec{x};\vec{t})$ and corresponding *probability density functions of the bounds* $f_{S\vec{x}}(\vec{x};\vec{t}), f_{I\vec{x}}(\vec{x};\vec{t})$.

The width of the *uncertainty area* is determined by the function

$$\Delta F_{\vec{x}}(\vec{x};\vec{t}) = F_{S\vec{x}}(\vec{x};\vec{t}) - F_{I\vec{x}}(\vec{x};\vec{t}).$$

For stochastic processes, this function is equal to zero; in total uncertainty $\Delta F_{\vec{x}}(\vec{x};\vec{t}) = 1$.

The *cut sets* t_1, t_2 *of a hyper-random process* $X(t)$ are *independent* if the corresponding two-dimensional probability densities factorize, i.e.,

$$\begin{aligned} f_{S\vec{x}}(x_1, x_2; t_1, t_2) &= f_{Sx_1}(x_1; t_1) f_{Sx_2}(x_2; t_2), \\ f_{I\vec{x}}(x_1, x_2; t_1, t_2) &= f_{Ix_1}(x_1; t_1) f_{Ix_2}(x_2; t_2), \end{aligned} \tag{8.1}$$

The *expectations of the bounds of the hyper-random process* $X(t)$ are defined by the expressions

$$m_{Sx}(t) = \mathrm{E}_S[X(t)], \;\; m_{Ix}(t) = \mathrm{E}_I[X(t)], \tag{8.2}$$

the *variances of the bounds* $D_{Sx}(t)$, $D_{Ix}(t)$ by

$$D_{Sx}(t) = \mathrm{E}_S\left[(X(t) - m_{Sx}(t))^2\right], \quad D_{Ix}(t) = \mathrm{E}_I\left[(X(t) - m_{Ix}(t))^2\right]. \tag{8.3}$$

and the *standard deviations* (*SD*) *of the bounds* by

$$\sigma_{Sx}(t) = \sqrt{D_{Sx}(t)}, \quad \sigma_{Ix}(t) = \sqrt{D_{Ix}(t)}. \tag{8.4}$$

The expectations of the bounds $m_{Sx}(t)$, $m_{Ix}(t)$ characterize the mean values of the hyper-random process $X(t)$ calculated for the upper and lower bounds of the distribution function. The variances of the bounds $D_{Sx}(t)$, $D_{Ix}(t)$, together with the standard deviations of the bounds $\sigma_{Sx}(t)$, $\sigma_{Ix}(t)$, characterize the degree of dispersion of this hyper-random process with respect to the corresponding expectations $m_{Sx}(t)$ and $m_{Ix}(t)$. It is easy to check that $m_{Sx}(t) \le m_{Ix}(t)$ and the ratio between $D_{Sx}(t)$ and $D_{Ix}(t)$ can take any value.

The *correlation functions of the bounds of a hyper-random process* are the functions

$$\begin{aligned} K_{Sx}(t_1, t_2) &= \mathrm{E}_S[X(t_1)X(t_2)], \\ K_{Ix}(t_1, t_2) &= \mathrm{E}_I[X(t_1)X(t_2)], \end{aligned} \tag{8.5}$$

and the *covariance functions of the bounds* are the functions

$$\begin{aligned} R_{Sx}(t_1, t_2) &= \mathrm{E}_S[(X(t_1) - m_{Sx}(t_1))(X(t_2) - m_{Sx}(t_2))], \\ R_{Ix}(t_1, t_2) &= \mathrm{E}_I[(X(t_1) - m_{Ix}(t_1))(X(t_2) - m_{Ix}(t_2))]. \end{aligned} \tag{8.6}$$

The covariance and correlation functions of the bounds are related as follows:

$$\begin{aligned} R_{Sx}(t_1, t_2) &= K_{Sx}(t_1, t_2) - m_{Sx}(t_1) m_{Sx}(t_2), \\ R_{Ix}(t_1, t_2) &= K_{Ix}(t_1, t_2) - m_{Ix}(t_1) m_{Ix}(t_2). \end{aligned} \tag{8.7}$$

The covariance functions of the bounds, the correlation functions of the bounds, and the *normalized covariance functions of the bounds*

$$r_{Sx}(t_1,t_2) = \frac{R_{Sx}(t_1,t_2)}{\sigma_{Sx}(t_1)\sigma_{Sx}(t_2)},$$
$$r_{Ix}(t_1,t_2) = \frac{R_{Ix}(t_1,t_2)}{\sigma_{Ix}(t_1)\sigma_{Ix}(t_2)} \tag{8.8}$$

characterize the dependence of the cut sets of the hyper-random process.

The *cut sets* t_1, t_2 *of the hyper-random process* $X(t)$ are *uncorrelated* if the associated covariance functions of the bounds $R_{Sx}(t_1,t_2) = R_{Ix}(t_1,t_2) = 0$, and *orthogonal* if the associated correlation functions of the bounds $K_{Sx}(t_1,t_2) = K_{Ix}(t_1,t_2) = 0$.

If there is no correlation

$$K_{Sx}(t_1,t_2) = m_{Sx}(t_1)m_{Sx}(t_2),$$
$$K_{Ix}(t_1,t_2) = m_{Ix}(t_1)m_{Ix}(t_2),$$

and if there is orthogonality

$$R_{Sx}(t_1,t_2) = -m_{Sx}(t_1)m_{Sx}(t_2),$$
$$R_{Ix}(t_1,t_2) = -m_{Ix}(t_1)m_{Ix}(t_2).$$

The notions of independence, noncorrelatedness, and orthogonality of the cut sets of a hyper-random process, as we see, are similar to those of a random process. If the cut sets of a hyper-random process are correlated, they are *dependent*. The converse is *not true in general*. If the cut sets are independent, they are uncorrelated. If the cut sets are orthogonal, they can be either dependent or independent and either correlated or uncorrelated. If the expectation of the upper (lower) bound of the cut set at t_1 and the expectation of the lower (upper) bound of the cut set at t_2 is equal to zero, or if the expectations of the upper and lower bounds of any of these cut sets is equal to zero, then the orthogonality of the cut sets implies their noncorrelatedness, and their noncorrelatedness implies their orthogonality.

8.2.3 Bounds of Moments

In order to describe hyper-random processes, other characteristics are used, similar to those describing hyper-random variables, in particular bounds of moments. The *bounds of the expectation* are

$$m_{sx}(t) = \mathrm{E}_s[X(t)] = \sup_{g \in G} m_{x/g}(t),$$
$$m_{ix}(t) = \mathrm{E}_i[X(t)] = \inf_{g \in G} m_{x/g}(t),$$

and the *bounds of the variance* are

$$D_{sx}(t) = \sup_{g \in G} D_{x/g}(t),$$
$$D_{ix}(t) = \inf_{g \in G} D_{x/g}(t).$$

The *bounds of the correlation function* are

$$K_{sx}(t_1, t_2) = \sup_{g \in G} K_{x/g}(t_1, t_2),$$
$$K_{ix}(t_1, t_2) = \inf_{g \in G} K_{x/g}(t_1, t_2),$$

and the *bounds of the covariance function* are

$$R_{sx}(t_1, t_2) = \sup_{g \in G} R_{x/g}(t_1, t_2),$$
$$R_{ix}(t_1, t_2) = \inf_{g \in G} R_{x/g}(t_1, t_2).$$

The bounds of the correlation function, the bounds of the covariance function, and the bounds of the expectation may correspond to different conditions g. Therefore, in general,

$$R_{sx}(t_1, t_2) \neq K_{sx}(t_1, t_2) - m_{sx}(t_1)m_{sx}(t_2),$$
$$R_{ix}(t_1, t_2) \neq K_{ix}(t_1, t_2) - m_{ix}(t_1)m_{ix}(t_2).$$

The *cut sets of a hyper-random process* $X(t)$ at t_1, t_2 are *uncorrelated under all conditions* if $R_{sx}(t_1, t_2) = R_{ix}(t_1, t_2) = 0$, and *orthogonal under all conditions* if $K_{sx}(t_1, t_2) = K_{ix}(t_1, t_2) = 0$.

Note that the concepts of noncorrelatedness and orthogonality of the hyper-random process *differ* from the ones presented here. The noncorrelatedness and orthogonality of the cut sets do not imply their noncorrelatedness and orthogonality under all conditions. In the general case, the noncorrelatedness and orthogonality of the cut sets under all conditions do not imply their noncorrelatedness and orthogonality.

It should be noted that the set of bounds of all moments defines the bounds of the distribution *ambiguously*.

8.3 Stationary and Ergodic Hyper-random Processes

Well known for stochastic processes, the concepts of stationarity and ergodicity can be generalized to hyper-random processes as well.

8.3.1 Stationary Hyper-random Processes

For hyper-random processes, a distinction is made between stationarity in the *narrow* and *broad* senses *for all conditions*, as well as between stationarity in the *narrow* and *broad* senses without reference to conditions.

A *hyper-random process* $X(t) = \{X_g(t) | g \in G\}$ is said to be *stationary in the narrow sense under all conditions* $g \in G$ if for all g its components $X_g(t)$ are stochastic processes that are stationary in the narrow sense.

One-dimensional conditional probabilistic characteristics of such process do not depend on the time (i.e., the argument t) and multidimensional conditional probabilistic characteristics depend on the difference in the argument values. In particular, the conditional distribution functions $F_{x/g}(x; t) = F_{x/g}(x)$ and the conditional probability density functions $f_{x/g}(x; t) = f_{x/g}(x)$.

A *hyper-random process* $X(t) = \{X_g(t) | g \in G\}$ is said to be *stationary in the broad sense under all conditions* $g \in G$ if for all fixed g the conditional expectation $m_{x/g}(t)$ does not depend on the argument t ($m_{x/g}(t) = m_{x/g}$) and the conditional correlation function $K_{x/g}(t_1, t_2)$ depends only on the duration τ of the interval $[t_1, t_2]$ and the conditions g: $K_{x/g}(t_1, t_2) = K_{x/g}(\tau)$. Note that in this case the conditional covariance function $R_{x/g}(t_1, t_2)$ also depends only on τ and g: $R_{x/g}(t_1, t_2) = R_{x/g}(\tau)$.

The bounds of the expectation $m_{sx}(t)$ and $m_{ix}(t)$ of a hyper-random process that is stationary in the broad sense under all conditions g are independent of time t, i.e., $m_{sx}(t) = m_{sx}$, $m_{ix}(t) = m_{ix}$, while the bounds of the correlation function

$$K_{sx}(\tau) = \sup_{g \in G} K_{x/g}(\tau), \quad K_{ix}(\tau) = \inf_{g \in G} K_{x/g}(\tau)$$

and the bounds of the covariance function

$$R_{sx}(\tau) = \sup_{g \in G} R_{x/g}(\tau), \quad R_{ix}(\tau) = \inf_{g \in G} R_{x/g}(\tau)$$

depend only on τ.

A *hyper-random process* $X(t)$ is said to be *stationary in the narrow sense* (*strictly*) if the bounds of its N-dimensional distributions for each N depend only on the durations of the intervals $[t_1, t_2], \ldots, [t_1, t_N]$ and do not depend on the position of these intervals on the t axis.

Hyper-random processes which do not satisfy this requirement are said to be *non-stationary in the narrow sense*.

The properties of stationary hyper-random processes are similar to the properties of stationary stochastic processes, in the sense that the bounds of the many-dimensional distribution function, and many-dimensional probability density functions of the bounds, do not depend on a shift in t. In addition, the above one-dimensional characteristics do not depend on the argument t, and the two-dimensional characteristics depend on the duration τ of the interval $[t_1, t_2]$, i.e.,

$$f_{Sx}(x;t) = f_{Sx}(x), \quad f_{Ix}(x;t) = f_{Ix}(x),$$
$$f_{Sx}(x_1, x_2; t_1, t_2) = f_{Sx}(x_1, x_2; \tau), \quad f_{Ix}(x_1, x_2; t_1, t_2) = f_{Ix}(x_1, x_2; \tau).$$

The following properties are intrinsic to the moment functions of the bounds of a stationary hyper-random process $X(t)$: the expectations of the bounds and the variances of the bounds are constants ($m_{Sx}(t) = m_{Sx}$, $m_{Ix}(t) = m_{Ix}$, $D_{Sx}(t) = D_{Sx}$, $D_{Ix}(t) = D_{Ix}$) and the correlation functions of the bounds $K_{Sx}(t_1, t_2)$, $K_{Ix}(t_1, t_2)$, the covariance functions of the bounds $R_{Sx}(t_1, t_2)$, $R_{Ix}(t_1, t_2)$, and the normalized covariance functions of the bounds $r_{Sx}(t_1, t_2)$, $r_{Ix}(t_1, t_2)$ do not depend on the position of the interval $[t_1, t_2]$ on the t axis, but only on its duration τ:

$$\begin{aligned}
&K_{Sx}(t_1, t_2) = K_{Sx}(\tau), \quad K_{Ix}(t_1, t_2) = K_{Ix}(\tau), \\
&R_{Sx}(t_1, t_2) = R_{Sx}(\tau), \quad R_{Ix}(t_1, t_2) = R_{Ix}(\tau), \\
&r_{Sx}(\tau) = R_{Sx}(\tau)/D_{Sx}, \quad r_{Ix}(\tau) = R_{Ix}(\tau)/D_{Ix}.
\end{aligned}$$

A *hyper-random process* $X(t)$ is said to be *stationary in the broad sense* if expectations of its bounds are constants ($m_{Sx}(t) = m_{Sx}$, $m_{Ix}(t) = m_{Ix}$) and the correlation functions of its bounds depend only on the duration τ of the interval $[t_1, t_2]$: $K_{Sx}(t_1, t_2) = K_{Sx}(\tau)$, $K_{Ix}(t_1, t_2) = K_{Ix}(\tau)$. In this case the covariance functions of the bounds depend only on the duration τ: $R_{Sx}(t_1, t_2) = R_{Sx}(\tau)$, $R_{Ix}(t_1, t_2) = R_{Ix}(\tau)$.

Note that the concepts of a hyper-random process that is stationary in the broad sense and a hyper-random process that is stationary in the broad sense under all conditions are *different notions*.

8.3.2 Spectral Description of Stationary Hyper-random Processes

The spectral representation of hyper-random processes often substantially facilitates their analysis, in particular when they possess stationary properties.

The *power spectral densities of the upper and lower bounds* (*energy spectra of the bounds*) of a stationary hyper-random process $X(t)$ are the functions $S_{Sx}(f)$, $S_{Ix}(f)$ related to the correlation functions of the bounds $K_{Sx}(f)$, $K_{Ix}(f)$ by

$$\begin{aligned}
S_{Sx}(f) &= \int_{-\infty}^{\infty} K_{Sx}(\tau) \exp(-\mathrm{j}2\pi f \tau) d\tau, \\
S_{Ix}(f) &= \int_{-\infty}^{\infty} K_{Ix}(\tau) \exp(-\mathrm{j}2\pi f \tau) d\tau,
\end{aligned}$$

$$K_{Sx}(\tau) = \int_{-\infty}^{\infty} S_{Sx}(f)\exp(j2\pi f\tau)df,$$
$$K_{Ix}(\tau) = \int_{-\infty}^{\infty} S_{Ix}(f)\exp(j2\pi f\tau)df.$$

The power spectral densities of the bounds of a hyper-random process possess similar properties to the power spectral density of a stochastic process.

Hyper-random white noise is a stationary hyper-random process $N(t)$ with zero expectations of the bounds and for which the power spectral densities of the bounds are constants, i.e., $S_{Sn} = N_S/2$, $S_{In} = N_I/2$, where N_S, N_I are constants.

The *correlation functions of the bounds of hyper-random white noise* are described by Dirac delta functions:

$$K_{Sn}(\tau) = N_S\delta(\tau)/2, \;\; K_{In}(\tau) = N_I\delta(\tau)/2.$$

Note that the same expressions describe the *covariance functions of the bounds of hyper-random white noise*.

The *conditional power spectral density* $S_{x/g}(f)$ of the hyper-random process $X(t)$ is defined as the Fourier transform of the conditional correlation function:

$$S_{x/g}(f) = \int_{-\infty}^{\infty} K_{x/g}(\tau)\exp(-j2\pi f\tau)d\tau,$$

where the function $K_{x/g}(\tau)$ is related to $S_{x/g}(f)$ by the inverse Fourier transform:

$$K_{x/g}(\tau) = \int_{-\infty}^{\infty} S_{x/g}(f)\exp(j2\pi f\tau)df.$$

The *bounds of the power spectral density of the hyper-random process* are the following functions:

$$S_{sx}(f) = \sup_{g\in G} S_{x/g}(f), \quad S_{ix}(f) = \inf_{g\in G} S_{x/g}(f).$$

Hyper-random white noise under all conditions is a hyper-random process $N(t)$ that is stationary under all conditions and for which the conditional expectations are equal to zero and the conditional energy spectra are independent of the frequency, i.e., $S_{n/g} = N_g/2$, where N_g is a constant depending, in general, on the condition g.

The conditional correlation function of such noise is described by the Dirac delta function: $K_{n/g}(\tau) = N_g\delta(\tau)/2$. Its conditional covariance function is described by the same expression.

8.3.3 Ergodic Hyper-random Processes

Some stationary hyper-random processes have ergodic properties.

A *stationary hyper-random process* $X(t) = \{X_g(t) | g \in G\}$ is said to be *ergodic under all conditions g* if all its stochastic components $X_g(t)$ are ergodic processes.

In particular, a *hyper-random process* $X(t)$ that is *stationary in the broad sense under all conditions* is said to be *ergodic in the broad sense under all conditions g*, if for all $g \in G$ the conditional expectation $m_{x/g}$ coincides with the *conditional time average*

$$\bar{m}_{x/g} = \lim_{T\to\infty} \frac{1}{T} \int\limits_{-T/2}^{T/2} x_g(t)dt \tag{8.9}$$

calculated for *any realization* $x_g(t)$ of the stochastic process $X_g(t)$, and the conditional covariance function $R_{x/g}(\tau)$ coincides with the *conditional autocovariance function*

$$\bar{R}_{x/g}(\tau) = \lim_{T\to\infty} \frac{1}{T} \int\limits_{-T/2}^{T/2} \left(x_g(t+\tau) - \bar{m}_{x/g}\right)\left(x_g(t) - \bar{m}_{x/g}\right)dt. \tag{8.10}$$

In this case the conditional correlation function $K_{x/g}(\tau)$ coincides with the *conditional autocorrelation function*

$$\bar{K}_{x/g}(\tau) = \lim_{T\to\infty} \frac{1}{T} \int\limits_{-T/2}^{T/2} x_g(t+\tau)x_g(t)dt.$$

On this basis, the conditional expectations, conditional covariance functions, and conditional correlation functions can be calculated by the formulas $m_{x/g} = \bar{m}_{x/g}$, $R_{x/g}(\tau) = \bar{R}_{x/g}(\tau)$, $K_{x/g}(\tau) = \bar{K}_{x/g}(\tau)$.

If the set G consists of one element, the ergodic hyper-random process degenerates into an *ergodic stochastic process*.

The *average bounds of an ergodic hyper-random process* $X(t) = \{X_g(t) | g \in G\}$ are the magnitudes

$$\bar{m}_{sx} = \sup_{g\in G} \bar{m}_{x/g}, \quad \bar{m}_{ix} = \inf_{g\in G} \bar{m}_{x/g},$$

the *autocorrelation function bounds* are the functions

$$\bar{K}_{sx}(\tau) = \sup_{g\in G} \bar{K}_{x/g}(\tau), \quad \bar{K}_{ix}(\tau) = \inf_{g\in G} \bar{K}_{x/g}(\tau),$$

and the *autocovariance function bounds* are the functions

$$\bar{R}_{sx}(\tau) = \sup_{g \in G} \bar{R}_{x/g}(\tau), \quad \bar{R}_{ix}(\tau) = \inf_{g \in G} \bar{R}_{x/g}(\tau).$$

The *average bounds of an ergodic hyper-random process* $\bar{m}_{sx}$, $\bar{m}_{ix}$ coincide with the *expectation bounds* m_{sx}, m_{ix}, and the *autocorrelation function bounds* $\bar{K}_{sx}(\tau)$, $\bar{K}_{ix}(\tau)$ likewise to the *correlation function bounds* $K_{sx}(\tau)$, $K_{ix}(\tau)$, the *autocovariance function bounds* $\bar{R}_{sx}(\tau)$, $\bar{R}_{ix}(\tau)$ likewise to the *covariance function bounds* $R_{sx}(\tau)$, $R_{ix}(\tau)$, and the *sample variance bounds* $\bar{D}_{sx} = \bar{R}_{sx}(0)$, $\bar{D}_{ix} = \bar{R}_{ix}(0)$ likewise to the *variance bounds* D_{sx}, D_{ix}.

All information about the characteristics of an *ergodic stochastic process* is contained in any one of its realizations. Therefore it is possible to calculate moments and other characteristics using any single realization. Unfortunately, a single realization is not sufficient to calculate characteristics of an *ergodic hyper-random process*. A *realization set* is required, i.e., one realization for each condition. This greatly complicates the calculations. However, it is *possible to use only a single realization* in the case when the hyper-random process exhibits the properties of stationarity and ergodicity *over an interval of finite duration*. Such hyper-random processes are discussed in the next section.

8.3.4 Fragmentary-Ergodic Hyper-random Processes Under All Conditions

Consider a hyper-random process $U(t) = \{U_h(t) | h = 1, 2, \ldots, H\}$ that is ergodic under all conditions $h = \overline{1, H}$, with *ergodic stochastic components* $U_h(t)$ described by the probability density functions $f_h(x)$. Suppose that the *components* $U_h(t)$ are *almost ergodic* over an *interval of duration* T, i.e., the characteristics of these components can be calculated in this interval with negligible error using a single realization.

A *hyper-random process* that is *fragmentary-ergodic under all conditions* is a hyper-random process $X(t) = \{X_g(t) | g = 1, 2, \ldots, G\}$ whose stochastic components $X_g(t)$ consist of *fragments of duration* T of the functions $U_h(t)$, $h = \overline{1, H}$ (see Figs. 8.2 and 8.3).

> *Each realization* of a hyper-random process that is fragmentary ergodic under all conditions carries statistical information about *all* its *stochastic components*. Therefore, *to calculate the characteristics of such a process,* any *single realization is sufficient.*

It should be noted that, for a fragmentary-ergodic *stochastic process* $X_g(t)$ (see Fig. 3.3), the order of the distributions $f_h(x)$ is *determinate*; and for a *hyper-random process that is fragmentary ergodic under all conditions* (see Fig. 8.3), when the conditions g are fixed, the order is also *determinate*. However, when the conditions are not fixed, the order is *uncertain*.

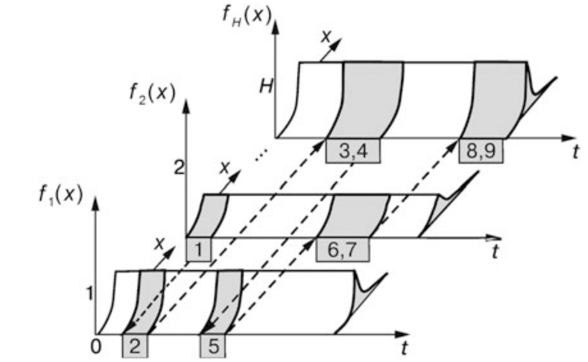

Fig. 8.2 Forming the one-dimensional probability density function of a *fragmentary-ergodic stochastic process* $X_g(t)$ for a fixed condition g from the conditional probability density functions of a hyper-random function $U(t) = \{U_h(t) | h = 1, 2, \ldots, H\}$ that is ergodic under all conditions

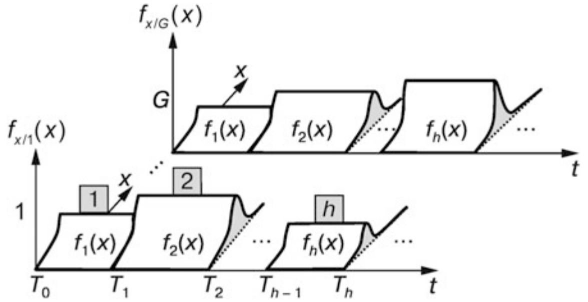

Fig. 8.3 One-dimensional conditional probability density functions $f_{x/g}(x; t)$ of a fragmentary-ergodic hyper-random process $X(t) = \{X_g(t) | g = 1, 2, \ldots, G\}$ with fragments described by the probability density functions $f_h(x)$, $h = 1, 2, \ldots, H$

8.4 Transformation of Hyper-random Processes

A transformation of a hyper-random process generates a new hyper-random process whose characteristics are defined by the characteristics of the initial process and the characteristics of the transformation operator.

On the basis of the methods used to calculate characteristics of transformed *random processes* under different linear and nonlinear, inertialess and inertial transforms, similar methods have been developed for calculating the characteristics of transformed *hyper-random processes*. Readers interested in this matter are referred to the monographs (Gorban 2011, 2014, 2017).

Here we note only the two main results concerning the effectiveness of different approaches for describing hyper-random processes:

- The relations describing conversions of *hyper-random vector variables* can be used to describe *inertialess conversions* of hyper-random processes;
- The main means for describing *inertial conversions* of hyper-random processes are the *conditional moments of the distribution* (*primarily, conditional expectations and conditional covariance functions*), *the bounds of these moments, and also the bounds of the power spectral density*.

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Chapter 9
Fundamentals of the Mathematical Statistics of Hyper-random Phenomena

Abstract The notion of a hyper-random sample and statistics of hyper-random variables are formalized. Estimators of the characteristics of hyper-random variables are examined. The notions of a generalized limit and a spectrum of limit points are introduced. Here we formalize the notions of convergence of hyper-random sequences in a generalized sense in probability and in distribution. Generalized law of large numbers and generalized central limit theorem are presented and peculiarities of them are studied. We present experimental results demonstrating the lack of convergence of the sample means of real physical processes to fixed numbers.

Mathematical statistics is usually associated with probability theory, and is often seen as a component of probability theory. Until recently, this view of statistics was fully justified and did not cause any serious objections. However, over time it became clear that mathematical statistics is not confined only to probability theory. It is applied, in particular, in the theory of hyper-random phenomena. It has been found that, for situations involving a large amount of data, hyper-random statistical models that take into account violations of statistical stability provide a much more adequate description of reality than their random statistical analogues.

The current chapter presents the basic mathematical statistics of hyper-random phenomena.

9.1 Hyper-random Samples

The basic concepts of the mathematical statistics of the theory of hyper-random phenomena are based on those of probability theory. We begin with the mathematical statistics of a *scalar hyper-random variable*, which is a set of scalar random variables $X_g = X/g$ observed under conditions $g \in G$: $X = \{X_g | \ g \in G\}$.

This chapter is based on material from the books (Gorban 2011, 2014, 2016, 2017)

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The *entire assembly* (*general population*) *of the hyper-random variable* $X = \{X_g | \ g \in G\}$ is the *infinite set of all its determinate realizations* (*sample elements or components*) observed *under all conditions* $g \in G$. This set can be either countable or uncountable.

This definition implies that the general population of the hyper-random variable X is the union of the populations of all its random components $X_g, g \in G$. The general population can be described by the *many-valued distribution function* $\tilde{F}_x(x)$ of the hyper-random variable X, the set of conditional distribution functions $F_{x/g}(x)$ ($g \in G$), the upper and lower bounds of the distribution function $F_{Sx}(x)$, $F_{Ix}(x)$, the moments of the bounds, the bounds of the moments, and other characteristics.

A set of members of the general population

$$\vec{x} = (x_1, \ldots, x_N) = \left\{x_{1g}, \ldots, x_{Ng} | g \in G\right\} = \left\{\vec{x}_g | g \in G\right\}$$

of the hyper-random variable X obtained for a finite number N of experiments under different *fixed* or *non-fixed conditions* $g \in G$ is called a *sample of the population*, and its elements $x_1, \ldots, x_N$ or $x_{1g}, \ldots, x_{Ng}$ are called the *sampling values* or *realizations*.

Without specifying a condition g, each sampling value $x_n \left(n = \overline{1, N}\right)$ is a *set of determinate values* (*set of numbers*), and when the condition g is specified, each sampling value x_{ng} is a *determinate value* (*number*).

One assumes that a sample $x_1, \ldots, x_N$ belongs to the hyper-random variable $X = \{X_g | g \in G\}$ described by the conditional distribution functions $F_{x/g}(x)$, $g \in G$ if it is obtained from a general population described under condition g by the distribution function $F_{x/g}(x)$.

An infinite set of samples $\vec{x} = (x_1, \ldots, x_N)$ of size N taken from a general population without specifying a condition g forms an N-dimensional hyper-random vector

$$\vec{X} = (X_1, \ldots, X_N) = \left\{X_{1g}, \ldots, X_{Ng} | g \in G\right\} = \left\{\vec{X}_g | g \in G\right\},$$

called a *hyper-random sample*, and an infinite set of samples $\vec{x}_g = \left(x_{1g}, \ldots, x_{Ng}\right)$ of size N taken from this general population under condition g forms an N-dimensional *random vector* (*random sample*) $\vec{X}_g = \left(X_{1g}, \ldots, X_{Ng}\right)$.

One generally assumes that all elements of a hyper-random vector are described by the same many-valued distribution function $\tilde{F}_x(x)$, and each component X_{ng} $\left(n = \overline{1, N}\right)$ of the random vector $\vec{X}_g$ corresponding to the specific condition g is described by the same single-valued distribution function $F_{x/g}(x)$ (or probability density function $f_{x/g}(x)$, Fig. 9.1a).

One usually assumes that the components X_n of the hyper-random sample $\vec{X}$ are *mutually independent under all conditions*. Then the conditional distribution function $F_{\vec{x}/g}(\vec{x})$ of the hyper-random sample $\vec{X}$ under conditions $g \in G$ factorizes: $F_{\vec{x}/g}(\vec{x}) = \prod_{n=1}^{N} F_{x/g}(x_n)$.

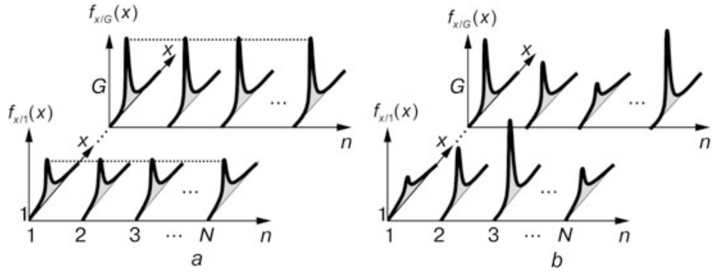


Fig. 9.1 Probability density functions of the random components $X_{1g}, \ldots, X_{Ng}, g = 1, \ldots, G$ of the hyper-random sample $\vec{X} = \{X_{1g}, \ldots, X_{Ng} | g = 1, \ldots, G\}$: homogeneous (**a**) and heterogeneous (**b**)

Besides the *homogeneous model of hyper-random sample* described above, one sometimes uses a *heterogeneous sample model* in which the random components X_{ng} of the hyper-random sample $\vec{X}$ have different distribution laws (Fig. 9.1b).

In the theory of hyper-random phenomena, a *statistic* is *any function* of the *hyper-random sample* $\vec{X}$, *random sample* $\vec{X}_g$ under a fixed condition $g \in G$, *determinate many-valued sample* $\vec{x}$, or *determinate single-valued sample* $\vec{x}_g$ under a fixed condition $g \in G$.

9.2 Assessing Characteristics and Parameters of a Hyper-random Variable

9.2.1 General Considerations

Using the general population of a hyper-random variable, it is theoretically possible to calculate various *exact determinate characteristics and parameters*, such as the conditional distribution functions $F_{x/g}(x)$, bounds of the distribution function $F_{Sx}(x)$, $F_{Ix}(x)$, conditional expectations $m_{x/g}$, expectations of bounds m_{Sx}, m_{Ix}, bounds of the expectation m_{sx}, m_{ix}, conditional variances $D_{x/g}$, variances of bounds D_{Sx}, D_{Ix}, bounds of the variance D_{sx}, D_{ix}, and so on.

Using certain statistics of realizations of the hyper-random variable, one can calculate *approximate evaluations* of the same characteristics and parameters, in particular, assessments of conditional distribution functions $F^*_{x/g}(x)$, bounds of distribution functions $F^*_{Sx}(x)$, $F^*_{Ix}(x)$, conditional expectations $m^*_{x/g}$, expectations of bounds m^*_{Sx}, m^*_{Ix}, bounds of the expectation m^*_{sx}, m^*_{ix}, conditional variances $D^*_{x/g}$, variances of bounds D^*_{Sx}, D^*_{Ix}, bounds of the variance D^*_{sx}, D^*_{ix}, and so on.

If the sample is *hyper-random*, then the assessments are *hyper-random estimators*, and if it is *determinate*, then the assessments are *determinate estimates*.

9.2.2 Forming Estimates

Estimates can be made in several steps. First, samples $x_{1g}, \ldots, x_{Ng}$ are formed separately for each condition $g \in G$. Using samples $\vec{x}_g = (x_{1g}, \ldots, x_{Ng})$ for all $g \in G$, one then calculates the conditional characteristic and parameter estimates, in particular, estimates of the conditional distribution functions $F^*_{x/g}(x)$, estimates of the conditional expectations $m^*_{x/g}$, estimates of the conditional variances $D^*_{x/g}$, and others.

From the conditional distribution functions $F^*_{x/g}(x)$ for all $g \in G$, one can calculate estimates of the distribution function bounds:

$$F^*_{Sx}(x) = \sup_{g \in G} F^*_{x/g}(x), \quad F^*_{Ix}(x) = \inf_{g \in G} F^*_{x/g}(x),$$

and estimates of the parameters describing these bounds, viz., estimates m^*_{Sx}, m^*_{Ix} of the expectations of the bounds, estimates D^*_{Sx}, D^*_{Ix} of the variances of the bounds, and so forth.

Using estimates of the conditional variables, one can calculate estimates of the corresponding variable bounds, for example, estimates of the expectation bounds $m^*_{sx} = \sup_{g \in G} m^*_{x/g}$, $m^*_{ix} = \inf_{g \in G} m^*_{x/g}$, estimates of the variance bounds $D^*_{sx} = \sup_{g \in G} D^*_{x/g}$, $D^*_{ix} = \inf_{g \in G} D^*_{x/g}$, etc.

When applying this technique, *certain difficulties can be expected in the first stage*, when the samples $\vec{x}_g$ for all $g \in G$ are formed, because at first glance, it is difficult to control and maintain the conditions g. The situation *is facilitated* by the fact that the ergodic property is inherent in a lot of actual samples and the calculation of a number characteristics *does not require information about the specific conditions under which the conditional characteristics have been obtained*.

> The most important thing is that, in the sample formation phase, all possible conditions g of the set G should be represented, and for every fixed condition g in the sample $\vec{x}_g$, *only the data corresponding to this condition g should be used*.

Typically, for actual phenomena occurring in the real world, in the case of a broad observation interval, the latter requirement can be easily satisfied because, although the conditions often vary continuously, they vary sufficiently slowly, and it is possible to evaluate the maximum number of elements N_s for which the conditions can be treated as practically constant (Fig. 9.2).

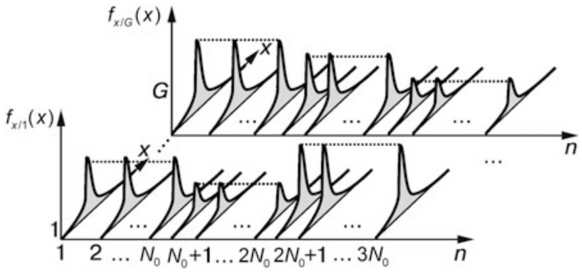


Fig. 9.2 Probability density functions of the random components $X_{1g}, \ldots, X_{Ng}$, $g = 1, \ldots, G$ of a heterogeneous hyper-random sample $\vec{X} = \{X_{1g}, \ldots, X_{Ng} | g = 1, \ldots, G\}$ when the statistical conditions vary slowly

> Therefore one can collect data over a broad observation interval (that is, essentially larger than N_s) without caring about what the statistical conditions are at any given time and in what way they alternate, and then separate the resulting data into a number fragments containing N_s consistent elements. Using these fragments, which represent the variable under different statistical conditions g, one can then calculate the required estimates. The main requirement for this technique is to collect the data for all possible observation conditions in G.

Of course, a number of questions arise. What are conditions under which hyper-random assessments converge to the exact characteristics and parameters? What type of parameters and characteristics are there? What are their distribution laws? The answers to these and other questions give the *generalized law of large numbers* and the *generalized central limit theorem.*

To understand this material, the reader should be familiar with some new mathematical concepts, such as the *generalized limit* and the *convergence of sequences in the generalized sense*.

9.3 Generalized Limit and the Convergence of Sequences in the Generalized Sense

Experimental investigations of actual processes of different physical nature over long observation intervals show (see Sect. 6.3) that real assessments do not tend toward any definite finite limits, or to infinity. By increasing the sample size, they fluctuate within certain ranges. Such behavior can be described mathematically by divergent sequences. So let us pause briefly on the mathematical apparatus developed to study divergent sequences.

9.3.1 Generalized Limit

According to classical concepts, the *numerical sequence* $x_1, x_2, \ldots, x_n$ is considered a *convergent sequence* if there is a limit $a = \lim\limits_{n\to\infty} x_n$ (see Sect. 4.2). If the limit exists then it is *unique* without fail. When the sequence has no limit, it is considered to be a *divergent sequence*.

Examples of divergent numerical sequences are $1, -1, 1, -1, \ldots$ and $1, 1, -1, 1, 1, -1, 1, 1, -1, \ldots$ (Fig. 9.3).

Note that here we have chosen for clarity examples of sequences with repetitive members. However, the reasoning here is valid for divergent sequences whose terms are not repeated.

From every infinite sequence one can form the set of *partial sequences* (*subsequences*) derived from the original sequence by discarding some of its members, while *maintaining the order of the remaining members*. It can then be shown that, when the sequence converges, all its partial sequences converge too. If the sequence diverges, then not all its partial sequences necessarily diverge. Some of them can converge to certain limits (*limit points*). The set of all limit points a_m, $m = 1, 2, \ldots$ of the sequence $x_1, x_2, \ldots, x_n$, also called *partial limits*, form the *spectrum of limit points* $\tilde{S}_x$. This is a generalization of the limit concept to any sequence, including divergent sequences. If the sequence converges, the spectrum of limit points consists of a single element (number), and if it is divergent, it consists of a set of numbers.

The spectrum of limit points can be described by the expression

$$\tilde{S}_x = \underset{n\to\infty}{\mathrm{LIM}}\, x_n, \tag{9.1}$$

where, unlike the conventional limit $\lim\limits_{n\to\infty}$, we use the symbol $\underset{n\to\infty}{\mathrm{LIM}}$ of the *generalized limit*.

The spectra of limit points, for example, of the sequences $1, -1, 1, -1, \ldots$ and $1, 1, -1, 1, 1, -1, 1, 1, -1, \ldots$, consist of two partial limits equal to -1 and $+1$.

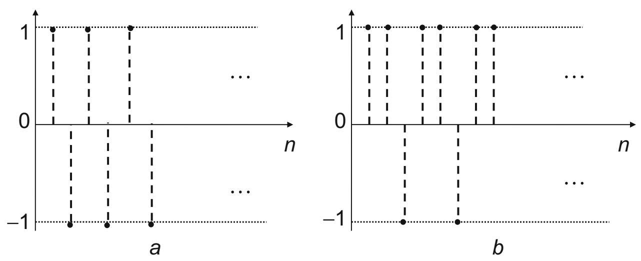


Fig. 9.3 Examples of the divergent sequences: $1, -1, 1, -1, \ldots$ (**a**) and $1, 1, -1, 1, 1, -1, 1, 1, -1, \ldots$ (**b**)

Expression (9.1) can be interpreted as the *convergence of the sequence to the spectrum of limit points*. The spectrum may be discrete, continuous, or mixed (discrete-continuous). If the spectrum forms a continuous interval, we say that the *sequence converges to the interval*.

A divergent sequence can be characterized not only by the spectrum of limit points, but also by the *set* (in general) *of measures* described by the *many-valued* (in general) *distribution function of the limit points*

$$\tilde{F}_x(x) = \underset{n \to \infty}{\mathrm{LIM}} \frac{m_n(x)}{n}, \tag{9.2}$$

where $m_n(x)$ is the number of terms of the sequence $x_1, x_2, \ldots, x_n$ that are less than x.

If the sequence converges in the usual sense to the number a, the distribution function of limit points is described by the unique distribution function $F_x(x)$ in the form of a unit step function at the point a (Fig. 9.4a) (then the measure is equal to one at the point a and zero at all other points).

If the sequence diverges (converges to a set of numbers, or in particular cases to an interval), the distribution function is either a single-valued non-decreasing function $F_x(x)$ that differs from the unit step function (Fig. 9.4b), or a many-valued function $\tilde{F}_x(x)$ (Fig. 9.4c) [in the *special case* it is described by a rectangle of unit height (Fig. 9.4d)]. Note that when the distribution function is *single-valued* and the spectrum of limit points is *discrete*, the distribution function is described by a *jump function* (Fig. 9.5).

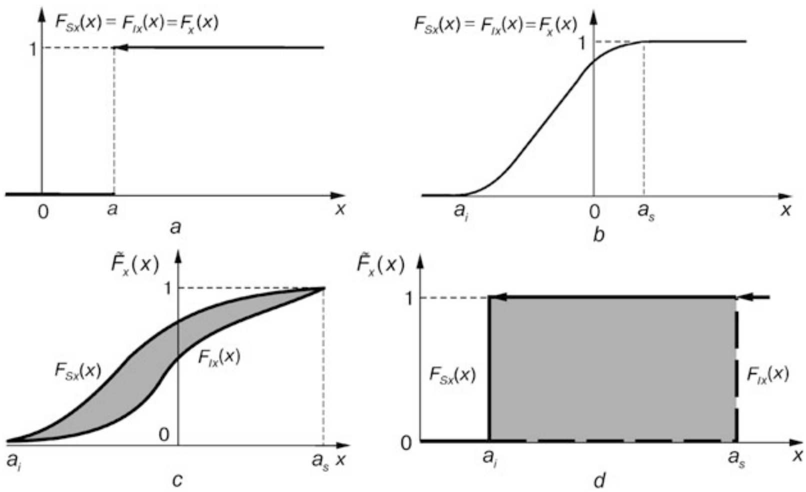


Fig. 9.4 Single-valued $F_x(x)$ (**a**, **b**) and many-valued $\tilde{F}_x(x)$ (**c**, **d**) distribution functions of the limit points and their bounds $F_{Ix}(x)$, $F_{Sx}(x)$ for sequences that converge to the number a (a) and to the interval $[a_i, \ a_s]$ (**b**–**d**)

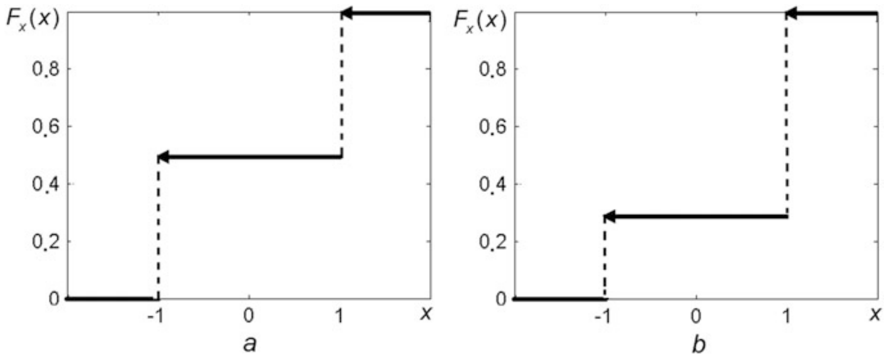


Fig. 9.5 Distribution functions of the limit points $F_x(x)$ of the sequences $1, -1, 1, -1, \ldots$ (**a**) and $1, 1, -1, 1, 1, -1, 1, 1, -1, \ldots$ (**b**)

Using the terminology of the theory of hyper-random phenomena, we can say that the *spectrum of the limit points of a numerical sequence* can be:

- A *number* (interpreting by the set of real numbers with unit measure at the point $x = a$ and zero measure at all other points) (Fig. 9.4a),
- A *random variable* (Fig. 9.4b),
- A *hyper-random variable* (Fig. 9.4c) [in the degenerate case, an *interval variable* (Fig. 9.4d)].

In other words, the *numerical sequence* may converge to a *number* or to a *set of numbers* (in particular cases, to an *interval*). If it converges to a set of numbers, the spectrum of limit points may be either a *random variable* or a *hyper-random variable*.

9.3.2 Convergence of Sequences of Hyper-random Variables

By analogy with the convergence of a *sequence of random variables* (see Sect. 4.2) the concept of *convergence in the generalized sense* of a *sequence of hyper-random variables* is introduced in the theory of hyper-random phenomena. There are different forms of convergence, viz., *in distribution function, in mean-square, almost surely* (*with probability one*), *and in probability* (*in measure*). Let us consider the convergence of a sequence in the generalized sense in probability and in distribution function.

Suppose we have a sequence of hyper-random variables $\mathrm{X} = \{X_1, \ldots, X_N\}$ and a hyper-random variable X, where $X_n = \{X_{ng} | \; g \in G\}$ $(n = \overline{1, N})$ and $X = \{X_g |\; g \in G\}$. For all $X_1, \ldots, X_N$ and X, there are distribution functions

$$\tilde{F}_{x_1}(x) = \left\{F_{x_1/g}(x) \middle|\ g \in G\right\}, \ldots, \tilde{F}_{x_N}(x) = \left\{F_{x_N/g}(x) \middle|\ g \in G\right\}$$

and $\tilde{F}_x(x) = \left\{F_{x/g}(x) | g \in G\right\}$. Then the sequence of hyper-random variables X *converges in the generalized sense* to the hyper-random variable *X in probability* ($P\{|X_N - X| > \varepsilon\} \to 0$) if for all conditions $g \in G$ and $\varepsilon > 0$, when $N \to \infty$,

$$P\left\{\left|X_{Ng} - X_g\right| > \varepsilon\right\} \to 0, \tag{9.3}$$

i.e., for all $g \in G$, the random sequence $X_{1g}, \ldots, X_{Ng}$ converges in probability to the random variable X_g.

The sequence of hyper-random variables X *converges in the generalized sense* to the hyper-random variable *X in distribution* ($\tilde{F}_{x_N}(x) \to \tilde{F}_x(x)$) if for each point x where $F_{x/g}(x)$ is continuous and for all conditions $g \in G$, when $N \to \infty$,

$$F_{x_N/g}(x) \to F_{x/g}(x). \tag{9.4}$$

As in the case of sequences of random variables, in the case of sequences of hyper-random variables convergence in distribution is *weaker* than convergence in probability, i.e., a sequence of hyper-random variables that converges in probability also converges in distribution. The converse is not always true.

> It follows from the definitions that, like a numerical sequence, a *hyper-random sequence* can converge to a *number* (a determinate variable whose distribution function is a unit step function), to a *random variable*, or to a *hyper-random variable*.

It is obvious that a *random sequence* can also converge to a *number*, to a *random variable*, or to a *hyper-random variable*.

It is interesting to trace the various representations of convergence of sequences (Table 9.1) in the framework of different disciplines (see Sects. 4.2 and 9.3.1).

In classical mathematical analysis, convergence concerns numerical sequences and a limit can only be a number. Probability theory extends the idea of convergence to random sequences. According to probability theory, a random sequence can converge either to a number or to a random variable.

Table 9.1 Convergence concept in different disciplines

Discipline	Sequence	Limit (spectrum of limit points)		
Mathematical analysis	Numerical	Number		
Probability theory	Random	Number	Random variable	
Theory of hype-random phenomena	Numerical	Number	Random variable	Hyper-random variable
	Random			
	Hyper-random			

The theory of hyper-random phenomena introduces the concept of a generalized limit. This greatly expands the scope of the concept of convergence. It becomes clear that numerical, random, or hyper-random sequences can converge in the generalized sense to a number, a random variable, or a hyper-random variable.

Using the techniques for description of divergent sequences, we can consider a *generalized law of large numbers for random sequences*.

9.4 Generalized Law of Large Numbers

In the case of a random sequence in typical interpretation for probability theory, the law of large numbers is described by (4.4) (see. Sect. 4.3), which says that the average m^*_{xN} of the random samples $X_1, \ldots, X_N$ converges in probability to some number m_x that is a conventional limit of the average m_{xN} of the expectations $m_{x_1}, \ldots, m_{x_N}$ of the random variables $X_1, \ldots, X_N$.

Examining the proof of this assertion (which we shall not present here), we find that *there is no assumption* in the proof that the average m^*_{xN} of the random samples and the average m_{xN} of the expectations have the conventional limits. This means that the sequences $\{m^*_{xN}\} = m^*_{x1}, \ldots, m^*_{xN}$ and $\{m_{xN}\} = m_{x1}, \ldots, m_{xN}$ *may not have limits in the conventional sense*, i.e., the sequences *may be divergent*.

But if they do not converge in the conventional sense, they *can converge in the generalized sense to many-valued variables*, i.e., to random or hyper-random variables.

Hereafter, following the above-mentioned convention about designations of single-valued and many-valued variables and functions, the single-valued limits of the sequences $\{m^*_{xN}\}$ and $\{m_{xN}\}$ will be denoted by m^*_x and m_x, and many-valued ones in the same manner, but with tilde, viz., $\tilde{m}^*_x$ and $\tilde{m}_x$.

Whether the considered limits are single-valued or many-valued, according to (4.3), when the sample size N increases, the *sample mean* m^*_{xN} *gradually approaches the average of the expectations* m_{xN}.

When $N \to \infty$, there are two possibilities:

Case 1 The variable m^*_{xN} converges to the *single-valued* average of the expectations m_x (number).

Case 2 The variable m^*_{xN}, becoming a many-valued variable $\tilde{m}^*_x$ in the limit, converges in the general sense to a *many-valued variable* $\tilde{m}_x$.

Case 1 is the idealized case considered in probability theory [see, for example, (Olkin et al. 1994; Devor 2012; Gnedenko 1988; Gorban 2003)]. In this case, the limit m_x of the average of the expectations is described by the distribution function $F_{m_x}(x)$, which is a unit step function at the point m_x. The distribution function $F_{m^*_{xN}}(x)$ of the sample mean m^*_{xN} tends to it when $N \to \infty$ (see Fig. 9.6a).

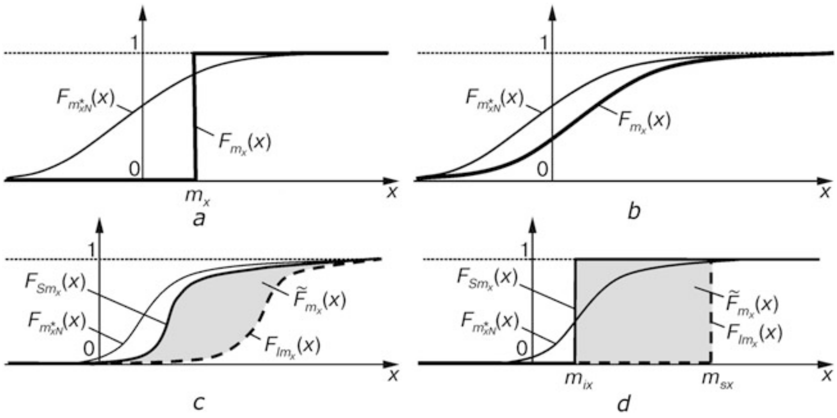


Fig. 9.6 Formation of the limiting distribution function $\tilde{F}_{m_x^*}(x)$ of the sample mean in the case of a random sequence: the limiting sample mean and the limiting average of the expectations are a number (**a**), a random variable (**b**), and a hyper-random variable (**c**, **d**) (**c** is the general case and **d** is a special case)

Case 2 is more realistic. Here the limiting sample mean $\tilde{m}_x^*$ and the limiting average of the expectations $\tilde{m}_x$ are described by the many-valued spectra $\tilde{S}_{m_x^*}$ and $\tilde{S}_{m_x}$, respectively. In this case there may be two variants:

Case 2.1. The limit of the sample mean $\tilde{m}_x^*$ and the limit of the average expectations $\tilde{m}_x$ are *variables of random type*. Then the spectra $\tilde{S}_{m_x^*}$ and $\tilde{S}_{m_x}$ are characterized by the *single-valued distribution functions* $F_{m_x^*}(x)$ and $F_{m_x}(x)$ (see Fig. 9.6b).

Case 2.2. The limit of the sample mean $\tilde{m}_x^*$ and the limit of the average expectations $\tilde{m}_x$ are *variables of hyper-random type*. Then the spectra $\tilde{S}_{m_x^*}$ and $\tilde{S}_{m_x}$ are characterized by the *many-valued distribution functions* $\tilde{F}_{m_x^*}(x)$ and $\tilde{F}_{m_x}(x)$ (see Fig. 9.6c).

Since the *convergence in distribution* of a sequence of *random variables* is weaker than *convergence in probability*, in Case 2.1, the limiting distribution function $F_{m_x^*}(x)$ coincides with the limiting distribution function $F_{m_x}(x)$.

For *hyper-random variables*, convergence of the sequence *in distribution* is also weaker than convergence *in probability*. Therefore, in Case 2.2, the limiting distribution function $\tilde{F}_{m_x^*}(x)$ coincides with the limiting distribution function $\tilde{F}_{m_x}(x)$. In this case, the lower bound $F_{Im_x^*}(x)$ of the limiting distribution function $\tilde{F}_{m_x^*}(x)$ coincides with the lower bound $F_{Im_x}(x)$ of the limiting distribution function $\tilde{F}_{m_x}(x)$, and the upper bound $F_{Sm_x^*}(x)$ of the limiting distribution function $\tilde{F}_{m_x^*}(x)$ coincides with the upper bound $F_{Sm_x}(x)$ of the limiting distribution function $\tilde{F}_{m_x}(x)$.

The uncertainty area located between the specified bounds is shaded in Fig. 9.6c. It can be shown (Gorban 2017) that, if the distribution function describing the spectrum of the sequence of averages of determinate values is many-valued, then the corresponding uncertainty area is continuous. So the *uncertainty area of the distribution function* $\tilde{F}_{m_x}(x)$ *is continuous.*

The interval in which the sample mean m^*_{xN} fluctuates when $N \to \infty$ is described by the lower bound m^*_{ix} when the function $F_{Sm^*_x}(x)$ begins to rise from zero, and the upper bound m^*_{sx} when the function $F_{Im^*_x}(x)$ reaches unity. Naturally, these bounds coincide with the corresponding bounds m_{ix}, m_{sx} of the functions $F_{Sm_x}(x)$, $F_{Im_x}(x)$: $m^*_{ix} = m_{ix}$, $m^*_{sx} = m_{sx}$. These bounds can be either finite or infinite.

Note that Case 2.2 includes the special case when the limiting sample mean $\tilde{m}^*_x$ and the limiting average of the expectations $\tilde{m}_x$ are of *interval type* (Fig. 9.6d).

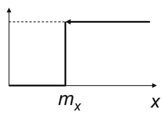
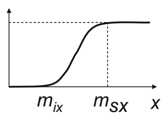
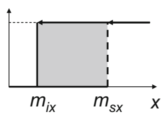
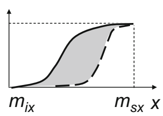
Systematizing the results of the present section, we may note the following:

> The sample mean m^*_{xN} of a random sample can converge to a *number* m_x (finite or infinite) or *fluctuate within a certain interval* $[m_{ix}, m_{sx}]$.

In the latter case, we shall say that there is *convergence of the sample mean to the interval*.

> Theoretically, the limit of the sample mean $\tilde{m}^*_x$ and the limit of the average of the expectations $\tilde{m}_x$ can be *numbers*, *random variables*, *intervals*, or *hyper-random variables*. The spectra $\tilde{S}_{m^*_x}$ and $\tilde{S}_{m_x}$ can be *numbers* or *intervals*. The limiting distribution functions $F_{m^*_x}(x)$ and $F_{m_x}(x)$ can be of *unit step type*, *single-valued functions*, or *many-valued functions with a continuous uncertainty area* (Table 9.2).

Table 9.2 Types of limiting parameters and distribution functions of random sequences

Limiting sample mean $\tilde{m}^*_x$ and limiting average of expectations $\tilde{m}_x$	Spectra $\tilde{S}_{m^*_x}$ and $\tilde{S}_{m_x}$	Limiting distribution functions $\tilde{F}_{m^*_x}(x)$ and $\tilde{F}_{m_x}(x)$
Number	Number	
Random variable	Interval	
Interval	Interval	
Hyper-random variable with continuous uncertainty area	Interval	

Convergence of the sample mean to a number is not corroborated by experiments, while convergence to an interval is corroborated by many them. We shall discuss the question of the type of limiting distribution function after studying the generalized central limit theorem.

9.5 Generalized Central Limit Theorem

Here we consider the *central limit theorem for a random sequence* (see Sect. 4.4). Using the technique devised to obtain (4.6), a more general statement can be proven: if the conditions specified in the Lindeberg–Feller theorem are satisfied, the difference between the distribution function $F_{m^*_{xN}}(x)$ of the sample mean m^*_{xN} and the Gaussian distribution function $F(x/m_{xN}, D_{xN})$ described by (4.5) converges uniformly to zero:

$$\lim_{N\to\infty}\left[F_{m^*_{xN}}(x) - F\left(x/m_{xN}, D_{xN}\right)\right] = 0. \tag{9.5}$$

There is a significant difference between (4.6) and (9.5). The expression (4.6) implies that the sample mean m^*_{xN} has a single-valued limiting distribution function $F_{m^*_x}(x)$ to which the distribution function $F_{m^*_{xN}}(x)$ tends when $N \to \infty$, and there is a *single-valued* Gaussian limiting distribution function $F_{m_x}(x) = F(x/m_x, D_x)$ to which the distribution function $F(x/m_{xN}, D_{xN})$ tends, where m_x and D_x are the expectation and the variance of the limiting distribution function, respectively.

The formula (9.5), on the other hand, allows the given limiting distribution functions to be *many-valued*. The many-valuedness of the limiting distribution function to which the function $F(x/m_{xN}, D_{xN})$ tends is stipulated by the many-valuedness of the expectation and (or) variance. Therefore, in the expression $\tilde{F}_{m_x}(x) = \tilde{F}\left(x/\tilde{m}_x, \tilde{D}_x\right)$ representing the limiting distribution function of the average of the expectations, the many-valued parameters $\tilde{m}_x$ and $\tilde{D}_x$ appear. In general these parameters are hyper-random variables. Therefore the function $\tilde{F}\left(x/\tilde{m}_x, \tilde{D}_x\right)$ is a hyper-random function. It can be interpreted as a set of single-valued Gaussian distribution functions. Each of these is described by a single-valued expectation $m_x \in \tilde{m}_x$ and variance $D_x \in \tilde{D}_x$.

The relation

$$F_{m^*_{xN}}(x) \to \tilde{F}\left(x/\tilde{m}_x, \tilde{D}_x\right)$$

follows from (9.5), implying that there is convergence in distribution of the sequence of determinate functions $F_{m^*_{xN}}(x)$ to the hyper-random function $\tilde{F}\left(x/\tilde{m}_x, \tilde{D}_x\right)$. In other words, the many-valued limiting distribution functions $\tilde{F}_{m^*_x}(x)$, $\tilde{F}\left(x/\tilde{m}_x, \tilde{D}_x\right)$ are described by identical sets of single-valued conditional distribution functions.

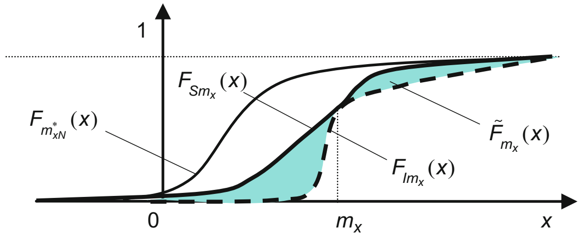


Fig. 9.7 Formation of the limiting distribution function $\tilde{F}_{m_x^*}(x)$ of the sample mean of the random variable when the expectation m_x is a number and the variance $\tilde{D}_x$ is a many-valued variable

When $\tilde{m}_x = m_x$ and $\tilde{D}_x = D_x$ (i.e., both parameters are numbers) and $D_x = 0$, the limiting Gaussian distribution function $F_{m_x}(x) = F(x/m_x, D_x)$ is the unit step function shown in Fig. 9.6a by the bold line; and when $\tilde{m}_x = m_x$ and $\tilde{D}_x = D_x$ are numbers but $D_x \neq 0$, this distribution function is described by the single-valued Gaussian curve shown in Fig. 9.6b by the bold line.

When the limiting expectation $\tilde{m}_x$ or the limiting expectation $\tilde{m}_x$ and limiting variance $\tilde{D}_x$ are many-valued variables, the limiting distribution function $\tilde{F}_{m_x}(x)$ is a many-valued function. In Fig. 9.6c–d, it is displayed by the shaded areas.

When the limiting expectation is a number ($\tilde{m}_x = m_x$) and limiting variance $\tilde{D}_x$ is a many-valued variable, the limiting distribution function $\tilde{F}_{m_x}(x) = \tilde{F}\left(x/m_x, \tilde{D}_x\right)$ has a specific feature: it consists of two areas touching at a single point. The abscissa of the contact point is equal to the expectation m_x (shaded areas in Fig. 9.7).

9.6 Experimental Study of the Convergence of the Sample Mean

The theoretical research presented in Sects. 6.4 and 6.5 indicates that with increasing sample size the sample means are *not necessarily normalized* (i.e., they do not necessarily take on the Gaussian character) and tend to a certain fixed value. This result is quite different from the conclusion from classical probability theory. It raises a very important question: how do the actual sample means behave?

To answer this question, we return to the investigation of the mains voltage oscillations and the radiation intensity of the pulsar PSRJ 1012+5307 in the X-ray frequency band (see Sects. 6.3.1 and 6.3.7) and present some results of additional experimental studies of the above processes. The point about these examples is that the variation of the mains voltage is one of the most unstable and the radiation intensity of the pulsar is one of the most stable.

9.6.1 Experimental Study of Mains Voltage Oscillations

Additional studies of mains voltage fluctuations were carried out using the 60-h record (Fig. 1.8a) on which calculations were carried out in Sect. 6.3.1. Studies

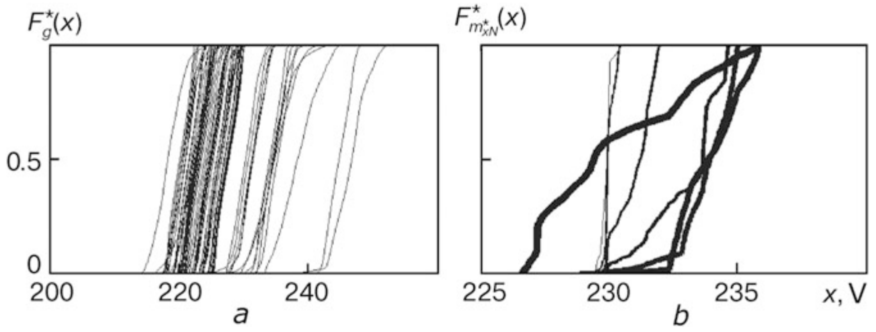


Fig. 9.8 Estimates of the distribution functions of mains voltage oscillations on 64 contiguous observation intervals (**a**) and estimates of the distribution function of the voltage sample mean $F^*_{m^*_{xN}}(x)$ for sample sizes $N = 2^r$, $r = 8, 10, 12, 14, 16, 18, 20$ (**b**) (the line thickness increases with the value of the parameter r)

consisted in calculating and analyzing estimates of the distribution functions of the voltage fluctuations $F^*_g(x)$ on adjacent observation intervals, each lasting about one hour ($g = \overline{1, 64}$) (Fig. 9.8a), and also the estimate of the distribution function of the sample mean $F^*_{m^*_{xN}}(x)$ (Fig. 9.8b).

It can be seen from Figs. 1.8a and 9.8a that the quality of the electricity is low. The voltage varies between $x_i = 215$ V and $x_s = 255$ V. There is no tendency for stabilization of the sample mean (see Fig. 1.8b), implying a clear statistical stability violation in this process.

The curves of the distribution functions $F^*_g(x)$ corresponding to different values of the parameter g differ essentially from one another (primarily by their location) (see Fig. 9.8a), and this confirms the claimed *nonstationarity* of the oscillations.

The calculation results for the estimate of the sample mean distribution function $F^*_{m^*_{xN}}(x)$ for exponentially growing sample size (see Fig. 9.8b) show that $F^*_{m^*_{xN}}(x)$ *does not tend to any given limiting distribution function* $F_{m_x}(x)$, and the sample mean m^*_{xN} *does not tend to any given limiting value* m_x.

On the basis of the curves for the estimate of the distribution function of the sample mean $F^*_{m^*_{xN}}(x)$ for small values of the parameter r (8 and 10) (see Fig. 9.8b), we may with some level of skepticism conclude that it tends to a Gaussian distribution with decreasing variance, as probability theory would predict. However, for large values of r (starting from 10 to 20), the assumed trend is not confirmed.

When the sample size increases, the variance of the sample mean m^*_{xN} sometimes increases (for values of r from 8 to 14 and from 18 to 20) and sometimes decreases (for r from 14 to 18). In general, as one moves from small to large sample sizes, the variance does not manifest any tendency to go to zero, as would have been predicted by probability theory (see Fig. 9.6a), but in fact increases, even by a significant factor (the range of the sample mean increases approximately from 1 to 8 V).

It follows from these results that the *distribution function of the sample mean tends to a many-valued function* $\tilde{F}_{m_x}(x)$ *of general form* (see Fig. 9.6c). When data is processed from other recordings of mains voltage fluctuations, we obtain the same result.

9.6.2 Experimental Study of Pulsar Intensity Fluctuations

Additional studies of pulsar radiation intensity PSRJ 1012+5307 were conducted using a 16-year record, for which the calculations of the statistical instability parameters γ_N^* and Γ_N^* were carried out in Sect. 6.3.7. The results of these studies are presented in Fig. 9.9.

In contrast to the curves of the distribution functions $F_g^*(x)$ calculated for mains voltage fluctuations, which differ essentially from each other (see. Fig. 9.8a), similar curves $F_g^*(x)$ calculated for the pulsar (Fig. 9.9a) almost coincide. This indicates that *there are no apparent violations of statistical stability*.

However, the curves of the estimate for the distribution function of the sample mean $F_{m_{xN}^*}^*(x)$ calculated for different sample sizes (see Fig. 9.9b) demonstrate that the sample mean m_{xN}^* *does not converge to any given limiting value* m_x and that the distribution function $F_{m_{xN}^*}^*(x)$ *does not converge to any given limiting distribution function* $F_{m_x}(x)$.

On the basis of the curves describing the distribution function $F_{m_{xN}^*}^*(x)$ for small values of r (from 8 to 13), we may conjecture that it converges to a Gaussian distribution function with decreasing variance, as predicted by probability theory. However, calculations for large values of r (starting from 13 to 15) do not confirm the assumed trend. The distribution function of the sample mean $F_{m_{xN}^*}^*(x)$ becomes explicitly non-Gaussian. At first, when the sample size increases (r changes from 8 to 13), the variance of the sample mean m_{xN}^* decreases. However, going further still, the variance remains practically unchanged. The range of the sample mean stays near the same level (approximately 0.04 V).

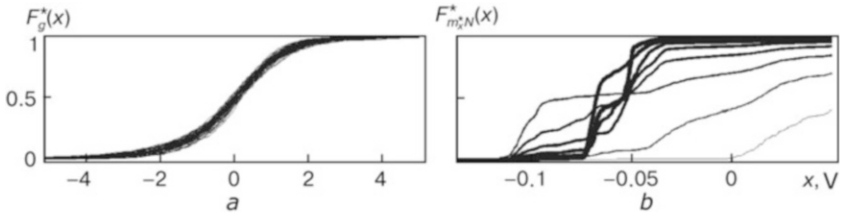


Fig. 9.9 Estimates of the distribution functions of pulsar radiation $F_g^*(x)$ on 64 contiguous observation intervals (**a**) and estimates of the distribution function of the voltage sample mean $F_{m_{xN}^*}^*(x)$ for sample sizes $N = 2^r$, $r = \overline{8, 15}$ (**b**) (line thickness increases with increasing values of the parameter r)

The shape of the curves presented in Fig. 9.9b suggests the convergence of the distribution function $F^*_{m^*_{xN}}(x)$ to a many-valued function $\tilde{F}_{m_x}(x)$, as shown in Fig. 9.7.

Studies of the distribution functions of the sample means of many other processes show that, when the data volume is large, *there is no convergence tendency* of the estimate $F^*_{m^*_{xN}}(x)$ of the distribution function of the sample mean toward any specific distribution law, and in particular toward a Gaussian distribution with variance that tends to zero.

Thus, experimental studies of actual physical processes show that for a *small data volume* one observes *trends of normalization and stabilization of sample means*, but no such tendencies for *large amounts of data*.

The change in the behavior of the sample means can be explained by a violation of statistical stability by the actual processes over long observation intervals. These violations lead to restrictions in the accuracy of measurement of real physical quantities, so we now turn to this issue.

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Chapter 10
Assessing Measurement Accuracy on the Basis of the Theory of Hyper-random Phenomena

Abstract A number of measurement models are considered. The point determinate—hyper-random measurement model is examined. It is shown that the error corresponding to this model is in general of a hyper-random type that cannot be represented by a sum of random and systematic components. For hyper-random estimators, the notions of 'biased estimator,' 'consistent estimator,' 'effective estimator,' and 'sufficient estimator' are introduced. The concept of critical sample size of the hyper-random sample is determined. We describe a measurement technique corresponding to the determinate—hyper-random measurement model. It is shown that, under unpredictable changes in conditions, the classical determinate—random measurement model poorly reflects the actual measurement situation, whereas the determinate—hyper-random measurement model adequately represents it.

10.1 Hyper-random Measurement Models

Modern metrology is based on the following:

- The ideal value of a physical quantity is determinate, unique, and does not change during the measurement time,
- The measurement does not change its characteristics during the measurement,
- Statistical conditions are constant during the measurement time,
- The result of a concrete measurement is unique.

None of these items is reasonable, to put it mildly. All actual physical objects and physical quantities describing them are liable to change over time (except perhaps some universal constants). *Everything is changing: the object of measurement* (*measurand*), *the measure, and the measurement conditions*. No measurement is carried out instantaneously. All measurements are made over some time interval. Therefore the *measurement result is an average value representing the various*

This chapter is based on material from the books (Gorban 2011, 2014, 2016, 2017)

I.I. Gorban, *Randomness and Hyper-randomness*, Mathematical Engineering,
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states of the measured object, the different states of the measure, and different measurement conditions over this interval.

Of course, it is very convenient to represent the measurand by a determinate, unique, and unchanging value, and the measurement result by a random variable. But this primitive model does not reflect the many nuances of a real situation. The first step to improve the model is to assume that, not the estimator Θ^*, but rather the measurand has a random character. Figure 10.1 shows schematically the distribution functions $F_\theta(\theta)$ and $F_{\theta^*}(\theta)$ of the measurand and the estimator corresponding to such a model. This measurement model can naturally be called *random–random*.

In this case, the bias of the estimator is $\varepsilon_0 = m_{\theta^*} - m_\theta$, where m_{θ^*} is the expectation of the estimator Θ^*, m_θ is the expectation of the random measurand Θ, and σ_θ and σ_{θ^*} are the standard deviations of the measurand Θ and its estimator Θ^*, respectively.

More refined than determinate—random and random–random models are *determinate—hyper-random* (Fig. 10.2a) and *random—hyper-random* (Fig. 10.2b) *models*. In the first, the measurand is described by a determinate model, and in the second by a random model. In both cases the estimator is represented by a hyper-random variable.

In the figure, $F_{S\theta^*}(\theta)$ and $F_{I\theta^*}(\theta)$ are the upper and lower bounds of the distribution function of the hyper-random estimator Θ^*; ε_{S0} and ε_{I0} are the biases of the upper and lower bounds of the distribution function of the hyper-random estimator with respect to the measurand (if the measurand is determinate, then $\varepsilon_{S0} = m_{S\theta^*} - \theta$, $\varepsilon_{I0} = m_{I\theta^*} - \theta$, and if it is random, then $\varepsilon_{S0} = m_{S\theta^*} - m_\theta$, $\varepsilon_{I0} = m_{I\theta^*} - m_\theta$); $m_{S\theta^*}$, $m_{I\theta^*}$ are the expectations of the upper and lower bounds of the hyper-random estimator; and $\sigma_{S\theta^*}$, $\sigma_{I\theta^*}$ are the standard deviations of the appropriate bounds of the hyper-random estimator. The uncertainty area of the hyper-random estimator is shaded.

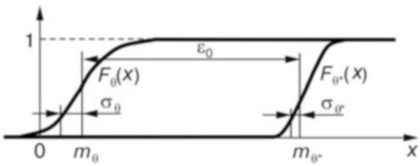

Fig. 10.1 The random—random measurement model

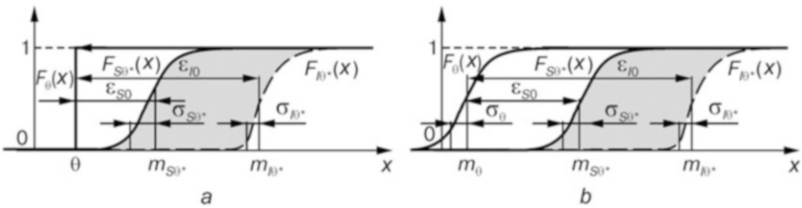

Fig. 10.2 The determinate—hyper-random (**a**) and random—hyper-random (**b**) measurement models

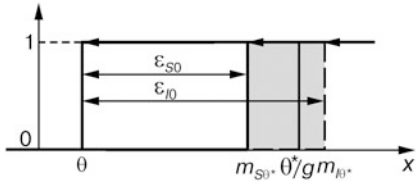

Fig. 10.3 The determinate—interval measurement model

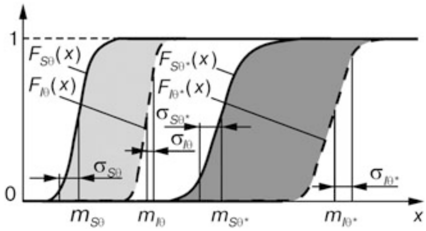

Fig. 10.4 The hyper-random—hyper-random measurement model

A particular case of the determinate—hyper-random measurement model is the *determinate—interval model* (Fig. 10.3). In this case, the measurand is considered as a determinate value, and the estimator as an interval variable. The uncertainty area of the interval variable is shaded in the figure.

The next step toward a realistic model of reality would be a *hyper-random—hyper-random measurement model* (Fig. 10.4), in which the measurand and its estimator are both represented by hyper-random models.

Here we shall consider in detail the *determinate—hyper-random measurement model*. Note that the most common hyper-random—hyper-random measurement model is studied in the monographs (Gorban 2011, 2014).

10.2 Point Hyper-random Estimator of a Determinate Measurand

10.2.1 Formulation of the Problem

Consider the estimation of a *determinate value* θ by observation of the *hyper-random variable* $X = \{X_g | g \in G\}$. A *point hyper-random estimator* Θ^* is a statistic that is a function of a sample $\vec{X} = (X_1, \ldots, X_N)$ of sample size N from a general hyper-random population. The estimator Θ^* can be represented as a set of *random estimators* $\Theta_g^* = \Theta^*/g$ corresponding to different conditions $g \in G$: $\Theta^* = \left\{\Theta_g^* | g \in G\right\}$, where Θ_g^* is a function of the random sample $\vec{X}_g = \vec{X}/g$. The estimate θ^* of the hyper-random estimator Θ^* can be represented by a set of determinate values $\theta_g^* = \theta^*/g$, corresponding to different conditions $g \in G$: $\theta^* = \left\{\theta_g^* |\ g \in G\right\}$.

The *accuracy of a point estimator* can be characterized in different ways depending on the task. In general, it is characterized by the *hyper-random error* $Z = \{Z_g | g \in G\} = \Theta^* - \theta = \{\Theta_g^* - \theta | g \in G\}$. For a *fixed condition* g, the parameter characterizing the accuracy of the *random estimator* Θ_g^* is the expectation of the square of the *random error* $Z_g = \Theta_g^* - \theta$ [mean square error (see 5.1)]:

$$\Delta_{z_g}^2 = \mathrm{E}\left[\left|\Theta_g^* - \theta\right|^2\right].$$

The accuracy of the *estimate* θ_g^* obtained under a *fixed condition* g is characterized by the *determinate error* $z_g = \theta_g^* - \theta$.

Under *varying conditions*, the accuracy is characterized by the interval containing the values $\Delta_{z_g}^2$, $g \in G$. The error can be either positive or negative. Therefore, the *upper limit* of the considered interval is described by the expression

$$\Delta_z^2 = \max\left[\Delta_{Sz}^2, \Delta_{Iz}^2\right],$$

where $\Delta_{Sz}^2 = \mathrm{E}_S\left[|\Theta^* - \theta|^2\right]$, $\Delta_{Iz}^2 = \mathrm{E}_I\left[|\Theta^* - \theta|^2\right]$ are mean square errors calculated using the upper $F_{S\theta^*}(\theta)$ and lower $F_{I\theta^*}(\theta)$ bounds of the estimator distribution function.

The accuracy of the estimator is also characterized by the *bounds of the mean square error*

$$\Delta_{iz}^2 = \inf_{g \in G} \mathrm{E}\left[\left|\Theta_g^* - \theta\right|^2\right], \quad \Delta_{sz}^2 = \sup_{g \in G} \mathrm{E}\left[\left|\Theta_g^* - \theta\right|^2\right]$$

and the roots of these magnitudes Δ_{iz}, Δ_{sz}, which for simplicity we shall call the *error bounds*.

10.2.2 Biased and Unbiased Estimators

The hyper-random estimator Θ^* *of the determinate value* θ is *unbiased* (under all conditions $g \in G$) if for all $g \in G$ the expectation $m_{\theta^*/g} = \mathrm{E}\left[\Theta_g^*\right]$ of the conditional random variable Θ_g^* calculated on a sample set of *any finite size N* equals the estimating value θ: $m_{\theta^*/g} = \theta$. Otherwise the estimator is *biased*. The amount of bias (*systematic error*) under the condition g is described by the expression $\varepsilon_{0/g} = m_{z/g} = m_{\theta^*/g} - \theta$.

Note that, in the case of a bias, the *expectations* $m_{\theta^*/g}$ and the *bias* $\varepsilon_{0/g}$ are usually considered to be *independent of the sample size N* (as for the determinate—random measurement model, see Sect. 5.2.2).

The bounds Δ^2_{Sz}, Δ^2_{Iz} and Δ^2_{iz}, Δ^2_{sz} can be represented as follows:

$$\Delta^2_{Sz} = m^2_{Sz} + \sigma^2_{Sz}, \quad \Delta^2_{Iz} = m^2_{Iz} + \sigma^2_{Iz}, \tag{10.1}$$

$$\Delta^2_{iz} = \inf_{g \in G}\left[m^2_{z/g} + \sigma^2_{z/g}\right], \quad \Delta^2_{sz} = \sup_{g \in G}\left[m^2_{z/g} + \sigma^2_{z/g}\right], \tag{10.2}$$

where $m_{Sz} = m_{S\theta^*} - \theta = \varepsilon_{S0}$, $m_{Iz} = m_{I\theta^*} - \theta = \varepsilon_{I0}$ are expectations of the error bounds that coincide with the estimator biases for the upper and lower bounds of the distribution function;

$$\sigma^2_{Sz} = \mathrm{E}_S\left[(Z - m_{Sz})^2\right] = \sigma^2_{S\theta^*}, \quad \sigma^2_{Iz} = \mathrm{E}_I\left[(Z - m_{Iz})^2\right] = \sigma^2_{I\theta^*}$$

are the variances of the error bounds that coincide with the variances of the estimator bounds; and $\sigma^2_{z/g} = \sigma^2_{\theta^*/g} = \mathrm{E}\left[\left(\Theta^*_g - m_{\theta^*/g}\right)^2\right]$ is the conditional error variance, which coincides with the conditional estimator variance (Fig. 10.5).

The error z *under changing conditions* is described by the inequality

$$\varepsilon_{S0} - k\sigma_{Sz} < z < \varepsilon_{I0} + k\sigma_{Iz} \tag{10.3}$$

(Fig. 10.6), while the interval containing the measurand θ (confidence interval) is described by the inequality

$$\theta^* - \varepsilon_{I0} - k\sigma_{I\theta^*} < \theta < \theta^* - \varepsilon_{S0} + k\sigma_{S\theta^*}$$

(Fig. 10.5), where k is a constant determining the degree of confidence.

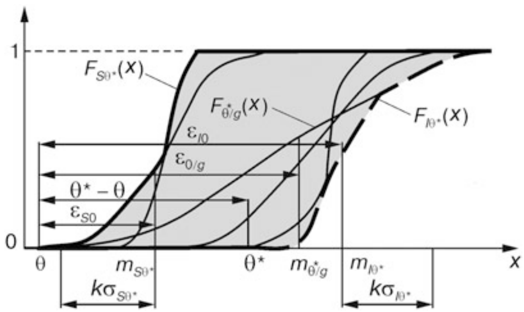

Fig. 10.5 Some conditional distribution functions $F_{\theta^*/g}(x)$ (*thin lines)* for different conditions g, together with the *upper* $F_{S\theta^*}(x)$ (*bold solid line*) and *lower* $F_{I\theta^*}(x)$ (*bold dotted line*) bounds of the distribution function

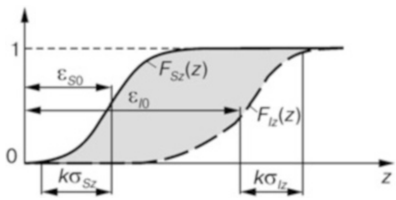

Fig. 10.6 Model of the measurement error

If the distribution functions of the estimators Θ_g^* do not intersect and, with increasing expectations of the estimators $m_{\theta^*/g}$, their variances increase (case (a) distribution in the classification of Sect. 7.2.4) or decrease (case (b) distribution), this interval is determined by the bias bounds $\varepsilon_{i0} = \inf_{g \in G} \varepsilon_{0/g}$, $\varepsilon_{s0} = \sup_{g \in G} \varepsilon_{0/g}$ and the standard deviation bounds $\sigma_{i\theta^*}$, $\sigma_{s\theta^*}$.

For a case (a) distribution

$$\theta \in [\theta^* - \varepsilon_{s0} - k\sigma_{s\theta^*}, \ \ \theta^* - \varepsilon_{i0} + k\sigma_{i\theta^*}],$$

and for a case (b) distribution

$$\theta \in [\theta^* - \varepsilon_{s0} - k\sigma_{i\theta^*}, \ \ \theta^* - \varepsilon_{i0} + k\sigma_{s\theta^*}].$$

10.2.3 Consistent and Inconsistent Estimators

The hyper-random estimator Θ^* *of a determinate parameter* θ is *consistent* if, *under all conditions* $g \in G$, its random estimators Θ_g^* converge in probability to this value:

$$\lim_{N \to \infty} P\left\{\left|\Theta_g^* - \theta\right| > \varepsilon\right\} = 0,$$

where $\varepsilon > 0$ is any small number. A hyper-random estimator that does not satisfy this condition is said to be *inconsistent* .

Note that a consistent hyper-random estimator satisfies very stringent requirements. Under all conditions, such an estimator has the convergence property, and in addition, under all conditions, it converges to the *same value* (a number describing the measurand).

According to the results of experimental studies, because of statistically unpredictable changes in conditions, real estimators do not have the convergence property. With increasing sample size, they *do not tend to a unique number, but to a set of numbers*. Therefore, actual estimators are well described by *inconsistent hyper-random estimators* characterized by a spectrum of limit points (see Sect. 9.3).

10.2.4 Measurement Error

In the *determinate—random measurement model*, the error is a *random variable* described by *systematic and random components*, and characterized by two parameters: the *bias* ε_0 and the *standard deviation of the estimator* σ_{θ^*}. In the *determinate—hyper-random measurement model*, the error is a *hyper-random variable*. It

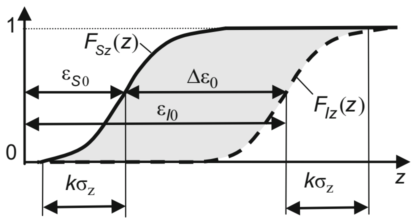

Fig. 10.7 Model of the measurement error in the particular case when the bounds of the distribution function differ only in expectation

has an *uncertainty area* and is described by the inequality (10.3), in which there are four parameters $\varepsilon_{S0}, \varepsilon_{I0}, \sigma_{Sz}, \sigma_{Iz}$ defining the location and size of the uncertainty area on the error axis (Fig. 10.6).

In the *particular case* when the form of the bounds of the distribution function is identical (then $\sigma_{Sz} = \sigma_{Iz} = \sigma_z$) and the expectations of the bounds are different, the error Z can be represented by an additive model $Z = \mathrm{E}_0 + V$ with two components: the *uncertainty component* E_0, characterizing the location and extent of the uncertainty area, and the *random component* V, with zero expectation, characterizing the shape of this area.

The uncertain (statistically unpredictable) error component E_0 can be characterized by an *interval value* $\Delta\varepsilon_0 = [\varepsilon_{S0}, \varepsilon_{I0}]$ and the random error component V by the standard deviation of the error σ_z. The uncertain component E_0 can itself be divided into two components (Fig. 10.7): the *systematic error* $\varepsilon_0 = \varepsilon_{S0} = m_{Sz}$, characterizing the beginning of the uncertainty area, and the *interval error* $[0, \Delta\varepsilon_0]$, characterizing the length of the uncertainty area, where $\Delta\varepsilon_0 = \varepsilon_{I0} - \varepsilon_{S0}$. So the total error has three components: the *systematic* ε_{S0}, *random* V, and *interval* $[0, \Delta\varepsilon_0]$ components.

Note that, even in the case when the standard deviations of the bounds are different ($\sigma_{Sz} \neq \sigma_{Iz}$), it makes sense to assign the *systematic error*, determining it as $\varepsilon_0 = \varepsilon_{S0} = m_{Sz}$.

Thus, in the determinate—random and determinate—hyper-random measurement models, the errors are of different types and are described by different parameters. In the first case, the error has a *random nature* and contains *systematic* and *random* components. In the second case, it has a *hyper-random nature* and in general *cannot be decomposed into components*. In particular cases, the *error can be represented by three components*: *systematic*, *random*, and *interval*.

10.2.5 Efficient and Sufficient Estimators

The hyper-random estimator Θ_e^* *of the determinate parameter* θ is *efficient* (*under all conditions* $g \in G$) if for all $g \in G$ the mean square error $\Delta^2_{z_{eg}}$ of the random estimator $\Theta^*_{eg} = \Theta^*_e/g$ is smaller than the mean square error $\Delta^2_{z_{ig}}$ of any other estimator $\Theta^*_{ig} = \Theta^*_i/g$ of the same sample size. In other words, for all $g \in G$,

$$\mathrm{E}\left[\left(\Theta^*_{eg} - \theta\right)^2\right] < \mathrm{E}\left[\left(\Theta^*_{ig} - \theta\right)^2\right], \quad i = 1, 2, \ldots$$

The magnitude $\mathrm{E}\left[\left(\Theta^*_g - \theta\right)^2\right]$ is not in general the variance of an estimator. It is the variance $\sigma^2_{\theta^*/g}$ for unbiased estimators. For such estimators, the efficiency condition can be written as $\sigma^2_{\theta^*_e/g} < \sigma^2_{\theta^*_i/g}$, $i = 1, 2, \ldots$ for all $g \in G$.

The hyper-random estimator Θ^* *of the determinate parameter* θ is *sufficient* (*under all conditions* $g \in G$) if for all $g \in G$ the N-dimensional conditional probability density $f_{\vec{x}/\theta^*,g}(x_1, \ldots, x_N)$ of the sample of the hyper-random variable X is independent of the parameter θ, i.e., the estimator contains all useful information about θ contained in the sample.

10.3 Statistical Measurement Under Unpredictably Changing Conditions

10.3.1 Initial Assumptions

Consider the measurement of a measurand under *unpredictably changing conditions*. As noted above, any measurement method is based on a number of assumptions. We shall suppose that the measurand θ is a *scalar determinate* that does *not change during the measurement*, and that the measurement results are of *hyper-random nature* and adequately described by a *hyper-random sample* $(X_1, \ldots, X_N)$ of the hyper-random variable X. During the measurement process, the statistical conditions *change unpredictably*. They *change slowly*, so the observation interval can be divided into G fragments of equal duration corresponding to nearly constant statistical conditions. The sample elements are taken with uniform step. The number of elements corresponding to a fragment is N_s. Under fixed statistical conditions g $(g = \overline{1, G})$ the elements of a random sample $\left(X_{1g}, \ldots, X_{N_s g}\right)$ are *uncorrelated* and have the *same unknown* distribution law $F_g(x)$ and the *same unknown* variance $D_{x/g}$.

The *determinate sample* $(x_1, \ldots, x_N)$ is a *realization* of the *hyper-random sample* $(X_1, \ldots, X_N)$, and the *determinate sample* $\left(x_{1g}, \ldots, x_{N_s g}\right)$ under conditions g is a *realization* of the random sample $\left(X_{1g}, \ldots, X_{N_s g}\right)$.

We propose that, over the interval $m_{x/g} \pm 3\sigma_{x/g}$, the distribution functions $F_g(x)$ $(g = \overline{1, G})$ do *not intersect*, where $m_{x/g}$ and $\sigma_{x/g} = \sqrt{D_{x/g}}$ are respectively the mean and the standard deviation of the random sample $\left(X_{1g}, \ldots, X_{N_s g}\right)$. Note that this assumption is not very important. It just makes it easier to find the bounds of the distribution function $F_{Sx}(x)$, $F_{Ix}(x)$ of the hyper-random quantity X by selecting the conditional distribution functions with the minimum and maximum means m_{ix}, m_{sx}.

The point estimate θ^* of the measurand is the average of the estimates of the expectations of the bounds m_{Sx}^*, m_{Ix}^* formed on the basis of the sample $(x_1, \ldots, x_N)$:

$$\theta^* = \left(m_{Sx}^* + m_{Ix}^*\right)/2. \tag{10.4}$$

10.3.2 Statistical Measurement

Taking into account the assumptions, the measurement of a measurand θ can be represented by the following steps:

(1) Measuring the measurand N times and forming as a result the sample $(x_1, \ldots, x_N)$;
(2) Using the technique described in Sect. 6.1.3, calculating the number of samples N_s corresponding to the interval of statistical stability with respect to the average τ_{sm};
(3) Dividing the sample into fragments containing N_s items;
(4) Calculating the expectation estimate $m_{x/g}^*$ for each fragment g ($g = \overline{1, G}$);
(5) Determining the estimates m_{Sx}^*, m_{Ix}^* of the bound expectations $\left(m_{Sx}^* = m_{ix}^* = \inf\limits_g m_{x/g}^*,\ m_{Ix}^* = m_{sx}^* = \sup\limits_g m_{x/g}^*\right)$ and the numbers of fragments g_S, g_I corresponding to the expectations of the upper and lower bounds;
(6) Calculating the estimates of the standard deviations of the upper and lower bounds σ_{Sx}^*, σ_{Ix}^* for fragments g_S, g_I;
(7) Calculating the estimate θ^* by (10.4);
(8) Calculating the bounds of the confidence interval θ_i, θ_s, taking into consideration the fact that the systematic error $\varepsilon_0 = m_{S\theta^*} - \theta$ and the standard deviation of the distribution bounds are reduced $\sqrt{N_s}$ times due to the averaging of the data, i.e.,

$$\begin{aligned} \theta_i &= \theta^* - \varepsilon_0 - \left(m_{Ix}^* - m_{Sx}^*\right) - k\sigma_{Ix}^*/\sqrt{N_s}, \\ \theta_s &= \theta^* - \varepsilon_0 + k\sigma_{Sx}^*/\sqrt{N_s}; \end{aligned} \tag{10.5}$$

(9) Calculating the bounds of the measurement error interval

$$\begin{aligned} z_i &= \varepsilon_0 - k\sigma_{Sx}^*/\sqrt{N_s}, \\ z_s &= \varepsilon_0 + \left(m_{Ix}^* - m_{Sx}^*\right) + k\sigma_{Ix}^*/\sqrt{N_s}. \end{aligned}$$

Note that, in the case of a correlation between the items, the value N_s should be changed on T_s/τ_c samples, where T_s is the duration of the fragment containing N_s samples and τ_c is the correlation interval.

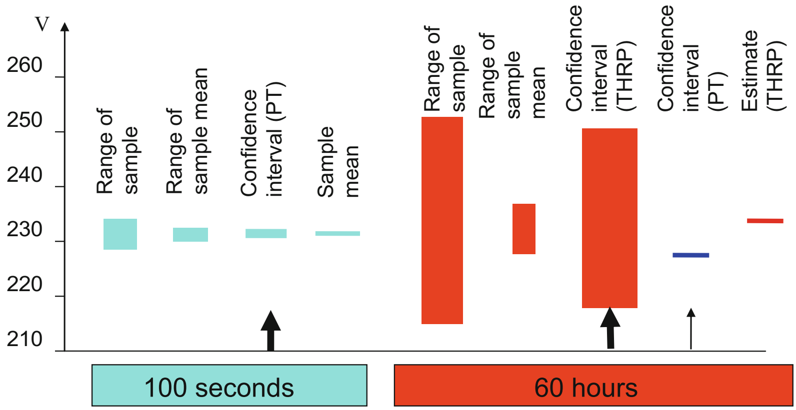

Fig. 10.8 Calculation results for the parameters characterizing the city mains voltage over 100-s and 60-h observation intervals, using calculation techniques based on probability theory (PT) and the theory of hyper-random phenomena (THRP)

As an example, Fig. 10.8 shows the calculation results for the parameters characterizing the mains voltage at the end of 100-s and 60-h observation intervals. The left side of the figure, corresponding to the 100-s observation interval, repeats Fig. 5.5 which was obtained using the classic determinate—random measurement model based on probability theory (see Sect. 5.3). The right side of Fig. 10.8, corresponding to the 60-h observation interval, presents the parameters obtained using the determinate—hyper-random measurement model based on the theory of hyper-random phenomena (except the parameter marked with a thin arrow).

For the 60-h observation interval, the sample range and the range of the sample mean are obtained from the data of Figs. 1.8a and 9.8b. The confidence interval (THRP) marked with a bold arrow and the estimate (THRP) are calculated using the technique described in the present section. The confidence interval (PT) marked by a thin arrow is calculated using the classic technique described in Sect. 5.3.

The results shown in the figure for the 100-s and 60-h observation intervals differ considerably. The parameters on the left side of the figure reflect the state of the electrical supply network under the *specific statistical conditions* that occurred during the relevant 100-s observation interval. The parameters on the right side (except for the one marked by a thin arrow) represent the state of the network for the *varying set of statistical conditions* that succeeded one another *unpredictably* during the relevant 60-h observation period. The parameter marked by a thin arrow characterizes the state of the network for the *set of different but very specific statistical conditions* that succeeded each other over the same 60-h period of observation.

For the *100-s observation interval, the most informative parameter is the confidence interval calculated using the classic technique of probability theory,* and for the *60-h interval, it is the confidence interval calculated using the technique based on the theory of hyper-random phenomena* (in Fig. 10.8 these parameters are marked by two bold arrows).

For the 60-h observation interval, the confidence interval, with width 50 mV and average value 229.4 V (calculated in accordance with probability theory and indicated in the figure by a thin arrow) is *not informative*, because it takes into consideration the *concrete sequence* of changes in the conditions which, in the next 60-h observation interval, is likely to be something quite different. The confidence interval, with width 33 V and average 233.5 V (calculated using the theory of hyper-random phenomena and marked by a thick arrow) contains useful practical information about the average dynamics of the voltage changes in the power supply.

The loss of useful information in the first case and the fact that it is kept in the second arise because, when there are violations of statistical stability, the classic determinate—random measurement model reflects the real situation with considerable distortion, while the determinate—hyper-random measurement model is able to represent it adequately. It follows from the above example that ignoring the violation of statistical stability can lead to absurd results, and in particular, to an unjustifiable overstatement of measurement accuracy estimators by factors of hundreds or more. The conclusion is obvious:

> *When statistical stability is violated, the determinate—random measurement model and the measuring techniques based on it cannot be used.* In this case, *other models and measurement techniques must be used,* and in particular the *determinate—hyper-random measurement model and techniques based on it, which take into consideration the violations of statistical stability.*

Note that the technique described in the present section is one of a number of modifications based on the determinate—hyper-random measurement model. On the basis of this model, other modifications can be developed, taking into account different specific features of the concrete measurement conditions.

10.4 Critical Size of the Hyper-random Sample

With increasing size N of the hyper-random sample $(X_1, \ldots, X_N)$, the error in the hyper-random estimator Θ^* of the determinate measurand θ often decreases. However, unlike a random estimator, it never tends to zero when $N \to \infty$. This feature can be illustrated by an example similar to the one given in Sect. 5.4.

Suppose the measurand θ is determinate and the estimator Θ^* is a hyper-random variable. The elements of the hyper-random sample $(X_1, \ldots, X_N)$ are independent. The statistical conditions change slowly and this allows us to divide the observation interval into G fragments of identical length corresponding to nearly constant statistical conditions. The elements of the sample are taken with uniform step. In any fragment, the number of samples is N_s.

The distribution law of the random elements $X_{1g}, \ldots, X_{N_s g}$ under the fixed condition g is fixed. Under different conditions g, the distribution laws of the elements are different, but all of them have the same variance D_x and differ from each other only in the expectation value. The estimate θ^* is described by (10.4).

Then, taking into consideration (10.2), the error bounds can be written as

$$\Delta_{iz} = \sqrt{\varepsilon_i^2 + D_x/N_s},$$
$$\Delta_{sz} = \sqrt{\varepsilon_s^2 + D_x/N_s},$$

where $\varepsilon_i^2 = \inf\limits_{g \in G} \left[\varepsilon_{0/g}^2\right]$ and $\varepsilon_s^2 = \sup\limits_{g \in G} \left[\varepsilon_{0/g}^2\right]$ are the lower and upper bounds of the square of the bias.

The dependence of the error bounds Δ_{iz}, Δ_{sz} on the defining parameters is shown in Fig. 10.9. The dotted lines represent the lower error bounds and the solid ones the upper error bounds. Thicker lines correspond to large values of the variance D_x. It is clear from the figure that, with increasing sample size N_s, the upper bound of the error Δ_{sz} tends to $\varepsilon_s = \varepsilon_0 + \Delta\varepsilon_0$ (ε_0 is the systematic error or bias and $\Delta\varepsilon_0$ is the length of the uncertainty area). Therefore, even if we make the unlikely assumption that the value N_s tends to infinity, the determining upper bound of the error Δ_{sz} will never be less than the value $\Delta\varepsilon_0 \neq 0$. When the bias ε_0 is negligible, the magnitude $\Delta_{sz} \to \Delta\varepsilon_0 \neq 0$.

So with the determinate—hyper-random measurement model, we can explain the *inability in practice to achieve infinitely high accuracy*, even with an unlimited amount of data.

As in the case of a random sample (see Sect. 5.4), there is no sense in increasing the size of the hyper-random sample unless it leads to a noticeable increase in measurement accuracy. For a hyper-random estimator there is a *critical sample size* N_{0s} that can be defined as the sample size N_s at which the upper bound of the square of the bias ε_s^2 is much greater than the estimator variance D_x/N_s. Taking into account

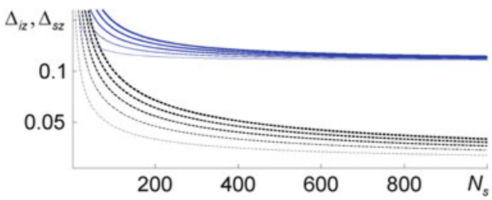

Fig. 10.9 Dependence of the error bounds Δ_{iz}, Δ_{sz} on the sample size N_s and the variance D_x for the determinate—hyper-random measurement model: $\varepsilon_0 = \varepsilon_{S0} = 0.01$, $\Delta\varepsilon_0 = 0.1$, and $D_x = 0.2\,;0.4\,;0.6\,;0.8\,;1$

the standard recommendations (GOST 2006) (see Sect. 5.4), it follows from this definition that the critical volume is

$$N_{0s} \simeq 64 D_x / \varepsilon_s^2.$$

To summarize this chapter we should pay attention to the following:

(1) *Real estimators of physical quantities are statistically unstable* (*inconsistent*).
(2) The *limited measurement accuracy of physical quantities is explained by the statistical instability* (*inconsistency*) *of real estimators*.
(3) The classic determinate—random measurement model *ignores* the *statistical instability* (*inconsistency*) *of real estimators*, while the determinate—hyper-random measurement models *take it into consideration*. Therefore the *hyper-random measurement models describe the measurement procedure more adequately than the classic determinate—random model*.
(4) *In general, the hyper-random error cannot be decomposed into components*, and in particular, into systematic and random components.
(5) *In a particular case* (coincidence of the form of the bounds of the estimator distribution function), the *hyper-random error can be represented as the sum of three components: systematic, random, and interval*. With increasing sample size, the random component is reduced, but the *systematic and uncertain components do not change*.

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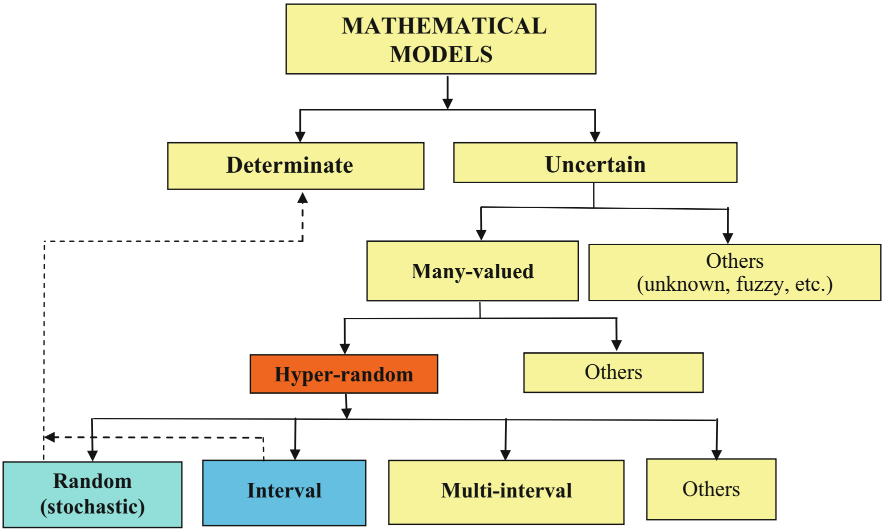
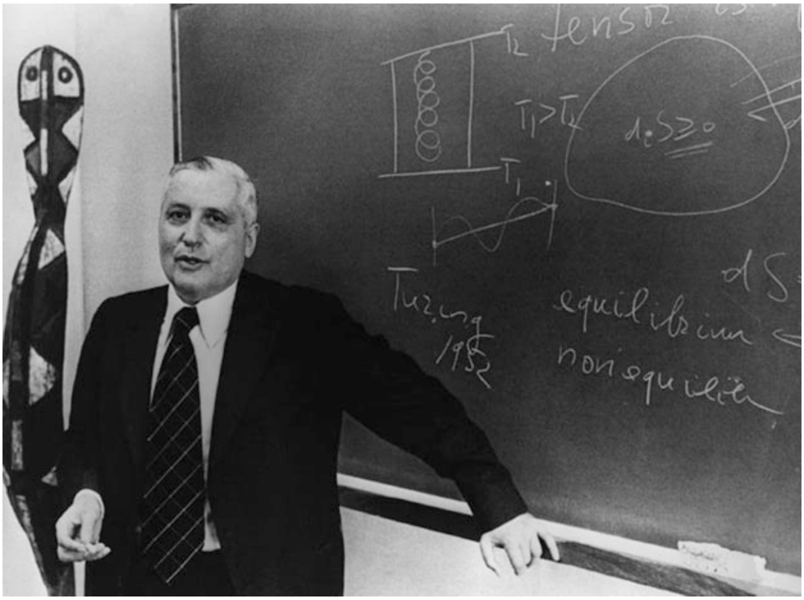
Part V
The Problem of Adequate Description of the World



Classification of mathematical models



Viscount Ilya Romanovich Prigogine, Belgian physical chemist, Nobel laureate (1917–2003) (Source: http://www.thefamouspeople.com/profiles/images/ilya-prigogine-4.jpg)

I.R. Prigogine:

> The discovery of nonequilibrium structures, as is well known, was accompanied by a revolution in the study of trajectories. It was found that the trajectories of many systems were unstable. This means that we can make a reliable prediction only for short time intervals. The brevity of these same intervals (also called the temporal horizon or Lyapunov exponent) means that after a certain period of time the trajectory inevitably eludes us, i.e., we lose information about it. This, incidentally, is another reminder that our knowledge is only a small window on the universe, and that because of the instability of the world, we should refuse even the dream of exhaustive knowledge. Looking through the window, we can certainly extrapolate the available knowledge beyond our vision and guess what could be the mechanism that controls the dynamics of the universe. However, we should not forget that, although in principle we can know the initial conditions at an infinite number of points, the future nevertheless remains fundamentally unpredictable. (Prigogine 1991).

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Chapter 11
Determinism, Uncertainty, Randomness, and Hyper-randomness

Abstract We investigate different ways to produce an adequate description of the real physical world. Here we discuss the reasons for using the random and hyper-random models. We present the classification of uncertainties. We also discuss an approach leading to a uniform description of the various mathematical models (determinate, random, interval, and hyper-random) by means of the distribution function. A classification of these models is proposed. We examine the causes and mechanisms at the origin of uncertainty, marking out reasonable areas for practical application of random and hyper-random models.

11.1 Conceptual Views of the Structure of the World from the Standpoints of Determinism and Uncertainty

Scientists have been debating about the underlying principles of the universe for centuries. Some suppose that everything in the world is intrinsically determinate, some are sure that it is random. And there are those who hold still other views.

11.1.1 Laplace's Determinism

The supporters of *determinism* believe that any physical system can be adequately described by differential equations.

In the case of full knowledge of the initial data (laws, relations, and initial conditions), the *behavior of the given physical system is completely predictable*. This is the essence of *P. S. Laplace's determinism* (Laplace 1982), the bases of which were laid down by *R. Descartes*, *T. Hobbes*, *B. de Spinoza*, *I. Newton*, *G. W. Leibniz*, *J. Bernoulli*, *P. T. baron d'Holbach*, and other scientists and philosophers of the sixteenth to eighteenth centuries.

This chapter is based on material from the books (Gorban 2011, 2014, 2016, 2017).

I.I. Gorban, *Randomness and Hyper-randomness*, Mathematical Engineering,
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G. W. Leibniz, for example, wrote: “The present always hides the future in its bosom, and any given state is naturally explainable solely in terms of the one immediately preceding it”.

In his famous book “The art of conjecturing”, J. Bernoulli (1713) concludes with the following words (Bernoulli 1986): “This, finally, causes the apparently singular corollary: if observations of all events be continued for the entire infinity (with probability finally turning into complete certitude), it will be noticed that everything in the world is governed by precise ratios and a constant law of changes, so that even in things to the highest degree casual and fortuitous we would be compelled to admit as though some necessity and, I may say, fate.”

According to P. S. Laplace everything complies with determinate laws, and the current state of the world is completely determined by its previous states, i.e., it is predetermined. This view was supported by many physicists, including *A. Einstein*, who once remarked that “God does not play dice with the Universe”.

11.1.2 Stochastic Approach

The supporters of a *stochastic* (*probabilistic*) *world order*, while not usually rejecting the determinate approach, focus their attention on the phenomenon of statistical stability, providing additional opportunities for predicting the relative frequency of mass events and averaged quantities.

In the twentieth century, thanks to the carefully argued classical works of *R. A. Fisher*, *H. Cramer*, *C.R. Rao*, and others in probability theory, regarding the accuracy of prediction and measurement, it was concluded that the *larger the sample number, the greater the accuracy*. Theoretically, it was proven that (Tikhonov and Kharisov 1991; Van Trees 2004; Gorban 2003):

> *In the case of estimator convergence* (*consistency*) *and unlimited increase in sample size, there are no fundamental limitations on the accuracy of prediction and measurement.*

Unfortunately, this optimistic conclusion is not confirmed in practice. Engineers and physicists know that in many cases, by increasing the sample size, it is possible to improve the accuracy of measurement and prediction, but not indefinitely: sooner or later, other factors will limit the possibility of further increasing the accuracy.

Supporters of the mathematical theory of interval analysis and the physico-mathematical theory of hyper-random phenomena explain this effect by the presence of *uncertainty*.

11.1.3 Interval Approach

There are two versions of the *interval approach*. One was developed in the framework of probability theory for the problem of evaluating statistical intervals. It was initiated by *P. S. Laplace* and later developed (1927) by *E. B. Wilson* (Hunsaker and Lane 1973). The conclusions regarding the potential accuracy of measurement and prediction following from this work are typical of the stochastic approach:

> *There are no fundamental limitations on accuracy when the amount of data is unlimited.*

The other direction of the interval approach is purely mathematical. The phenomenon of statistical stability plays no role here. It was first formulated in the 1960s, with the need to account for rounding errors in digital computers. Its development led to the mathematical theory of interval analysis. The subject of this theory is the interval characterized by *lower and upper bounds*.

Development of the interval analysis theory is associated with the names of *R. E. Moor* (Moor 1966), *Yu. I. Shokin* (Shokin 1981), and many other scientists. Over the past half century, many interesting results have been obtained in the field of interval mathematics (see, for instance, Shary 2010; Dobronets 2004). As compared with the stochastic approach, a different conclusion is drawn regarding the potential accuracy of measurement and prediction:

> *Even with an unlimited volume of data, the accuracy is theoretically limited.*

Note that this conclusion is not based on the physics of real phenomena. It follows from the initial mathematical ideas.

11.1.4 Hyper-random Approach

On the basis of experimental studies that demonstrate the absence of convergence of relative frequencies and sample means, the supporters of the *physico-mathematical theory of hyper-random phenomena* uphold the *hyper-random concept of world building*.

If we admit the non-ideal nature of the phenomenon of statistical stability and accept the physical hypothesis that hyper-random models provide an adequate description of real physical phenomena, we draw a different conclusion from the one suggested by probability theory. According to the theory of hyper-random phenomena:

The accuracy of measurement and prediction are limited in principle, even when the amount of data available is unlimited.

Thus, on the key question of the potential accuracy of measurement and prediction, the conclusions of *the mathematical theory of interval analysis and the physico-mathematical theory of hyper-random phenomena actually coincide*.

11.2 Fundamental Questions

Clearly, the real world is ruled by both deterministic and non-deterministic laws. It is interesting to consider the interpretation of this fact by *Max Planck* and *Erwin Schrodinger* (Schrodinger 1944). Their positions are close. Deterministic laws were considered by M. Planck as a *dynamic type of law*, and by E. Schrodinger as an *order*, while non-deterministic laws were considered by the former as a *statistical type of law*, and by the latter as a *disorder*.

E. Schrodinger made the following observation[1] (Schrodinger 1944): "It appears that there are two different 'mechanisms' by which orderly events can be produced: the 'statistical mechanism' which produces 'order from disorder' and one producing 'order from order'. For an unbiased mind, the second principle seems simpler and more likely. Without doubt, this is so. That is why physicists were so proud to have found the first principle ('order from disorder'), which is actually followed in Nature, and which explains such an enormous variety of natural phenomena, and in particular, their irreversibility."

The above-mentioned 'mechanisms' are shown schematically by the two solid white arrows in Fig. 11.1. We may add the obvious 'mechanism' that produces '*disorder from disorder*' (dotted white arrow), *but can order cause disorder* (red solid arrow)?

This is one of a long list of fundamental questions relating to *uncertainty*. It also includes the following: What is the role of uncertainty in the real world? What is the nature of uncertainty. What kinds of mathematical models provide an adequate description of reality?

On the most fundamental level, we must ask why the random models used to describe real physical phenomena are so widespread and why the interest in hyper-random models is growing. Is this so because there are many parameters and characteristics of real natural phenomena that we do not know and we have no better way to clothe our ignorance than the random and hyper-random models? Or

[1]Translated from the Russian edition.

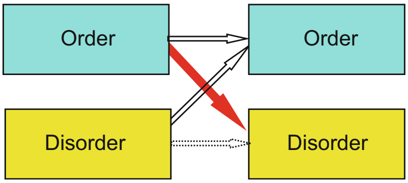

Fig. 11.1 Schematic representation of the 'mechanisms' producing order and disorder

is the world really based on the random and hyper-random principles, so that it is only by using the random and hyper-random models that we can in fact give an adequate description of reality?

Certainly, these questions are very profound. It would hardly be possible to give exhaustive answers. But let us nevertheless try to understand them. We thus start with the terminology, classification of uncertainties, and classification of common mathematical models.

11.3 Parameters of Physical Systems

The visual environment is constantly changing, and this is manifested by the way its properties and the properties of the entities making it up—different physical systems—are themselves changing. The state of these entities is characterized by *physical* and *nonphysical quantities*. Physical quantities are *measurable values*, in contrast to nonphysical ones. The concept of a physical quantity proposed by a standard is defined in Sect. 1.6. In this definition we find the dual physical and mathematical natures of the concept of physical quantity: on the one hand this concept is interpreted as a physical property, and on the other, it is characterized quantitatively as a mathematical object.

The physical quantity can be described by various mathematical models which, in contrast to the physical quantity, we shall call the magnitudes or *parameters* (or again the *state coordinates*), supplemented where necessary by explanatory words characterizing their specificity. The parameters can be *scalar* (one-dimensional) or *vectorial* (multidimensional).

We shall distinguish *determinate* and *uncertain* (*indeterminate*) *parameters*. These are usually considered as a function of time. A determinate parameter takes a concrete value at any fixed time. In the scalar case, this value is described by a number (natural, real, or complex) and in the vector case, by a vector (a set of natural, real, or complex numbers). A parameter described by an infinite vector will be called a *characteristic*. A determinate characteristic is represented by a single-valued determinate function. An uncertain parameter, unlike a determinate one, does not assume a concrete value at a fixed time, and an uncertain characteristic is not described by a concrete determinate function. In particular cases, an indeterminate characteristic may be a stochastic process, each realization of which is a determinate process.

Note that the terms 'determinate' and 'indeterminate' are used here, rather than 'deterministic' and 'nondeterministic'. We use these terms because the word 'deterministic' suggests that there is an inevitable consequence of antecedent sufficient causes, and the word 'nondeterministic' suggests that the uncertainty comes from the possibility for some events to occur uncaused, or at least partially uncaused, while the words 'determinate' and 'indeterminate' remain neutral about why things may be different in different runs of the process.

Note that the division of the parameters and characteristics of real systems into determinate and uncertain ones is not quite correct. There is usually only one realization. When it is not many-valued, there is no way to assess whether a parameter or characteristic is determinate or uncertain.

Another thing here is the models. When we build a mathematical model, we can assume that some of its parameters (or characteristics) are determinate, while others are uncertain. *Models* that do not contain uncertain parameters will be called *determinate*, while those that do contain such parameters will be called *uncertain* or *indeterminate*. *Systems* adequately described solely by determinate models will be referred to as *determinate, while those* represented by both indeterminate and determinate models, or just indeterminate models, will be called uncertain or indeterminate. In the following, we shall say that:

> *An adequate description of the given entity implies full compliance of the model with regard to the determinate and indeterminate properties of the entity.*

Therefore, when studying such properties of physical quantities, we shall make no distinction between *real physical quantities* and their *adequate models*.

11.4 Classification of Uncertainty

The notion of uncertainty is not as obvious as it seems at first glance. Indeed, it is not always possible to formulate precisely what is meant here. There are many similar concepts that are very close in meaning. They include, for example, *ambiguity, multiple values, randomness, inaccuracy, inadequacy, multiple meaning, chaotic states, fuzziness, etc*. Some of these concepts are vague. Others are formalized, but they are based on different model representations, so it is difficult to establish the relationships between them, e.g., between the concepts of *randomness* and *chaos*.

In this connection, it is no simple matter to give a systematic overview of the various concepts of uncertainty. However, one of the most successful among the known classifications is the one presented in the monograph (Bocharnikov 2001) (Fig. 11.2). Although many important concepts, in particular, the concepts of interval variable, multi-interval variable, hyper-random phenomena, etc., are absent from this review, it is made quite clear that randomness, many-valuedness, and uncertainty are not identical concepts. *Randomness* is a special case of

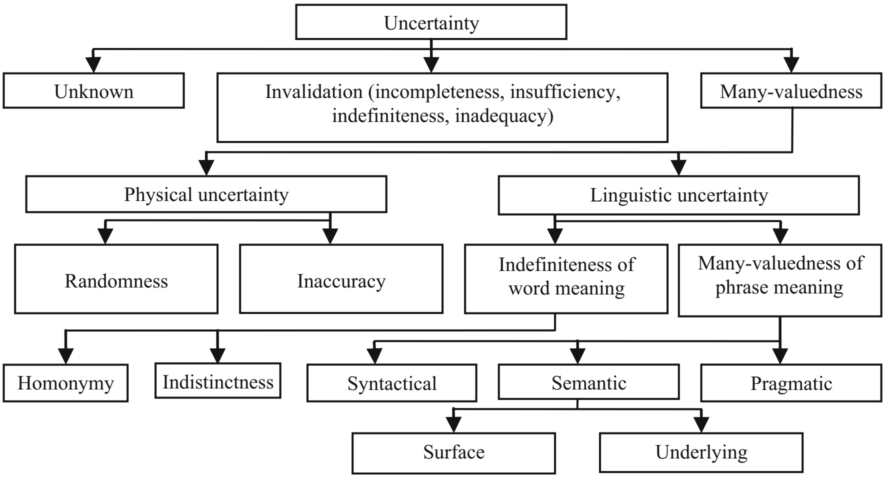

Fig. 11.2 Classification of uncertainty according to V. P. Bocharnikov (2001)

many-valuedness (*multiple meaning*) and the latter is a special case of uncertainty. The concepts of randomness, many-valuedness, and uncertainty may refer to different *mathematical objects* (*phenomena*), namely *events*, *variables*, and *functions* (*processes*).

11.5 Uniform Description of Models

The distribution function provides a uniform description of different types of models, in particular the determinate, random, interval, and hyper-random models. Recall that the important characteristics of the *many-valued distribution function* $\tilde{F}(x)$ of the hyper-random variable are its *upper* and *lower bounds*, defined analytically as follows (Fig. 9.4):

$$F_{Sx}(x) = \sup_{g \in G} F_{x/g}(x), \quad F_{Ix}(x) = \inf_{g \in G} F_{x/g}(x).$$

Between the bounds of the distribution function there is an *uncertainty area* formed by a number of curves $F_{x/g}(x)$, $g \in G$. If these curves densely fill the space between the bounds of the distribution function, the uncertainty area is *continuous* (shaded band in Fig. 9.4c), otherwise it is *discontinuous*.

A *degenerate case of a hyper-random variable is the random variable*. For the random variable X, the bounds of the distribution function coincide with its distribution function $F(x)$: $\tilde{F}(x) = F_S(x) = F_I(x) = F(x)$, and the uncertainty area shrinks to a line (Fig. 9.4b).

The determinate value (*constant*) a can be regarded approximately as a random variable X, whose distribution function $F(x)$ has the form of a *unit step* at the point a: $F(x) = \text{sign}[x - a]$ (Fig. 9.4a), where $\text{sign}[x - a]$ is described by (2.6).

The interval variable determined by the interval $[a_i, \ a_s]$ can be regarded as a hyper-random variable X, whose upper bound is described by a unit step at the point a_i: $F_{Sx}(x) = \text{sign}[x - a_i]$ and whose lower bound is described by a unit step at the point a_s: $F_{Ix}(x) = \text{sign}[x - a_s]$ (Fig. 9.4d). The uncertainty area of the interval variable is continuous.

If $a_i \to -\infty$ and $a_s \to \infty$, then the step associated with the upper bound of the distribution function is situated at minus infinity, and the step associated with the lower bound at plus infinity. The interval variable with these distribution bounds can be regarded as a *variable that is completely undefined in this interval*.

The multi-interval variable (Shary 2010) consisting of a set of disjoint intervals has a *discontinuous uncertainty area*. Obviously, the determinate magnitude (constant) (Fig. 9.4a) involves no uncertainty. It is characterized by a concrete value a. The interval variable (Fig. 9.4d) describes the uncertainty characterized by the interval bounds. The random variable (Fig. 9.4b) describes another type of uncertainty, which can be described approximately by the expectation (or mode) and the slope of the curve of the distribution function (which is characterized by the variance or standard deviation).

It should be noted that, *in any interval model and in any random model, determinism is not completely eliminated.*

It is present in them, but only on a higher level, i.e., it is manifested in the form of determinate parameters (interval bounds, expectation, variance, etc.) and determinate characteristics (distribution function, distribution density, etc.). Clearly, in a random pattern, determinism plays a more significant role than in an interval model.

The hyper-random variable takes into account two types of uncertainty, one of which is typical for the interval variable, while the other is typical for the random variable.

In the *hyper-random model, determinism is present* in the form of determinate parameters (moments of bounds, bounds of moments, etc.) and determinate characteristics (conditional distribution functions, conditional probability densities, bounds of the distribution function, etc.), too. However, the role of determinism in this model is *essentially less important than in a random model.*

If we rank the various models according to the role that determinism plays in them, they can be arranged as follows: determinate, random, hyper-random, multiinterval, and interval.

It follows from this brief digression that *uncertainty is tending to squeeze determinism out*. However, this does not mean that it will be fully displaced in the future. Whatever uncertainty model is used, *it is impossible to avoid the determinate variables and functions*. Ultimately, all models are described by determinate means.

There is another important point here. On the one hand, there is no gulf between determinate and uncertain phenomena, as one might imagine. As can be seen from Fig. 9.4:

Determinate, random, interval, and multi-interval variables, as well as determinate and random events, can be regarded as degenerate cases of the hyper-random variable. It is obvious that random functions can be regarded as degenerate hyper-random functions, and determinate functions as degenerate random or degenerate hyper-random functions.

On the other hand, there is a difference between the determinate, random, interval, multi-interval, and hyper-random models. It is associated with the different proportions of determinism and uncertainty in these models.

11.6 Classification of Mathematical Models

Particularities of the distribution function can serve as a basis for classifying mathematical models. Considering uncertainty as an alternative to determinism and taking into account the considerations outlined in the previous section, we may suggest the classification presented in Fig. 11.3.

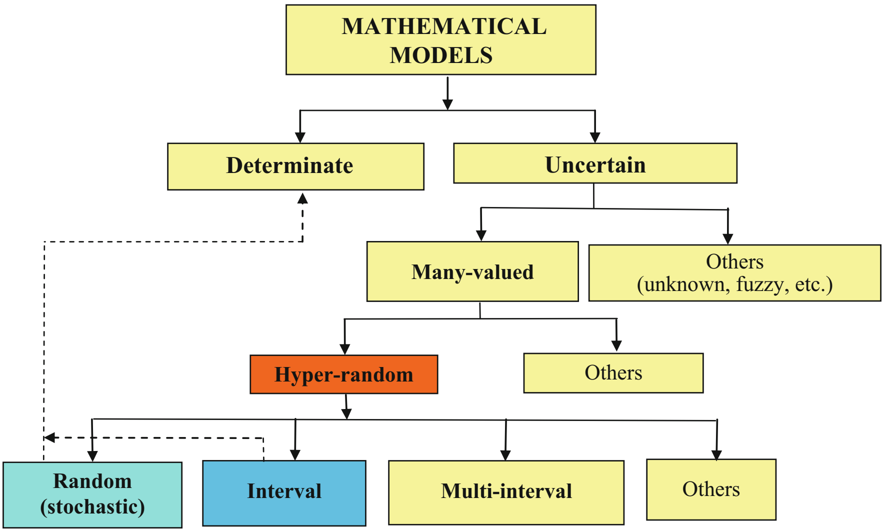

Fig. 11.3 Classification of mathematical models

The classification under many-valued models implies, in general, not only uncertainty which exists on the level of the set of single-valued realizations (as, for example, in classical random models), but also *uncertainty which exists on the level of a single many-valued realization*. In the latter case it is assumed that the realization of a physical quantity is described, not by a number, but by a set of numbers (by a many-valued variable),[2] and the realization of the physical process is described by a many-valued function.

11.7 Formation of Uncertainty

In Sect. 11.2, questions were raised about the reasons why different types of uncertainty models are used, and about their adequacy for describing real physical phenomena. In the present section, we examine these questions.

11.7.1 Formation of Uncertainty from a Sequence of Determinate Variables

Recall that, by an *adequate description*, we understand *full accordance of the model in properties of uncertainty and determinism to the actual object being modeled* (see Sect. 11.3). Note that, when studying the uncertainty properties of physical quantities, this adequacy allows one to replace the actual physical quantities by their adequate models without any loss.

We shall make the following three assumptions:

1. There are physical quantities that are adequately described by determinate values (numbers).
2. The phenomenon of statistical stability exists objectively and has a non-ideal character which is manifested through the violation of the convergence of the sample means of any physical processes.
3. Actual physical quantities are formed by taking the union (averaging) of infinitely many different physical quantities, so any physical quantity can be regarded as the limit of a sequence of sample means.[3]

[2]Strictly speaking, such an interpretation of the physical quantity contradicts the standard (GOST 1970).

[3]Exceptions are perhaps the physical constants, such as the speed of light and a small number of other physical constants that are accepted as unchanging by convention.

Note that the first assumption does not require knowledge of the determinate values (numbers). It fixes only the possibility of an adequate description of the physical quantities by certain concrete numbers. This is not as obvious as it might seem at first glance. All physics apart from quantum mechanics is based on the single-valuedness paradigm. But is the world unequivocal in reality? This is not an easy question, and we cannot answer it now. By making the first assumption, we follow the conventional line, understanding that the issue is not completely solved and requires more careful study.

The first assumption can be regarded as a *physico-mathematical hypothesis* which links the real physical world and the abstract mathematical model. The other two assumptions are *physical hypotheses* confirmed by the results of experimental investigations. Building on these hypotheses, we shall *prove the existence of physical quantities adequately described by interval and hyper-random variables*.

It follows from the hypotheses that there are actual physical quantities that are adequately described by the spectrum of limit points of a determinate sequence x_1, $x_2, \ldots, x_N$ $(N \to \infty)$. *Theoretically*, the spectrum of this sequence $\tilde{S}_x = \underset{N\to\infty}{\mathrm{LIM}}\, x_N$ can be a *determinate value* (*finite or infinite number*), a *random variable*, an *interval variable*, or a *hyper-random variable with continuous uncertainty area* (see Table 9.2). This should be considered as a list of *possible candidates*.

If the spectrum is a *finite number*, then the sequence converges. This option does not satisfy the second hypothesis, so it should be excluded from the list of candidates. Theoretically, if the spectrum is represented by an *infinite quantity*, this option does not contradict the assumptions. However, the existence of infinitely large physical quantities is very debatable, so this candidate must also be excluded from consideration.

It was shown in Sect. 1.6 that *probability is an abstract mathematical concept* that cannot, in principle, be measured with strictly zero error (due to the limited nature of the phenomenon of statistical stability). Therefore, random variables must also be excluded from consideration as not being in accord with reality and the second hypothesis.

If the spectrum is a non-degenerate hyper-random, the distribution function $\tilde{F}(x)$ is a many-valued variable (see Fig. 9.4c, d).

As a result, only an *interval variable* and a *hyper-random variable with continuous uncertainty area* remain on the list. The result presented in the present section can be interpreted as a theoretical justification for the following:

- *Order can cause disorder* (see Fig. 11.1).
- *In the real world there are physical quantities adequately described by uncertain variables, but of only two types: interval variables and hyper-random variables with continuous uncertainty areas.*
- *These physical quantities are formed from a sequence of physical quantities that are adequately described by determinate variables.*
- *Such physical quantities are formed as a result of the violation of convergence* (*the non-ideal nature of the statistical stability phenomenon*).

11.7.2 Causes of Statistical Stability Violation in the Real World

There are many factors causing violations of statistical stability in the real world. One of them is the *delivery of substance, energy, and* (*or*) *data from the outside*. Such a flow into an *open system* generates and feeds statistically unstable nonequilibrium flicker noise (see Sect. 6.2.2).

Special attention should be paid to the fact that:

Broadband statistically stable noise after linear low-pass filtering may be statistically unstable.

A typical example of such filtering is integration. By integrating the process $X(t)$, one obtains a process $Y(t)$ whose power spectral density $S_y(f)$ depends on the power spectral density $S_x(f)$ of the initial process. The relationship between them is given by the well known expression

$$S_y(f) = \frac{S_x(f)}{4\pi^2 f^2}.$$

As a result of integration, *stationary statistically stable noise* types corresponding to the range from white to pink noise become *nonstationary statistically unstable noise* types in the brown–black area of the spectrum (see Fig. 6.2).

In the process of wave propagation, the energy is damped down. The attenuation depends on the frequency. In the ocean, for example, the damping of sound energy is usually described by a law $\zeta = nf^m$ in dB/km, where $n < 1$, f is a frequency, and $m \in [1, 2]$.

It follows from this that:

As a result of natural space damping, emitted statistically stable noise types close to white noise become statistically unstable in the ocean.

11.7.3 Formation of Uncertainty in the Context of Nonlinear Transformations

Statistically unstable noise may occur as a result of various *nonlinear transformations*. In some *nonlinear transformations*, many-valuedness (which is a kind of uncertainty) arises. In this case, the uncertainty area of the distribution function $\tilde{F}(x)$ may be discontinuous.

As examples, one may consider nonlinear transformations described by the functions $x = \pm\sqrt{a}$ and

$$x = \text{Arcsin}\ a = (-1)^n \arcsin a + \pi n, \tag{11.1}$$

where $\sqrt{a}$ is the arithmetic root, a is a real number, and $n = 0\,,\ \pm 1\,,\ \pm 2\,, \ldots$. In the first case the spectrum of values is described by two conditional distribution functions in the form of unit step functions at the points $x = -\sqrt{a}$ and $x = \sqrt{a}$, and in the second, by a countable set of distribution functions in the form of unit step functions at the points described by (11.1). In both cases the *uncertainty area is discrete* and hence discontinuous.

As another example, it is well known that the process after detecting the amplitude-modulated signal has spectral components corresponding to the envelope of the radio signal in its spectrum. If the power spectral density of the envelope is described by a function of type $1/f^{\beta}$, where $\beta \geq 1$, then after suppression of the high-frequency carrier, the filtered process is statistically unstable.

11.7.4 Indicators of Statistical Stability Violations

We thus see that there are many causes of statistical stability violations. They cannot always be linked with specific features of the process. Some typical indicators of stability violations are listed in Table 11.1. The relevant sections are given in square brackets.

Table 11.1 Signs of statistical stability violations

No.	Indicator	Violation with respect to: Average	SD
1.	*Of spectrum*		
	– Fast rise in the spectral power density of the process with decreasing frequency f (Sect. 6.2.3)	+	+
	– Low-frequency narrowband process (Sect. 6.2.5)	+	+
	– High-frequency narrowband process (Sect. 6.2.5)	–	+
2.	*Of correlation*		
	– High correlation of samples of a low-frequency process (Sects. 6.2.4 and 6.2.5)	+	+
	– High correlation of samples of a high-frequency process (Sect. 6.2.5)	–	+
3.	*Of moments*		
	– Statistically unpredictable fluctuations in the process mean	+	+
	– Statistically unpredictable fluctuations in the process standard deviation	–	+
	– Absence of the mean in the distribution (Sect. 6.2.6)	+	+
	– Absence of the variance in the distribution (mean exists) (Sect. 6.2.6)	–	+
4.	*Of probability density function* (*PDF*) *and distribution function* (*DF*)		
	– Obvious location instability of the sample PDF (or DF) (Sects. 9.4 and 9.5)	+	+
	– Obvious form instability of the sample PDF (or DF) when the location is stable (Sects. 9.4 and 9.5)	–	+
	– Existence of ‘heavy tails’ (Sect. 6.2.6)	+	+
5.	*Of backbone*		
	– Formation of the process with elements of low-frequency filtration (Sects. 11.7.1 and 11.7.2)	±	±
	– Formation of the process with elements of nonlinear transformation (Sect. 11.7.3)	±	±

11.8 Using Different Types of Models

Research indicates that the *interval* and *hyper-random models* can in theory *adequately* reflect the non-determinate properties of the surrounding world. *Random models* do this only *approximately*.

> This circumstance, however, does not mean that the random and other simple models are useless.

Of course, this is not so. Incomplete accordance of these models to the modeled objects becomes apparent only for *large sample sizes*.

Sample sizes are often *small*. Then the error in the description of actual objects by random models and some other simple models is negligible. These models are easier to use than the interval or hyper-random models and are therefore *preferable* in many cases.

The need for more complex *interval* and *hyper-random models* arises when the *limited character of the phenomenon of statistical stability becomes apparent, usually for large observation intervals and large sample sizes*.

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Epilogue

Summing up our considerations of all these issues we would like to draw attention to the following key points. Statistical stability is a *physical phenomenon*, which manifests itself in the *stability of the related frequency of the actual mass events, sample means, and other statistics*. At present, there are two theories to describe this: *probability theory* and the *theory of hyper-random phenomena*. *Probability theory* is based on the *assumption that the statistical stability phenomenon is perfect*, or put another way, *statistics have the convergence property* (*estimators are consistent*). *The theory of hyper-random phenomena* is based on the *assumption that the phenomenon of statistical stability is not ideal*, or put another way, *statistics do not have the convergence property* (*estimators are inconsistent*).

Numerous experimental studies of *actual phenomena of different physical nature* indicate that the *statistics generated by real samples do not show any tendency to convergence*. A trend towards convergence is observed only when the sample volume is small. When the sample volume is large, such a trend is not observed. The results of these experimental studies corroborate the doubts expressed by many eminent scientists about the possibility of describing mass physical phenomena by stochastic (probabilistic) models (see the quotes on the introductory pages to each part of the book). Apparently, the physical world is actually subject to three *types of law*:

- *Determinate*,
- *Statistically predictable* (*random, stochastic,* or otherwise *probabilistic*), and
- *Statistically unpredictable*.

The *presence of statistically unpredictable laws explains the limited accuracy of any statistical measurement of physical quantities* and *the limited accuracy in forecasting real events*.

The recorded violation of convergence of the relative frequency of an actual event means that the basic concept of probability theory, viz., *the probability, is an abstract mathematical concept which does not have a physical interpretation*. This

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fact indicates that it may be necessary to review the postulates of a number of physical disciplines, in which the probability concept and convergence play a key role, in particular, *statistical mechanics*, *statistical physics*, and *quantum mechanics*.

It should be borne in mind that, when the *sample size is small*, statistically unpredictable laws *have practically no influence* on the measurement results of the physical quantities. *This means that one can use the classical models and statistical methods of probability theory to solve many practical tasks*.

For *large sample sizes*, when the *violation of statistical stability is evident*, the application of classical probability models leads to *unacceptably large measurement errors*. *It is then advisable to use the approach proposed by the theory of hyper-random phenomena*. Note that hyper-random models may be useful, not only in the case of large samples, but also for small samples, in particular for the simulation of various physical phenomena under conditions where the small amount of statistical material prevents one from obtaining high quality assessments of the parameters and characteristics.

Unlike random models, hyper-random models can be used theoretically to describe physical phenomena for both large and small observation intervals, and for both large and small samples. However, hyper-random models are more complicated than random ones. Therefore, all things being equal, random models should be preferred, and *hyper-random models only used when random models cannot provide an adequate description of reality*.

* * *

The material presented here provides an initial understanding of the physical phenomenon of statistical stability and methods of description. For readers wishing to obtain a deeper and systematized understanding of *mathematical probability theory*, we can recommend the books (Devor 2012; Devor and Kenneth 2012; Feller 1968; Loéve 1977; Ventsel 1963; Gnedenko 1988; Gorban 2003), while those seeking a detailed acquaintance with *experimental studies of the phenomenon of statistical stability and the methods developed from the theory of hyper-random phenomena* are referred to the monographs (Gorban 2011, 2014, 2017).

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